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SportMont

ORIGINAL SCIENTIFIC PAPER

Performance Profiling in Handball Using Discriminative Variables and its Practical Applications

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Abstract

Performance profiles of teams performance highlight areas of weaknesses and strengths for coaches to inform their decision-making on how to spend their limited training time with athletes. This study used a stepwise discriminative analysis approach comparing one successful team's (TEAM) performances through five consecutive seasons against a) other top four teams (TOP4) and b) teams with a final rank between 5th and eight (LOW) in a semi-professional league. The predictive model created was used to set forth a performance profile for the selected team. A total of 95 matches of the TEAM's matches from the last five seasons are in the analysis. The objective was to create a performance profile with relevant performance indicators selected based on the discriminant analysis results of the selected TEAM and discuss its practical applicability. For matches against other TOP4 teams, the predictive model created consisted of three variables; legal stops, blocked shots and 9 m shots, classifying 72.6% correctly. The LOW ranked teams model had six variables and correctly classified 94.4% of cases (assists, blocked shots, legal stops, the goalkeeper saved shots, 2-minute exclusion, and shot efficiency). The selected variables are presented in Table 4, with medians and a 95% confidence interval of the median as a team performance profile. The profile provides the coaches with two models containing values that can serve as a reference for this team's performance. The profile of this TEAM's performances during the last five seasons generally aligns with the variables associated with success in other studies in female handball.

Keywords: *team performance profile, performance indicators, discriminative analysis, female handball*

Introduction

Team handball (hereafter handball), like other team sports, represents a complex adaptive system of two seven-a-side teams with clear intentions, competing under official rules which act as informational task constraints upon the players' behaviour, limiting the possible movement solutions (Button et al., 2021). The game is physically demanding, intermittently fast, with frequent transitions, duels, and scoring (Kniubaite et al., 2019) for 60 minutes of playtime split into two periods. Changes to the official rules by the game's governing body have facilitated development in this direction. As an example, the need to always identify a goalkeeper specifically was removed in 2016, paving the way for faster goalkeeper substitutions for an extra offensive player. Further adjustments to the restart after conceding a goal have also been made to make the game even faster (IX. Rules of the Game: Indoor Handball, 2022). Within each team, the sport places position-specific physical demands on players (Luteberget & Spencer, 2017; Foretic et al., 2022) as their roles and places on the court differ. A recent systematic review on physical demands in handball reports that wings and backs cover more distance than piv-

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ots. On the other hand, pivots perform the most body contacts during a match. Wing players run more fast breaks, while backs perform more throws than wings and pivots (García-Sánchez et al., 2023). Handball play differs between sexes, as female players cover more distance during matches at a higher relative workload than males, who perform more high-intensity strength-related actions (Michalsik & Aagaard, 2014).

In such a fast and dynamic game, it is valuable to know which variables contribute to success more than others to direct efforts in training. Discriminative analysis of game-related statistics is a standard method in sports performance analysis that allows this discrimination. The method constructs a model able to classify performances based on as few variables as possible and a high percentage of correct classifications. Discriminant function analysis allows the creation of a prediction model of group membership based on a collected data set for two or more groups (O'Donoghue, 2009). Regarding female handball, the method has primarily been used to discriminate between winners and losers in semi-professional (Þorgeirsson et al., 2022a) and international level (de Paula et al., 2020) handball. Other uses include comparisons of handball player's body composition to those of athletes from other ball sports (Mala et al., 2015) and for handball talent identification for example (Fernández-Romero et al., 2016; Naisidou et al., 2017). Being a starter or non-starter is another marker of success where discriminatory analysis was used at the youth club level, analysing anthropometry, physical performance measurements and handball skills tests (Saavedra et al., 2020). The primary purpose of performance analysis is to enhance performance (O'Donoghue, 2009), but engaging in the practice can help coaches to enhance their coaching practices (Martin et al., 2018). It can prove challenging to align research aims with the aims of coaches primarily concerned with their team performances. Performance profiles can be used to present typical characteristics of performers through selected performance indicators associated with success. Regularly, papers are published presenting different profiles of handball players describing anthropometry and position-specific demands (Vila et al., 2012), performance levels (Moss et al., 2015) and technical activity (Michalsik et al., 2015).

Previous work on females has attempted to analyse the discriminative variables between winning and losing at the elite level (Saavedra et al., 2018) and the semi-professional level (Þorgeirsson et al., 2022a, 2022b). One cross-sectional research investigated female handball on the domestic semi-professional level. It highlighted goalkeeper saved shot efficiency and total shot efficiency as discriminating between winning and losing matches, classifying 87% correctly. To our knowledge, no performance profiles for a handball team have been published using game-related statistics. Performance profiles can be an accessible information template from performance analysts to the coaches working with players. There are several known barriers for coaches to engage in performance analysis; time and cost are primary, as well as technical know-how and the integration into practice itself (Martin et al., 2018). Therefore, a model containing the most relevant performance indicators with numbers that can be used as reference values might prove valuable. Until recently, most have focused on the elite level and compared several major championships (Saavedra et al., 2018; de Paula et al., 2020). Therefore, the objective is to create a performance profile of one selected team's performances (TEAM) based on performance indicators from a discriminative analysis. Specifically, by analysing the performance of one successful female team using discriminative analysis of the game-related statistics from five successive seasons (2018/19 - 2022/23).

Methods

Participants

The team selected (TEAM) for this analysis represents one of the top four (TOP4) teams in the final ranking in all five seasons (2018-2023) in the Icelandic semi-professional league. During this period, the team was under the charge of the same coach. All 95 matches played by the TEAM in the last five seasons of the top semi-professional league in Iceland were included in the analysis (eight-team top league with one lower league). The data set comprises 22 variables describing the efficiency of shots; goalkeeper saves, and the frequency of offensive and defensive-related events. Game-related statistics were obtained from the statistics company HBStatz, the official statistical partner of the Icelandic Handball Federation. The data used is all publicly available online https://hbstatz.is/OlisDeildKvennaLeikir.php. The TEAM had decent success in these five seasons, winning the national championship (2019 and 2023) and the cup competition (2019, 2022), in addition to winning the league twice (2018-2019 and 2022-2023). A more detailed description of the team's performances against other TOP4 and LOW-ranked teams can be viewed in Table 2.

Procedures

During the handball matches, the data was entered by a trained person into an online web-based platform tailored explicitly for handball. The data is then exported into an Excel spreadsheet for computation of efficiency variables by one of the authors (Ó.S.) and subject to a random check by one of the authors (S.Þ.). The information in this research is available online in the public domain; therefore, no informed consent was necessary. The dependent variable in the study was the selected female team compared to a) the other three teams in the top four final league rank and b) LOW ranked teams (5th position and lower). The independent variables was the game-related statistics listed in Table 1. This method has been used with similar studies in the Icelandic handball league (Þorgeirsson et al., 2022a, 2022b).

Statistical analysis

This is structure-oriented comparative research of handball game-related statistics (Pfeiffer & Perl, 2006) using discriminatory analysis to identify variables discriminating between the selected team's performance and different quality opponents. A summary of the team's league performances regarding points earned, goals scored and conceded against TOP4 and LOW ranked teams are in Table 2. Discriminant analysis was performed using the sample-splitting method depending on the final league rank, comparing the TEAM to other TOP4-ranked teams and then separately against other LOW ranked teams (5th to eighth). Wilks's lambda (λ) was used to measure the deviations within each group to the total deviations. The sample included variables that minimized the value of λ, assuming the value of F was great-

Table 1 Definitions of the Game-Related Statistics

Variable	Definition
Shots	Percentage of converted shots relative to the number of shots made.
6 m shots	Percentage of converted shots at 6 m relative to the number of shots made. The shot is from a zone outside the 45° angle from the left and the right.
7 m shots	Percentage of penalties (7 m) converted relative to the number of penalties taken.
9 m shots	Percentage of converted shots at 9 m relative to the number of shots made. The shot is from a backcourt player either (a) over or through the defence, or (b) after a breakthrough but with a defence player in front.
Wing shots	Percentage of converted shots from the wing area relative to the number of shots made. The shot is from a zone within the 45° angle from the left and the right without a defence player in front.
Fast-break shots	Percentage of shots converted in a fast-break situation (rapid switch from defence to attack without the defence organized) relative to the number of shots made in this situation.
Breakthrough shots	Percentage of shots converted in a breakthrough situation relative to the number of shots made in this situation (a) from a backcourt player after breakthrough in the 9 m zone without a defence player in front, (b) from the pivot after a 1:1 situation, (c) from the left or right back after a breakthrough of a 1:1 situation.
Yellow cards	Yellow cards received by each player and/or coaching staff member.
Red cards	Red cards received by each player and/or coaching staff member.
2-min exclusions	Number of 2-minute suspension received by players and/or coaching staff.
Assists	Number of passes from one offensive player to another leading directly to a goal scored.
Technical fouls	Number of turnovers made by the offensive team where the ball is awarded to the defence due to a foul in offence.
Blocked shots	Number of blocked shots by a defender
Legal stops	Number of fouls committed by a team without getting penalized with a 2-minute exclusion or a red card
Steals	Number of turnovers in favour of the defence due to actions of anticipation and snatching the ball.
GK shots	Percentage of shots stopped relative to the number of shots made by the attackers.
GK 6 m shots	Percentage of 6 m shots stopped relative to the number of shots made by the attackers.
GK 7 m shots	Percentage of penalties (7 m) stopped relative to the number of penalties taken by the attackers.
GK 9 m shots	Percentage of 9 m shots stopped relative to the number of shots made by the attackers.
GK wing shots	Percentage of shots stopped in the wing area relative to the number of shots made by the attackers.
GK breakthrough shots	Percentage of shots stopped in breakthrough situations relative to the number of shots made by the attackers.

er than F=3.84 for inclusion. The next step was to combine the variables pairwise. A new variable was added if λ was greater than F. Wilks lambda, canonical correlation, and the percentage of correctly classified matches (the team against oppositions, either TOP4 or LOW ranked teams) was calculated. Table 4 presents the team's performance profile (James et al., 2005). An alpha level threshold of .05 was set for statistical significance. The statistical analysis was performed with the software package IBM SPSS Statistics (Version 27.0; IBM, Armonk, NY, USA).

Results

Table 2 presents the teams league performances with the number of matches, goals scored and conceded on average

Table 2. The Teams League Performance Statistics in Total, Against TOP4 Teams and Lower Ranked Teams

Statistics	Opposition	Season				
		2018-2019	2019-2020	2020-2021	2021-2022	2022-2023
	All					
Number of league matches		21	18	14	21	21
Final league position		1st	2nd	3rd	2nd	2nd
Points earned on average (%)		81	80.5	60.5	71.5	85.5
	TOP4					
Number of league matches		9	8	6	9	9
Points earned on average (%)		61	50	25	55.5	66.5
Goals scored per match (M)		25.22	22.33	22.33	25.67	26.56
Goals conceded per match (M)		21.11	19.3	28.4	25.11	16.21

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Table 2. The Teams League Performance Statistics in Total, Against TOP4 Teams and Lower Ranked Teams

Statistics	Opposition	Season				
		2018-2019	2019-2020	2020-2021	2021-2022	2022-2023
	LOW					
Number of league matches		12	10	8	12	12
Points earned on average (%)		96	100	87.5	91.5	100
Goals scored per match (M)		26.92	29.6	29.63	27.75	31.17
Goals conceded per match (M)		18.83	18.5	20.88	21.17	22.08

Note. M = mean, Seasons 2019-2020 and 2020-2021 fewer matches were played because of COVID19 restrictions

and points earned (%) provided for each of the five seasons, in total and against the TOP4 teams, and LOW ranked teams separately.

Table 3 presents the discriminatory analysis comparing the team selected against its opponents according to the final league rank of TOP4 or LOW ranked teams. It includes Wilks's lambda, the canonical correlation and the predictive model's percentage of correctly classified cases.

Table 3 Discriminative Analysis of the TEAMs Performance against TOP4 and LOW ranked teams separately.

Model´s statistics	Period 2018-2023	
	Against TOP4 (n = 76)	Against LOW (n = 94)
Correctly classified (%)	70.2	94.4
Wilks lambda	0.702	0.250
Canonical correlation	0.546	0.866
Selected variables	Legal stops	Assists
in order of importance	Blocked shots	Blocked shots
	9 m shots	Legal stops
		GK saved shots
		2-minute exclusion Shots

Note. Seasons 2019-2020 and 2020-2021 fewer matches were played because of COVID19 restrictions; a GK = Goalkeeper

Against other TOP4 teams, the models correctly classified 70.2% of TOP4 teams with three selected variables and 94.4% against the LOW-ranked team's performances using six variables. The TOP4 model included legal stops, blocked shots and 9 m shot efficiency. The model for performances against LOW ranked teams contained assists, blocked shots, legal stops, GK saved shots, 2-minute exclusions and total shot efficiency.

Table 4. The performance profile of the TEAM according to the opponent's quality based on league performances from the last five seasons.

Predictive models	Team		TOP4		LOW	
	Median	CI	Median	CI	Median	CI
TOP4						
Legal stops (#)	19	17-22	13	11-16		
Blocked shots (#)	4	3-6	2	1-3		
9 m shots (%)	42	37-50	36	28-38		
LOWER						
Assists (#)	12	11-13			5	4-6
Blocked shots (#)	4	3-5			1	1
Legal stops (#)	23	21-24			14	13-15
GK saved shots. (%)	40	39-44			27	24-30
2 min exclusion (#)	1	1-2			2	2-3
Shots (%)	61	57-63			43	39-46

Note. # = frequency of events, CI = 95% Confidence interval of the median

Table 4 presents the performance profile of the selected team using the selected variables as performance indicators presenting the median and the 95% confidence interval of the median.

Discussion

This research explored the discriminant variables for one top four TEAM against two groups of opponents according to final league rank in a semi-professional league. The objective was to create a performance profile of a female handball team and discuss how the information obtained with this method could inform the practices of handball coaches in semi-professional leagues. These results highlight how different sets of variables emerge as discriminant depending on the quality of the opponents. Therefore, presenting the medians and 95% confidence intervals of the medians of the relevant performance indicators holds the potential to be a valuable reference for coaches in match preparation, during and post-match analysis.

Offensively, only 9 m shot efficiency was included in the model against the TOP4 teams. The 9 m shot had already been highlighted in previous work on the 2012 Olympics (Milanović et al., 2018) and again during the final minutes in balanced matches at this level for males and females as characteristic of successful performances (Þorgeirsson et al., 2022b). The selected TEAM has this characteristic in common with other successful teams. Total shot efficiency, which had been identified as discriminant in previous work on semi-professional league (Þorgeirsson et al., 2022a) and also international level (Saavedra et al., 2018), was only found to be discriminant in the LOW model. Furthermore, in the LOW model, assists also emerged as discriminant, in accordance with a study on the elite level for unbalanced and very unbalanced matches (de Paula et al., 2020). Perhaps this tendency of successful teams creating more assists has more to do with matches being unbalanced rather than the performance level, even though the exact definition of what counts as an assist in handball can be seen as vague.

Defensively, the first two variables named in the TOP4 model are blocked shots and legal stops. While legal stops physically stop the offense and interrupt the offensive build-up, blocked shots may lead to a quick turnover creating chances for a fast break. So does stealing the ball, which was not found to be discriminating for the TEAM, even though it is important in recent research for balanced matches on the female elite level, just as the blocked shots (de Paula et al., 2020). Interestingly, legal stops were not a deciding factor in this female league, even though it was for the male league (Laxdal & Ivarsson, 2023). Actively seeking legal stops in the female league may not yield better results, but this TEAM's defense style can be considered an aggressive characteristic. However, this aggressive but still legal style (only 1-2 2-minute exclusions per match) may have the desired effect on the opponents. However, that is very difficult or even impossible to observe using statistics. In addition to blocked shots and legal stops found in the TOP4, the LOW model also identified the goalkeeper's total shot saved efficiency and 2-minute exclusions were included. The goalkeeper's efficiency was expected as it aligns with a previous study comparing male and female handball across a single season(Þorgeirsson et al., 2022a). Finally, five out of nine variables in the two models are defense-related signals key TEAM characteristics (blocked shots and legal stops in both models).

This study's limitations include i) the outcome-based nature of this data does not provide insight into the process and action leading to the events. ii) the results represent only one TEAM's performance, and their profiles may vary. iii) The variables used for shots and goalkeeper saves are entered as efficiency values; therefore, the weighted importance of each cannot be assumed from this analysis. iv) The models produced discriminate only between the team's performance and the opponents without regard to the outcome of the games. v) Goalkeepers fastbreak shot save could not be included in the analysis because of too many missing data points.

Potential directions for research in this field are to get qualitative feedback from coaches on performance profiles like the one produced here to improve its structure. Furthermore, to perform an advanced analysis of the processes behind statistics using dynamic process-oriented data. It could also be interesting to explore if key actions for defense could be presented in a performance profile for different teams to compare.

Practical implications

The selected variables used as performance indicators for the TEAM are actions performed under match conditions (constraints). In preparation for matches, the different profiles of the opponents of different qualities can shape the tactical preparation for each match. To effectively train these actions, creating a representative training environment is important. We, therefore, recommend that coaches manipulate task constraints for the desired actions. This could include constraining the starting players (more quality) during training against their teammates (less quality) in preparation for matches. These constraints could target the player's defensive actions specifically and force the players to explore different solutions and find new successful ways to block shots and perform legal fouls. Since stealing the ball has been related to success in female handball, the coach could create specific incentives in training to steal the ball to add that variable to their defensive profile. This might direct coaches to play more full court (as opposed to half) in match preparation to enable representative decision-making by players. This approach could prepare the team for competition against a quality opponent by developing problem-solving abilities in a representative context. Changing the game's rules (task constraint) even a little bit can significantly impact the player's emergent movement and team synergies (Araújo et al., 2022). These recommendations align with those made by authors of similar research on elite-level female handball, emphasising representative task design for learning (de Paula et al., 2020). In matches, the performance profile can serve as reference values for the TEAM's performance and post-match evaluation.

Conclusions

Performance profiles can provide a statistic-based knowledge of the TEAM's performances. Should coaches decide to engage, they can use the model to reflect and inform their coaching practice and philosophy. The defensive profile of the team analysed was a key characteristic. In match preparation, targeting the preferred actions specifically using task constraints in representative handball training (e.g., rules, space, and time constraints) is recommended.

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Conflict of Interest
The authors declare that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Effects of Acute Caffeine Administration on Pituitary–Testicular Hormonal Responses and Muscle Damage Biomarkers following Muscular Endurance in Well Resistance–Trained Males

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Abstract

The purpose of this study was to investigate the effects of acute caffeine administration on pituitary-testicular hormonal (prolactin (PRL), follicle-stimulating hormone (FSH), luteinizing hormone (LH), and total testosterone (TT)) responses and muscle damage biomarkers (creatine kinase (CK), creatine kinase-MB, myoglobin, and troponin) following muscular endurance test. In addition, total repetitions to failure were determined. Eleven well resistance-trained males (29.11±3.21 years) performed two consecutive trials (7 days apart). Using a double-blind, placebo-controlled, randomized crossover design, participants administered either 5 mg/kg of caffeine or a placebo 1 hour prior to the test. The test consisted of 3 sets × repetitions to failure at 70% of 1-RM, with 90 s recovery interval, in each bench press, biceps curl, shoulder press, leg press, back squat, and leg extension. Repetitions were counted in each set in each exercise. Blood samples were collected 1 h after the end of each trial from each participant to measure the aforementioned study parameters. Results indicated that caffeine had positively effect on FSH (p=.046) and TT (p=.049) but had negatively effect on CK (p=.012) compared to the placebo. However, no significant (p>0.05) differences were found in PRL, LH, CK-MB, myoglobin, and troponin between trials. Total repetitions to failure were significantly (p=.013) greater in caffeine trial (334.38±9.70) than in the placebo (309.07±9.43). In conclusion: 5 mg/kg of caffeine administered 1 h prior to muscular endurance test performed by multiple resistance exercises had beneficial effect on FSH and TT probably due to affecting androgen receptors. However, the values of biomarkers of muscle damage were similar in both trials. Hence, male strength athletes may consider using this dose pre-training as an effective ergogenic aid during muscle endurance training.

Keywords: *adenosine receptors, androgen receptors, catecholamines, hypothalamus, prolactin*

Introduction

Resistance exercise enhances muscle function (Heavens et al., 2014) by increasing myofibrillar recruitment (Fragala et al., 2011; Machado et al., 2012) as a result of mechanical stress, metabolic demands, and endocrine activity (Heavens et al., 2014; Raastad et al., 2000). Mechanical stress represents a primary stimulus for hormonal response and, in turn, acclimatization to resistance training (Glover et al., 2022). Contraction cycle requires accelerated physiological demands during resistance exercise to restore protein breakdown. Of relevance, hormonal activity plays a crucial role in gene expression (Machado et al., 2012; Heavens et al., 2014) and remodeling process of muscle proteins (Raastad et al., 2000; Wu & Lin, 2010).

Testosterone, a potent antagonistic catabolic hormone, is secreted from testes by two main hormones include follicle-stimulating hormone (FSH) that stimulates spermatogen-

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esis and luteinizing hormone (LH) which synthesizes testosterone (Willoughby et al., 2014; Glover et al., 2022). These hormones are secreted from pituitary and regulated by hypothalamus through release of gonadotropin-releasing hormone (GnRH) (Cadegiani et al., 2019), making hypothalamic-pituitary-gonadal (HPG) axis is a pivotal factor in the process of protein synthesis. Increased blood testosterone level strengthens muscular strength (Spiering et al., 2009; Stokes et al., 2013; Willoughby et al., 2014). However, GnRH elicits excretion of prolactin (PRL) (Willoughby et al., 2014), and high abnormal blood PRL level can affect androgen receptors (Spiering et al., 2009), contributing to protein degradation. All of these hormones maintain homeostasis of muscle function in resistance training, especially when myofibrils predispose to damage resulted from repetitive contractions (Heavens et al., 2014).

Repetitive eccentric and concentric contractions until momentary muscle failure (muscular endurance) can induce muscle damage. Its signs include sarcolemmas tears, myofibrillars disruption (Heavens et a., 2014), Z-discs mutilation, and increased permeability of muscle proteins into blood (Owens et al., 2019), such as creatine kinase (CK), myoglobin, and troponin (Machado et al., 2012). In this context, Heavens et al. (2014) found greater prolonged muscle damage markers included CK and myoglobin following descending pyramid (starting at 10 repetitions and descending 1 repetition as fast as possible) for squat, deadlift, and bench press, but in trained men compared in women. On the other hand, short period of recovery between sets can induce muscle damage and change hormonal response (Heavens et al., 2014; Smilios et al., 2013). For instance, high volume (3-5 sets × more than 10 repetitions) with short recovery (60-90 s) elevates muscle damage biomarkers and blood testosterone level (Smilios et al., 2013) probably to restore protein degradation. Further, submaximal intensity has more efficacy at triggering hormonal response (McCaulley et al., 2009). In this sense, Raastad et al. (2000) revealed that acute response of testosterone was higher in submaximal (70% of one repetition maximum (1-RM)) than in maximal (100% of 1-RM) intensity after resistance exercises. They also found no statistical differences in FSH and LH between both intensities. Stokes et al. (2013) demonstrated significant increased blood testosterone, but without response on PRL, between pre- and post-resistance exercises. These prior studies, however, have not used any ergogenic intervention, such as caffeine.

Caffeine is ingested before resistance exercise session to increase athletic performance and enhance more repetitions to failure (Grgic & Mikulic, 2021; Ruiz-Fernández et al., 2023) by reducing pain sensation and rating of perceived exertion (RPE) (Wu & Lin, 2010; Polito et al., 2019; Ferreira et al., 2021). Some prior studies (Green et al., 2007; Polito et al., 2019) have found significantly increased the number of repetitions to failure following caffeine ingestion compared to placebo. Conversely, other studies (Astorino et al., 2008; Duncan et al., 2013; Grgic & Mikulic, 2021) have failed to find the same caffeine's effects. Further, Ferreira et al. (2020) reported by review a contradiction in caffeine's effects regarding muscular endurance, especially lower extremities. A common alert on resistance exercise is that "Further research is merited on the topic of caffeine and muscular endurance". Most of these studies investigated a repetition to failure tests have consisted of 2-4 resistance exercises. To the best of knowledge, however, only one study conducted by Davis et al. (2012) has been investigated a muscular endurance with high volume (4 sets per exercise) of six exercises, but in healthy men. Surprisingly, they reported a significantly higher repetition to failure in only one (bench press) exercise.

On the other hand, caffeine plays an important role in hormonal changes, such as increased testosterone levels (Glover et al., 2022). In a study conducted on resistance-trained men who performed 8 exercises (3 sets of 10 repetitions at 75% of 1-RM), results showed no statistical difference in testosterone between caffeine and control (Wu & Lin, 2010). For muscle damage, a few studies (Machado et al., 2008; Soleimani et al., 2018) have demonstrated that caffeine had no effect on signs of muscle damage after resistance session. However, other studies reported that caffeine decreased oxidative stress but in brain rat (Vieira et al., 2017), and attenuated delayed-onset muscle soreness (DOMS) (Hurley et al., 2013). Due to the paucity of data in the literature regarding caffeine's effects on combination of hormonal response and muscle damage biomarkers following muscular endurance, a practical applicability needs further research.

Consequently, the primary purpose of the current study was, for the first time, to investigate the acute effects of caffeine administration on pituitary-testicular hormonal (PRL, FSH, LH, and TT) responses and muscle damage biomarkers (CK, CK-MB, myoglobin, and troponin) following muscular endurance test performed by multiple exercises in well resistance-trained males. The second aim was to determine the sum of repetitions to failure in each exercise and the total repetitions after caffeine and placebo administration. The hypothesis of the study was that caffeine may enhance performance by attenuation of pain sensation during performance.

Methods

Participants

Eleven well resistance-trained males (age: 29.11±3.21 years, height: 180.64±4.41 cm, body mass: 92.55±3.80 kg, BMI: 28.18±2.76 kg/m2, training experience: 7.82±1.25 years) volunteered to participate in the study. The inclusion criteria were considered as follows: regular resistance training (5 times per week) for at least 5 years prior to the study; familiar to perform successful all exercises used in the study; low habitual coffee (1-2 cup per day, roughly ≤100 mg per day); and aged between 25-35 years. Participants who had consumed anabolic steroids (one month prior to the study), ergogenic aids (one week prior to the study), and smokers were not allowed to engage. Each participant was informed about the protocol of the study as well as a possible problem resulting from ingestion of caffeine such as numbness, and thus, provided written informed consent to participate. The study was approved by the Al-Ahliyya Amman University Ethics Committee (Code: FES-18G-280-2023). All the study procedures were in accordance with the latest version of the Declaration of Helsinki.

Experimental design

Intervention order (caffeine or placebo) was randomly assigned to participate. The study used a double-blind crossover design, which neither examiners nor participants alert of intervention order. All participants were instructed not to engage strenuous exercise 48 hours prior to trials, maintain their dietary pattern during the days between trials, not to consume coffee after 8:00 PM the night of the trial, and not to ingest breakfast meal on the day of the trial. Roughly 500 ml of water were drunk by participants during each trial to avoid throat dehydration, which may affect the performance. The study trials were performed in at 20-22°C and 46-49% relative humidity.

Experimental protocol

Data collection occurred over 4 days with different interval periods (started on 2/7/2023). On the first day, participants were asked to measure their demographic characteristics and subjected to a 1-RM test for bench press, biceps curl, shoulder press, leg press, back squat, and leg extension. After 5 days, blood samples were collected from each participant as baseline measurements (Table 1). After 3 days, participants were randomly assigned to either caffeine intake (CAF trial) or placebo (7 days apart). The caffeine and placebo were administered 1 h prior to muscular endurance test (Polito et al., 2019). During this time, participants were seated in the Gym room (talking or reviewing social media by mobile). After that, participants performed a prior warm-up (10 repetitions at 50% of own 1-RM) in each exercise. Then, participants were ordered to perform the test. The muscular endurance test consisted of 3 sets × repetitions to failure in bench press, biceps curl, shoulder press, leg press, back squat, and leg extension at 70% of 1-RM, with a short rest interval (90 s) and 1 s cadence of each concentric and eccentric contraction (Polito et al., 2019). Momentary muscle failure was considered when the participant could not preserve the success extension and flexion. All participants were given verbal encouragement throughout the test to complete as many repetitions as possible. At the end of each trial, blood samples were collected from each participant to measure the study parameters.

Table 1. Baseline measurements (Mean ± SD).

	Baseline values	**Reference range**
PRL (ng/ml)	11.94±1.81	4.04–15.2
FSH (IU/L)	7.05±2.82	1.5–12.4
LH (IU/L)	5.36±2.67	1.7–8.6
TT (ng/ml)	6.51±3.56	2.49–8.82
CK (U/L)	166.82±4.53	38–197
CK-MB (U/L)	14.95±2.24	Up to 25.0
Myoglobin (μg/L)	51.88±3.82	<90
Troponin (ng/ml)	0.10±0.11	0.001–0.16

PRL: Prolactin; FSH: Follicle-stimulating hormone; LH: Luteinizing hormone; TT: Total testosterone; CK: Creatine kinase; CK-MB: Creatine kinase-MB

One repetition maximum

At the beginning, participants subjected a warm-up with a single set (10 repetitions) in aforementioned exercises at 50% of estimated 1-RM, with 30 s rest interval. Subsequently, a 1-RM test for each exercise was started. Determined 1-RM ensued according to the methods of Willoughby et al. (2014). 1-RM constituted the maximum weight lifted once with proper technique and success repetition. Pilot test showed no difference in 1-RM bench press (t=1.71, p=.211), biceps curl (t=0.44, p=.610), shoulder press (t=0.36, p=.492), leg press (t=-0.46, p=.693), back squat (t=0.43, p=.236), and leg extension (t=0.76, p=0.812).

Intervention administration

5 mg/kg of caffeine (Florida Supplement Caffeine Capsule, Nutrix Research, USA) that swallowed by 250 ml of water (18°C) or placebo (dextrose) was provided to participants in an identical capsule 1 h prior to muscular endurance test. These capsules were coded and prepared by a nutritionist who was not involved in the study. The selected caffeine dose was chosen due to its role in elevation of plasma caffeine concentration (Graham, 2001). One week later, participants repeated the same procedure with other intervention.

Blood samples

Blood samples (6 ml) were collected from each participant 1 hour following the completion of each trial to measure the study parameters (pituitary-testicular hormones and muscle damage biomarkers). Serum samples were used for all parameters analysis and centrifuged at 3500-4000 rpm for 10 min. PRL, FSH, LH, were analyzed by (Cobas6000-REI, Switzerland), CK and TT were analyzed using (Snibe- immunoassay, Shenzhen, China), CK-MB was analyzes using (S1/CobasC311-REI, Switzerland), and the analysis of myoglobin and troponin were done by (Immunolyte, 210, USA). The reference ranges of parameters were found in Table 1.

Statistical analysis

The Shapiro-Wilk test was applied to verify the distribution of the data. All the analysis variables were normally distributed ($p>0.05$). Two-way ANOVA (CAF/PLA × sets) with repeated measurements was used to analyze differences between the sets in each exercise within-group and between-group. One-way ANOVA with repeated measurements was used to test differences between the sum of repetitions within-group and between-group. Bonferroni post-hoc was used to determine a significant difference. A Paired sample t test was used to test between-trials differences in the total number of repetitions to failure and to test between-trials differences in the study parameters. SPSS version 23.0 (Statesoft, Tulsa, USA) was used for all analyses. Descriptive statistics are presented as Mean±Standard deviation. The level of significance was set at $p\leq0.05$.

Results

In regard to study parameters, results indicated no significant differences in PRL (t=-1.164, p=.271), LH (t=-0.630, p=.543), CK-MB (t=1.849, p=.094), myoglobin (t=0.681, p=.511), and troponin (t=1.796, p=.103) between trials. However, caffeine had a positive effect in FSH (t=2.283, p=.046) and TT (t=2.236, p=.049), and a negative effect in CK (t=3.071, p=.012) compared to the placebo (Table 2).

Table 2. Pituitary-testicular hormonal responses and muscle damage biomarkers in both trials. Data were analyzed using paired sample t test.

	Caffeine	Placebo
PRL (ng/ml)	4.77±2.55	4.91±2.46
FSH (IU/L)	6.63±3.06 *	6.49±2.18
LH (IU/L)	15.09±2.76	14.66±3.74
TT (ng/ml)	7.88±2.06 *	7.21±2.90
CK (U/L)	280.09±5.97	271.82±6.27 *
CK-MB (U/L)	20.23±1.53	19.91±1.43
Myoglobin (μg/L)	78.77±3.72	76.64±4.48
Troponin (ng/ml)	0.11±0.01	0.12±0.02

Note. Significance level was set at p<0.05. Values expressed as Mean±SD. *: Significant difference between caffeine and placebo. PRL: Prolactin; FSH: Follicle-stimulating hormone; LH: Luteinizing hormone; TT: Total testosterone; CK: Creatine kinase; CK-MB: Creatine kinase-MB

For muscular endurance, Table 3 illustrates the repetitions in each of the sets performed in each exercise in each trial. There were no statistical differences in the repetitions to failure of each set between caffeine and placebo ($p>0.05$). Within-group, however, some different repetitions throughout the sets in all exercises was occurred. In this regard, there was a significant difference in all exercises between the 1st and 2nd sets ($p<0.05$), 1st and 3rd sets ($p<0.05$), and between 2nd and 3rd sets ($p<0.05$). Notably, in caffeine trial, no significant difference was found between 2nd and 3rd sets in only back squat. ($p=.302$).

Table 3. The number of repetitions in each set of resistance exercises in both trials. Data were analyzed using two-way ANOVA

	Bench press			Biceps curl			Shoulder press		
	Set 1	Set 2	Set 3	Set 1	Set 2	Set 3	Set 1	Set 2	Set 3
Caffeine	27.36 ±1.91 2,3	22.55 ±1.51 3	17.45 ±2.97	22.73 ±3.85 2,3	19.27 ±2.00 3	15.55 ±1.97	22.09 ±2.21 2,3	18.09 ±3.76 3	13.64 ±2.69
Placebo	26.00 ±2.00 2,3	21.27 ±1.56 3	16.73 ±2.33	21.45 ±2.02 2,3	18.55 ±2.42 3	14.82 ±1.99	20.36 ±4.25 2,3	16.82 ±2.04 3	13.36 ±3.12
	Leg press			**Back squat**			**Leg extension**		
	Set 1	**Set 2**	**Set 3**	**Set 1**	**Set 2**	**Set 3**	**Set 1**	**Set 2**	**Set 3**
Caffeine	23.91 ±2.07 2,3	20.45 ±2.50 3	16.55 ±3.01	17.64 ±5.61 2,3	15.73 ±3.35	15.73 ±3.35	20.00 ±2.10 2,3	15.00 ±4.73 3	11.45 ±1.04
Placebo	21.36 ±2.20 2,3	19.09 ±2.34 3	15.27 ±2.45	17.91 ±4.58 2,3	14.73 ±1.95 3	10.18 ±3.47	17.91 ±1.97 2,3	13.91 ±5.70 3	9.36 ±1.03

Note. Significance level was set at p<0.05. Values expressed as Mean SD. 2: significant different to 2nd set; 3: significant different to 3rd set.

Table 4 shows the sum of repetitions of the sets in each exercise and the total repetitions to failure in each trial. Within-group, Bonferroni post hoc test revealed significantly ($p<0.05$) differences between bench press and the other exercises after both caffeine and placebo administration. The test also showed that leg press had significantly ($p<0.05$) greater repetitions than in shoulder press, back squat, and leg extension in both trials. In addition, the sum of repetitions in upper body exercises were significantly ($p<0.05$) more than in lower body exercises in caffeine and placebo trials. Between-group, results revealed no significant ($p>0.05$) differences in bench press, biceps curl, and shoulder press, but showed significantly ($p<0.05$) caffeine's effect in leg press, back squat, and leg extension compared to the placebo. In addition, caffeine had significantly ($p<0.05$) higher sum of repetitions in upper and lower body exercises than in the placebo. Last, total repetitions to failure were significantly ($t=9.41$, $p=.013$) higher after caffeine administration compared to the placebo.

Table 4. Sum of repetitions in each exercise and total repetitions to failure in both trials. Data were analyzed using one-way ANOVA for sum of repetitions between- and within-group and paired sample t test for total repetitions

	Bench press	Biceps curl	Shoulder press	Leg press	Back squat	Leg extension	Total
Caffeine	67.36±4.47 a	57.55±3.52 c	53.82±3.96	60.91±3.93 b	48.29±4.19	46.45±3.91	334.38±9.70 *
	Sum of repetitions (178.73±6.73) @, *			Sum of repetitions (155.65±5.98) *			
Placebo	64.00±4.30 a	54.82±3.46 c	50.54±4.42	55.72±3.41 b	42.82 ± 3.61	41.17±5.88	309.07±9.43
	Sum of repetitions (169.36±7.11) @			Sum of repetitions (139.71±6.44)			

Note. Significance level was set at p<0.05. Values expressed as Mean±SD. a: significant different to other 5 exercises; b: significant different to shoulder press, back squat, and leg extension; c: significant different to back squat and leg extension. @: significant difference between upper and lower body exercises. *: significant difference between caffeine and placebo.

Discussion

The present study investigated the acute effects of caffeine on pituitary-testicular hormonal responses and muscle damage biomarkers following muscular endurance test performed by multiple resistance exercises. The total repetitions to failure was also determined compared to the placebo. The highlight findings of the current study were as follow: 1) acute caffeine (5 mg/kg) administration 1 h prior to muscular endurance test increased positively concentration of FSH and TT in well resistance-trained males; 2) muscle damage biomarkers were similar in caffeine and placebo; 3) the total repetitions to failure were more after caffeine administration compared to the placebo.

In regard to hormonal response, there is a consensus that resistance exercise could increase testosterone concentration (Spiering et al., 2009; Smilios et al., 2013; Stokes et al., 2013). This increment depends on mechanical stress associated with heavy loading workout and adaptation to resistance training (Kraemer & Ratamess, 2005; Heavens et al., 2014). The participants in the present study have experience of a minimum 7 years, and the experimental protocol consisted of high intensity workout, which explain the elevation of TT. In addition, increased testosterone level is linked to the movement velocity (Heavens et al., 2014). Of relevance, HPG axis is activated during slow movement velocity (Lee et al., 2002), and submaximal intensity with high volume and short recovery (60-90 s) have been shown for triggering the HPG axis (Kraemer & Ratamess, 2005; McCaulley et al., 2009). In this context, Smilios et al. (2013) revealed no statistical difference in testosterone level either after submaximal or maximal movement velocity (3:3 vs 1:1). However, the previous study was conducted without caffeine intervention. In the present study, the velocity applied during all of the sets was 1 s for each concentric and eccentric contraction. Probably, several factors may change testosterone concentration, such as the number of sets, the number of exercises, the nature of selected exercises (upper or lower body muscles), and the time of recovery interval. In the present study, the number of exercises is greater than the number in the prior studies. Biologically, testosterone increases as a result of secretion FSH, LH, and sex hormone-binding globulin (SHBG) (Spiering et al., 2009; Wu & Lin, 2010; Stokes et al., 2013). In the current study, FSH was significantly higher in caffeine trial compared to the placebo. However, compared to baseline measurements, FSH and LH decrease in both trials but still within normal range. Although testosterone secretion is not related to the values of FSH and LH since these hormones' values are within normal range, future research is needed to elucidate these findings. Actually, investigation of the acute caffeine effects in resistance exercise has been cited as a further research need.

In addition, activation of large muscle mass can induce high testosterone response (Kraemer & Ratamess, 2005). The participants in the present study have a great muscle mass depended on their BMI. Incorporation of lower muscles during resistance training can augment blood testosterone level 1 hour after the end of session (McCaulley et al., 2009). In the current study, the participants predisposed to perform leg press, back squat, and leg extension and these exercises along with large muscle mass may explain the increment of TT in both trials compared to baseline measurements. The higher the blood testosterone level after caffeine administration may also, in part, be attributed to greater heavy load (more repetitions to failure) (McCaulley et al., 2009), which is caused by reduction in RPE (Ferreira et al., 2021).

In resistance training, myofibrillar anabolic rate should be higher than catabolism when exercising repetition to failure to synthesis satellite calls and thus repair damaged fibers (Heavens et al., 2014). This process, however, requires high blood testosterone concentration. On the other hand, physical stress can induce elevation in blood PRL levels (Spiering et al., 2009), predisposing muscle fiber to challenge between repair of damaged sarcomeres and physiological demands for restoring testosterone. In the present study, the value of PRL in both trials has reached just to upper limits of normal range. The explanation of this response may be attributed to increased dopamine levels because of the study's protocol. Of relevance, caffeine could elicit catecholamines during resistance exercise which raise free fatty acids (Wu & Lin, 2010), facilitating high energy expenditure during performance.

For muscle damage, CK levels were higher after caffeine administration compared to the placebo. What is more important, the values of CK in both trials and baseline measurements are within the acceptable criterion in athletes. The other muscle damage biomarkers measured in the current study were also within normal range. Hence, there is difficult to explain that caffeine had a positive effect on damage biomarkers since all the biomarkers values in both trials were in the near upper limits. The explanation of this result may, in part, be attributed to the participant's high adaptation phase. Some studies have demonstrated that caffeine had no effect on muscle damage biomarkers. For instance, AbuMoh'd et al. (2021) demonstrated that 6 mg/kg of caffeine had no beneficial effects in attenuation of muscle damage biomarkers. However, the protocol in that study consisted of resistance exercises (3 sets × 10 repetitions, with 2 min recovery interval) followed by an incremental treadmill test in long-distance runners. Machado et al. (2008) reported that 4.5 mg/kg of caffeine had no effect in CK, lactic dehydrogenase (LDH), aspartate aminotransferase (AST), and alanine aminotransferase (ALT) following resistance exercises compared to placebo, but in male soccer players. Soleimani et al. (2018) also found no significant effect in CK and serum high sensitivity C-reactive protein (hs-crp) after caffeine supplementation (2 g per day for 14 days) in overweight students. These prior studies did not measure muscular endurance and the population were not resistance trained athletes. Basically, the elevations of biomarkers following exercise are usually higher in untrained and/or recreationally trained players than competitive athletes.

In agreement with prior studies (Green et al., 2007; Davis et al., 2012; Ferreira et al., 2021; Ruiz-Fernández et al., 2023), caffeine had beneficial effect on muscular endurance. The explanation of this result might be attributed to the biologic role of caffeine in antagonizing adenosine receptors (Duncan et al., 2013; Ferreira et al., 2021), reducing PRE and pain sensation (Owens et al., 2019; Ferreira et al., 2021). In this sense, Duncan et al. (2013) reported lower RPE after ingestion of 5 mg/kg of caffeine compared to placebo. However, they suggested that pain sensation was significantly higher in lower body (deadlift, and back squat) compared to bench press and prone row. In the present study, although the total repetitions in the lower body exercises (leg press, back squat, and leg extension) were more in the caffeine trial compared to the placebo, the total repetitions in the lower muscles were significantly less than the upper muscles in both caffeine and placebo trials, agreeing with the prior study. In addition, the result of the current study may also be caused by the nature of muscle fibers. In fact, the lower body muscles have a greatest motor unit size than the upper muscles, but the type of lower muscle fibers are, in anatomy nature, slow oxidative, potentiating participants intolerant to repeat more repetitions. Hence, repeated eccentric and concentric contractions until momentary muscle failure per set making the number of repetitions per set and the number of sets per exercise decrease.

Further, the biomarkers in the present study were measured only 1h after the end of each trial. So that a one limitation of the present study was that muscle damage biomarkers were not measured after 24 hours. Thus far, the ergogenic effect of caffeine on muscular endurance remains uncertain. Importantly, the discrepant evidence in the previous studies regarding muscular endurance probably due to the type of contraction (isometric, isotonic, or isokinetic), the amount of dose, the nature of participants (elite or untrained-resistance exercise), habitual or inhabitual caffeine intake, gender, and the number of sets per exercise. Hence, further research is needed to measure these variables at several points.

Conclusion

Administration of 5 mg/kg of caffeine 1 h prior to muscular endurance test performed by multiple resistance exercises enhances significantly hormonal response, namely follicle-stimulating hormone and total testosterone and increases total number of repetitions to failure in 11 well resistance-trained males compared to the placebo. However, the findings do not support the acute caffeine prevent muscle damage biomarkers following muscular endurance. Hence, male strength athletes may consider using this dose pre-training as an effective ergogenic aid before muscle endurance exercise.

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Declaration of conflicting interests
The Author declares that there is no conflict of interest with any financial organization regarding the material discussed in the manuscript.

Research ethics and athlete consent
The study was approved by the Al-Ahliyya Amman University Ethical Committee (FES-18G-280-2023). All study procedures and potential side effects of caffeine consumption, such as temporary tachycardia, hypertension, and numbness, were explained, and participants provided written informed consent.

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SportMont

ORIGINAL SCIENTIFIC PAPER

Differences in Technical and Tactical Efficiency between Winning and Defeated Water Polo Teams: A Comparative Analysis

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Abstract

Notation analysis is basically the gathering and analyzing of information that has been gained from observing performance in a competitive situation. The aim of this paper was to determine and explain the results and differences in technical and tactical efficiency in men's water polo between winning and defeated teams. The sample of the entities included 31 matches from the men's tournament held at the Olympic Games in Tokyo 2021. The sample of variables covered 18 defense and attack parameters of efficiency. Using a t-test for independent samples led to a result that the winning and losing teams differed significantly in four variables. Significant differences of winning teams in the realization with an equal number of players was probably stemmed from having better skills in achieving optimal conditions in preparation and performance. The reasons for the superiority of the winning teams in the part of defensive actions can probably be found in better and more coordinated actions of all defensive players in exclusions, blocking the ball and goalkeeper shot saves. These findings encourage coaches to improve players' skills by providing optimal conditions for the part of the technical-tactical solutions in defense and attack.

Keywords: *men water polo players, match analysis, technical indicators, tactical indicators*

Introduction

Notation analysis is basically the gathering and analyzing of information that has been gained from observing performance in a competitive situation. Success in any sport, including water polo, depends on several factors such as morphological structure, psychomotor abilities, cognitive abilities, conative characteristics, physiological and functional characteristics, technical and tactical knowledge, theoretical knowledge of water polo players and other (Hraste, 2021). Playing tactics in water polo is one of the most important segments because the outcome of the game depends greatly upon it (Hraste, 2021). According to Hraste (2021) there are numerous tactical possibilities in the defense and attack phase in water polo. In the defense phase, there are pressing, zone and combined defense, while in attack, the team can rely on a quick transition, outside shot, play with one or two center forwards, etc. The team will develop a style of play according to their fitness and technical capabilities (Hraste, 2021). In addition to adjusting the tactics according to one's own abilities, the tactics are also adjusted depending on the opponent, in a way of attempting to annul the opponent's advantages and take advantage of the disadvantages. Several investigations showed that a difference in the level of water polo players has a relevant impact on the occurrence of technical and tactical indicators especially in relation to even, counterattack, and power play situations. (Argudo, Ruiz, & Alonso, 2008; Lupo, Condello, & Capranica, 2012; Garcia-Marin, Iturriaga, & Manuel, 2017a; Garcia-Marin, Iturriaga, & Manuel, 2017b; Hraste, Jelaska, & Clark, 2020). In men´s games, eight game-related statistics, i.e. shots, extra player shots, 5 m-shots and assists (offensive efficiency), blocked shots, goalkeeper-blocked shots, goalkeeper-blocked extra player shots and goalkeeper-blocked 5 m shots (de-

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fensive efficiency) were considered (Escalante, Saavedra, Mansilla, & Tella, 2011) to distinguish between winning and losing teams in the final phase of the 2008 Olympic Games held in Beijing. In other studies, the offensive performance indicators i.e. centre goals, power play goals, counterattack goals, assists, offensive fouls, steals, blocked shots and won sprints, as well as defensive indicators i.e. goalkeeper blocked shots, goalkeeper-blocked inferiority shots and goalkeeper-blocked 5 m shots (Escalante et al., 2013) were used to distinguish between performances in international championships and their relationship with the phases of competition. In the research of Argudo, Ruiz and Alonso (2009) at 10 World Championships in the men's water polo between the winning team and defeated team, the authors found differences in the determination of the action, precise passing and shooting. Differences between two different levels in men's water polo emerged in the frequency of occurrence of counterattack and power play actions, the duration of even situations, the mean number of players directly involved during power play actions, the mean number of the passes during even and power play actions, the frequency of occurrence of the shots during counterattack and power play actions, the frequency of occurrence of goals during even actions, the frequency of occurrence of shots originating from different zones of the court, and the type of shots performed (Lupo, Tessitore, Minganti, & Capranica, 2010). One more research was conducted on how much certain factors affect the game of water polo, i.e. which elements of the game are the key to achieving success in men's water polo. Takagi et al. (2005), based on data from 108 matches from the 2001 World Cup in water polo, factorized the structure of both men's and women's water polo games and found that out of 32 variables, only two determine the winner of a water polo match: the ability to realize counterattacks and players more, and success in blocking and rescuing from the opponent's shots in a game with one less player. Observing the differences between the winning and losing water polo teams led to a conclusion that top level winning teams homogeneously distributed their shot opportunities at the second offensive line with balanced efficacy, creating variability and uncertainty in their opponents' defensive action (Canossa et al., 2020). Hraste, Jelaska and Stipić (2022) found and explained eight of eighteen technical-tactical variables that differentiate top women's winning and losing teams (goals in a situation with an equal number of opposing players, shots from the counterattack, goals from the counterattack, man-up goals, goalkeeper shot save, opponent's shots on goal, blocks and swimming for the ball).

Looking at previous research on the technical-tactical structure of differences between defeated and winning teams, it can be noticed that there was inconsistency in the decisive factors. The reasons for the above can be found in the diversity of research methodologies and, above all, in the dynamism of the water polo game in space and time. Namely, water polo is a sport in which frequent changes in the rules of the game and the development of the energy component of the game lead to changes in technical and tactical efficiency. Because of all of the above, continuous research into the technical-tactical structure of the water polo game is required.

The aim of this paper is to determine and explain the results and differences in technical and tactical efficiency in men's water polo. There is an assumption that top men water polo players would differ in some variables of technical and tactical efficiency.

Material and methods

Participants

The sample of the entities included matches from men's tournament held at the Olympic Games in 2021. The results of matches of teams that lost more than 2 games with 8 goals difference and more (South Africa and Kazakhstan), were excluded from the sample of the entities. So called outliers were left out of the overall tournament statistics in order to get relatively homogeneous participating teams (Hungary, Greece, Italy, Australia, USA, Japan, Spain, Serbia, Montenegro and Croatia). One match that ended in a draw was also excluded from the entity sample. At the Olympic Games held in Tokyo, 42 matches were played in the men's tournament, and for the purposes of this research, 31 matches were analysed.

Measures

The sample of variables included 18 parameters of efficiency: total number of shots (TS), shots in a situation with an equal number of opposing players (SE), goals in a situation with an equal number of opposing players (GE), shots from fouls (SF), goals from fouls (GF), number of penalties (TP), goals from the penalties (GP), shots from the counterattack (SC), goals from the counterattack (GK), man-up shots (MS), man-up goals (MG), goalkeeper shot save (GS), opponent's shots on goal (OS), blocks (BL), stolen balls (SB), swimming for the ball (SB), exclusions (E) and lost balls (LB). Variables from previous research (Lupo et al., 2012; Graham & Mayberry, 2014; García-Marín & Iturriaga, 2017) were partly used to shape the set of technical and tactical efficiency parametars. Four top water polo experts participated in the selection of all 18 statistical parameters, who believed that the selected parameters cover the entirety of the water polo game very well. The selected variables appear for the first time in a design scientific paper covering men's water polo.

Procedures

The data were collected from the official records of the Total Waterpolo platform which are tracked during the playing of water polo games. Upon a request of the authors, Total Waterpolo sent their data of the observed matches. The data can also be found at the following link: https://total-waterpolo.com/tokyo-2020-mens-olympic-water-polo-tournament/. Official staff registered all of the collected data. Reliability of the data was tested by additional reviewing of 10 matches. The reviewing was done by two independent water polo experts. Each frequency of variable for each group of players was collected and compared to data from the official records. Reliability coefficients for single data was calculated as a ratio of reviewed observed frequencies and official record frequencies.

Statistical Analysis

For the purposes of this study, basic statistical parameters in the form of arithmetic mean (AM), median (MED), minimum score (MIN), maximum score (MAX), standard deviation (SD), skewness (SK) and kurtosis (KUR) were calculated, and a t-test for independent samples was calculated. The level of statistical significance was set to 5%. Data were processed using software system Statistica ver. 13.2. (Dell Inc., Tulsa, OK, USA).

Results

Table 1 shows basic descriptive parameters of the variables of situational efficiency of the winning teams (mean, median, minimum score, maximum score, standard deviation, skewness and kurtosis).

Table 1. Descriptive statistics: arithmetic mean (AM), median (MED), minimum score (MIN), maximum score (MAX), standard deviation (SD), skewness (SK) and kurtosis (KUR) for the variables of situational efficiency of the winning teams

VAR	AS	MED	MIN	MAX	SD	SK	KUR
TS	30,00	29,50	24,00	35,00	3,17	-0,06	-1,00
SE	17,83	18,00	12,00	23,00	3,39	0,03	-1,24
GE	5,37	6,00	1,00	10,00	2,27	-0,19	-0,33
SF	0,83	1,00	0,00	3,00	0,91	0,64	-0,79
GF	0,30	0,00	0,00	1,00	0,47	0,92	-1,24
TP	1,33	1,00	0,00	5,00	1,37	1,23	1,56
GP	1,20	1,00	0,00	5,00	1,30	1,23	1,40
SC	0,97	1,00	0,00	3,00	0,93	0,62	-0,44
GC	0,67	1,00	0,00	3,00	0,76	1,17	1,66
MS	8,90	8,50	3,00	15,00	2,76	0,23	-0,38
MG	4,87	5,00	2,00	9,00	1,80	0,29	-0,54
GS	10,77	10,00	7,00	17,00	2,71	0,72	-0,43
OS	19,30	18,50	11,00	28,00	4,32	0,32	-0,55
BL	4,93	5,00	0,00	11,00	2,32	0,12	0,50
STB	6,07	6,00	1,00	13,00	2,48	0,59	0,95
SWB	2,10	2,00	0,00	4,00	1,21	0,04	-1,15
E	14,03	14,00	7,00	20,00	3,48	-0,12	-0,70
LB	10,33	9,00	5,00	21,00	3,74	0,82	0,62

Legend TS - total number of shots, SE - shots in a situation with an equal number of opposing players, GE - goals in a situation with an equal number of opposing players, SF - shots from foul, GF - goals from foul, TP - number of penalties, GP - goals from the penalties, SC - shots from the counterattack, GC - goals from the counterattack, MS - man-up shots, MG - man-up goals, GS - goalkeeper shot save, OS - opponent's shots on goal, BL - blocks, STB - stolen balls, SWB - swimming for the ball, E – exclusions, LB - lost balls

Table 2 shows basic descriptive parameters of the variables of situational efficiency of the defeated teams (mean, median, minimum score, maximum score, standard deviation, skewness and kurtosis).

Table 2. Descriptive statistics: arithmetic mean (AM), median (MED), minimum score (MIN), maximum score (MAX), standard deviation (SD), skewness (SK) and kurtosis (KUR) for the variables of situational efficiency of the defeated teams

VAR	AS	MED	MIN	MAX	SD	SK	KUR
TS	30,57	30,50	24,00	42,00	4,17	0,79	0,71
SE	17,13	17,00	10,00	26,00	3,45	0,34	0,42
GE	3,33	3,00	1,00	8,00	1,83	0,85	0,21
SF	1,13	1,00	0,00	5,00	1,17	1,40	2,79
GF	0,10	0,00	0,00	1,00	0,31	2,81	6,31
TP	1,00	0,00	0,00	9,00	1,91	2,89	10,14
GP	0,80	0,00	0,00	7,00	1,56	2,68	8,19
SC	0,57	0,00	0,00	2,00	0,73	0,90	-0,47
GC	0,37	0,00	0,00	2,00	0,61	1,50	1,33
MS	10,53	10,00	4,00	19,00	3,78	0,46	-0,76
MG	4,13	4,00	0,00	10,00	2,26	0,87	0,86
GS	9,00	9,00	3,00	16,00	2,85	0,39	0,45
OS	21,27	21,50	12,00	29,00	3,71	-0,36	0,40
BL	2,57	2,50	0,00	6,00	1,55	0,26	-0,64
STB	6,17	5,00	1,00	15,00	3,31	0,94	0,73
SWB	1,87	2,00	0,00	4,00	1,22	0,03	-1,22
E	11,83	12,00	6,00	20,00	2,90	0,49	0,74
LB	10,23	10,00	3,00	17,00	3,27	0,13	-0,02

Legend: TS - total number of shots, SE - shots in a situation with an equal number of opposing players, GE - goals in a situation with an equal number of opposing players, SF - shots from foul GF - goals from foul, TP - number of penalties, GP - goals from the penalties, SC - shots from the counterattack, GC - goals from the counterattack, MS - man-up shots, MG - man-up goals, GS - goalkeeper shot save, OS - opponent's shots on goal, BL - blocks, STB - stolen balls, SWB - swimming for the ball, E – exclusions, LB - lost balls

Table 3 presents the results of the t-test for independent samples. According to the results from table 3, in men's water polo, the winning and losing teams differ statistically and significantly in four of eighteen situational variables: goals in a situation with an equal number of opposing players, goalkeeper shot save, blocks and exclusions.

Table 3. T-test for independent samples (AS DEF - arithmetic mean of defeated men's teams; AS WIN - arithmetic mean of winning women's teams; t value; p-level of significance)

VAR	AS DEF	AS WIN	t value	p
TS	30,57	30,00	0,59	0,56
SE	17,13	17,83	-0,79	0,43
GE	3,33	5,37	-3,83	0,00
SF	1,13	0,83	1,11	0,27
GF	0,10	0,30	-1,97	0,05
TP	1,00	1,33	-0,78	0,44
GP	0,80	1,20	-1,08	0,29
SC	0,57	0,97	-1,86	0,07
GC	0,37	0,67	-1,68	0,10
MS	10,53	8,90	1,91	0,06
MG	4,13	4,87	-1,39	0,17
GS	9,00	10,77	-2,46	0,02
OS	21,27	19,30	1,89	0,06
BL	2,57	4,93	-4,65	0,00
STB	6,17	6,07	0,13	0,90
SWB	1,87	2,10	-0,74	0,46
E	11,83	14,03	-2,66	0,01
LB	10,23	10,33	-0,11	0,91

Legend: TS - total number of shots, SE - shots in a situation with an equal number of opposing players, GE - goals in a situation with an equal number of opposing players, SF - shots from foul GF - goals from foul, TP - number of penalties, GP - goals from the penalties, SC - shots from the counterattack, GC - goals from the counterattack, MS - man-up shots, MG - man-up goals, GS - goalkeeper shot save, OS - opponent's shots on goal, BL - blocks, STB - stolen balls, SWB - swimming for the ball, E – exclusions, LB - lost balls

Discussion

In this research the winning team on average scored more than two goals in a situation with an equal number of opposing players. The reasons for the observed statistically significant difference can most likely be sought in better selection and higher quality of players of the winning teams. Namely, the higher quality of the players probably collectively contributes to a better preparation of all the actions that precede a shot, as well as the realization itself (Hraste et al., 2022). The goalkeepers of the winning teams had almost two more saves than the goalkeepers of the losing teams. The goalkeeper is a very important link within the team. The significant role of the goalkeeper in the overall result of a water polo match has already been confirmed in a previous study (Escalante et al., 2013). A good goalkeeper makes a team better, but also having players in the field with a well-placed defense certainly helps the goalkeeper save more shots. Also, players from the winning teams blocked almost twice as many shots on goal as players from the losing teams which represents a statistically significant difference which was also determined in the research of Escalante et al. (2011). The reason for this can also be sought in a better quality of individuals. Just as with the execution of the shot, a better player performs all the necessary actions to set up a high-quality and successful block. It is interesting to note that the winning teams had statistically significantly more expulsions than the defeated teams. The number of exclusions is a seemingly "negative" variable of situational efficiency. However, it must be considered that good teams do not want to concede easy field goals (Hraste, 2021). They prefer to make an exclusion, so they defend with one less player. Given a good goalkeeper and well-placed blocks, this tactic is clearly successful (Hraste, 2021). The dominance of winning teams in the realization with an equal number of players probably stemmed from better skills in achieving optimal conditions in the preparation and implementation of all offensive actions (Escalante et al., 2011; Hraste et al., 2022). The reasons for the superiority of the winning teams in the part of defensive actions can probably be found in better and more coordinated actions of all defensive players in blocking the ball, reducing the opponent's shots and goalkeeper shot save. This research confirmed some previous research in differentiating top male water polo players according to match outcomes (Takagi et al., 2005; Lupo et al., 2010).

In the previous chapter, the significant difference observed between winning and losing teams in the variables: goals in a situation with an equal number of opposing players, goalkeeper shot save, blocks and exclusions is partially consistent with some research (Canossa et al., 2020; Escalante et al., 2011; Escalante et al., 2013; Hraste et al., 2022) but different from other studies (Argudo et al., 2008; Lupo et al., 2012; Garcia-Marin et al., 2017b). The reasons for the observed discrepancies can probably be found in different research methodologies, evolution of the technical-tactical and energy structure of water polo players, and especially in the fact that the water polo rules have been continuously changing over the past twenty years. The changes in water polo rules that were adopted in 2019 had the goal of making water polo faster and more dynamic, and based on the aforementioned characteristics, to reduce the gap between the so-called "big" and "small" national teams. The existence of statistically significant differences in only four variables and their absence in the remaining fourteen variables can most likely be attributed to the very equal quality of the ten national teams that were observed, which is mostly indicated by the results of the matches. The opinions of water polo experts and the results of the matches suggest that the changes in the water polo rules justified the goal and that today's water polo research is different compared to the research until 2019 (Takagi et al., 2005; Argudo et al., 2008; Lupo et al., 2010; Escalante et al., 2011; Lupo et al., 2012; Escalante et al., 2013; Garcia-Marin et al., 2017b).

This study has some limitations. The first limitation can be found in the unequal distribution of the total number of analyzed matches in different phases (preliminary phase, n=19; quarter-final/semi-final phase/play-offs for 5th to 8th place/bronze medals/gold medals, n=12). Another limitation can be found in the preliminary stage, in which some matches often happened with some teams with no longer having the possibility to advance to the next round, which could have affected some of the statistics related to the game.

In future research, it is necessary to continue working with this type of analysis, adding variables that would give an even better insight into the structure and differences of technical-tactical activities (e.g. the number and effectiveness of shots with regard to the position on the field and/or the role in the game).

Conclusion

This research showed that the technical and tactical indicators of the top men's water polo competitions vary in relation to the outcome of the match. The statistically significant higher number of goals scored by the winning teams during the actions of an equal number of opposing players suggests better skills in preparation and execution. Better indicators of winning teams in three defensive actions confirmed their overall superiority. Selected defensive and offensive variables indicated the need for a good balance between the two groups of variables. All defensive and offensive factors related to the performance of the winning team showed that more goals were combined with defensive tasks in terms of goalkeeper shot save, blocks and exclusions. Notation analysis is a great tool for water polo coaches, fitness coaches and sports scientists to be aware of the real requirements of the game. In one hand, these findings encourage coaches to improve the player's skills in providing optimal conditions for the execution of shots, and on the other hand to enhance the ability to cover and guard the direct striker playing in different internal and external positions in the game.

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Conflict of Interest
The authors declare that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Validity and Reliability of Wearable Devices during Self-Paced Walking, Jogging and Overground Skipping

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Abstract

Wearable technology can track unusual exercise, providing data for improving fitness. The aim of the study was to determine validity and reliability during walking, jogging, and skipping. Eighteen volunteers completed 5 min self-paced activities interspersed with 5 min rest. Variables and devices were step count (Garmin Instinct), estimated energy expenditure (Garmin Instinct, Polar Vantage M2), and heart rate (Garmin Instinct, Polar Vantage M2, Polar OH1, Polar Verity Sense). Validity measures were mean absolute percent error (MAPE) and Lin's Concordance (CCC), and reliability were coefficient of variation (CV), and intraclass correlation (ICC). Thresholds were MAPE ≤5%, CCC≥0.90, CV≤10%, ICC≥0.70. Garmin Instinct step count during skipping was not considered valid (MAPE=90.2%, CCC=0.008) or reliable (CV=6%, ICC CI=0.4). Energy expenditure during skipping was not valid or reliable in the Garmin Instinct (MAPE=28%, CCC=0.27; CV=19%, ICC=0.61) or the Polar Vantage M2 (MAPE=19%, CCC=0.57; CV=13%). While the Polar Vantage M2 was reliable for estimated energy expenditure during walking and jogging activities, wrist-worn devices (Garmin Instinct, Polar Vantage M2) were neither valid nor reliable in returning estimated energy expenditure during overground skipping. From a wider perspective, wearable device algorithms for estimating energy expenditure should continue to be refined until they return the same level of accuracy as what is currently observed for heart rate, and to a lesser extent step count. Skipping may be an excellent unusual activity for testing wearable devices.

Keywords: *wearable activity tracker, unusual exercise, accuracy, consistency*

Introduction

Wearable technology, particularly activity trackers, have sprung up in popularity in recent years. These wearable devices are designed to monitor and track physical activity, including steps taken, estimated energy expenditure, and heart rate (Bunn, Navalta, Fountaine, & Reece, 2018). Systematic reviews consistently report that using wearable activity trackers leads to improvements in weight loss, waist circumference, and body mass index (Yen & Chiu, 2019). Wearing an activity tracker and receiving daily feedback on progress can led to increased physical activity and reduced sedentary behavior (Jakicic et al., 2016). However, the boundless enthusiasm for wearable devices should be tempered, as the benefits of using activity trackers may depend on motivation and level of engagement with the device (Cadmus-Bertram, Marcus, Patterson, Parker, & Morey, 2015).

Wearable technology has been increasingly used to monitor and track physical activity beyond traditional exercise activities, such as in water-based sports, dancing, and video gaming. Activity trackers have been proposed to obtain metrics during water-based activities such as kayaking and canoeing (Umek & Kos, 2018; Liu, Wang, Qiu, Zhang, & Hao,

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2021). Similarly, wearable sensors can be used to assess activity during both social and jazz dancing activities (Stančin & Tomažič, 2022), making it a useful tool for dancers who must perform repetitive leaping motions to track their progress and improve their performance. Furthermore, virtual reality gaming has gained popularity as an alternative form of exercise, and wearable technology such as virtual reality headsets and motion capture devices can be used to monitor and track physiological responses during these activities (Cao, Xie, & Chen, 2019). Overall, wearable technology has shown potential in monitoring and tracking physical activity in a variety of unusual exercise activities, providing useful data for individuals to optimize their performance and improve their overall fitness levels. However, further research is needed to explore the accuracy and reliability of these devices in diverse settings and unusual activities.

Bounding straight to the issue, overground skipping is a non-traditional exercise that has not been investigated to date in the context of wearable technology devices. Most humans transition from walking to running as a faster method of travel and exercise (Navalta, Davis, Carrier, Sertic, & Cater, 2021), however, running requires greater energy expenditure than walking (Harrell et al., 2005). Some people, particularly children, perform skipping as a transitory movement as an alternative to walking and running (Navalta et al., 2021). The available literature centers on rope skipping, rather than overground skipping (Verdel et al., 2022; Yongmao & Yuxin, 2023). One study evaluated skipping in 20-sec bouts as obtained from a novel smart patch (Verdel et al., 2022), and a recent investigation was conducted on rope skipping and deep learning algorithms (Yongmao & Yuxin, 2023). However, overground skipping is different in that the action involves forward propulsion that is absent in rope skipping, which may be responsible for half of the energy cost of walking (Gottschall & Kram, 2005). Thus, the evaluation of wearable devices to return valid and reliable measurements during overground skipping should be conducted.

The purpose of this investigation was to determine the validity and reliability of wearable technology devices during an overground skipping activity. While many studies are designed to determine validity, there is a need to also obtain reliability metrics (Carrier, Barrios, Jolley, & Navalta, 2020). Previous research has shown wearable devices to have varied levels of validity, with heart rate generally being acceptable, energy expenditure estimation unacceptable, and step count falling in between (Bunn et al., 2018). Based on previous literature and testing performed in our laboratory, it was hypothesized that during overground skipping, step count would be reliable but not valid, estimates of energy expenditure would be neither valid nor reliable, and heart rate would be both valid and reliable. These findings are important, as this is the first study to report validity and reliability of wearable devices during an overground skipping activity.

Methods

Participants

Eighteen participants (female n=10, male n=8, transgender, intersex, or other n=0) provided written informed consent that was approved by the Institutional Review Board (IRB approval# UNLV-2022-80) and volunteered for the study. A power analysis was performed using pilot data, indicating the need for at least eleven participants (coefficient of determination r^2=0.57, correlation *r* effect size=0.755, a=0.05, b=0.80). Participants were screened and deemed not to require medical clearance to complete exercise according to the American College of Sports Medicine preparticipation health screening recommendations. The mean demographic information included: age=26±8 years, body mass=73.3±11.9 kg, height=66.4±3.5 cm, and Body Mass Index=25.6±3.0 kg•m^{-2}. Participants self-identified ethnic group included African American (n=1), White (n=10), Hispanic (n=3), Polynesian (n=1), and Southeast Asian (n=3).

Device Setup

Demographic information was obtained and input into the criterion and wearable devices prior to testing each participant. Participants were outfitted with criterion devices (K5 [COSMED, Rome, Italy] for energy expenditure; H10 [Polar Electro, Kempele, Finland] for heart rate) and wearable devices (described below), and a secure Bluetooth connection was confirmed. In all cases, devices were affixed to the body according to manufacturer recommendations. Briefly, the K5 was secured on the back of the participant, the H10 attached around the chest. The biceps-worn experimental devices were the Polar OH1 (Polar Electro, Kempele, Finland) and Polar Verity Sense (Polar Electro, Kempele, Finland), placed on the right and left biceps each. The wrist-worn experimental devices were the Garmin Instinct (Garmin Limited, Olathe, Kansas) and Polar Vantage M2 (Polar Electro, Kempele, Finland), placed on the right and left wrist each. Two of the same models of each experimental device were used simultaneously so that concurrent reliability could be obtained (Pinedo-Jauregi, Garcia-Tabar, Carrier, Navalta, & Camara, 2022).

The chest-worn (H10) and biceps-worn devices (OH1, Verity Sense) were connected via Bluetooth to an iPad mini (Apple Inc., Cupertino, CA) with the PerformTek application (Valencell, Inc., Raleigh, NC) which returns heart rate of all connected devices on a single csv file. The wrist-worn devices (Instinct, Vantage M2) collected information directly onto the device and were downloaded at a later time using recommended applications (Instinct through the Garmin Connect application and Bluetooth connection; Vantage M2 through the Polar FlowSync desktop application and Polar Flow online database).

Exercise Bouts

After confirming connection to the criterion and experimental devices, participants performed a self-paced walk back and forth through an indoor hallway along a 61-meter track, followed by 5 minutes (min) of seated rest. All devices were reset during the period of rest (i.e., new activity recording session with the same demographic and anthropometric data in the watch; not a factory reset). Participants then completed 5 min of self-paced running back and forth through the same indoor hallway, followed by 5 min of seated rest. All devices were again reset during the period of rest. Finally, participants performed 5 min of self-paced overground skipping back and forth through the same indoor hallway. Step count was collected via manual clicker step counters by two independent observers during the overground skipping bout only. The step count was the arithmetic mean of the observer's manual counts. A step was defined as the completion of a unilateral stride before transferring motion to the contralateral side of the body. The protocol is similar in terms of timing and administration to other investigations conducted by the laboratory group (Montes & Navalta, 2019; Navalta et al., 2019; Montes, Tandy, Young, Lee, & Navalta, 2022).

Devices

Polar H10: Although the use case specific to overground skipping has not been determined, the Polar H10 chest strap has been shown in other settings to have acceptable reliability (Speer, Semple, Naumovski, & McKune, 2020) and to be valid compared to electrocardiography (Gilgen-Ammann, Schweizer, & Wyss, 2019). The Polar H10 is an electrocardiogram-based heart rate sensor secured around the chest at the level of the xyphoid process. The H10 contains plastic electrodes on the underside of the strap that detects heart rate. The sensor materials include acrylonitrile butadiene styrene (ABS), ABS plus glass fiber (ABS+GF), polycarbonate, and stainless steel, and the strap material is composed of 38% polyamide, 29% polyurethane, 20% elastane, 13% polyester, and silicone prints. The H10 has a sampling frequency of 1000 Hertz (Hz).

Polar OH1: The Polar OH1 is a photoplethysmography (PPG) device that uses an optical heart rate sensor worn on the upper arm. The OH1 is an optical heart rate sensor worn on the upper arm. The sensor materials include ABS, ABS+GF, poly(methyl methacrylate) (PMMA), and steel use stainless (SUS) 316. The device was positioned so that the sensor was on the underside of the armband and firmly against the skin. The OH1 has a sample rate of 135 Hz.

Polar Verity Sense: The Polar Verity Sense is also a PPG device worn on the upper arm. The sensor materials include ABS, ABS+GF, PMMA, and SUS 316. The device was positioned with the sensor on the underside of the armband and firmly against the skin. The Verity Sense has a sample rate of 135 Hz.

Garmin Instinct: The Garmin Instinct is a PPG device. The physical size is 46 x 46 x 12.5 millimeters (mm) with a mass of 45 grams (g). The optical heart rate sensor is worn on the wrist and employs Garmin ElevateTM heart rate sensor technology. The sample rate is unknown.

Polar Vantage M2: The Polar Vantage M2 is a PPG device. The physical size is 45 x 45 x 15.3 mm with a mass of 52 g. The optical heart rate sensor is worn on the wrist and employs Precision PrimeTM heart rate sensor technology. The Vantage M2 is reported to have a sample rate of 135 Hz.

Statistical Analysis

Reported measures associated with validity include mean absolute percent error (MAPE), Lin's Concordance Correlation Coefficient (CCC), and the mean absolute error (MAE). The equations for these metrics were input into an Excel spreadsheet (Microsoft Excel for Mac version 16.66.1, Redmond, WA). For validity thresholds our laboratory group has used a MAPE value ≤ 5%, and a CCC ≥ 0.90 (J. W. Navalta et al., 2020). A device meeting both thresholds was considered evidence in support of validity. The Bland-Altman analysis was used to determine agreement. Bias and limits of agreement were determined using the blandr analysis in jamovi (version 2.3.19.0).

Reported measures associated with reliability include the coefficient of variation (CV), and two-way mixed model with absolute agreement intraclass correlation coefficient (ICC). The CV was determined using Excel (Microsoft Excel for Mac version 16.66.1, Redmond, WA), and the ICC (single measures) using SPSS Statistics (IBM SPSS Statistics, version 28.0.1.0, Chicago, IL). For non-laboratory settings a threshold of ≤ 10% for CV, and ≥ 0.70 for ICC, with a lower bound of the 95% CI ≥ 0.70 has been used (Navalta et al., 2019). A device meeting both thresholds was considered evidence in support of reliability.

Results

Step Count

Step count was obtained only from the Garmin Instinct during the skipping trial because the other devices did not return the measure. Manual step count was not obtained from one participant and was not recorded from the Garmin Instinct for two other participants. Thus, validity measures reflect data points from 30 distinct bouts, and reliability measures from 16 pairs (see table 1). Based on the predefined definition of a step, the device did not meet any of the threshold measures for validity and overestimated the step count by nearly double (see Table 1). The Garmin Instinct met the reliability thresholds for CV and ICC but was not considered reliable because the lower end of the 95% CI did not meet the established threshold.

Table 1. Step count during the self-paced skipping bout. Criterion (manually counted steps), and validity and reliability measures of the Garmin Instinct.

	Validity	Data Points	Average	MAPE (%)	Bias	Limits of Agreement	CCC
Skipping	Manual Count	30	507.3 (32.9)				
	Garmin Instinct	30	957.2 (139.9)	90.2	–499.8	181.6 to –718.5	0.008
	Reliability	Data Points	Device 1	Device 2	CV (%)	ICC	95% CI
	Garmin Instinct	16	933.7 (165.3)	1000.9 (103.4)	6.0	0.74	0.40 to 0.90

Average is the arithmetic mean (standard deviation) measurement of step count. MAPE = mean absolute percent error, CCC = Lin's Concordance Correlation Coefficient, CV = coefficient of variation, ICC = Intraclass correlation coefficient, CI = Confidence interval of the ICC. Values noted in bold and italics meet the predetermined threshold for validity or reliability.

Energy Expenditure

There was no missing energy expenditure data from the criterion device or any of the wearable activity trackers. Validity and reliability measures for wrist-worn devices (Garmin Instinct, Polar Vantage M2) during walking, jogging, and skipping are provided in Table 2. The biceps-worn devices did not return an estimate of energy expenditure. Neither the Garmin Instinct nor the Polar Vantage M2 met the predetermined validity thresholds when returning estimates of energy expenditure across any bout of exercise. The Polar Vantage M2 met the reliability thresholds during self-paced walking and jogging, but not overground skipping.

Table 2. Energy expenditure during self-paced walking, jogging, and skipping bouts. Criterion (COSMED K5), and validity and reliability measures of the Garmin Instinct, and Polar Vantage M2.

	Validity	Data Points	Average	MAPE (%)	Bias	Limits of Agreement	CCC
Walking	COSMED K5	36	26.2 (5.6)				
	Garmin Instinct	36	37.1 (13.6)	57.0	−10.9	19.3 to −41.0	-0.06
	Polar Vantage M2	36	37.3 (9.0)	46.7	−11.1	6.1 to −28.3	0.15
	Reliability	Data Points	Device 1	Device 2	CV (%)	ICC	95% CI
	Garmin Instinct	18	38.7 (15.4)	36.5 (12.4)	14.3	0.75	0.45 to 0.90
	Polar Vantage M2	18	37.9 (10.1)	36.7 (8.7)	6.0	0.89	0.74 to 0.96
Jogging	Validity	Data Points	Average	MAPE (%)	Bias	Limits of Agreement	CCC
	COSMED K5	36	54.6 (12.2)				
	Garmin Instinct	36	64.6 (14.8)	26.0	−10.0	15.4 to −35.4	0.42
	Polar Vantage M2	36	61.1 (13.1)	18.1	−6.6	12.0 to −25.2	0.63
	Reliability	Data Points	Device 1	Device 2	CV (%)	ICC	95% CI
	Garmin Instinct	18	68.2 (14.3)	64.0 (12.6)	10.9	0.59	0.18 to 0.82
	Polar Vantage M2	18	64.0 (12.6)	60.3 (12.8)	4.8	0.90	0.76 to 0.96
Skipping	Validity	Data Points	Average	MAPE (%)	Bias	Limits of Agreement	CCC
	COSMED K5	36	59.6 (14.3)				
	Garmin Instinct	36	57.4 (19.6)	28.1	2.2	42.6 to −38.3	0.27
	Polar Vantage M2	36	64.5 (17.4)	18.6	−4.9	23.1 to −32.9	0.57
	Reliability	Data Points	Device 1	Device 2	CV (%)	ICC	95% CI
	Garmin Instinct	18	60.9 (19.3)	56.1 (20.2)	19.1	0.61	0.21 to 0.83
	Polar Vantage M2	18	71.8 (17.5)	59.7 (14.7)	13.1	0.83	0.60 to 0.93

Average is the arithmetic mean (standard deviation) measurement of kilocalories expended. MAPE = mean absolute percent error, CCC = Lin's Concordance Correlation Coefficient, CV = coefficient of variation, ICC = Intraclass correlation coefficient, CI = Confidence interval of the ICC. Values noted in bold and italics meet the predetermined threshold for validity or reliability.

Heart Rate

Heart rate measures were not obtained from the Polar OH1 for one participant during the walking trial. There was no missing heart rate data from the criterion device or any of the other wearable activity trackers. All devices met the preestablished validity thresholds during the self-paced walking trial (see Table 3). One wrist-worn device (Garmin Instinct) and one biceps-worn device (Polar Verity Sense) met the thresholds for reliability during walking (see Table 3) and can be considered both valid and reliable during this type of exercise.

Table 3. Heart rate during self-paced walking, jogging, and skipping bouts. Criterion (H10), and validity and reliability measures of wrist-worn (Garmin Instinct, Polar Vantage M2) and biceps-worn (Polar Verity Sense, Polar OH1) devices.

	Validity	Data Points	Average	MAPE (%)	Bias	Limits of Agreement	CCC
Walking	Polar H10	36	104.5 (17.1)				
	Garmin Instinct	36	105.5 (16.6)	2.8	−1.0	7.0 to −9.1	0.97
	Polar Vantage M2	36	104.7 (14.2)	3.3	−0.2	13.0 to −13.4	0.91
	Polar Verity Sense	36	104.3 (16.7)	0.7	0.3	2.2 to −1.7	0.99
	Polar OH1	35	104.5 (18.0)	2.1	0.5	14.9 to -13.9	0.91
	Reliability	Data Points	Device 1	Device 2	CV (%)	ICC	95% CI
	Garmin Instinct	18	107.3 (16.5)	106.6 (15.9)	2.2	0.96	0.91 to 0.99
	Polar Vantage M2	18	105.0 (11.1)	106.0 (16.4)	3.3	0.78	0.50 to 0.91
	Polar Verity Sense	18	105.9 (16.3)	105.6 (16.0)	0.3	0.99	0.99 to 1.0
	Polar OH1	17	106.3 (18.7)	107.0 (16.2)	2.7	0.842	0.62 to 0.94
Jogging	Validity	Data Points	Average	MAPE (%)	Bias	Limits of Agreement	CCC
	Polar H10	36	154.9 (13.9)				
	Garmin Instinct	36	147.2 (15.8)	5.9	7.7	32.8 to −17.5	0.55
	Polar Vantage M2	36	148.2 (15.1)	4.4	6.7	26.1 to −12.7	0.69
	Polar Verity Sense	36	151.7 (13.7)	2.0	3.1	10.3 to −4.0	0.94

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Table 3. Heart rate during self-paced walking, jogging, and skipping bouts. Criterion (H10), and validity and reliability measures of wrist-worn (Garmin Instinct, Polar Vantage M2) and biceps-worn (Polar Verity Sense, Polar OH1) devices.

Jogging	Polar OH1	38	152.6 (13.7)	1.5	2.3	4.9 to −0.3	0.98
	Reliability	Data Points	Device 1	Device 2	CV (%)	ICC	95% CI
	Garmin Instinct	18	148.5 (14.3)	150.5 (11.4)	3.3	0.71	0.38 to 0.88
	Polar Vantage M2	18	153.5 (14.1)	145.8 (12.4)	4.1	0.67	0.30 to 0.86
	Polar Verity Sense	18	153.1 (13.9)	151.6 (14.1)	0.9	0.96	0.89 to 0.98
	Polar OH1	18	153.3 (14.1)	153.2 (14.0)	0.1	0.99	0.99 to 1.0
Skipping	Validity	Data Points	Average	MAPE (%)	Bias	Limits of Agreement	CCC
	Polar H10	36	170.6 (17.5)				
	Garmin Instinct	36	133.3 (25.5)	21.7	37.3	84.7 to −10.1	0.16
	Polar Vantage M2	36	156.3 (24.0)	8.4	14.3	48.4 to −19.8	0.53
	Polar Verity Sense	36	168.7 (17.1)	1.1	2.0	4.7 to −0.8	0.99
	Polar OH1	36	167.3 (18.6)	1.9	3.3	16.1 to −9.5	0.92
	Reliability	Data Points	Device 1	Device 2	CV (%)	ICC	95% CI
	Garmin Instinct	18	136.8 (22.8)	132.5 (28.6)	9.5	0.49	0.04 to 0.77
	Polar Vantage M2	18	170.1 (17.2)	147.2 (21.2)	10.6	0.67	0.30 to 0.87
	Polar Verity Sense	18	170.1 (17.2)	169.6 (16.9)	0.4	0.99	0.98 to 0.99
	Polar OH1	18	168.8 (18.3)	168.2 (19.1)	0.4	0.99	0.98 to 0.99

MAPE = mean absolute percent error, CCC = Lin's Concordance, CV = coefficient of variation, ICC = Intraclass correlation coefficient, CI = Confidence interval. Values noted in bold and italics meet the predetermined threshold for validity or reliability.

During both self-paced jogging and self-paced skipping, the biceps-worn devices (Polar OH1, Polar Verity Sense) met all preestablished thresholds for validity and reliability (see Table 3). Thus, the Polar OH1 and the Polar Verity Sense can be considered valid and reliable during jogging as well as overground skipping.

Discussion

The purpose of this investigation was to determine the validity and reliability of wearable technology devices to return step count, energy expenditure, and heart rate during walking, jogging, and overground skipping. It was hypothesized that, during overground skipping, step count would be reliable but not valid, energy expenditure would be neither valid nor reliable, and heart rate would be both valid and reliable. First, the hypothesis for step count was partially supported because step count during self-paced skipping was neither valid nor reliable. Second, the hypothesis for energy expenditure was supported because estimated energy expenditure during self-paced skipping was neither valid nor reliable. Third, the hypothesis for heart rate was supported for biceps-worn devices (valid and reliable) but not wrist-worn technology (neither valid nor reliable).

The conclusions drawn from step count obtained during skipping are obtained from a single wrist-worn wearable device and should be considered with caution. The device did not meet the validity thresholds, likely because of how a step was defined in the current investigation (as a complete unilateral stride before transferring motion to the contralateral side of the body). The definition did not account for the foot tap down during a stride that many participants performed during the skipping motion, and likely accounted for the very large bias that was observed. It may be useful for future researchers interested in step count during overground skipping to account for the foot tap, as it appears at least one wearable device registers this action. It is proposed that users of consumer activity trackers may utilize this knowledge to their advantage, to increase step count during wellness program challenges by performing overground skipping as their preferred method of transportation, which would conceivably double the step count compared to the more mundane method of walking. The practical implications of this finding should not be skipped over.

Heart rate was observed to be both valid and reliable in all devices during the walking bout. When wrist-worn devices (Garmin Instinct, Polar Vantage M2) are considered, the finding did not extend to activities of greater relative intensity, being considered neither valid nor reliable during jogging and overground skipping. This finding aligns with a study using a different wrist-based device, the Garmin fēnix 5, which reported moderate validity measures during sitting and walking, but poor validity measures during increased intensities of exercise (Duking et al., 2020). However, an earlier Polar wrist-worn model, the Polar Vantage V, was reported to have acceptable MAPE values for heart rate during running and cycling at low and high self-selected intensities (Hajj-Boutros, Landry-Duval, Comtois, Gouspillou, & Karelis, 2023). The specific wrist-based devices used in the current investigation have little published literature outside of conference abstracts, many of which are from our laboratory group. The same holds true for one of the biceps-worn devices, the Polar Verity Sense, in that published validation literature currently exists only in abstract form (Bodell et al., 2021). The other biceps-worn device, the Polar OH1, is reported to have acceptable validity during treadmill (MAPE between 0.2 and 1.9%) and cycle exercise (MAPE between 0.6 and 3.9%) (Muggeridge et al., 2021), spin bike activities (mean bias less than 1 bpm) (Hettiarachchi, Hanoun, Nahavandi, & Nahavandi, 2019), front crawl swimming at various intensities (ICC between 0.72 and 0.96)(Olstad & Zinner, 2020), and endurance sports (difference from criterion < 5%), as well as acceptable reliability during endurance sports (ICC = 0.99) (Hermand, Cassirame, Ennequin, & Hue, 2019). Based on the results of the current investigation, biceps-worn devices may be preferrable for individuals who are interested in obtaining both valid and reliable heart rate measures in non-traditional activities such as overground skipping.

It has been noted that estimated energy expenditure in wearable devices during exercise validation studies is generally poor (Bunn et al., 2018). As described with heart rate above, the specific devices used in the current investigation have not been reported in the published literature, and the same holds true for energy expenditure. The existing literature details opportunities for improvement when energy expenditure is considered. Physical activity energy expenditure in wearable devices including the Garmin vívofit, Jawbone Up24, and Fitbit Flex, compared to doubly labeled water were deemed unacceptable (MAPE range for 12 devices 19.4% to 100.2%) (Murakami et al., 2019). Estimated energy expenditure returned from the Garmin vivo HR+ and Fitbit Charge 2 were evaluated against COSMED units (K4, K5) during progressive exercise tests to volitional fatigue (treadmill, cycle ergometer) and MAPE did not meet validity threshold (18.9% to 43.5%) (Reddy et al., 2018). During trail running, the Hexoskin biometric shirt did not return correlated energy expenditure measures compared to the COSMED K4 (r = -0.058) (Tanner et al., 2016). The Apple watch 6 (14.9% to 47.8% MAPE), Polar Vantage V (15.6% to 34.6% MAPE), and Fitbit Sense (17.8% to 45.1% MAPE) all displayed unacceptable MAPE during walking, running, resistance exercises, and cycling (Hajj-Boutros et al., 2023). The current investigation adds to the volume of literature, in that estimated energy expenditure obtained from Garmin Instinct and Polar Vantage M2 wrist-worn devices during self-paced walking, jogging, and overground skipping are similarly poor when validity measures are considered. While the current investigation reports the Polar Vantage M2 to satisfy reliability thresholds during walking and jogging, this is of little consolation if the validity assumptions are violated.

The current investigation is not without limitations or hurdles, which are inherent in wearable technology validation studies (Navalta & Bunn, 2023). As mentioned previously, the predetermined definition for what constitutes a step during the overground skipping motion likely led to the device returning poor validation measures. Future studies employing overground skipping while obtaining step count measures should take this factor into account. Another limitation is that the order of the exercise bouts was constant, with walking first, then jogging, and ending with overground skipping. Because the intent of the study was to obtain concurrent validity and reliability measures, it is believed that this approach is acceptable. However, it is unknown whether scheduling the overground skipping trial last affected mechanics of the skill, and whether potential fatigue may have induced extraneous movements that decreased validity and or reliability measurements in certain devices. Future studies employing various exercise modalities should not skip over randomizing the order of exercise bouts.

In conclusion, the modality of overground skipping is an area of research that represents boundless possibilities. In the context of the current investigation, several findings are reported on specific wearable devices for the first time. Regarding heart rate, the Garmin Instinct, Polar Vantage M2, Polar Verity Sense, and Polar OH1 met all thresholds to be considered both valid and reliable during self-paced walking. Only the biceps-worn devices (Polar OH1, Polar Verity Sense) met the heart rate thresholds for self-paced jogging and overground skipping. The Garmin Instinct was the only device to return step count, and it was neither valid nor reliable during overground skipping according to our predetermined definition of a step. While the Polar Vantage M2 was reliable for energy expenditure during walking and jogging, wrist-worn devices (Garmin Instinct, Polar Vantage M2) were neither valid nor reliable in returning estimated energy expenditure during overground skipping. From a wider perspective, wearable device algorithms for estimating energy expenditure should continue to be refined until they return the same level of accuracy as what is currently observed for heart rate, and to a lesser extent step count. Overground skipping may be an excellent unusual exercise for the continued testing of these measures as leaps and bounds are made in the development wearable activity devices.

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Conflict of Interest
The authors declare that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Early Gains in Motor Learning Measured Through Two Coordination Tests: A Retrospective Analysis of Gender Differences

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Abstract

Motor skills can be improved through rapid on-the-job training or slower multi-session learning. The objective of this study was to determine the rapid learning differences between male and female university students during the execution of two motor coordination tests. Available data from 716 female and 331 male college students were retrospectively analyzed. The female participants had a mean age (±SD) of 19.6 (±1.55) years, while the male participants recorded a mean age of 19.8 (±1.87) years. Data were collected using two motor coordination tests, each performed in triplicate. The statistical method used in this analysis was mixed-model ANOVA. The interaction effect of gender and number of attempts was statistically significant for both motor coordination tests ($F=12.446$; $p<0.01$; $\eta^2p=0.13$ & $F=11.169$; $p<0.01$; $\eta^2p=0.01$). Post-hoc testing showed that males performed better at the tasks in all three runs, and both genders improved their performance in subsequent trials. However, females showed a larger relative improvement from trial to trial than did males. The two coordination tests yield similar results. The observed differences in improvements in the coordination tests may be attributed to different motor learning strategies and cognitive processing between the sexes.

Keywords: *motor adaptation, gross motor ability, neuroplasticity*

Introduction

Motor learning is defined as a set of processes associated with practice that leads to relatively permanent changes in the capability for motor response (Schmidt, 1988). Acquiring new motor skills necessitates learning the process, which is associated with the activity of different brain regions, and having the ability to transfer accomplished movements to new conditions and task variants (Seidler, 2010). Such a hierarchical organization of motor control means that no movement is performed in a completely identical manner (Profeta & Turvey, 2018). Outcomes of the motor learning process can be derived from the progress, stability, consistency, and adaptability observed in motor performance. When motor performance embodies all these qualities, it exhibits the characteristics of a well-honed motor skill (Magill & Anderson, 2013). Coordination, which rests at the core of every movement, is a complex, multistructural, and qualitative motor ability. It is influenced by the mechanism for the regulation of movement, that is, its subordinate mechanism for structuring movement, and it permeates almost all motor abilities (Bruton & O'Dwyer, 2018). A high level of coordination allows movements to be controlled and adjusted in real-time to meet performance goals, and it is achieved by triggering a motor program developed based on previously assimilated abilities (Iorga, Jianu, Gheorghiu, Crețu, & Eremia, 2023). The more developed coordination a person has, the better and more successful he will be in sequencing movements or actions, i.e., his ability of motor planning will be at a higher level, which will then help solve motor tasks or problems (Kimura, Yokozawa, & Ozaki, 2021). Some of the factors influencing the acquisition of new motor skills include: 1) relevant prior knowledge and skills; 2) attentiveness, concentration, and distractibility; 3) interests and acquired pref-

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erences; 4) motivation and competitiveness; 5) self-confidence and optimism; 6) other aspects of temperament and personality; 7) enthusiasm and energy level; 8) fatigue and anxiety (Howe, Davidson, & Sloboda, 1998).

Differences in coordination between genders are noticeable from early childhood (Adriyani, Iskandar, & Camelia, 2020; Morley, Till, Ogilvie, & Turner, 2015). Males outperform females in fundamental motor skills during young adulthood (Díaz, Rojas, & Morera, 2015). Movements involving precision, which require coordination, are more strongly genetically determined in women, whereas movements involving strength and speed are more genetically determined in men (Iorga et al., 2023). Significant differences between men and women were found in accuracy (Moreno-Briseño, Díaz, Campos-Romo, & Fernandez-Ruiz, 2010), but there were no significant differences in the phase of adaptation between the sexes. This suggests that learning mechanisms may contribute differently in men and women. Motor learning consists of two types. Fast learning is defined as motor knowledge acquired through a single training session or activity, whereas slow motor learning is achieved through consecutive training sessions over time (Costa, Cohen, & Nicolelis, 2004). A recent systematic review highlighted the need for research on motor skill acquisition to develop new strategies for motor learning (Newell, 2020). Sex-based differences in motor skills have been the subject of numerous studies (Díaz et al., 2015; Gromeier, Koester, & Schack, 2017; Kokštejn, Musalek, & Tufano, 2017; Dinkel & Snyder, 2020; Zheng, Ye, Korivi, Liu, & Hong, 2022), however the nuances of fast motor learning, especially among young adults, remain relatively unexplored. Many studies have concentrated on preschool or school children, focusing on motor skill differences but not specifically on the rapid acquisition of these skills. While differences in both gross and fine motor skills have been identified (Rodrigues, Ribeiro, Barros, Lopes, & Sousa, 2019), scant evidence on how minor disparities in childhood may influence the acquisition of motor skills in young adulthood. For such a population, it is essential to account for both biological and environmental differences, the latter stemming from diverse societal expectations for boys and girls. Furthermore, a significant portion of existing research does not clearly distinguish between fast and slow motor learning, often broadening its scope to include a range of age groups. The importance of understanding fast motor learning becomes evident when considering its applications in athletic training, rehabilitation, or daily activities requiring swift adaptability. This is particularly relevant for young adults, such as university students, who frequently face new motor challenges. However, empirical insights on gender-specific teaching methodologies for this age group are limited. Addressing this gap, this study delves into the differences in fast motor learning between male and female university students. The goal is to inform about how motor skills are taught across genders, while also investigating the inherent neuromotor differences. If there are neuromotor differences in fast motor acquisition between sexes, such differences could have exists an impact on how males and females are taught motor skills to accommodate different strategies. This study explored the sex-based differences in coordination improvement using repeated motor tests. It operates on the premise that the repeated administration of these tests, typically three times, triggers short-term motor learning, subsequently affecting test scores by influencing their improvement rates. Therefore, the main goal of this study was to determine the fast-learning differences between male and female university students when performing two motor coordination tests.

Methods

Participants

This study was conducted retrospectively using motor coordination test data routinely collected during physical education classes at the university level over the period of ten years. The results of two tests measuring motor skills (crawling and jumping over obstacles and the backward polygon test) were processed to determine whether there was a sex difference in polygon performance speed in each attempt depending on gender.

The sample consisted of 716 female and 331 male first-year physiotherapy students. The mean age (±SD) for the female participants was 19.6 (±1.55) years, while for the male participants, the mean age was 19.8 (±1.87) years. The use of retrospective data was reviewed and approved by the Ethics Committee of the university where the data were collected (KL:602-03/22-18/386; URBR:251-379-10-22-02). Mean male and female participants' height was 181.33 (±7.04) and 165.87 (±17.0) cm respectively, while average mass was 80.5 (±13.3) and 62.6 (±12.7) kg, respectively. Participants with physical or mental disabilities that could potentially influence the test results were excluded. Students who were professional athletes were excluded from the study.

Procedure

Before any of the tests, the students warmed up properly and a demonstration of both tests was provided. Two experienced researchers, a kinesiologist and physiotherapist, conducted the tests. A hand stopwatch was used to monitor test timing. Two standardized coordination tests used in student population at elementary, high school and university levels with high levels of reliability, homogeneity and sensitivity were conducted – MBKPOP (crawling and jumping over obstacles) and MREPOL (backwards polygon test) (Neljak, Sporiš, Višković, & Markuš, 2012). The values of the test reliability level are satisfactory under the assumption that the reliability of 0.90 is satisfactory (Hopkins, 2000).

Measurements

The first test (MBKPOP) consisted of four obstacles spaced equidistantly at a distance of 10 m. The participants were required to jump over the 1st and the 3rd obstacle, crawl under the 2nd and the 4th obstacle, turn back, and then jump over and crawl back to the starting line.

The second test (MREPOL) consisted of two obstacles, the 1st one at 3 m and the 2nd one at 6 m from the starting point. The participants had to scuttle backward with their hands and feet, move across the 1st obstacle and under the 2nd obstacle, and then scuttle back 3 m to the finish line. The goal of both tests was to run an obstacle course as quickly as possible.

The participants performed both tests three times; they were given verbal instructions and visual demonstrations but were not permitted a trial run. The participants had 5 min between test attempts to rest.

Statistical analysis

Before proceeding with formal statistical analysis, the assumptions of normality of residuals, homoscedasticity, sphericity, and absence of extreme outliers were verified. A mixed-model ANOVA was used to examine differences in coordination. After analysis of the interaction model, post-hoc paired sample t-tests were conducted to compare test attempts segregated by sex. To quantify the effect size, both the partial eta-squared and Cohen's d tests were used. Cohen's guidelines were used to interpret effect size (Cohen, 1988). Bootstrapping was used to simulate a 95% confidence interval for Cohen's d. To adjust for possible type I error inflation, the Benjamini-Hochberg correction was applied to p values (Benjamini & Hochberg, 1995). The probability of a type I error was set at 5% ($p<0.05$). All analyses were performed using R statistical software, version 4.2.1 developed by R Core Team and the R Foundation for Statistical Computing.

Results

Table 1 presents the data from both motor coordination tests for all three attempts in both sexes. The MBKPOP test was performed on 662 females and 316 males, and the MREPOL test was performed on 716 females and 331 males. This difference in test sample size was due to erroneous testing and time constraints. A mixed-model ANOVA (Table 2) showed a significant interaction between gender and test attempts (Figure 1 and 2). According to the partial eta-squared statistic, the effect size was medium ($\eta^2p=0.13$) for the MBKPOP test and

Table 1. Descriptive statistics of coordination tests

Variables	$\bar{x}$	SD	Min	Max	n
Females					
MBKPOP 1	19.24	3.85	8.25	32.8	662
MBKPOP 2	17.53	3.41	8.63	32.05	
MBKPOP 3	17.04	3.48	9.05	30.06	
MREPOL 1	12.58	2.76	6.55	22	716
MREPOL 2	11.68	2.47	6.1	21.5	
MREPOL 3	11.42	2.36	6.37	19.78	
Males					
MBKPOP 1	15.88	2.9	9.08	27.4	316
MBKPOP 2	14.65	2.87	9.22	25.6	
MBKPOP 3	14.25	2.98	7.64	24.2	
MREPOL 1	9.3	1.86	5.92	15.02	331
MREPOL 2	8.69	1.77	5.43	16.23	
MREPOL 3	8.52	1.74	8.33	14	

Note. MBKPOP: crawling and jumping over obstacles; MREPOL: backward polygon test. x: mean; SD: standard deviation; n: sample size

Table 2. Mixed-model ANOVA

	Test attempt			Gender			Interaction		
	F	p	η^2p	F	p	η^2p	F	p	η^2p
MBKPOP	453.42	<0.01†	0.317	190.96	<0.01†	0.164	12.446	<0.01†	0.13
MREPOL	246.29	<0.01†	0.189	450.56	<0.01†	0.302	11.169	<0.01†	0.01

Note. MBKPOP: crawling and jumping over obstacles; MREPOL: backward polygon test. F: F ratio; η2p: partial eta squared (effect size); †p < 0.05

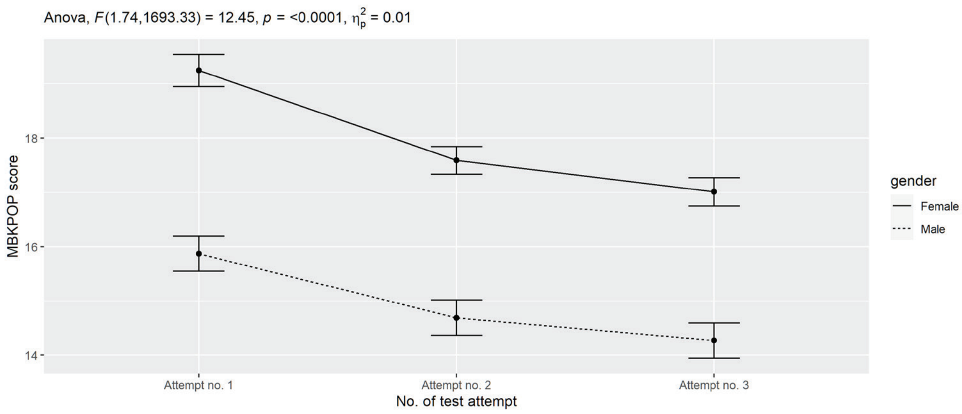

FIGURE 1. Interaction plot of MBKPOP (crawling and jumping over obstacles) motor test
Note: Error bars represent 95% confidence intervals; F: F ratio; η^2p: partial eta squared (effect size)

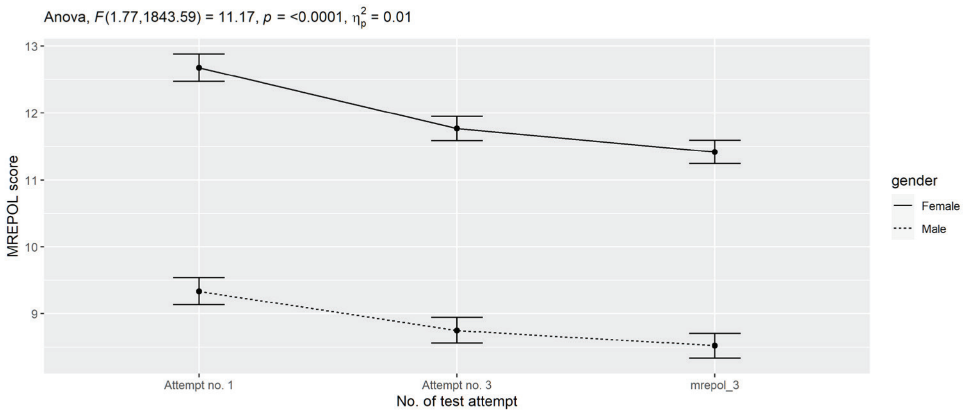

FIGURE 2. Interaction plot of MREPOL (backwards polygon test) motor test
Note: Error bars represent 95% confidence intervals; F: F ratio; η^2p: partial eta squared (effect size)

small (η^2p=0.01) for the MREPOL test. Post-hoc paired t-tests (Table 3) showed that female and male participants improved the MBKPOP and MREPOL test scores in subsequent tests.

For females, Cohen's d showed that the effect size of the difference between the first and second MBKPOP tests was large (d=0.86), and the difference between the second and third tests was small (d=0.36). The effect size of the difference between the first and second MREPOL tests was medium (d=0.57) and the difference between the second and third tests was small (d=0.30), similar to the previous test.

Table 3. Post-hoc paired t-test

	Female				Male			
	t	p	d	95% CI	t	p	d	95% CI
MBKPOP 1-2	22.0	<0.01†	0.857	0.76; 0.95	12.4	<0.01†	0.7	0.55; 0.88
MBKPOP 2-3	9.35	<0.01†	0.364	0.28; 0.45	4.41	<0.01†	0.249	0.13; 0.4
MREPOL 1-2	15.3	<0.01†	0.571	0.5; 0.66	8.43	<0.01†	0.464	0.33; 0.61
MREPOL 2-3	7.88	<0.01†	0.295	0.22; 0.37	3.52	<0.01†	0.194	0.09; 0.3

Note. MBKPOP: crawling and jumping over obstacles; MREPOL: backward polygon test. t: t – statistic; d: Cohen's d (effect size); 95% CI:95% confidence interval; †p< 0.05

For males, the effect size between the first and second MBKPOP tests was medium (d=0.70), and the difference between the second and third tests was small (d=0.25). The effect size of the difference between the first and second tests was medium (d=0.46), and the difference between the second and third tests was virtually negligible (d=0.19).

Discussion

This study demonstrated a sex-based difference in the progression of test scores; males showed greater progress between the first and second measurements, whereas females displayed consistent progress across all three measurements.

The initial phase of motor skill learning typically develops quickly across various experimental procedures and then slows as additional demands are placed in subsequent sessions. This pattern has also been observed in children older than 10 years who begin displaying adult-like learning patterns (Adi-Japha, Berke, Shaya, & Julius, 2019). Motor learning is largely driven by practice, and the acquisition of new skills is reflected in novel combinations of muscle activation that improve performance. The absence of performance improvement through repetition indicates the retention phase of motor skills (Constantino Coledam, 2020). Adapting to a new motor task requires appropriate actions in response to incoming information, such as the environment (e.g., a moving target) or sensor inputs from the body. Bianco et al. (Bianco et al., 2020) suggested that females tend to favor cautious cognitive processing, while males lean towards a more reactive and faster cognitive process.

From this perspective, it can be inferred that men employed their existing motor skills and capabilities to perform optimally during their initial attempts, leaving minimal room for substantial progress in subsequent measurements. In contrast, female participants approached the tests more cautiously, relying on learning from each run, thereby achieving relatively more improvements across runs.

There is little experimental data concerning the differences in motor learning between genders especially in young adult population. One study comparing children aged 7 and 8 years old using Movement Assessment Battery for Children Test found difference in acquisition of motor skills in subtest scores concerning manual dexterity and ball skills but no difference in total impairment and balance scores (Junaid & Fellowes, 2006). Another study comparing adults averagely aged 39.2 (SD=13.5) found expected difference in throwing accuracy between genders, no difference in adaptation to constraints imposed by the researchers, but it showed that women had retained larger after effects adaptation after experimental constraints were removed, indicating that females exhibit a deeper internal recalibration of their motor-visual relationship in this specific task (Moreno-Briseño et al., 2010). With regards to our study this implies that while men might have an initial performance advantage due to factors like strength or prior exposure, women seem to exhibit a faster rate of improvement, possibly due to a combination of deeper cognitive engagement, a propensity towards spatial alignment in motor learning, and the specific demands of the coordination tests. This difference in learning strategy and adaptation showcases the importance of individualized training approaches and the potential to harness these inherent strengths across genders.

There are more studies examining gender differences in coordination in general, but they have mixed results. For instance, (Lyakh, 2021) in a study comparing 90 parameters of motor coordination, found negligible sex differences in most of the parameters among kickboxing and taekwondo athletes aged 18-27 years. However, Stelmach et al. (Stelmach, Rydzik, & Ambroży, 2020) found that male swimmers aged 14-16 years demonstrated higher coordination efficiency than their female counterparts. The findings of the current study revealed significant differences in the coordination tests between male and female students, echoing the results of Stelmach et al.'s study, although their focus was on adolescents. It is posited that physiological factors such as strength and training years could contribute to higher coordination efficiency in males than in females. This study draws a comparison with a student group aged 9-10 years, considering the absence of similar studies on young individuals.

This study offers a unique contribution to the assessment of the student population by bridging the gap in existing studies. Variations in repeated motor task approaches are likely best elucidated by neuromotor and physiological differences, as supported by research conducted by (Roivainen, Suokas, & Saari, 2021).

Limitations and future recommendations

It was challenging to ascertain bias in this research based on the timing of data collection. Nonetheless, the findings will undoubtedly contribute to a better understanding of the sex-based differences in coordination between students. However, it is crucial to recognize the inherent limitations of this study. Considering the results of previous research, anthropometric characteristics, and data on prior involvement in any form of nonprofessional athletic activities of students were not included in this research. In future research, it would be beneficial to consider the anthropometric measurements of participants and their previous participation in athletic activities. Future research should include more tests that cover coordination and retention measurements and additional tests that indicate changes in movement efficiency, such as the analysis of oxygen consumption, heart rate, or muscle activity. In addition, a lack of age-appropriate tasks was noticed, which warrants designing age-appropriate tasks. It is also suggested that participants be given a brief questionnaire as a manipulation check, which would help assess their comprehensive understanding of the coordination task, level of focus during the performance, emotional state (e.g., potential fear of increasing speed), and overall motivation.

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Conflict of Interest
The authors declare that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

The Effects of Student-centred Learning Methods and Motivational Climate on Dance Learning

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Abstract

Student-centred pedagogy is based on constructivist and democratic principles where students are encouraged to develop their reflective and critical thinking and sense of responsibility. The objectives of this study was to: (1) determine which of the three student-centred teaching methods (individual, couple or team) will best influence students in learning dances; (2) analyse the impact of motivational climate on different student-centred teaching methods during the learning of the dances among female and male students. The learning of dance structures from the Adriatic dance zone has been analysed because of its applicability to the Physical Education curricula. The pool of subjects consisted of 30 female and 28 male kinesiology students (aged 21–24), divided into three groups according to their preferred learning methods: individual, couple and team. Three experts evaluated the performance of dancers according to the criteria determined beforehand, in detail. For the assessment of motivational climate in Physical Education, Motivational Climate on Physical Education Scale (MCPES) scale was used with subscales: Autonomy, Social relatedness, Task involving climate and Ego involving climate. According to a two-way ANOVA with the independent variables of gender and teaching methods, and Tukey post-hoc test, the most successful students preferred the couple method in learning linđo, and individual method in learning quatro passi. Significant differences were found in Task climate subscale between the couple and team-teaching method. Further investigation of teaching methods in PE classes is needed to confirm optional student-centred method defined by gender, age, and the type of motor skills.

Keywords: *Adriatic dance zone, gender differences, preferred teaching methods*

Introduction

The implementation of dance structures in PE classes can effectively introduced schoolchildren to the creativity-enhancing proces (Neville & Makopoulou, 2021), which is in accordance with modern society, and the education which fosters lifelong exercise for health and the quality of life. Dance, as art in motion and stage art, is a physical activity with numerous advantages in educational process both among children and youth. The implementation of dance structures in PE curricula has a positive effect on the development of both motor and health status, and can develop students' creativity, sensitivity, emotions, originality, inclusion, and understanding between the genders (Stivaktaki, Mountakis & Bournelli, 2010; Mattsson & Lundvall, 2015; Gibbs, Quennerstedt & Larsson, 2017; Neville & Makopoulou, 2021; Mattsson & Larsson, 2021). Effective learning and teaching are a constant goal of the pedagogues and educators at all educational levels. Two basic approaches in which educational goals could be reached are teacher-centred or student-centred approaches (Serin, 2018; Matsuyama, et al., 2019). Teacher-centredness refers to the traditional communication of knowledge to students in a learning environment, in which the teacher has the primary responsibility (Mascolo, 2009), when knowledge is solely disseminating from a teacher to a student. Student-centredness strives to develop critical thinking and problem-solving instructions among active students in which they expand their skills and understanding.

Teaching styles are extensively investigated in PE classes (Jaakkola, & Watt, 2011; Kolovelonis, Goudas & Gerodimos, 2011; Chatoupis, & Vagenas, 2018; Karlefors & Larsson, 2018; Fernández & Espada, 2021.), but only a few studies have an-

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alysed the dance teaching styles in PE (Pitsi, Digelidis & Papaioannou, 2015; Cuellar-Moreno, 2016). Therefore, available information analysing the effects of different student-centred teaching styles during both the learning and evaluating dances in PE classes is limited. Teacher–learner relationship in the learning process could be observed and analysed through the reproductive and productive categories. The difference between these two category styles is marked by the share of learner decision-making in classes. According to previous studies, command style, as teacher-centred method, and practice and inclusion styles, as student-centred method, are mostly used in PE classes (Chatoupis, 2018; Fernández & Espada, 2021).

The motivational climate in PE classes affects motivation, personal experience of students, and their attitudes towards physical activity (Moreno, Jiménez, Gil, Aspano & Torrero, 2011; Gil-Arias, Claver, Práxedes, Villar & Harvey, 2020). The authors Soini, Liukkonen, Watt, Yli-Piipari and Jaakkola (2014) developed the tool for assessing the motivational climate in Physical Education. In the process of developing a Motivational Climate on the Physical Education Scale (MCPES scale) the authors used four dimensions that presented social relatedness, perceived autonomy, and task- and ego-involving climate (Standage, Duda & Ntoumanis, 2003; Liukkonen, Barkoukis, Watt & Jaakkola, 2010.).

The teaching goals while learning dances can be achieved with different approaches, learning styles and methods. But question is which kind of learning method or learning environment is the most effective and in the same time the most acceptable for students. The assumption is that the active participation of students in class will encourage their critical reflection and independent problem solving. Especially, the content of dances in PE classes has a prominent place in the development of student's creativity (Neville & Makopoulou, 2021), and inclusiveness. Also, for a dance, learned in PE, to have a lifelong effect, learning method that contribute most to a motivating and positive environment should be put in the focus of modern teaching. Therefore, it is important to gain insight in usage of different student-centred teaching methods during learning dances and ensure optimal motivational climate for realization educational goals.

The objectives of this study are to: (1) determine which of the three student-centred teaching methods (individual, couple, or team) will best influence students in the learning of the dances; (2) analyse the impact of motivational climate on different student-centred teaching methods during the learning of the dances among female and male students.

Methods

Participants

The research was conducted on a sample of 30 female and 28 male subjects, third-year male and female students of kinesiology aged 21 to 24, studying full-time at the Undergraduate Study of Kinesiology, the Faculty of Kinesiology, the University of Split, taking the course named Theory and Methods in Dances.

Study protocol

At the beginning of the learning process, the method of teaching, and the evaluation of dances were chosen independently, by the students. According to their choice three study groups were formed: a) Individual (N=7; 5 female students, and 2 male students); b) Couple (N=17; 10 female students, and 7 male students); c) Team (N=34; 15 female students, and 19 male students). Before the beginning of the study, it was determined whether the students had already had any previous experience in folk dances, and those having such experience did not participate in the study. Then, the participants from all groups were given basic information and instructions regarding: a) basic information on the study and the study objectives, b) basic information on the questionnaire being conducted and the way to fill out the questionnaire, as well as anonymity in the interpretation of results. The participation in the study was voluntary, and the written consent was obtained.

Two dances from the Adriatic dance zone (linđo from Dubrovnik and quatro passi form the island of Korčula) have been chosen for this investigation because of their applicability for Physical Education curricula. Uniform evaluation criteria were set prior to the performance and evaluation of the subjects, without the possibility of additional agreements or mutual consulting (Božanić & Miletić, 2012; Grčić, Miletić & Kuzmanić, 2015). In the first phase of the experiment, basic steps and movements were demonstrated and explained to all students in the same way. Then each student was asked to decide on how they wanted to proceed both with the dance practicing, and during the evaluation of the after-learning process. Three different teaching models of dance training were offered to students. The students who chose the individual teaching method were oriented towards the independent practice of dance steps, and they were evaluated during independent performance. The students who chose the couple teaching method were instructed to practice dance steps in couples, and they were evaluated during couples' performance. The students who chose the team-teaching method were instructed to practice dance steps in groups of five to eight students, and they were evaluated during team performance. Regardless of which learning method was implemented, the judges always evaluated the students individually and according to pre-defined, equal criteria. The basic dance steps in all teaching methods (individual, couple, or team) were the same. An overview of preferred teaching styles is presented in Graph 1. In the second phase, the students in all three groups were continuing to learn dances firstly in slower rhythm followed by counting, than in regular rhythm followed by counting. Then the dances were demonstrated to students followed by original music, when student where connecting basic dance parts in final choreography. Students' were encouraged to practice final choreography in their preferred teaching methods: individual, couple or team).

Measurement

The assessment was carried out by three independent experts in order to determine the objectivity of the tests for evaluating dance structures, in accordance with a previous study conducted on a student population (Božanić & Miletić, 2011; Castillo & Espinosa, 2014, Miletić 2022). The evaluation was implemented by breaking down the traditional ethno-choreography into basic parts, and consequently evaluating each dance part independently by grading it with mark 2 if performed correctly, mark 1 if performed partially correctly, and mark 0 if performed incorrectly (Božanić & Miletić, 2011; Miletić 2022). Final maximal score for each choreography was 10 points.

Motivational Climate on Physical Education Scale (MCPES) questionnaire was implemented according to a study of Soini, Liukkonen, Watt, Yli-Piipari and Jaakkola (2014), in which the questionnaire scales were reduced from 45 to 18 in validation, the latter determined by 4 factors: Autonomy factor; Social relatedness factor, Task involving climate factor and Ego involving climate factor. The five items from the Autonomy dimension represent a chance to choose among different activities in a PE lesson, and examine the opportunities that PE provides to support students' independence, free choices, and the extent to which they can intervene in shaping a lesson (Topatsi et al., 2022). The five items from the Task-involvement dimension represent effort and self-improvement, and examine the participant's effort regarding personal improvement, and the perception that mistakes are a part of the learning process. The four items from the Ego-involvement dimension represent normative comparison and examine the presence of competitive climate in the lessons, and the sense of superiority over classmates. The four items from the Social relatedness dimension represent the students' unity in PE classes, and explore the development of team spirit, unity, and collaboration between the students to resolve difficult situations during a lesson. According to Soini, Liukkonen, Watt, Yli-Piipari and Jaakkola (2014), the theoretical contents of the dimensions of motivation climate were as follows: (1) Task-involving climate – trying one's best, mistakes were seen as part of the learning process, learning new things, progress in one's own skills; (2) Ego-involving climate – competing in relation to others, comparison in relation to others, showing superiority in relation to others, importance of succeeding more than others, normative comparison; (3) Autonomy support freedom – to make choices, possibility to make choices, possibility to affect the way PE lessons are run, possibility to affect common issues; (4) Social relatedness – supported working together as a team, cohesion during classes, pulling together, class united when practicing. Each item was rated on a five-point Likert-type scale ranging from 1 (strongly disagree) to 5 (totally agree). This study was approved by the Ethical Board of the Faculty of Kinesiology, the University of Split (approval number: 2181-205-02-05-23-006, 15, February, 2023).

Statistics

Methods for data analysis were selected according to the aims of the research. In order to determine the influence of different student-centred learning methods on successful dance performance, and to analyse the influence of a motivational climate on dance performance, especially according to gender, a two-way analysis of variance (two-way ANOVA) was used with the independent variables of: a) gender (male and female students), and b) teaching methods (individual, couple, team). The Tukey post-hoc test was used to determine the significant differences among means. For analysing the objectivity of dance experts, and in order to assess the internal consistency of the MCPES subscales, Cronbach's alpha was calculated. Statistica 13.0 (TIBCO Software Inc, USA) was used for all analyses and a p-level of 95% was applied.

Results

According to students' preferred teaching styles presented in Fig. 1, most of the students (58.62%), chose team-teaching style. Second preferred teaching style is couple (29.31%), and the fewest percentage of students chose individual teaching style (12.97%). Male students will often choose team-teaching style when learning dances, while female students more often choose individual and couple teaching style.

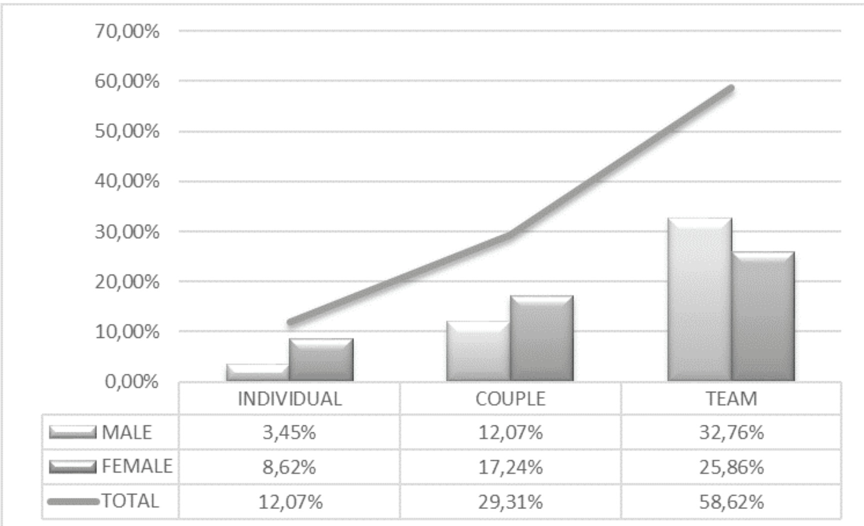

FIG 1. Student preferences in teaching methods (in percentages)

Results of Two-way analysis of variance (two-way ANOVA) with the independent variables of gender and teaching methods and Tukey Post Hoc Test (Table 1 and Table 2) were presented according to Pitsi, Digelidis & Papaioannou, (2015).

Table 1. Internal Consistency of the scales

	α-Cronbach
Task climate (I)	.73
Social relatedness (II)	.81
Autonomy (III)	.72
Ego Climate (IV)	.75
Linđo dance (L)	.98
Quatro passi dance (Q)	.98

Table 2. Mean values and standard deviations (M ± SD) of the three teaching methodes groups (males and females separately) in MCPES subscales (I, II, III and IV) and dance performance (L and Q) and main results of Two-way analysis of variance (two-way ANOVA) with the independent variables of gender and teaching methods and Tukey Post Hoc Test.

	Individual			Couple			Team			Total N=58
	male n=2	female n=5	total n=7	male n=7	female n=10	total n=17	male n=19	female n=15	total n=34	male n=28 female n=30
I	2.90±0.14	2.72±0.67	2.77±0.56	2.94±0.22	3.12±0.54b	3.05±0.44a	2.73±0.55	2.21±0.81	2.50±0.71	2.79±0.47 2.60±0.80
II	3.88±0.88	3.95±0.54	3.93±0.57	4.00±0.65	4.35±0.56	4.21±0.60	3.95±0.72	3.83±0.89	3.90±0.79	3.96±0.68 4.03±0.76
III	3.70±0.71	4.48±0.30	4.26±0.54	4.09±0.38	4.52±0.29	4.34±0.39	4.15±056	4.37±0.64	4.25±0.60	4.10±0.52 4.44±0.49f
IV	2.25±1.06	3.05±0.27	2.82±0.62	2.54±0.73	2.35±0.83	2.43±0.77	2.57±0.69	2.22±0.85	2.41±0.77	2.54±0.69 2.40±0.82
L	9.00±0.47	7.80±1.76	8.14±1.56	9.33±0.88de	8.37±1.49	8.76±1.33c	6.47±2.58	5.93±2.20	6.24±2.40	7.37±2.53 7.06±2.19
Q	9.17±0.24	8.73±1.59	8.86±1.32c	8.29±1.33	7.40±1.71	7.76±1.58	7.14±1.52	7.04±1.66	7.10±1.56	7.57±1.55 7.44±1.72

a: p<0.05 from group total, b: p<0.05 from group females, c: p<0.05 from group total, d: p<0.05 from group male, e: p<0.05 from group female, f: p<0.05 from males total.; I - MCPES subscale Task climate (rang 1-5); II - MCPES subscale Social relatedness (rang 1-5); III - MCPES subscale Autonomy (rang 1-5); IV - MCPES subscale Ego climate (rang 1-5); L - Linđo folk dance performance (rang 0-10); Q - Quatro passi folk dance performance (rang 0-10)

Internal consistency reliability presented with Cronbach alpha coefficients, for the four MCSPES sub-scales (Table 1) were all above 0.70 and ranged from 0.72 (Autonomy) to 0.81 (Social relatedness). According to the values of Cronbach's alpha, calculated for assessing reliability for linđo and quatro passi dances variables, both tests have shown good measuring characteristics of objectivity (0.98) which indicated clearly determined and transparent criteria of the evaluation of dance structures. The highest mean scores (Table 2) of the participants scored on the MCPES sub-scales, for all three groups (individual, couple, team), have been found for the Autonomy dimension, and the lowest mean scores have been found for the Task climate (team) and Ego climate factor (couple and team) in total sample of subjects. According to the Two-way analysis of variance, the Autonomy support dimension had statistical significance F(1.52)=7.354 (p=0.009) for the gender factor. According to the Tukey post-hoc test, significant differences were obtained between male and female students from the overall sample (p=0.015). For the teaching method factor, according to the Two-way analysis of variance, the Task climate dimension had statistical significance F(2,.52)=4.855 (p=0.012). Tukey post-hoc test shows significant differences between couple- and team research group (p=0.011) in favour of the couple group. Furthermore, Tukey post-hoc test shows significant differences between the female couple group and female team group in favour of the female couple group (p=0.008).

In performing linđo dance, the results have shown a significant interaction with the teaching method factor F(2,52)=9.998 (p=0.000). According to the results of the Tukey post-hoc test, significant differences have been found between the total sample of subjects in couple- and team research groups (p=0.034) in favour of the couple group; and in male couple- and female team group (p=0.010) in favour of the male couple group. The differences between the teaching methods in linđo dance performance, separately for each gender, are presented in Fig 2.

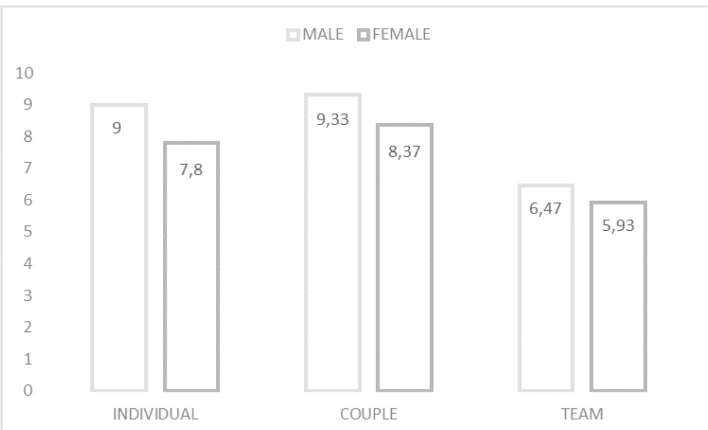

FIG 2. Differences between teaching methods in Linđo dance performance, separately by gender

In performing the quatro passi dance, the results have shown a significant interaction with the teaching method factor F(2,52)=3.985 (p=0.025). According to the results of the Tukey post-hoc test, significant differences have been found between the total sample of subjects in individual- and team research groups (p=0.024) in favour of the individual group. The differences between the teaching methods in quatro passi dance performance, separately for each gender, are presented in Fig 3.

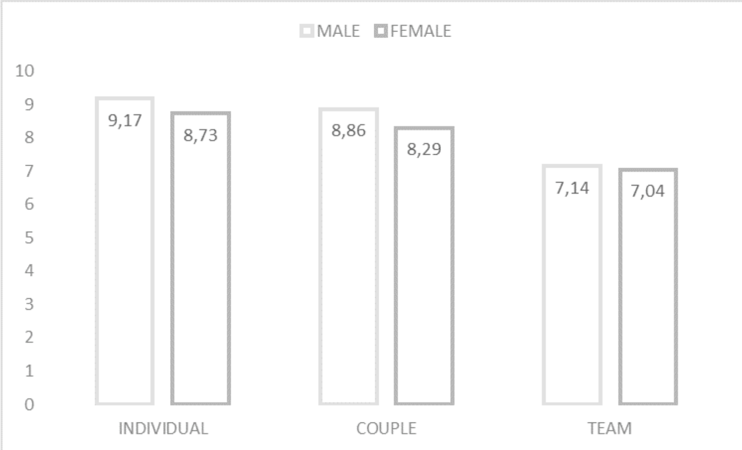

FIG 3. Differences between teaching methods in Quatro passi dance performance, separately by gender

Discussion

Current study has confirmed that the determined criteria for the evaluation of linđo and quatro passi dances were an adequate tool for the objective evaluation of dance skills among beginners. According to the Two-way analysis of variance (two-way ANOVA) with the independent variables of gender and teaching methods and Tukey Post Hoc Test results obtained in this study, only the subscales of the MCPES questionnaire, Task climate, and Autonomy factor have significant roles in the teaching of the dances.

The student-centred learning method in PE assumes an environment in which students can express their preferences and make their own decision regarding creation of PE lessons content. Therefore, the high values of the Autonomy subscale on dance lessons with both genders in this research can be connected to the possibility that students choose the method of dance practicing, as well as the environment in which they dance skills will be assessed. As expected, no differences were founded between the groups (individual, couple or team) because each of three methods was the student choice. Student-centred learning generally has a good effect on the MCPES Autonomy subscale, regardless of the dance skill that students subsequently demonstrate. At the same time, a significantly greater effect on student–centred learning method was recorded on the female population. On a sample of student population from previous studies, there were differences in teaching styles and the Autonomy factor, as male students from the self-check group felt greater autonomy than those from reciprocal and command groups (Pitsi, Digelidis & Papaioannou, 2015). The self-check style could be correlated the most with student-centred learning, therefore, these results are not in line with this study and gender specifics during student-centred learning should be investigated further.

According to second independent variable in two –way ANOVA, teaching method, students in couple group have significantly better results in Task climate subscale then students in team group. The results obtained on the female population especially contribute to this task involving learning climate. Since the students achieved better dance results in couple group then in team group, it is possible that high task involvement climate influenced the better results in dance. When it came to the dance activity, female students felt more competent than male students in making decisions, choosing content and generally in influencing the flow of work in class. The results are congruent with a study by Quested & Duda (2010) conducted on a sample of dancers, according to which dancers' perceptions of autonomy support significantly and positively predicted autonomy and relatedness satisfaction. Further studies of motivation climate and dance performance are necessary to confirm those assumptions.

Student-cantered autonomy in the teaching process, as one of three dimensions of social environment, affects the improvement of intrinsic motivation due to the ability of students to choose certain contents and methods in the teaching process, and to make decisions themselves (Black & Deci, 2000). Self-determination theory (SDT), significantly contributes to understanding the cognitive, emotional, and behavioural patterns related to student progress (Ryan & Deci, 2020), especially in PE. According to SDT, autonomy, competence, and relatedness are the key constructs of psychological well-being and optimal functioning. The environment that will contribute to meeting students' needs in autonomy, competence, and relatedness is the one that promotes a sense of student satisfaction. On the contrary, the environment that limits and restricts students' achievements in need for autonomy, competence, and relatedness will negatively affect the overall feeling of satisfaction (Hagger & Chatzisarantis, 2007; Ryan & Deci, 2020). In this study, female students have achieved the best results in individual teaching method and, with regards to male students, the highest scores have been obtained in team group regarding Autonomy factor. Therefore, significant differences between genders have been found in the creation of positive motivation climate regarding autonomy support freedom, manifested as the possibility to make choices, and the possibility to affect the way PE lessons are run. The differences obtained between the couple- and team-teaching methods regarding Task-involving climate show that the students in couple research group, especially female students, experienced the couple method as the most convenient one for learning, by means of which they can learn new things, and make progress in their skills, and by means of which mistakes will be seen as a part of the learning process. The fact that students are in a position to choose the teaching method that suits them best during learning and evaluation creates a student-centred teaching environment in which students are more active and satisfied during the teaching process (Vansteenkiste, Simons, Soenens, & Lens, 2004). When students are encouraged to develop their own perception in a classroom climate, they develop their individual responsibility. The idea of giving responsibility to students, allowing them to act effectively, and stimulating reflective and critical thinking in the classroom enrich the democratic society (Soini, 1918).

Students in couple research group have achieved the best results in performing linđo dance, and students in individual research group have achieved the best results in performing quatro passi dance. Most of the students choose team group both for practicing and evaluation. The learning group consisting of 5 to 8 students has probably been the one in which dance competences are the least obvious. It is possible that mostly female students choose team group when they self-assess their performance as worse than the performance of the others. Further investigation of the teaching methods correlation, motivation and level of performance is needed to confirm these conclusions. In this study, the lowest value of task climate and dance performance has been detected in team group. On the other hand, students in individual- and couple groups are more successful in dance performance and less numerous. Individual- and couple teaching methods are a more logical choice of motivated students who intend to improve they dance performance while practicing individually, or in a couple, according to teacher's instructions. Students' possibility to participate in the choosing of the teaching method contributes to the student-centred educational process as a whole.

Although studies usually favour more inventive and less traditional teaching styles in physical education (Koloveloni, Goudas & Gerodimos. 2011; Jaakkola & Watt, 2011; Cuellar-Moreno, 2016; Gokhan, 2012). The obtained results suggest that an optimal teaching method cannot be generalised. Cuellar - Moreno (2016) studied the effects of the command and mixed style on primary-school students, and the main conclusion of this study was that the combination of teaching styles, as opposed to using only a traditional, and reproductive teaching styles, contributed to more varied and positive PE teaching, strengthening students' attention capacity, satisfaction, and appropriate behaviour, while also enabling a proper development of motor skills. Therefore, there is a need to analyse the influence factor, such as age, gender, and the type of activity, in future studies related to the influence of different teaching methods on the learning process itself and on the quality of motor performance.

Study limitation

The limitation of this study is in the small samples of participants in some research subgroups. The essence of student-centred learning is that students can choose learning method. In order for the results to be relevant and in accordance with the set goal of the research, the groups had to be formed according to the choice of the students. The fact that very few students choose an individual method while learning folk dances is important information in the organization of a dance PE lesson. At the same time, small samples make it impossible to generalize the results and limit the conclusions. Further research is needed, on a larger sample of subjects and especially on a sample of school-children, because dance related content used in this study had been intended for beginners and the school programme. Conducting a study on a sample of school-children in PE classes should confirm the most efficient student –centred method while learning dances.

Conclusion

Student-centred learning generally has a good effect on the creation of positive motivation climate regarding autonomy support freedom, manifested as the possibility to make choices and significantly greater effect on student–centred learning method was recorded on the female population. Couple teaching method has a better effect then team teaching method in creation of positive task – involving climate, especially among female students who experienced the couple method as the most convenient one for learning folk dances. In student-centred methods of learning, teachers intend to create positive environment in classes in which students actively participate throughout developing critical thinking and problem solving intentions while reaching the goals of the learning process (Noltemeyer, Palmer, James & Petrasek, 2019; Granero-Gallegos, Baños, Baena-Extremera & Martínez-Molina, 2020). This study confirms that implementation of student-centred methods while teaching of the folk dances in PE classes can affect the way PE lessons are run. Students have achieved the best results in the learning of the Adriatic dance zone dances in individual- and couple-teaching method, and most students preferred to practice dances in team group. Very few students choose the individual method in teaching of the dances, and the small sample of subjects limits the possibility of drawing conclusions for individual subgroups. Further studies are necessary to investigate gender specifics during student-centred learning and motivation climate with dance performance to confirm present results in order to develop student-centred education in PE classes.

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Conflict of Interest
The author declares that there are no conflicts of interest.

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ORIGINAL SCIENTIFIC PAPER

Comparison of Body Compositions among Endurance, Strength, and Team Sports Athletes

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Abstract

Regular monitoring and evaluation of body composition determine an athlete's output, such as their probability of winning competitions. The purpose of this study was to analyze the differences in athletes' body composition among several sports categories. The data collected were on 251 athletes aged 20 to 30 years during the pre-competition period. The subjects were divided into three groups: endurance, power, and team sports. SECA 515mBCA was used to analyze the athletes' body composition, which included their body fat percentage (BFP), fat mass (FM), fat-free mass (FFM), skeletal muscle mass (SMM), total body water (TBW), and extracellular water (ECW). The results showed that all body composition parameters were lowest in the endurance category. Fat composition was lowest in the endurance category but did not differ significantly between categories in male athletes. Meanwhile, in female athletes, there was a significant difference in FM between sports categories ($p=0.009$). FFM, SMM, TBW, and ECW differed significantly between categories ($p<0.05$). In female athletes, these variables were higher in the team sports category, as opposed to the power category among male athletes. This study concluded that there were differences in lean body mass composition between sport categories, while fat mass did not differ between sport categories in male athletes. This can be attributed to the differences in training programs and needs of each sport category in maximizing performance. These findings can provide advice to sports professionals that each sport category has its own body composition specifications.

Keywords: *healthy lifestyle, athlete, sports category*

Introduction

Assessing athletes' body composition is an important step in optimizing their performance (Toselli, 2021). The parameters used to evaluate body composition in sports include fat mass at the molecular level and cellular levels (such as body cell mass), intracellular or extracellular fluids, and muscle mass from the tissue level (Campa et al., 2021). Body composition can change according to training and physical activity. Each sport requires a certain type of exercise and activity, which can affect the athlete's body composition. As a result, body composition standards depend on the type of sport (Thomas et al., 2016). Based on differences in athletes' phase angles within a sport among different game roles, sport modalities are divided into three categories: endurance, power/velocity, and team sports (Campa et al., 2022). Study results by Campa et al. (2019) found differences in the body composition of endurance and power sport athletes in a healthy population.

Body composition is associated with increased cardiorespiratory fitness and strength. Athletes have a lower fat mass percentage than non-athletes of the same age. Monitoring fat mass (FM) in athletes is important because excess fat mass in the body reduces performance and decreases the athletes' level of achievement in sports (Chathuranga & Perera, 2022).

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Several studies have analysed the effects of body composition, especially body fat percentage (BFP) and fat-free mass (FFM), on athlete performance. A study by Esco et al. (2018) revealed that high BFP and low FFM are associated with decreased performance and becoming easily fatigued. High lean soft tissue mass and skeletal muscle mass (SMM) composition in the power category can be advantageous to meet the strength and power requirements in achieving maximum performance (Lukaski & Raymond-Pope, 2021). Other studies have also shown that total body water (TBW) and extracellular water (ECW) composition are associated with strength and power in the power and team sport categories (Silva et al., 2014). These findings support the importance of monitoring athletes' body composition. Monitoring an athlete's body composition is important to support regional and national sporting achievements. However, the different BFP, FM, FFM, SMM, TBW, and ECW compositions among the endurance, power, and team sports categories are unknown. The purpose of this study was to analyze the differences in athletes' body composition between the categories of endurance, power, and team sport athletes. This study is expected to be a reference of body composition for sports clubs, institutions, and regional and national agencies in each sport category. This study's results can help the athletes in these organizations achieve higher productivity, and it can provide recommendations for dietary and nutrient intake for each sport category.

Methods

Study participants

This cross-sectional study was conducted on 251 highly trained athletes (132 males and 119 females). Data were taken during the pre-competition period for the largest national competition. Subjects were selected based on the following inclusion criteria: (i) aged between 20-30 years old; (ii) athletes who trained and quarantined in East Java; (iii) had no medical conditions or injuries; (iv) were willing to sign informed consent forms. The subjects were divided into three groups based on their sport modality: endurance, power, and team sports. The endurance sports category consisted of 16 athletes (marathon, open water swimming, and windsurfing athletes), and the power sports category consisted of 142 athletes (athletic, badminton, judo, karate, martial arts, roller skating, sprint, swimming, taekwondo, weight lifting, and wushu athletes), and the team sports category consisted of 93 athletes (basketball, football, futsal, handball, hockey, indoor volleyball, and kick volleyball athletes). This study obtained permission from the Indonesian National Sports Committee (KONI) of East Java (426/153/601.5/2022) and was approved by the Health Research Ethics Committee, Faculty of Public Health, Universitas Airlangga (181/EA/KEPK/2022).

Body composition measurements

Data were collected directly from the subjects before breakfast in the morning. The data collected for this study included age, gender, type of sport, weight (kg), height (cm), body fat percentage (BFP; %), fat mass (FM; kg), fat free mass (FFM; kg), skeletal muscle mass (SMM; kg), total body water (TBW; l), and extracellular water (ECW; l). Body height was measured using a portable stadiometer (SECA 213, SECA, Hamburg, Germany) with an accuracy level of 0.1 cm. Waist circumference was measured using measuring tape (SECA 208, SECA, Hamburg, Germany) with a 0.1 cm accuracy level at the mid-point between the lower costal margin and the anterior superior iliac crest. The measurements were repeated three times and the two measurements with the closest values were averaged as the result. Body mass index was also calculated by dividing the body weight in kilograms by the body height in square centimeters. Body composition was measured using BIA (SECA 515 mBCA, SECA, Hamburg, Germany). Body composition is the measurement that describes each compartment's components in the body, consisting of fat mass and non-fat mass. Fat mass refers to all the fat components' weight in the body, while non-fat mass includes bones, body fluids, muscles, and the non-fat parts of the human body.

Statistical analysis

The statistical analysis was computerized using the Software Statistical Packet for Social Science (SPSS) IBM SPS 21. All data are expressed as means and standard deviation (SD) after normality data distribution was calculated using the Kolmogorov-Smirnov test. Since the data distribution was not normal, the non-parametric statistic was applied. The Kruskal-Wallis test was used to compare the characteristics and body composition data between groups. The Mann-Whitney test was used to differentiate between groups for the post-hoc test, with a 95% significance level.

Results

Based on Table 1, there were significant differences in height, weight, and body composition components (FM, FFM, SMM, TBW, and ECW) between the groups of female athlete ($p < 0.01$). Female endurance athletes had lower body fat than females in other categories. Their FM composition showed no difference between the endurance and power categories, though there were significant differences between female endurance and team athletes, as well as between female power and team athletes. Furthermore, female athletes' FFM, SMM, TBW, and ECW compositions were higher in the team sports category compared to the other sport categories (FFM vs endurance: $p=0.007$; FFM vs power: $p=0.001$; SMM vs endurance: $p=0.006$; SMM vs power: $p=0.001$; TBW vs endurance: $p=0.008$; TBW vs power: $p=0.002$; ECW vs endurance: $p=0.012$; ECW vs power: $p=0.004$).

Table 2 presents the characteristics and body composition of male athletes. There were significant differences in weight, BMI, FFM, SMM, and TBW between groups ($p<0.01$). There were no significant differences in height, BFP, or FM in male athletes with different sports categories ($p>0.05$). Just like the female athletes, BFP and FM were lower in the male endurance athletes compared to the male athletes in the other categories. The FFM, SMM, TBW, and ECW compositions were highest in the power category but not significantly different from the team sport category ($p>0.05$). Meanwhile, those same compositions were significantly different between the endurance and power athletes, as well as the endurance and team sport athletes ($p<0.05$). In addition, the endurance and team sport athletes' BMIs were significantly different when compared to the power athletes compared to endurance ($p=0.011$) and to team sport ($p=0.040$). Female and male athletes' body composition mass in general are shown in Figure 1.

Table 1. Characteristics and body composition of female athletes

Variable	Endurance (n = 6)	Power (n = 69)	Team Sport (n = 44)	p-value (Kruskal-Wallis)
	Mean ± SD	Mean ± SD	Mean ± SD	
Age (years)	23.67±1.75	24.94±2.70	23.82±2.12†	0.049^
Weight (kg)	48.74±5.23	54.04±7.73	59.38±9.31*†	0.000^
Height (m)	1.57±0.06	1.58±0.05	1.66±0.08*†	0.000^
BMI (kg/m2)	19.64±1.74	21.52±2.55	21.18±1.86	0.172
BFP (%)	19.72±3.99	21.06±4.85	22.43±4.30	0.166
FM (kg)	9.68±2.66	11.63±4.38	13.54±4.30*†	0.009^
FFM (kg)	39.05±3.85	42.41±4.38	45.83±5.98*†	0.001^
SMM (kg)	17.44±2.41	19.46±2.64	21.39±3.32*†	0.000^
TBW (l)	28.43±2.93	31.07±3.35	33.57±4.67*†	0.003^
ECW (l)	11.70±1.13	12.75±1.49	13.83±2.17*†	0.000^

Note. BMI: body mass index); BFP: body fat percentage; FM: fat mass; FFM: free fat mass; SMM: skeletal muscle mass); TWB: total body water; ECW: extracellular water; SD: standard deviation; ^ Significantly different between groups ($p < 0.05$); * Significantly different from the endurance group ($p < 0.05$); † Significantly different from the power group ($p < 0.05$)

Table 2. Characteristics and body composition of male athletes

Variable	Endurance (n = 10)	Power (n = 73)	Team Sport (n = 49)	p-value (Kruskal-Wallis)
	Mean ± SD	Mean ± SD	Mean ± SD	
Age (years)	25.20±2.89	25.44±2.41	24.45±1.45†	0.063
Weight (kg)	61.88±7.92	69.89±12.90*	68.39±8.61*	0.080
Height (m)	1.70±0.05	1.70±0.06	1.73±0.08†	0.135
BMI (kg/m2)	21.29±2.34	24.07±3.76*	22.61±1.71†	0.010^
BFP (%)	12.01±5.66	13.05±6.49	12.71±3.83	0.733
FM (kg)	7.76±4.58	9.81±6.94	8.91±3.72	0.457
FFM (kg)	54.12±4.52	60.08±6.78*	59.47±5.97*	0.018^
SMM (kg)	25.85±2.79	29.35±3.81*	28.87±3.22*	0.020^
TBW (l)	38.99±3.55	43.60±5.18*	42.95±4.51*	0.019^
ECW (l)	15.22±1.37	16.93±2.26*	16.70±2.05*	0.056

Note. BMI: body mass index); BFP: body fat percentage; FM: fat mass; FFM: free fat mass; SMM: skeletal muscle mass); TWB: total body water; ECW: extracellular water; SD: standard deviation; ^Significantly different between groups ($p < 0.05$); *Significantly different from the endurance group ($p < 0.05$); †Significantly different from the power group ($p < 0.05$)

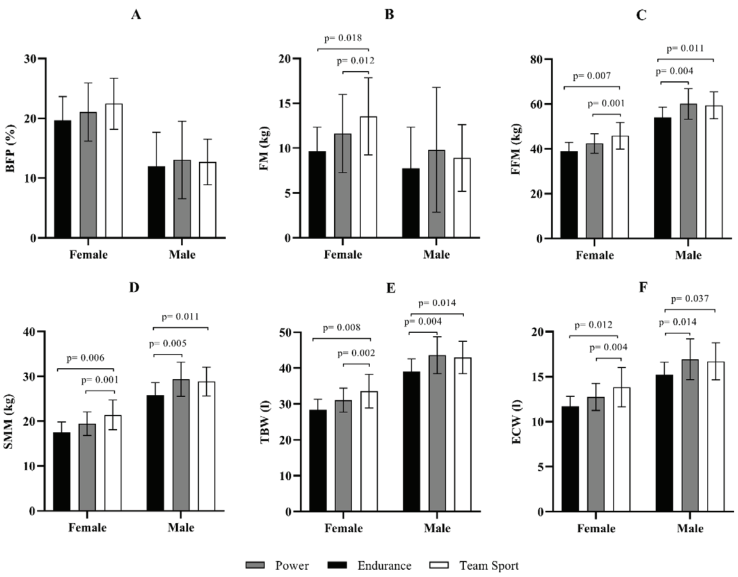

FIGURE 1. a. Body fat percentage (BFP); b. Fat mass (FM); c. Fat free mass (FFM); d. Skeletal muscle mass; e. Total body water (TBW); f. Extracellular water (ECW) in endurance (black symbols), power (grey symbols), and team sport (white symbols) on female and male athletes

Discussion

The main finding of this study is that all body composition parameters were lowest in the endurance category in female and male athletes. FM composition in female athletes was highest in the team sport category and differed from the other categories. FFM, SMM, TBW, and ECW compositions differed between categories. In female athletes, they were higher in the team sports category. Meanwhile, in male athletes, they were higher in the power category. In male athletes, the high FFM and SMM composition in the power category may correlate with the high BMI in this category compared to other sport categories. Body mass index cannot distinguish between FM and FFM, where athletes tend to have lower BFP and higher FFM, making it possible that high BMI is not due to high fat mass, but high FFM (Esco, et al. 2018). Body composition parameters are an essential part of sports performance and a prerequisite for reaching elite performance levels (Kutáč & Sigmund, 2017).

Previous studies have shown comparative results of body fat composition between different sports categories. This study showed no difference in BFP and FM in male athletes between categories, but it found lower BFP and FM in male and female endurance athletes. This is in line with a previous study that found a similar BF between the power and endurance categories (Trinschek et al., 2020). Studies have shown that minimal BFP is desirable in runners because excess adipose tissue causes increased muscle effort during accelerating strides, leading to higher energy expenditure at the same speed (Mooses & Hackney, 2016). This result is in line with other studies that also measured the body composition of male basketball athletes, where the average BFP was 11.69% (Gerodimos et al., 2005). Based on a previous study in male soccer athletes, low BFP (<12%) and high FFM can improve sprint performance (Ishida et al., 2021). Physiologically, moderate exercise (60-65% VO2max) increases fat oxidation. There is some research evidence to suggest that adaptations to endurance training, such as increased capillary angiogenesis and mitochondrial biogenesis, can increase fat oxidation (Holloszy & Coyle, 2016). This is one of the factors that contribute to fat mass being lower in the endurance category than in the other categories, though it is still not significantly different.

As expected, this study found higher FFM and SMM in the power and team sport categories. This is in line with a previous study that found female and male athletes in the power and team sport categories had a higher FFM and SMM composition than endurance athletes (Campa et al., 2022). Even for male athletes, there were significant differences in BMI, FFM, and SMM between sport categories, with the highest FFM and SMM found in the power category. Increasing FFM and SMM can be advantageous in sports that require strength and power (Lukaski & Raymond-Pope, 2021). Those in the power category had higher SMM than those in the endurance category during all training periods. However, in the power category, SMM increased from general to specific during the pre-competition phase (Trinschek et al., 2020). Good performance in the power category is achieved when an athlete has greater muscle mass (Hasan et al., 2021). Athletes with high-performance levels have high FFM values and low BFP (Nikolaidis, et al., 2015). In the team sport category, the FFM enhancement can help to improve jumps and sprints because excessive body mass can negatively impact these skills (Ishida et al., 2021). Previous studies have shown a relationship between BFP, FFM, and SMM with athletes' sprint performance (Dopsaj et al., 2020). Controlling body composition is important in sprinting. It must be done to maintain the body's efficiency in adapting during the training process to optimize competitive performance. Increased muscle mass helps athletes maximize strength, speed, acceleration, and agility (Shaw & Mujika, 2017).

High FFM in power and team sports is necessary because these sports are typically anaerobic. Power and team sport athletes have higher anaerobic power than endurance athletes, which is related to training adaptation and greater body strength (Degens et al., 2019; Campa et al., 2022). High-intensity intermittent sports predominantly use anaerobic resources because decisive actions rely on powerful movements (Glaister, 2005). Power athletes perform resistance and strength training with the main focus on muscle hypertrophy (Slater & Phillips, 2011). Muscle mass is an essential factor in sports that requires muscle strength or power (Garthe et al., 2013). Power is used to perform attacks or explosive movements that are integrated with reactions, intramuscular or intermuscular coordination, and timing to be effective in the application of certain techniques (Slimani et al., 2017). During the power athletes' performance, anaerobic contributions appear to be particularly important although other sources also contribute significantly to the overall work performed (Degoutte et al., 2003).

A previous study showed that those in the endurance category had the lowest muscle mass compared to other categories. The results of a study conducted by Campa, et al. (2019) also show that endurance athletes had less body fluids (TBW and ECW) than other athletes, which also plays a role in their lower body weight. This is in line with the increase in fluid quantity associated with heavier body mass, especially in power and team sport athletes.

Differences in body composition are influenced by several factors, such as the type of exercise and body activity during training and competition (Nepocatych et al., 2017). Previous research by Rejeki et al. (2023) also showed that the type of exercise had a positive effect on improving body composition. Previous studies have shown a relationship between body composition parameters and aerobic and anaerobic capacity (Akinoğlu & Kocahan, 2019). The mechanisms that are commonly found in sports are body morphological adaptations triggered by training components (training intensity or load) adjusted according to specific sports (Dopsaj et al., 2018). In addition, genetics, age, training status, competitive career stage, season, and sporting demands should be analyzed to interpret athletes' body composition. A previous study showed that body composition is influenced by age, sex, genetics, and ethnicity (Winkler et al., 2015). A study by Srhoj et al. (2002) predicted muscle mass was higher in East Asian athletes than in West Asian athletes. Based on those statements, this study may have different results from other countries due to the specific genetic and ethnic factors of Southeast Asia, especially Indonesia. Body composition is also influenced by external factors, one of which is the training season. During training season, increased FFM and SMM and decreased BFP have been reported (Gonzales-Rave et al., 2011). During training season, an increase in muscle strength and power has also been observed, as measured by squat, bench press, and countermovement jump tests (Marques, et al., 2008). The body composition profile for athletes from various sports categories during pre-season training can be referenced when monitoring and interpreting their body (Cullen et al., 2020). Other studies have also found that the body composition of elite athletes is different from that of amateur athletes (Slimani et al., 2017). Additionally, an athlete's body composition is influenced by the athlete's nutritional intake and position. The quality of an athlete's performance will be effective if they receive a good combination of proper training and nutritional intake (Hasan et al., 2021).

This study's strength is that it is the first study to compare athletes' body composition based on the three categories of endurance, strength, and team sports in the pre-competition period. A limitation of this study is the uneven distribution of samples, especially in the endurance category. Further research still needs to be done with a larger sample size, and by comparing trends in body composition changes in each periodization based on sports categories and in correlation with athlete performance.

Conclusion

In conclusion, BFP did not differ between sport categories, but FM differed between sport categories in female athletes. FFM, SMM, TBW, and ECW composition differed between categories. In female athletes, these variables were higher in the team sports category, while they were highest in the power category for male athletes. The results showed that all body composition parameters were lowest in the endurance category in female and male athletes. These differences can be influenced by differences in the training programs and needs of each category in maximizing performance. These findings can serve as advice to sports professionals that each sport category has its own body composition specifications.

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Conflict of Interest
The authors declare no conflicts of interest related to this manuscript.

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SportMont

ORIGINAL SCIENTIFIC PAPER

Experiences and Attitudes of Parents about Children's Free Time at Children's Sports Playgrounds

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Abstract

The importance of children's stay at children's playgrounds is important for the overall development of children. Numerous studies support children's stay at playgrounds to develop social, cognitive, and psychomotor skills. The goal of this study was to determine the attitudes of parents of early and preschool children about children's stay at playgrounds and parents' experiences with the equipment of children's playgrounds. Also, this study aims to correlate factors that affects children´s visit to playground. For the research, a questionnaire was constructed that contains units that examine the equipment of children's playgrounds, attitudes about the frequency of visits to children's playgrounds, and attitudes about the importance of visiting children's playgrounds. The research is carried out online and includes the population of parents of children of early and preschool age in the Republic of Croatia. The results of this research show the proximity of children's playgrounds, from 100 to 250 meters, to the respondent's place of residence. In all regions, the greatest availability is space for jogging and walking, while the least available space is for table tennis. Multiple regression analysis showed a statistically significant correlation of R=0.45 between visiting children's playgrounds and parents' attitudes and opinions about children's playgrounds. Our results show that the distance of children´s playgrounds largely depends on their attendance, but their equipment is not important for children's physical activity.

Keywords: *children's development, benefit of playgrounds, preschool age, physical activity*

Introduction

Play is an important activity for the child that develops numerous competencies and influences the child's development as a driver of active action in physical, social, cognitive and emotional aspects (Berk, 2004; Wood, 2013). One of the key problems of globalization is urbanization, which has an impact on shaping the environment in which we live (Luo, 2022; Safari & Knaftani, 2022). Therefore, it is important to see the role of the environment as a factor of well-being for the child and to design children's playgrounds as a space that encourages children to have fun but also to develop skills and knowledge (De Alvarenga et al., 2018; Yapici et al., 2019; Franklin at al., 2022). Numerous authors emphasize the importance of the equipment in children's playgrounds (Gray & Feldman, 1997; Valentine & McKendrck, 1997; Kan-Kilic, 2021; Sasakom et al., 2022,) and the role of the elements, the safety of children, the natural environment and hygiene as factors that influence attendance. . children's playgrounds (Barbour, 1999; Fjotroft, 2004; Eager et al., 2021; Lou, 2022).

Basic characteristics of children's playgrounds

According to Brownell (2022), the awareness of educational and local policy about the role of children's playgrounds in the overall development of children determines the equipment, number, and structure of children's playgrounds. Kan-Kılıç (2021) believes that the social awareness and responsibility of adults towards the physical and psychomotor de-

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velopment of children is an important predictor in the planning and construction of children's playgrounds. Dyment & O'Connell (2013) emphasize the importance of the structure of children's playgrounds and the equipment that defines the structure. The results of quantitative research, which Dyment & O' Connell (2013) conducted by video monitoring the frequency of visits to parts of children's playgrounds, point to reflecting on the elements that contribute to challenges in the development of children's competencies as well as children's safety. As an additional factor for the safety of children's health, they cite protection from natural sources of heat or cold, soft surfaces that mitigate falls, and the economic use of natural materials in the design of the elements of the children's playground. Therefore, Vlosky & Shape (2005) point to a minimal health risk in using natural materials in the construction of play elements on children's playgrounds (Ernst, 2017). Furthermore, the authors point out that the protection of children's health on children's playgrounds is an important criterion when designing playground equipment. The risk contributed by inadequate equipment (fragile materials, not resistant to weather conditions) and its use in children's playgrounds can endanger the safety and health of children (Eager et al., 2021; Sasakom et al., 2022). According to (Aklıbaşinda et al., 2018; Nasar & Holloman, 2022; Safari & Knaftaki, 2022) the green surface of children's playgrounds in urban areas enables the development of various group games such as football, field hockey, etc. elements, natural materials of elements and flowers and plants as characteristics of ecological diversity.

Development of children's competencies on children's playgrounds

The playground concept (Wolley, 2008) should have a structure that encourages children to develop numerous competencies such as motor, social, emotional, and cognitive. Hazlehurst et al. (2022) believe that children's physical activity on children's playgrounds contributes to children's mental health. According to studies (Bauman, 2004; Fjørtoft, 2004), physical activities on natural grounds have a greater impact on children's physical and motor development than activities on unequipped playgrounds, therefore it is necessary to provide children on playgrounds with elements that will encourage children to grab, climb, crawl, crawl, and swing. In addition to physical activities, children on playgrounds interact with other children and thus develop social skills (Dymentai & O'Connel, 2013) Interaction with other children on playgrounds contributes to self-regulation, and awareness of their physical capabilities in relation to others and encourages motivation in terms of setting limits and challenges (Miller et al., 2017) It is children of mixed age groups who interact with each other on children's playgrounds, who are successful in developing the aforementioned competencies because they support each other and the older are role models for the younger (Gray & Feldman, 1997; Parrott and Cohen, 2022).

Storli and Sandseter (2019) emphasize the connection between functional play and the equipment of children's playgrounds. The results of the research indicate the importance of the elements of children's playgrounds, which play a role in shaping children's perception of play and the functional structure of the game, with which children test the child's function and thereby develop cognitive abilities.

Parents attitudes about children's free time att children's playgrounds

A children's playground is a place where parents and children spend part of their free time. The possibility of socializing with other parents and creating friendships shows children the parental role as a model that realizes and builds social relationships (Leff et al., 2005). In this sense, children's playgrounds also have an educational role (Luo, 2022). In addition to socialization and interpersonal relationships, the authors highlight numerous reasons why parents and children visit children's playgrounds (Dyment & O'Connell, 2013; Aklibasinda et al., 2018). Parott & Cohen (2021) point out that parents have a greater preferance for playgrounds based on nature and natural materials. Parental motivation for staying at children's playgrounds includes awareness of the importance of physical activity in reducing obesity in children, which is a common phenomenon nowadays.

According to Franklin-Luther & Volk (2022), the time parents spend with their children on playgrounds and the reason for parents' participation in children's activities on playgrounds coincides with the emotional connection between children and parents. Excessive care and fear of injury can be factors that contribute to the reasons for children's absence. on children's playgrounds. The results of the Franklin-Luther & Volk (2022) research, which included 100 parents and their children observed playing on children's playgrounds, indicate that the representation of free play by children without the proximity of parents and semi-free play - play that is supervised and controlled by parents. However, the research does not include the age of the children, which is important because the psychomotor development of children of early and preschool age is growing, and the presence of parents is necessary. Also, it is necessary to look at the overall context, which includes the equipment of the playground, the proximity of the family home, and the protective factors found on the playgrounds. In the survey of parents' opinions of children's playground Carson et al. (2010) determined the three categories of visiting playground: satisfaction/services, safety, and sidewalks/parks. In recent decades, scholarly interest in children's recreational activities in parks has grown. The emotional demands and reactions of kids to their activities in a park's play areas, however, have received comparatively little attention. Additionally, little is known about how parents view and interact with children's play areas. In the research of Chen et al. (2020), the purpose of the qualitative research is to investigate parents' perceptions of this specific playing environment. With the help of participant observations, interviews, and research of online reviews, the paper discovers that taking children to parks helps parents reminisce about their own youth and experience emotional healing. According to the research, play areas serve as "building blocks" for family life and childhood as well as being used by kids to play. Babaev et al. (2019) supports playground's social and cultural potential as well as their function in the planning of practical leisure for a contemporary family. The results of research of parents attitudes about importance of playgrounds, indicate that playgrounds are intended for leisure and enjoyment of children and parents. The authors come to the conclusion that it is necessary to research, develop, and promote the positive experience of modern playgrounds for the organization of socially justifiable and diverse family leisure, contributing to the spiritual enrichment and physical improvement, the horizons broadening, and the realization of the creative potential of the individual, taking into account the importance of the theme playground's socio-cultural activities.

Methods

Participants

The research was conducted on a sample of 220 parents, of which 205 were women, 82% of whom were employed, and 15 men, of whom 96% were employed in the Republic of Croatia. The criteria for selecting respondents included parents who have early and preschool age (0-7 years). A significant number of parents declared that they have two or three children, 151 of them, 56 parents have only one child, while 13 have four or more children. Of these, 154 are children between the ages of three and five, followed by 46 parents who have a child between the ages of six and seven and 20 with children between the ages of zero and two. The average age of the respondents was 35±6. During data collection, an equal number of respondents from all regions of the Republic of Croatia was tried but it wasn´t reached. The largest number of respondents (n=138) was from the region of Slavonia, followed by the region of central Croatia (n=70) and the region of Dalmatia (n=12). All respondents agreed to the research and were familiar with the conducted research. During the research, the respondents' anonymity was respected and they could opt out of filling out the questionnaire at any time.

Measuring procedures

For the research, a questionnaire was constructed according to the Designing for Children's Play in Public Open Spaces model (Wolley, 2008) and adapted to the needs of this research. The first part of the questionnaire consists of seven independent demographic variables (gender, age of parents, place of residence by region, number of children, age of children, employment of father and mother). The next part of the questionnaire consisted of eleven questions related to the content found in children's playgrounds near the respondent's place of residence. Respondents had multiple answers to the above questions. The last part of the questionnaire consisted of thirty-two variables of parents' experience and attitudes about the use of children's playgrounds. Dependent variables are scaled according to the Likert scale: 1-completely disagree, 2-disagree, 3-agree, 4-completely agree. The survey questionnaire contains satisfactory metric characteristics, which were determined according to Cronbach's alpha, which is 0.85. The questionnaire was compiled in a Google form and delivered to kindergarten directors, who then forwarded it to parents.

Statistical analysis

After the collected data, statistical processing was performed using TIBCO Statistic Version 14.0.0.15 software (TIBCO Software Inc, USA). The data were processed using descriptive statistics to determine the arithmetic means of the responses according to the regions of residence. After that, a multiple regression analysis was applied to determine the connection between the frequency of visiting children's playgrounds and the predictor variables of parents' opinions and attitudes about children's playgrounds. Statistical significance was set at 0.05.

Results

The obtained results were processed using basic descriptive statistics to see if there are differences according to the location and attendance of children's playgrounds in the region of residence (Table 1). Descriptive statistics did not reveal a big difference in arithmetic means among the three regions. The respondents declared that the location of children's playgrounds is extremely close to them, from 100 to 250 m from their place of residence, and is located near residential buildings. It can be concluded that availability of children's playgrounds in the three macro-regions is satisfactory considering the answers received. Regardless of the proximity of children's sports playgrounds, attendance is only a few times a week.

Table 1. Location and frequency of children's playgrounds with regard to the region of residence.

Variable	Slavonia N=138		Dalmatia N=12		Central Croatia N=70	
	M	St.Dv.	M	St.Dv.	M	St.Dv.
Specify the approximate distance of the children's playground from your home.	3,26	1,49	3,75	1,35	3,76	1,49
I visit the children's playground (with parents, only child, sometimes alone and sometimes with parents).	1,12	0,49	1,16	0,57	1,00	0
A year ago, the children's playground was visited by the child (how often).	2,45	0,91	2,72	1,04	2,64	1,00
We spend time on the playground (indicate the approximate hourly rate).	1,73	0,60	1,16	0,49	1,78	0,54
A children's playground is located.	2,51	1,13	2,16	0,98	2,82	1,19

Legend: M-mean, St.Dv.- standard deviation

Table 2. shows the basic descriptive parameters of the arithmetic mean (M) and standard deviation (St.Dv.) about the region of residence of the respondents. Respondents from all regions found extremely low availability of sports content near their place of residence. In all regions, according to arithmetic averages, the most available space for walking and running is available (Slavonia – 3.05; Dalmatia – 3.25; Central Croatia – 3.06). The highest availability of a concrete football field according to the arithmetic mean of 3.25 is expressed in the region of Dalmatia with a standard deviation of 0.96. The variable A tennis court is easily accessible for children has the lowest accessibility in all regions according to arithmetic means (Slavonia – 1.69; Dalmatia - 1.25; Central Croatia -1.81).

Table 2. Available sports content with regard to the region of residence.

Variable	Slavonia N=138		Dalmatia N=12		Central Croatia N=70	
	M	St.Dv.	M	St.Dv.	M	St.Dv.
A football (concrete) field is easily accessible for children.	2,65	0,97	3,25	0,96	2,57	1,07
A football (grass) field is easily accessible for children.	2,71	0,97	2,25	1,28	2,66	1,08
A basketball court is easily accessible for children.	2,59	1,01	2,25	1,21	2,47	1,04
A handball court is easily accessible for children.	2,03	0,99	1,91	1,08	2,17	0,94
Free space for cycling or rollerblading is easily accessible to children.	2,60	1,01	2,58	1,24	2,62	1,11
A tennis court is easily accessible for children.	1,69	0,86	1,25	0,45	1,81	0,97
There is an easily accessible space for children to run and walk.	3,05	0,97	3,25	1,05	3,06	1,05
Space for free sports activities is easily accessible to children	2,91	0,93	2,92	1,61	2,84	1,06

Legend: M-mean, St.Dv.- standard deviation

Multiple regression analysis (Table 3.) showed a statistically significant correlation of R=0.45 between the criterion variable of visiting children's playgrounds and the predictor variables of parents' attitudes and opinions about children's playgrounds. The model is explained with 20% of the variance and is significant at the $p < .00$ level. Using the backward method of excluding predictor variables that least affect the criterion variable, a model of six variables that significantly affect attendance at children's playgrounds was obtained. The predictor variable of parents' awareness of spending free time on children's playgrounds contributes the most to the criterion variable, whose partial regression coefficient (Beta) is -.27. The negative correlation indicates that increasing parents' awareness of the importance of spending free time at children's playgrounds will not lead to an increase in attendance.

Table 3. Regression analysis.

Variable	Beta	B	t	p
Specify the approximate distance of the children's playground from your home.	-0,14	-0,04	-2,22	0,03
We spend time on the playground.	0,15	0,10	2,37	0,02
The children's playground is additionally equipped.	-0,23	0,00	-3,68	0,00
Children of early and preschool age should stay on the children's playground only with supervision parents or elderly people.	-0,18	-0,11	-2,76	0,01
It is important that parents and educators (if the child goes to kindergarten) they exchange information about what has been done free time at children's playgrounds.	0,21	0,10	3,08	0,00
It is important that parents stay with the children on the children's playground during free time.	-0,27	-0,16	-3,88	0,00
$R=0,45$; $R^2=0,20$; Adj. $R^2= 0,18$; Std. Err. est: 0,36 ; $F=8,90$; $p<0,00$				

Legend: *R – multiple correlations, R^2 – coefficient of determination, Adj. R^2 – adjusted coefficient of determination, Std. Err. est – standard error of estimation, F – test value, p – value of significance level of the F test, B – unstandardized partial coefficient, Beta – partial standardized coefficient of regression, t – value of the t-test of partial regression coefficient, p-value – significance level.

Discussion

According to the World Health Organization (WHO, 2010), children should be physically active for a minimum of 60 minutes a day. This includes intense activities that increase heart rate and rapid breathing. It can be concluded that spending free time in children's sports playgrounds is not culturally represented. Parental perception of children's sports playgrounds, according to the frequency of use of children's sports playgrounds, shows that children's playgrounds are not perceived as having multiple benefits for the child or there are aggravating circumstances that prevent regular use of children's sports playgrounds. Non-use of children's sports playgrounds can initiate other forms of leisure sports activities organized as shorter sports programs in kindergartens or other forms of structured and organized sports activities for children.

Going to children's playgrounds often depends on the parents and their capabilities. Table 1 shows that most parents go to playgrounds together with their children, mostly in central Croatia, where there are no discrepancies between the answers. In the regions of Slavonia and Dalmatia, there is a certain variation and sometimes children go to children's playgrounds independently. Franklin-Luther and Volk (2022) emphasize the existence of a culture of fear among parents, and they are more protective and prevent the child from playing independently so as not to injure himself. The consistency corresponds to central Croatia, related to the constant accompaniment of children to sports playgrounds, which is logical due to the size of the city and the higher level of perception of danger, but also the presence of fear. Previous research (Eager et al., 2021; Sasakom et al., 2022) confirms that the fear and perception of the safety of children's playgrounds, but also the neighborhood, is a key factor in the frequency of children spending their free time on playgrounds. Children of primary school age in the central Croatian region are also more inclined to spend their free time in some form of structured activities, unlike children from the Dalmatian region (Matijašević, 2022). Connecting this with this research, a different culture of spending free time between the two macro-regions is evident, which starts already at preschool age and continues during school days.

Sports children's playgrounds, in addition to being intended for fun, should be an incentive for the development of motor and functional abilities and cognitive development. Playgrounds are not only a few devices but also the accompanying content that is located nearby. Along with children's playgrounds, the highest availability of space for running and walking among all three regions is visible, while the lowest availability is space for table tennis (Table 2). In Dalmatia, the possibility of accessibility of a concrete football field has been expressed in comparison to other regions. Taking into account the cultural preferences of the Dalmatian region towards football as a sport, the presence of concrete football fields is expected. Football in Dalmatia is more than just a sport. He is the bearer of the sports culture of Dalmatia, integrated into the daily activities of children and young people. It is representative of Dalmatian identity and an expression of rivalry and macro-regional affiliation (Tsai, 2021).

Previous research by Nasar and Holloman (2022) points out that more than 87% of parents recognize the importance of children's playgrounds and all the natural elements that surround them. Insufficient free time for parents and the increased employment of mothers (82% of working mothers) due to the modern lifestyle leads to lower attendance at playgrounds, regardless of their awareness.

In this research, the equipment of children's playgrounds is negatively related to their attendance. Additional equipment for children's playgrounds will not lead to an increase in attendance. This confirms previous research by Maccket (2004), which indicates that the level of physical activity of children does not change with regard to the equipment of the playground. Additional equipment on sports grounds will not be a predictor of children's physical activity, but parents play an important role in physical activity. Any intervention in preschool depends on the involvement of parents (Franklin-Luther and Volk, 2022).

Through regression analysis, a negative correlation between the distance of children's playgrounds and the attendance of children's playgrounds was determined. Increasing the distance will decrease their attendance. According to UNICEF (2021), the availability of children's sports playgrounds is extremely important so that the child can realize the primary need for play. A macro-regional analysis determined the satisfactory proximity of children's playgrounds in the Republic of Croatia. Thus, their availability does not represent a problem of lower attendance determined by descriptive indicators.

Conslusion

This research showed equal availability of children's playgrounds in all macro-regions, but insufficient attendance. There is an insufficient number of facilities for table tennis in all macro-regions. Dalmatia has a marked availability of concrete football facilities, which is expected considering the preferences towards football. Further regression analysis established the connection between children's playground visits and parents' attitudes about the importance of children's playground visits. Increasing parents' awareness of the usefulness of children's playgrounds will not lead to an increase in the number of playgrounds being cut down. The distance of children's playgrounds largely depends on attendance, but their equipment is not essential for children's physical activity.

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Conflict of Interest
The author declares that there are no conflicts of interest.

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ORIGINAL SCIENTIFIC PAPER

The Intervention Program effect on the Quality of Children's Body Posture at Elementary Education Level

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Abstract

The goal of this research was to determine the impact of an intervention program containing music-accompanied sport and dance activities on the quality of pupils' body posture at the elementary level of education in physical and sports education classes. The research sample comprised Σ 102 pupils at the primary level of education, 53 of whom were boys with a decimal age of 8.83±0.69 and 49 were girls with a decimal age of 8.77±0.68. They were subjected to testing according to the Klein and Thomas method modified by Mayer. The output results of the posture diagnostics confirmed the significant effect of targeted exercises on the overall posture of boys with a substantial effect (Z=-6.131, p<0.05, r=0.60 – substantial effect). The greatest influence in the boys' group was confirmed in the general spine curvature in the sagittal plane (Z=-5.209, p<0.05, r=0.51 – substantial effect). In the girls' group, the targeted movement program was confirmed to have a substantial effect on overall body posture (Z=-5.793, p<0.05, r=0.59). In all other segments, it was significantly manifested with a medium effect. It was concluded that none of the participants had flawless posture. After the application of the intervention program, good posture was demonstrated by 62.2% of boys and 61.1% of girls. In the output test, the posture of 37.7% of boys and 38.8% of girls was incorrect. None of the pupils was diagnosed with very bad posture after the application of the intervention program. The relationship between BMI and posture assessment of the respondents was assessed. The results of the correlation of BMI values (r=0.007, p>0.05) and overall body posture did not show statistical dependence in both sexes. The intervention program including music-movement and dance elements had a significant effect on improving the posture quality of both boys and girls. The program supported by a musical piece was perceived positively by the pupils for its variety and emotionality. The program should be applied in the physical education classes at a frequency of minimum 2 times a week.

Keywords: *physical education, music-movement and dance program, body posture, younger school age*

Introduction

Sedentary lifestyle, physical inactivity, deteriorating physical fitness of children and its negative effects were pointed out by several studies even before the Covid-19 pandemic (HBSC-Slovakia, 2011; Israel & Holdoš, 2020). The global pandemic and the distance learning related to it put the children again in front of computers, isolating them and limiting their possibilities for organized physical activities and sports in schools. This significant decrease in physical activity due to the pandemic had an adverse effect on children's health, including the quality of their posture (Galmes-Panades & Vidal-Conti, 2022; Rozim, Daubnerová, Slováková & Bendíková, 2022). Proper posture is a concomitant phenomenon of physical and mental health maintained by the activity of muscles and nerves. It is defined by an upright posture, symmetrical development of muscles, and natural curvature of the spine in the form of cervical and lumbar lordosis, thoracic kyphosis, and adequate muscle tension. Such posture, however, is not only a mechanical assembly of individual parts of the human body, but also an expression of behaviour and attitude towards life (Hnízdil et al., 2005).

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When children start school, there is a change in their locomotor system, affected by straining the organism by sitting. Out of the total number of hours of the daily routine, children spend one third of their time at school in a static position. This fact starts to manifest itself as soon as children enter school by weakening the supporting locomotion system (Zukowska et al., 2014). Already in the first primary education stage (6-10 years), children have incorrect posture habits, which, if not compensated, can later lead to degenerative changes in their spine, often accompanied by pain and limited opportunities to assert themselves in life (Halmová, 2000). There is an increasing tendency of incorrect body posture and incorrect bendability of the spine, which can also be seen in the results of the study of Kratěnova et al. (2005), evaluating 3520 children from the Czech Republic aged 7, 11 and 15. In this study, they ascertained that while 33.1% of 7-year-olds had poor posture, it increased to 40.6% with 11-year-olds, and 40.9% with 15-year-olds. As the children grow older, the percentages get higher and higher. A whole range of factors contribute to incorrect posture (visual impairment, hearing defects, blocked airways, delayed mental development); external factors mainly include incorrect sitting, prolonged standing, and inappropriate movement habits. The majority of posture faults can be ascribed to muscle balance disorders or to muscle imbalance. According to the National Report on Youth Policy in the Slovak Republic for the Council of Europe, in the category of 0-18-year-old children and adolescents there is currently an increase in chronic health disorders that require intensive preventive care (Sobihardová et al., 2005). Factors determining body posture can be positively influenced by properly chosen physical activity and thus support appropriate development. However, more important than the correction of detected faulty posture is early and effective diagnosis, as a result of primary prevention. (Malá et al., 2023). Physical education at school plays a fundamental role in the development of physical literacy of children. The concept of physical and sports education aims to link it with health care, with a healthy lifestyle and to increase its attractiveness; it places greater emphasis on freedom for schools and teachers in choosing the content of lessons (Antala & Labudová, 2008). In order to achieve the aforementioned goal of physical education, it is necessary to offer pupils attractive and varied content with the possibility to experience success, which creates a basic precondition for a quest for another activity. Studies (Kouli, Rokka, Mavridis, & Derri, 2009; Owen, Smith, Lubans, Ng, & Lonsdale, 2014) point to the fact that fun and enjoyment in physical education classes can influence the development of positive attitudes towards physical activity. In connection with the above stated, it is important to note, the only activity of pupils based on their interest in a varied exercise program can lead to spontaneous involvement in the offered activities, which also leads to the development of movement skills or physical fitness. In an effort to exert a preventive effect and at the same time make exercise activities more attractive in the physical and sports education classes, intervention exercise programs are being created with a focus on health-oriented exercises and at the same time on increasing the level of movement skills. Health-oriented exercises are mostly aimed at strengthening the internal stabilization system, improvement of coordination, flexibility, balance and physical condition. The effect of compensatory exercises can significantly influence the level of mobility of the spine and alleviate the manifestations of muscle imbalance (Kanásová & Broďáni, 2007). It is believed that such programs for pupils should also include a program containing music and dance activities or aerobics.

The positive effect of an aerobic dance program on physical fitness and intrinsic motivation of primary school students is described in a study by Koli et al. (2009). Pantelić et al. (2019) confirmed the effect of an experimental dance program on children's motor coordination skills. Fong et al. (2018) studied the available literature on the effectiveness of dance interventions. They examined dance interventions lasting >4 weeks that included physical health outcomes and had a comparison group with a movement program. They compared different dance genres and structured movement interventions in a sample of 1,276 participants. Meta-analyses showed that the dance interventions significantly improved body composition, blood biomarkers, musculoskeletal function, and cardiovascular function. There are no programs specifically examining the impact of dance programs on body posture. Similary, Mishenko et al. (2023) draws attention to the insufficient number of studies in the given area.

The benefit of the study is the expansion of potential means in school physical education affecting the quality of children's posture through attractive music-movement programs.

The objective of this study was to determine the impact of an intervention program containing musical movement and dance tools on the quality of elementary school students' posture.

Methods

Subjects

The research group consisted of 102 primary school pupils, of which 53 were boys with a decimal age of 8.83±0.69 and 49 girls with a decimal age of 8.77±0.68. They were pupils of the third and fourth grades of elementary school. The average weight of the boys was 35.92±4.79 kg, average height 136±4.19 and BMI 19.36±1.99 kg/m2. The average weight of the girls was 32.62±4.22 kg, average height 134.39±4.44 cm and BMI 18.04±2.04 kg/m2.

Pupils participated voluntarily with written parental consent. Ethical approval was obtained from the ethical committee at the Matej Bel University under project number 1522/2022. Measurements were carried out in accordance with the ethical standards of the Declaration of Helsinki.

Measurement organization

The research took place in the school year 2021/2022 at an elementary school in Banská Bystrica. After a well-thought-out organization of the research with the aim of achieving the goal, in February initial diagnosis was carried out (V1), of somatometric indicators and a diagnosis of body posture were evaluated by a standardized test routinely used in schools. To evaluate and classify body posture, the Klein and Thomas method, modified by Mayer (in Hošková, & Matoušová, 2005; Bartík, 2005) was used, in which 5 body segments were evaluated visually: head and neck posture, chest shape, abdominal shape and pelvic tilt, overall curvature of the spine, shoulder height, and posture of shoulder blades (Table 1). Each dimension in the range of 1-4 points was graded. For the final typology, the probands were classified into 4 qualitative categories and the corresponding degrees of body posture. This method was employed in recent studies (Bendíková et al., 2020; Rozim et al., 2022). Participants were assessed by two trained persons (physiotherapists) and a consensus was reached in case of disagreement in the assessment.

Table 1. Evaluation of body postures

Evaluation of body postures		
I.	Correct body posture	5 points
II.	Goo d (almost correct) body posture	6 – 10 points
III.	Bad body posture	11 – 15 points
IV.	Incorrect body posture	16 – 20 points

Experimental Program

Subsequently, a movement program aimed at promoting health, perceived through the pupils' correct body posture, was included in the compulsory physical and sports education curriculum (supported by studies: Gao, Zhang & Stodden, 2013; May et al., 2019; Mischenko et al., 2020; Andrieieva et al., 2021; Kashuba et al., 2021) twice a week for 45 minutes, for 18 weeks, a total of 36 lessons. Each lesson of the exercise program included music-movement and dance-based activities (Table 2), and contained 10 minutes of warm-up, 25 minutes of main part and 5 minute Cool-Down. The music was played at a soft rhythm at 20-128 beats per minute (bpm). The main part included 30 minutes of continuous movement and dance activities accompanied with music, such as aerobic, dances, dance games, exercise on the wall bars, exercise on the Overball and Stability ball, skipping rope games. The intensity of the main program was 60% -75% of the maximum heart rate with a rhythm of 130-135 bpm in the first six weeks. After six weeks, from week seven to week eleven, the intensity was increased to 75%-85% of the maximum heart rate (music at 140-150 bpm). In the following seven weeks, the intensity of the program was variable, every subsequent lesson was reduced to 60%-70% of the maximum heart rate, and remaining weeks increased to 75-85% of the maximum heart rate. In the last part of the lessons, the participants were given a 5-minute cool down. Educational process was managed by a qualified teacher with years of experience. Subsequently, output testing (V2) of the same indicators was performed.

Table 2. Movement program plan for 1 month (sample)

Lesson	Warm-up (10min.)	Main Part (30 min.)		Cool down (5min.)
		A part	B part	
1	Warm-up	Stability ball	Dance games	stretching
2	Warm-up	Low Aerobic	Dance- modern	stretching
3	Warm-up	Stability ball	Dance- folk	stretching
4	Warm-up	Dance Aerobik	Dance - modern	stretching
5	Warm-up	Over ball	Rope Skipping	stretching
6	Warm-up	Exercise on the wall bars	Dance games	stretching
7	Warm-up	Skipping Rope games	Dance- folk	stretching
8	Warm-up	Low Aerobic	Dance Aerobik	stretching
Σ	80min.	240 min/month		40 min.

Statistics

The paired Wilcoxon signed-rank test was used to determine the significance of the differences between the input (pre-test) and output (post-test) measurements in the studied posture indicators. The coefficient r (Corder - Foreman, 2009) was used to calculate the effect size, which was interpreted according to the minimum threshold values as follows: r=0.10 - small effect, r=0.30 - medium effect, r=0.50 - large effect (Cohen, 1988). The relationship between the total point score of posture in the sets of girls and boys was determined by Spearman's correlation coefficient r, which was interpreted according to the minimum threshold values as follows: r=0.10 - small (weak) relationship, r=0.30 - moderate (medium strong) relationship, r=0.50 – large (strong) relationship (Cohen, 1988). In the study, we worked with ordinal types of data, therefore it was not necessary to verify the normality of their distribution and non-parametric statistical tests were used. The probability of a type I error was set to α=0.05 in all statistical analyses. Statistical analysis procedures were carried out in accordance with the recommendations of the publication Pivovarniček (2021) using IBM® SPSS® Statistics v28 and Microsoft® Office Excel 2016 software.

Results

Boys

In the evaluation of the overall boys' posture, which is the sum of the marks for the assessed elements, the respondents achieved an average value of (12.23±1.8) points at the beginning of the research. These values correspond to incorrect posture. The best average score for boys (2.04) was achieved in the head and neck area (Table 3) both at the initial and final assessment (1.66). The most serious postural disorders in boys' set (68%) were recorded in the physiological curvature of the spine in the sagittal plane and in assessment of the abdomen and pelvic tilt. Hyperlordosis of the fourth qualitative degree was registered in the case of three boys.

The individual values varied from 8 to 16 points at the initial diagnosis. Perfect posture (below/up to 5 points) was not recorded in case of any pupil. Good (almost perfect) posture (from 6-10 points) was recorded in 13.2% of boys and faulty posture in up to 84.9% of boys (Table 4). Very bad posture was present in the case of one pupil (16 points).

After the application of the intervention program, there has been noted a statistically significant improvement in body posture in boys (p<0.05) in all investigated segments of the spine (Table 5). The exercise program had the greatest impact on improvement in spinal curvature in the sagittal plane (r=0.51 large effect size) as well as in overall body posture (r=0.60 large effect size). After the application of the intervention program, 62.2% of the pupils showed good posture.

One of the causes of incorrect body posture is overweight or obesity. 28.3% of the boys in the research group were overweight. Two boys had proven obesity exceeding the 97 percentile. Therefore, also the relationship between BMI and body posture assessment in the respondents was evaluated. However, the results of the correlation between BMI values ($r=0.007$, $p>0.05$) and overall body posture did not point to statistical dependence (Table 6).

Girls

In the evaluation of the overall girls' posture, which is the sum of the marks for the assessed particular elements, the girl respondents achieved an average value of 11.73±2.04. Such values already belong to the third qualitative level, i.e. faulty posture (Table 3). The variance of individual values at the initial assessment ranged between 8-17 points. Perfect posture (up to 5 points) was not recorded in any pupil. Good (almost perfect) posture (between 6-10 points) was recorded in 28.6% of girls and faulty posture in 69.4% of girls (Table 4). Very bad posture was present in the case of one pupil (17 points). As in the case of boys, we recorded the best point score (1.96 points) in the evaluation of the position of the head and neck both during the initial and final assessment (Table 3). During the initial assessment of girls' body posture, the area of the abdomen and pelvis was most involved in incorrect posture. Up to 65.3% of female pupils had the slightly or significantly protruding abdomen. In one case of proband, a sunken chest was diagnosed. In the case of three probands, a very bad curvature of the lumbar area, a hyperlordosis ranging up to the fourth qualitative level of body posture was found.

After the application of the intervention target program focused on the primary education pupils' health, a statistically significant difference has been noted ($Z=-5.793$ $p<0.05$, $r=-0.59$ - large effect) with a large effect on overall body posture, in individual segments of the spine with a medium effect size (Table 5). In the output (post-test) testing, 61.2% of female pu-

Table 3. Average score of body posture in individual segments of the spine

	Boys (n=53)		Girls (n=49)	
	X_1±SD	X_2±SD	X_1± SD	X_2±SD
Overall body posture	12.23±1.8	9.79±1.72	11.73±2.04	9.49±1.8
Head and neck posture	2.04±0.48	1.66±0.59	1.96±0.54	1.59±0.54
Chest shape	2.43±0.69	1.98±0.50	2.35±0.52	1.88±0.53
Abdomen and pelvic inclination	2.74±0.59	2.26±0.56	2.71±0.65	2.24±0.60
Total Spine curvature	2.75±0.59	2.21±0.57	2.59±0.64	2.08±0.49
Shoulder blades/ scapulae	2.25±0.43	1.68±0.55	2.12±0.48	1.69±0.51

x - arithmetical mean; SD- standard deviation; n- sample size

Table 4. Assessment of body posture according to Klein and Thomas modified by Mayer

	Boys (n=53)		Girls (n=49)	
Body posture	V_1	V_2	V_1	V_2
Ideal body posture	(n=0) 0 %	(n=0) 0 %	(n=0) 0 %	(n=0) 0 %
Good body posture	(n=7) 13.2%	(n=33) 62.2%	(n=14) 28.6%	(n=30) 61.2%
Poor body posture	(n=45) 84.9 %	(n=20) 37.7 %	(n=34) 69.4 %	(n=19) 38.8 %
Incorrect body posture	(n=1) 1.9%	(n=0) 0 %	(n=1) 2%	(n=0) 0%

V_1 – input, V_2 – output

Table 5. Statistical significance of differences after the application of the body posture intervention program

Boys (n=52)	Head and neck posture	Chest shape	Abdomen and pelvic inclination	Total Spine curvature	Shoulder blades/ scapulae	Overall body posture
Z-value	-4.472	-4.347	-4.642	-5.209	-4.973	-6.131
p-value	0.000	0.000	0.000	0.000	0.000	0.000
r (effect size)	-0.43	-0.42	-0.45	-0.51	-0.48	-0.60
	medium effect	medium effect	medium effect	large effect	medium effect	large effect
Girls(n=49)	**Head and neck posture**	**Chest shape**	**Abdomen and pelvic inclination**	**Total Spine curvature**	**Shoulder blades/ scapulae**	**Overall body posture**
Z-value	-4.025	-4.261	-4.600	-4.811	-4.379	-5.793
p-value	0.000	0.000	0.000	0.000	0.000	0.000
r (effect size)	-0.41	-0.43	-0.46	-0.49	-0.44	-0.59
	medium effect	medium effect	medium effect	medium effect	medium effect	large effect

Table 6. Relationship between BMI and overall body posture (in output measurements)

Correlations			boys		girls	
			Overal body posture	BMI	Overal body posture	BMI
Spearman's rho	Overall body posture	Correlation Coefficient	1.000	0.007	1.000	0.046
		p		0.959		0.751
		N	53	53	49	49
	BMI	Correlation Coefficient	0.007	1.000	-0.046	1.000
		p	0.959		0.751	
		N	53	53	49	49

BMI - Body mass index, kg/ m^2-kilogram per square meter

pils demonstrated good posture.

Average BMI values for girls were (18.05±2.04), which means normal weight at the 75 percentile level. 6.1% of female pupils were overweight and 6.1% of female pupils were obese. The remaining pupils (n=37) had a normal weight in the range of 25-75 percentile. The relationship between BMI and overall posture of female students was investigated (Table 6). Similar to boys, a statistically significant relationship between increased or reduced BMI and overall body posture among female pupils ($r=-0.046$, $p>0.05$) was not recorded.

Discussion

The research proves an increasing tendency in the occurrence of deviations from correct body posture of children at the primary level. The study results confirmed the occurrence of wrong body posture in 86.8% of boys and 71.4% of girls at the primary level of elementary school as part of the entrance diagnostic. Similar results in the incidence of postural disorders were also ascertained by Medeková (2009), wrong posture among students of The International Standard Classification of Education, level 1 occurred in up to 78% of boys and 70% of girls, muscle imbalance even among up to 90% of boys and girls. Kanásová and Broďáni (2007) pointed out the wrong posture among children at the primary level in 100% of the probands. The most risky dimension with the highest representation of students was the height of the shoulders and the position of the shoulder blades (58% of the students) and the overall curvature of the spine in 55% of the students.

Lafond et al. (2007) assessed the posture of children in Canada. They evaluated postural deviations in the sagittal plane and their results show that children aged 4-12 years are characterized by postural disorders in the sagittal plane, especially forward head, shoulders and pelvis posture, and knee deformities. The incidence of postural disorders is also reported by Penha, João, Casarotto, Amino and Penteado (2005), who studied 136 girls aged 7–10 years. The main postural deviations found were knock-knee, medial rotation of the hip, antepulsion, pelvic anteversion, knee hyperextension, lumbar hyperlordosis, valgus ankle, imbalanced shoulders, lateral pelvic inclination, scoliosis, trunk rotation, thoracic hyperkyphosis, winged scapula, shoulder protraction, abducted scapula, medial rotation of shoulders, and head tilt.

In his study, Wojtków, Szkoda-Poliszuk and Szotek (2018) points out the occurrence of abnormal posture in 42 percent of examined younger school age children who were diagnosed by using a photogrammetric method. In the research by Sedres et al., (2015) the prevalence of postural changes was 79.7% (n=47), of which 47.5% (n=28) showed frontal plane changes and 61% (n=36) sagittal plane changes. A significant association was found between the presence of thoracic kyphosis and female gender, practice of physical exercises only once or twice a week.

Černický, Ratulovská, Pavlíkova, Vomela and Klein, (2018) considers it appropriate that the elements of the back school methodology implementation should be included in the subject of physical education. Within the author's study, it was found that up to 80 out of 260 school-aged children had faulty or bad posture. Authors recommend, when creating habits of correct posture, to practice physical activity in a playful way with the use of suitable motivational means such as words, music, colourful and appealing aids. The incorporation of posture correction activities into the teaching process was also followed by Kashuba et al. (2020), 139 students aged 6 to 10 with hearing impairment were involved in the testing. Children with postural deviation were included in a transformative experiment, in which the implementation technology was proposed for them, taking into account the indicators of the body posture biometric profile. The entire block consisted of preventive modules (education of postural preventive exercises, yoga breathing exercises, exercises according to Gitman Pilates, exercises on the fit ball). The effect of the intervention program turned out to improve the biogeometric profile index of the respondents at the statistical level of $p<0.05$.

In our study an intervention program with music-movement and dance elements into the content of school physical education was also applied, two times per week. The results of our research confirmed the significant impact of targeted exercises on overall body posture among both boys ($Z=-6.131$, $p<0.05$, $r=0.60$ - large effect) and girls ($Z=-5.793$, $p<0.05$, $r=0.59$) with great effect. After the application of the intervention program, good body posture was demonstrated by 62.2% of boys and 61.1% of girls.

Similar results were observed in the research of authors Mischenko et al., (2020), who investigated the method of correcting posture disorders using health-improving story rhythmic gymnastics, which was included in the curriculum of physical education for 8 to 10 year old girls for a period of 15 to 20 minutes. It showed that 78.5% of the girls in the experimental group showed a stabilization of the spine curvature angle value in the frontal plane. In the control group, stabilization was recorded in only 28.5% of the girls. In the experimental group, only two (14.2%) girls had slight deviation in posture. The authors recommend the health-improving story rhythmic gymnastics method for wide use in the correction of posture. Grygus, Nesterchuk, Hrytseniuk, Rabcheniuk, & Zukow, (2020) conducted an experiment on a sample of 169 elementary school students with the objective of validating a dance program included in education with an effort to correct body posture. The effectiveness of the program for correction of posture disorders was confirmed (p>0.05).

The results of our study are consistent with the opinions of experts (Gao et al., 2013; May et al., 2019; Andrieieva et al., 2021; Kashuba et al., 2021), that dance and choreographic exercises can be one of the most effective means of child's body development, correct posture formation and posture disorders prevention. Hojker, Vaishnav, Staparski, and Dobravec (2023) examined the impact of an intervention preventive program on posture and back pain in primary education students. 179 fourth graders of four elementary schools were included in the study. The main age of participants was 9.2±0.5 years. Half of the classes were part of the posture hygiene program throughout the school year, the other half served as a control group. The class consisted of nine exercises: five activation exercises to strengthen the muscles of the chest, shoulder, girdle, back, abdomen and pelvis; two exercises to stretch the thoracic and lumbar fascia; one to stretch the deep anterior fascia and one meditation exercise. The exercise lasted 10 minutes. The study results showed a significant improvement in sitting and standing posture in the tested group, where 72% of the children had ideal posture. In contrast, only 16% of the students in the control group had ideal posture. Despite the important impact of posture on spine health and, in addition, the increasing incidence of back pain in young people, similar to our study, the authors point to the lack of specific studies on the impact of preventive programs on posture, which brings up a great need for research and practical work in this area.

Some authors also investigated the dependence of body posture and BMI index on movement activity. Brzek et al. (2016) report that obese children more frequently suffer postural disorders in the sagittal plane than children with a normal weight, which results in the development of postural disorders and various health problems at such an early age. Maciałczyk-Paprocka et al. (2017) found that, in a group of pupils aged 7–12 years, the probability of occurrence of wrong body posture was significantly higher among the obese than among the pupils with a normal body weight, namely among boys (p=0.042) and also among girls (p=0.007). In their study, Zanovitová, Zanovit and Bendíková (2011) found that almost 60% of girls and 55% of boys have incorrect posture. In addition, the authors reported a difference in posture between the children who play sports in their free time and those who do not. The results evidenced significantly in favour of physically active children. In our study, no significant relationship between the BMI index of the observed group of boys and girls on body posture was confirmed.

The strength of the study is the fact that the results obtained are unique, as there are few studies presenting the use of music-movement means in relation to posture quality at the primary school level of physical education. Another advantage is easy implementation into the teaching process at low costs.

Limitations of the study are body posture was assessed using a visual method, and the use of the posture assessment method is influenced by the subjective assessment of the researcher. Also, the absence of a control group to determine the effect of the program and to generalise the conclusions. Future studies use of objective non-invasive diagnostic devices with a high degree of validity and reliability.

Conclusion

The prevalence of postural disorders among children is high (Albertsen et al., 2018). It is precisely for this reason that it is necessary to exert a preventive effect on children's correct body posture even within the framework of school physical and sports education. As part of our study, a targeted compensatory program with dance elements was created, to be included in the lessons twice a week, aimed at promoting health from the viewpoint of proper pupils' body posture. The results of the output (post-test) assessment of posture confirmed the significant effect of targeted exercises on the overall posture of boys with a large effect (Z=-6.131, p<0.05, r=0.60 - large effect). The greatest effect among boys was confirmed in the area of the total anteroposterior curvature of the spine (Z=-5.209, p<0.05, r=0.51 - large effect). In the girls' set, a large effect of the targeted movement program on overall body posture was confirmed (Z=-5.793, p<0.05, r=0.59). In all other segments, it was significantly manifested with a medium effect. In conclusion, not that not a single student had flawless posture. After the application of the intervention program, 62.2% of boys and 61.1% of girls demonstrated good posture. In the output test, 37.7% of boys and 38.8% of girls had incorrect posture. Not a single student was diagnosed with very bad posture after the application of the intervention program.

The relationship between BMI and posture assessment of the respondents was evaluated. The results of the correlation of BMI values (r=0.007, p>0.05) and overall body posture did not indicate any statistical dependence among male or female individuals.

Based on the results found at younger school age, the necessity of active prevention and enhancement of pupils' movement aspect of pupils within school physical education is emphasized, as well as the frequency of movement activities of pupils within the daily regime. Appropriately determined and implemented movement activity in childhood is the best prerequisite for maintaining proper posture and eliminating the occurrence of subsequent functional disorders of the locomotor system in adulthood.

The intervention program including music-movement and dance elements had a significant effect on improving the posture quality of both boys and girls. The program supported by a musical piece was perceived positively by the pupils for its variety and emotionality. Based on the results, the program should be applied in physical education classes at least 2 times a week.

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Conflicts of interest

The authors declare that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Effects of Learning Alpine Skiing Techniques on Postural Stability

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Abstract

Alpine skiing is a sport and recreational physical activity which requires fine postural control to maintain balance in challenging conditions. Theoretically, balance dominates in alpine skiing, but coordinated action of the whole body of the skiers is equally important. The aim of this research was to determine the effects of experimental short-term program of intensive training of alpine skiing techniques to postural stability (on Biodex Balance System) of students. The sample is divided into an experimental (31 students, age 21.4±1.0 and body height 180.7±6.3 cm) and control group (34 students, age 20.6±0.8 and body height 180.3±6.8 cm). The results of ANCOVA within variables for the evaluation of postural stability show statistically significant effects of the applying experimental program in all applied variables at the level of significance p=.000. From the mean value results (M) it is obvious that the experimental group achieved better results compared to the identical tests applied to the control group. The results of this research show that learning to ski can improve the ability to maintain balance, especially if it is conducted under the expert supervision of a ski instructor, which can have the effect of reducing the risk of injury.

Keywords: *ski training, students, biodex, dynamic balance*

Introduction

Alpine skiing is a sport and recreational physical activity which requires fine postural control to maintain balance in challenging conditions Staniszewski, Zybko and Wiszomirska (2016) according to Noe and Paillard (2005). Balance is a key component in many functional activities of daily living and many recreational activities and and as an information component of movement is the motor ability to keep the body in a stable position at rest or in motion. In movement, it implies the ability to quickly form compensatory (corrective) movements that are necessary to return the body to a balanced position (Mujanović, Nožinović Mujanović, & Šabović, 2012). In skiing, we talk about dynamic balance, because it is about creating balance position during skating (movement) on skis (in all directions; left-right, forward-backward), (Lešnik & Žvan, 2007). In order for the body to remain in balance, the center of the body gravity must be projected onto the surface of the support (Staniszewski, Zybko, & Wiszomirska, 2016). Since the skier moves on skis, with which he is connected by a ski boot and constantly changes the direction of movement, so the projection of the center of body gravity moves and it is necessary to constantly put it in the right position at a certain time.

Theoretically, balance dominates in alpine skiing, but coordinated action of the whole body of the skiers where work of the legs, upper body and hands is equally important (Mujanović, Atiković, & Nožinović Mujanović, 2014)

Due to its construction, the boot (ski boot) imposes certain mechanical restrictions on the degree of freedom of movement of the ankle, which together with the knee joint and hip joint allow body movements in order to maintain postural stability. On the other hand, when a skier wears a boot and skis, the support surface increases, which makes it easier for him to maintain a stable position on skis and reduce the amplitude of body movements (Noé, Amarantini, & Paillard, 2009). Boots also provide an additional tactile stimulus, recognized by the skin receptor in the lower parts of the shins and feet, as well as a mechanical stimulus related to shin support along the entire length of the tibia in contact with the boot tongue (Staniszewski, Zybko, & Wiszomirska, 2016).

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In alpine skiing, the lower limb, together with visual perception, play a fundamental role in the transfer of stimuli from the external environment to the balance control system, and it is necessary for the foot mechanoreceptors in the ski boot to sense the shape of the ground through the rigid sole of the boot and through the skis in order to maintain ski control (Tchórzewski, Bujas, & Jankowicz-Szymańska, 2013).

Cigrovski, Franjko, Rupčić, Baković, and Matković (2017) state that available balance tests differ greatly not just in their commercial availability but also in a safe and easy way of testing and they claim that theirs research posits balance as the major feature that needs to be incorporated in the conditioning training of people who plan to participate in alpine skiing.

The authors (Ćurić et al., 2018) state that learning setup method must always contain precise settings that allow you to perform the tasks correctly. Ćurić (2020) concludes that there is a causal relationship between mastering skiing technique and motor abilities, or that an increase in the level of motor abilities, agility and static leg strength during learning skiing, if sufficiently trained, can affect the success of mastering skiing techniques.

Also, the effectiveness of the ski program has been confirmed (Staniszewski, Zybko & Wiszomirska, 2016; Şimşek, Arslan, Polat & Koca, 2020) on dynamic balance, but in others, it has been inconclusive (Camliguney, 2013; Cigrovski et al., 2017).

Since the results are not consistent, this research is being conducted and the aim of this research was to determine if there are effects of experimental short-term program of intensive training of alpine skiing techniques to postural stability of students, in order to improve the mastery of ski technique through a cause-and-effect relationship.

Methods

Participants

The test sample included 65 male students enrolled in the second and third year of study at the Faculty of physical education and sport University of Tuzla. The sample is divided into an experimental group (31 students, age 21.4±1.0 and body height 180.7±6.3 cm) and control group (34 students, age 20.6±0.8 and body height 180.3±6.8 cm). Students who participated in the study were healthy, without those excused from physical education for health reasons, and they all gave their written consent to participate in testing. The study was carried out according to the principles of the Helsinki Declaration on experimentation on living subjects (WMA, 2017) and the research was approved by the institutional ethics committee (No. 03-3120-9.2.4/18 from 30th May, 2018).

Measurements

For testing proprioceptive abilities, a stabilization platform was used to test the dynamic balance (Biodex Balance system SD, USA). This device consists of a movable platform that allows oscillations of 20º in all directions relative to the ground and 12 possible levels of stabilization. The platform is connected to a computer program (Biodex, v3.01, Biodex, Inc.), which allows an objective and reliable evaluation of postural stability. The reliability and validity (ICC=0.72-0.89) of this test are very high (Schmitz & Arnold, 1998; Hinman, 2000), and as such are considered exceptionally suitable for this study.

Respondents were first tested on a stabilization platform barefoot, in a natural position on two legs slightly bent at the knee joint and arms next to the body. Afterwards they were tested on a stabilization platform in ski boots, in a natural position on two legs slightly bent at the knee joint (compared to the construction of a boot similar to all models used) and arms next to the body. The place where the respondent stands is determined by the height of the subject. In order to maintain a balance position during testing, respondents were able to perform small body and arm movements. The respondents had to look in front of them, try to maintain a balance position with as few oscillations as possible and be as static as possible. The respondents were tested 3 times for 20 seconds at level 2 of stability (static) with 10 seconds of rest between attempts. The results are expressed as: Overall Stability Index (OSI), Anterior-Posterior Balance Index (APSI) and Medial-Lateral Stability Index (MLSI). Testing was performed with open eyes.

For the assessment of postural stability, the tests were balance on two legs – BTL and balance on two legs with ski boots – BSB. Staniszewski, Zybko and Wiszomirska (2016) used the same test on an AccuSway (AMTI, USA) stabilometric platform.

Procedure

The assessment of motor abilities selected for this research was conducted two days before and two days after the end of the experimental program. The experimental program was implemented on a daily basis during the period from 9 am to 4 pm in duration of 33 hours for 6 days (4 hours of training

Table 1. Design of experimental program elaborated on daily basis

Learning day	Design of experimental program
1st	Determining the initial knowledge of alpine skiing and forming homogeneous groups of skiers by the quality of knowledge; Overview of equipment and attachment and removal of skis; Walking with and without ski poles on the platform, turning around the tops and tails, falling and lifting; Climbing the slope, turning on a slope, falling and lifting; Gliding straight down the gentle slope (on flat terrain and over uneven terrain), with different ways of stopping (at the end of a slope, in a plow, by a transient step); The use of ski lifts; Traversing the slope Traversing the slope with sliding and stoping; Gliding wedge (speed control and stopping in the plow); Wedge turns.
2nd	Repeat alpine skiing technique elements from the previous day; Ski curves with wedge turns; Gliding wedge on a steeper slope (speed control and stopping); Wedge turns on a steeper slope; Parallel turn towards the slope (with sliding and carving the skis); Wedge parallel turn.
3rd	Repeat alpine skiing technique elements from the previous days; Advanced wedge turns with pole plant; Wedge parallel turns on a steeper slope; Parallel turns on a gentle slope.
4th	Repeat alpine skiing technique elements from the previous days; Parallel turns on a steeper slope; Basic parallel turns on a gentle slope.
5th	Repeat alpine skiing technique elements from the previous days; Wedge turns; Wedge parallel turns; Basic parallel turns.
6th	Free practice.

with instructors and 2 hours of free practice for 5 days and on the 6th day 3 hours of free practice).

The experimental program is designed for beginners to learn the basic techniques of alpine skiing and is included in the curriculum of skiing lessons at the Faculty of physical education and sport University of Tuzla. The program itself (Table 1) was precisely determined by the predetermined number of repetitions of a particular methodical exercise or the ski technique itself and was formed on the basis of current knowledge in the training of motor activity of alpine skiing. It has been proven that the number of repetition and training of a particular element of ski technique and the way it is presented affects the higher level of the adopted knowledge of alpine skiing (Grouios, Kouthouris & Bagiatis, 1993; Almåsbakk, Whiting & Helgerud, 2004).

During the period of experimental treatment, the control group performed the regular duties prescribed by the curriculum (training of their choice for a practical exam in judo, handball and dances).

Data analysis

To determine the effects of an experimental program on some motor abilities of students a univariate covariance analysis was applied (ANCOVA). ANCOVA is suitable for comparing two groups that are tested before and after the intervention. The results on the pre-intervention test are treated as a control covariate, i.e. the statistical removal of previously existing differences between the groups (Pallant, 2011). As a preliminary analysis (assumption) for ANCOVA, Levene's test was used to evaluate the equality of variances between the compared groups. Statistical significance was set at $p \leq 0.05$. All statistical analyses were performed with the software package for statistical data processing SPSS Inc. (2008).

Results

Preliminary testing tested the assumption of variance homogeneity where no perceived contingency was noted in the applied variables. The statistical significance of Levene's test in all variables is $p>0.05$, indicating that observed variance, two groups of respondents, are similar in these variables. This means that there are no significant differences between the variants and the zero hypothesis is accepted and we conclude that the condition of homogeneity is met. Therefore, differences in the size of the experimental treatment effect between the groups can be attributed to the differences due to the treatment.

Table 2. Leven's test of equality of variance for both groups of respondents in the variables for assessing motor skills (balance)

Dependent variable	F	df1	df2	p-value
BTLOSI	.87	1	63	.355
BTLAPSI	3.73	1	63	.058
BTLMLSI	.35	1	63	.558
BSBOSI	3.70	1	63	.059
BSBAPSI	3.58	1	63	.063
BSBMLSI	.27	1	63	.615

Legend: F - Fisher's F ratio; p – level of statistical significance; significance BTLOSI- Balance on two legs Overall Stability Index; BTLAPSI- Balance on two legs Anterior-Posterior Balance Index; BTLMLSI- Balance on two legs Medial-Lateral Stability Index; BSBOSI-Balance on two legs with ski boots Overall Stability Index; BSBAPSI- Balance on two legs with ski boots Anterior-Posterior Balance Index; BSBMLSI- Balance on two legs with ski boots Medial-Lateral Stability Index

Table 3. Results of the ANCOVA for both groups of respondents in the variables for the assessment of motor skills (balance)

Dependent		Control group		Experimental group		ANCOVA	
Variable		M	SD	M	SD	F	p
BTLOSI	I	3.30	1.82	2.15	.68	55.50	.000
	F	3.27	1.69	1.52	.45		
BTLAPSI	I	1.85	.77	1.50	.45	47.78	.000
	F	1.94	.80	1.10	.38		
BTLMLSI	I	1.86	1.07	1.29	.45	38.29	.000
	F	1.84	1.03	.89	.29		
BSBOSI	I	4.87	2.15	4.05	1.81	135.48	.000
	F	4.82	1.86	2.32	1.16		
BSBAPSI	I	3.41	1.54	2.97	1.55	133.70	.000
	F	3.48	1.46	1.56	.73		
BSBMLSI	I	2.24	1.10	2.30	1.18	59.56	.000
	F	2.34	1.22	1.28	.63		

Legend: I – initial testing; F – final testing; M–arithmetic mean; SD–standard deviation; F - Fisher's F ratio; p – level of statistical significance BTLOSI- Balance on two legs Overall Stability Index; BTLAPSI- Balance on two legs Anterior-Posterior Balance Index; BTLMLSI- Balance on two legs Medial-Lateral Stability Index; BSBOSI-Balance on two legs with ski boots Overall Stability Index; BSBAPSI- Balance on two legs with ski boots Anterior-Posterior Balance Index; BSBMLSI- Balance on two legs with ski boots Medial-Lateral Stability Index

The results of the univariate analysis of covariance within the variables for balance assessment in Table 3 show statistically significant effects of the applied program of intensive training of alpine skiing technique in all applied variables. The contribution to these effects that is visible in the balance assessment variables is at a statistically significant level of p=0.000. A review of the results of the mean values (M) in these variables shows that the subjects of the experimental group achieved better results compared to the identical tests applied to the control group.

Discussion

A review of the results shows that all subjects, control and experimental group, in the test with ski boots achieved higher coefficients of stability index compared to identical tests performed without ski boots, which is expected because the ability to control the balance of body position is further limited by ski boots that reduces the mobility of the ankle and requires the involvement of other muscles in order to maintain balance. All balance tests assessed the dynamic component of balance. On the other hand, it is also prominent that the results of the experimental group compared to the control group show lower coefficients of stability index (a lower score is better) at a statistically significant level, and it can be stated that the applied program of intensive training of alpine skiing in all applied variables produced statistically significant effects.

A study (Wojtyczek, Pasławska, & Raschner, 2014) similar to our study on physical education and sports students aged 20-22 aimed at evaluating changes in balance in the medio-lateral direction after the mandatory 7-day ski camp compared to gender and skiing experience, shows that there has been an improvement in maintaining balance after the ski camp. The authors state that the improvements in maintaining a balanced position can be explained by using many different exercises during the skiing week, in which lateral movements and back-and-forth movements appear, which are an integral part of the skiing curriculum.

A study by the authors (Camliguney, Ramazanoglu, Atilgan, Yilmaz, & Uzun, 2012) aimed at observing the effects of intensive ski training, in duration of 6 days, as a new set of skills on the dynamic and static balance of male and female athletes of physical education students and sports that have regular and formal sports training, the results show that there were no statistically significant changes in the static balance test after the program, but in the dynamic balance test there was an improvement in the results of both male and female students ski training group in relation to the control group. The authors state that there was an improvement in the results of the ski group subjects compared to the control group and in the test for estimating the dynamic-isometric strength of the legs. As possible causes of such changes, the authors stated that during skiing, balance and movement controls are the focus of skiing for the lower extremities. This is especially expressed when during training on skis the center of gravity approaches the ground, there are frequent dynamic and isometric contractions of the muscles, which leads to an increase in balance and strength of the legs.

In a study aimed at determining the extent to which several days of skiing and the level of technical skills affect the level of postural stability of skiers (Staniszewski, Zybko, & Wiszomirska, 2016) on students of the Faculty of Physical Education and Sports of the University of Warsaw, beginners and advanced skiers during 9-day ski training camp, it was found that subjects who were skiing beginners had statistically significant improvements in results measured during maintaining a balanced position in ski boots after ski camp. They also state that in attempts in which participants stood on one leg, balance improved mostly in beginners. In conclusion, the authors note that nine days of skiing has a positive effect on maintaining postural stability only in conditions that include measurements in ski boots, or in those conditions in which the program was implemented.

Camliguney (2013), in a study on adolescents in aim to determine the effects of 5-day ski training on dynamic balance and vertical jumping, points out that the program had an effect on stability in the medio-lateral direction while negatively affecting stability in the anterio-posterior direction. Despite the small increase in results, there was no statistically significant increase in performing vertical jumps on one and two legs. The author indicates a relatively short period of ski training as a possible reason for the negative and insufficient effect on some of the subjects' abilities.

Cigrovski et al. (2017), in their study tested participants, after 10 days of alpine skiing, with two balance tests. Participants results did not significantly different in one of the balance test, but in the other balance test they found significant differences in the four out of six variables. Also authors Şimşek, Arslan, Polat, and Koca (2020), in their study, examine the effect of 5 days (25 hours) of ski training on balance performance for individuals who have never had ski training before. They received significantly different results which shows that skiing activity can be recommended especially for the development of dynamic balance performance.

It is necessary to note that the mentioned research was carried out on different devices for measuring both static and dynamic balance, which may have an impact on the inconsistency (contradiction, inconsistency) of the results obtained in the research. By comparing our results with the research results mentioned above, we can say that the skiing program, conducted in this research, has a positive effect on movements in both, the frontal and sagittal plain. Different mechanisms of controlling balance in both plains, which are activated during skiing, could be cause of such an outcome.

The strengths of this study is demonstrated through the use of appropriate instrument to measure dynamic balance, and the application of a specific exercise program.

The limitations of this study a relatively small sub-sample size and that the control group participating in other sports that also can affected dynamic balance. Therefore, recommendations for further research could be: a larger sample size, excluding the control group from any programmed activities and perhaps extending the duration of the program.

In conclusion, a professionally guided skiing program, especially for beginner skiers, which lasts a sufficient period of time can lead to changes in motor skills, especially in balance, as shown by this study. Consequently, our research shows that it is necessary to include balance exercises through fitness training, and directly influence the development of those motor abilities that are crucial for mastering of the elements of alpine skiing techniques, and also result in as well as a preventive effect on the possibility of injury.

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Conflict of Interest
The authors declare that there is no conflict of interest.

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SportMont

ORIGINAL SCIENTIFIC PAPER

Relative Age Effects in The Selection of Representative Athletes of Kosovo National Team in Handball, Football, Basketball, Volleyball - Post Puberty Age

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Abstract

Taking into account the role that the selection of athletes has in the success of clubs as well as in the sports life of young people, this research aims to highlight the impact of the Relative Age Effects (RAE) in the selection of athletes for representative teams of Kosovo in collective sports and the difference between the respective groups. The study was designed as an observational cross-sectional study. The total sample of persons in this research was n=237 (118 male and 119 female) athletes of the Kosovo National Team of team sports. The athletes were divided into 4 main groups: Q1 (January-March), Q2 (April-June), Q3 (July-September) and Q4 (October-December) with a special emphasis on the Q1 and Q4 group (for all sports the birth dates of the age post-puberty were collected). The analysis of the data clearly indicates that young athletes born in Q1 and Q2 were significantly more selected for representative teams compared to those in the other two groups. The results demonstrate a significant relative age effect on athlete selection based on gender for the Kosovo National Team. Specifically, athletes from Q1 and Q2 were favored. Based on these findings, it is strongly recommended that the respective federations, sports trainers, and relevant sports authorities take proactive measures to prevent the "discrimination" experienced by young athletes.

Keywords: *team sports, dynamic of relative age, selection, quartile, success, potential*

Introduction

Talent identification and physical ability assessment is a common process in the selection of team sports athletes. The selection of these talents depends on and is influenced by physical and bio-psychological abilities such as speed, strength, coordination, power, aerobic and anaerobic capacity, but based on numerous researches, the date of birth of individuals turns out to be a determining factor for their sports "fate" (Jakobsson, Julin, Persson, & Malm, 2021). For the first time, RAE (Relative Age Effect) was mentioned in 1982 in the context of sports performance in the publication "Unknown Exclusion in Sports" published by "Athletics" (Jakobsson, Julin, Persson, & Malm, 2021). In the adolescent years, significant alterations occur in the anthropometric and physiological traits of both males and females (Arrieta, Unda, & Gil, 2016). Consequently, sports participants who are relatively older may experience an advantage in terms of being chosen for elite teams. This advantage grants them improved access to coaching and facilities, ultimately aiding in their skill development compared to their relatively younger counterparts (Smith et al., 2022). As the selection process intensifies, the focus on the distribution criterion diminishes, and the Relative Age Effect (RAE) is most pronounced during the final selection stage overseen by regional team

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coaches (Lagestad, Steen, & Dalen, 2018). Despite technical abilities potentially providing an early indicator, many coaches tend to blend technical skills and potential with early accomplishments, which, once more, might result from physical advantages and similar factors (Fuhre, Øygard, & Sæther, 2022). The durations of professional careers can be linked to numerous factors apart from the relative age effect (Nakata, 2017). In team sports, there is a tendency to favor younger athletes born towards the end of the year, as they could potentially be less developed and face disadvantages in terms of their physical attributes, physiology, and maturity compared to their peers born in the earlier months of the year (Víllora, Vicedo, & Cordente, 2015; Wattie, Schorer, & Baker, 2015; Unda et al., 2016). Consequently, a considerable number of promising talents may go unnoticed due to the disadvantage associated with their relative age (Musch & Grondin, 2001). The extent of the relative age effect is influenced by interactions occurring at different developmental stages (Smith, Weir, Till, Romann, & Cobley, 2018). Failure to address the relative age effect at a young age will lead to not only a selective and arbitrary choice of elite athletes but also have a detrimental impact on public health, as sport participation is linked to overall physical activity levels, both in the present and future (Jakobsson, Julin, Persson, & Malm, 2021). The findings indicate the presence of relative age effects (RAEs) in German elite adult male and female soccer. This notable RAE observed in the study arises from a higher representation of players born in the early months of the year (Götze & Hoppe, 2021). To be more precise, the impact was identified in swimming, which is an individual sport, as well as in team sports like basketball, soccer, and team handball (Lidor, Maayan, & Arnon, 2021). The primary motivation behind this study is the insufficient research on the influence of relative age effects in Kosovo across different sports. This research aims to highlight the impact of this phenomenon (RAE) in the selection of athletes for representative teams, recognition and reflection of this phenomenon in the selection of the representative teams of Kosovo in collective sports, and the difference between the respective groups.

Methods

Study Design and Participants

The study was designed as a cross-sectional observational study with a deductive approach, where the participants in this research were a total of n=237 (118 male and 119 female), selected from four representative teams (two teams for each gender) from the short list of players for each sport mentioned above: Handball 60 players (15 male and 15 female), Football 80 players (40 male and 40 female), Basketball 48 players (24 male and 24 female) and Volleyball 49 players (24 male and 25 female). The criterion for the selection of the players was the age of the players (post-puberty age) where the minimum required age was U16 and the maximum age was not to exceed U20, which means that senior teams were not included due to the age difference between players. It is worth noting that the age of the selected players is the same calendar year for each team. The team selection method was simple random sampling.

Data Collection

The method of collecting the data needed to do this research was through their collection in the field by visiting the respective federations of each collective sport that this research affects. Some of the data were received in physical form by the responsible persons of the respective federations, while another part of the data collection was in electronic form via the Internet on the official websites of these federations. The data collected includes the birth dates of the participating athletes and they were categorized into yearly quarters, following a method similar to the approach used in the study by (Lemoyne, Pelletier, Trudeau, & Grondin, 2021). The collection of this data lasted about 3 months (March, April, May - 2022). In this research there was no need for the usual procedure of testing individuals, this is for the sole reason that for the calculation of the necessary results, only the date of birth of the athlete selected by the respective representative was needed. The collection of this data was in full compliance with the Declaration of Helsinki and approved by the Physical Education Board of the Faculty of Physical Education and Sport, University of Pristina "Hasan Pristina" Nr. 1124, 27.12.2022.

Data analysis

The data were analyzed using the statistical package SPSS v23.0 (SPSS Inc., Chicago, IL, USA) whereas the data (tables) were stored in Microsoft Excel. The parameters treated in this research were the birth dates of the athletes. The collected data were of the "categorical" type and for the purpose of their analysis the transformation of the data was done, where the months January, February, and March (Q1) were defined with the value 1, the months April, May, and June (Q2) with the value 2, the months July, August and September (Q3) with value 3 and October, November and December (Q4) with value 4. Cross tabulations (contingency tables) were used to display the distribution of data between the "Sports, Gender and Quartile" variables, the Chi-square test of independence was used to determine whether there is a significant association between those variables, and the Chi-square goodness-of-fit test was used to compare observed data with expected data to determine if there is a significant difference between quartiles. The significance level was set at $p<0.05$.

Results

As can be seen in the presented Table 1 and Table 2, the Chi-Square test results (with a p-value of 0.233) indicate that there is no statistically significant association between Quartiles (Q1, Q2, Q3, Q4) and Sports (Handball, Football, Basketball, Volleyball) in the representation of athletes in the Kosovo National Team. This means that there are no significant differences among sports regarding the effect of relative ages on athlete representation. Additionally, Cramer's V value (0.128) suggests a weak association, which further supports the notion that the effect of relative ages does not differ significantly across different sports. In Table 3 and Table 4, the Chi-Square test results (with a p-value of 0.006) indicate that there is a statistically significant association between Quartiles (Q1, Q2, Q3, Q4) and Gender (Male and Female) in the representation of athletes in the Kosovo National Team. This means that there are significant differences between genders regarding the effect of relative ages on athlete representation. Furthermore, both Phi (0.228) and Cramer's V (0.228) values show a moderate association between Quartiles and Gender, reinforcing the significant

Table 1. Distribution of data between the Sports and Quartile variables and the Chi-square test of independence

			Q1	Q2	Q3	Q4	Total
Sport	Football	Count	28	21	20	11	80
		Expected Count	33.4	21.6	13.8	11.1	80.0
		% of Total	11.8%	8.9%	8.4%	4.6%	33.8%
	Basketball	Count	27	11	6	4	48
		Expected Count	20.1	13.0	8.3	6.7	48.0
		% of Total	11.4%	4.6%	2.5%	1.7%	20.3%
Sport	Volleyball	Count	21	13	5	10	49
		Expected Count	20.5	13.2	8.5	6.8	49.0
		% of Total	8.9%	5.5%	2.1%	4.2%	20.7%
	Handball	Count	23	19	10	8	60
		Expected Count	25.1	16.2	10.4	8.4	60.0
		% of Total	9.7%	8.0%	4.2%	3.4%	25.3%
Total		Count	99	64	41	33	237
		Expected Count	99.0	64.0	41.0	33.0	237.0
		% of Total	41.8%	27.0%	17.3%	13.9%	100.0%
X^2(9, n-237)=11.666, p=0.233, φc=0.128							

Crosstab; Quartile columns Q1–Q4.

Note: Q1 (January, February, and March), Q2 (April, May, and June), Q3 (July, August, and September), Q4 (October, November, and December), X^2- Chi-square value, n - number, p – p-value, φc - Cramér's phi (Cramér's V)

relationship found. Table 5 and Table 6, representation of the Chi-Square goodness-of-fit test indicates a significant association between quartiles and the representation of athletes in the different quartiles, the p-value is extremely small (0.000). The observed distribution of athletes across the quartiles (Q1, Q2, Q3, Q4) is significantly different from what would be expected if there were no relationship between the quartiles and athlete representation. Specifically, Q1 and Q2 are over-represented in the selection of athletes, while Q3 and Q4 are under-represented.

Table 2. Distribution of data between the Gender and Quartile variables and the Chi-square test of independence

			Q1	Q2	Q3	Q4	Total
Gender	Male	Count	62	25	15	16	118
		Expected Count	49.3	31.9	20.4	16.4	118.0
		% of Total	26.2%	10.5%	6.3%	6.8%	49.8%
	Female	Count	37	39	26	17	119
		Expected Count	49.7	32.1	20.6	16.6	119.0
		% of Total	15.6%	16.5%	11.0%	7.2%	50.2%
Total		Count	99	64	41	33	237
		Expected Count	99.0	64.0	41.0	33.0	237.0
		% of Total	41.8%	27.0%	17.3%	13.9%	100.0%
X^2(3, n-237)=12.353, p=0.006, φc=0.228							

Crosstab; Quartile columns Q1–Q4.

Note: Q1 (January, February, and March), Q2 (April, May, and June), Q3 (July, August, and September), Q4 (October, November, and December), X^2- Chi-square value, n - number, p – p-value, φc - Cramér's phi (Cramér's V)

Table 3. The Chi-square goodness-of-fit test between "Quartiles"

	Quartile		
	Observed N	Expected N	Residual
Q1	99	59.3	39.8
Q2	64	59.3	4.8
Q3	41	59.3	-18.3
Q4	33	59.3	-26.3
Total	237		
X^2(3, n-237)=44.300, p<0.000			

Note: Q1 (January, February and March), Q2 (April, May and June), Q3 (July, August, and September), Q4 (October, November, and December), X^2- Chi-square value, n - number, p – p-value.

Discussion

Based on the results of the data distribution of this research, the relatively large impact of the phenomenon (RAE) on Kosovo's representatives in the above-mentioned collective sports is clearly observed. The primary objective of this research was to examine the presence of RAE in the selection of players in the representative teams of the respective sports with special emphasis on the comparison of Q1 and Q4 groups. The secondary aim was to compare which gender is more affected and influenced by RAE. By analyzing the data, it was clearly seen that the young people born at the beginning of the year or Q1 and Q2 groups were definitely more selected in the representative teams than the young people of the other two groups, also in both sexes the young people of the Q1 group were the majority in the selection compared to group Q2 and Q4. The findings of our study affirm our hypothesis and are in agreement with previous studies that RAE is present in the selection process of young people in the respective representatives.

All representative teams in specific sports have not been affected at the same level by the phenomenon (RAE), but it is clearly observed that gr. Q1 (January-March) was significantly larger in the selection of players in the representative teams than gr. Q3 (July-September). Studies on Relative Age Effect (RAE) have indicated that athletes born in the earlier part of the year (i.e., the first semester) tend to possess an advantage in terms of anthropometric measurements, physiological attributes, technical skills, and tactical abilities (Gil et al., 2014; Arrieta, Unda, & Gil, 2016; Ibáñez, Mazo, Nascimento, & Rubio, 2018). Additionally, when considering the athletes who achieved medals in the competitions within the studied timeframe, it's noteworthy that over 50% of them were born in the initial quarter of the year, while there were no medalists born in the final quarter (Bezuglov et al., 2022). An interesting discovery is the limited impact of the Relative Age Effect (RAE) on both individual and team performance in youth levels, whereas its significance becomes more apparent in junior and senior categories (de la Rubia, Calvo, & Lorenzo, 2020). Moreover, it is clearly observed that the male gender is more affected by the phenome-non (RAE) than the female gender, wherein the crosstabulation and the chi-square test prove that. These findings corroborate those of other authors (Arrieta, Unda, & Gil, 2016) the results of which show that this effect was not found for female players. Furthermore, there is a common belief that the Relative Age Effect (RAE) tends to be less pronounced in female sports, primarily because of lower levels of competition among young girls (Delorme & Raspaud, 2009; Sedano, Vaeyens, & Redondo, 2015). These findings affirm the hypothesis that it was expected that the male gender is more affected by the phenomenon (RAE) than the female gender during the selection for the National Team. Moreover, the results indicated that the Relative Age Effect (RAE) became more noticeable with higher levels of competition (Sedano et al., 2015). The main results indicated that Relative Age Effects (RAEs) were present in both males and females across the Regional Talent Hubs. However, they were statistically significant only in the case of males in the England National Youth Teams (Kelly et al., 2021). The most significant Relative Age Effect (RAE) is observed among highly accomplished young athletes. This implies that athletes born earlier in the year face fewer chances to join elite sports organizations and benefit from top coaching, increasing the likelihood of them leaving the sport prematurely during adolescence (Bezuglov et al., 2022). Additionally, although the specific risk pattern may differ when comparing various samples, Relative Age Effects (RAEs) are consistently observed and anticipated, particularly during the transitional adolescent years. This is a period when differences in maturation levels are most pronounced (Musch & Grondin, 2001; Malina, Bouchard, & Bar-Or , 2003). The selection process for the sports academy thus seems to favor athletes with chronological higher age, i.e., a so-called relative age effect is present (Ek et al., 2019). Maturity status should be considered not only during the selection process but also as a guiding factor in training, helping to address and reduce the disparities arising from varying maturity levels (Toselli et al., 2022). Neglecting to tackle the relative age concern and its underlying causes could lead to children disengaging from physical activities altogether, which could have lasting adverse effects on public health in the long run (Jakobsson et al., 2021). Continuous education of coaches and the promotion of awareness regarding RAEs in sports remain imperative. This is crucial in preventing athletes from leaving sports prematurely and ensuring that all players have equitable chances to showcase their talents across various levels of competition (Lemoyne, Pelletier, Trudeau, & Grondin, 2021). One of the significant strengths of this study is its comprehensive analysis of a diverse sample of athletes, providing a well-rounded perspective on the phenomenon. Additionally, the study employed well-established statistical techniques to analyze the data, enhancing the reliability of the results. Furthermore, being one of the few research in this domain in Kosovo, this study fills a critical knowledge gap and serves as a foundational piece for future investigations into relative age effects. Despite its contributions, this study has some limitations that warrant consideration. The sub-sample used in this research, although carefully selected from representative teams, may still be relatively small for broader data generalization. While the results provide valuable insights into the specific context of the

Kosovo National Team, caution should be exercised when applying these findings to other regions or sports without further validation. One limitation is the potential for publication bias in the selected literature, as it may not encompass all relevant studies on the topic. Additionally, the scope of the research might not fully cover all team sports, and variations in relative age effects between different sports might not have been fully explored. This research lays the groundwork for future investigations in the field of relative age effects on athletes. To build upon this study, it is recommended that researchers conduct more longitudinal studies to better understand how relative age impacts athletes' long-term development and career trajectories in various team sports. Additionally, incorporating larger and more diverse samples from different sports and regions will enhance the generalizability of the findings.

Conclusions

Based on these findings, it can be concluded that the relative age effect appears to be distributed similarly across different sports, and there is no strong evidence to suggest that one sport is more affected by relative ages than the others. The results reveal that the relative age effect has a significant impact on the selection of representative athletes for the Kosovo National Team based on gender. There are notable differences in the representation of athletes in different quartiles between males and females, indicating that gender plays a role in how relative age influences athlete selection, where the male gender is slightly more affected by the effect of relative age. Furthermore, Q1 and Q2 are over-represented in the selection of athletes, while Q3 and Q4 are under-represented. This suggests that the relative age effect plays a role in the selection of athletes, with more athletes being selected from the younger quartiles (Q1 and Q2) compared to the older ones (Q3 and Q4). Based on these findings, it is strongly recommended that the respective federations, sports trainers, and relevant sports authorities take proactive measures to prevent the "discrimination" experienced by young athletes.

Authors' affirmation of compliance

The study was conducted in accordance with the Declaration of Helsinki, and approved by the Physical Education Management of the Faculty of Physical Education and Sport, University of Pristina "Hasan Pristina, December 2022. Nr. 1124, 27.12.2022

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Conflict of Interest

The author declares that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Comparative Analysis of Anthropometric Characteristics and Body Composition among Elite Female Futsal Players in Montenegro, North Macedonia and Croatia

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Abstract

Anthropometric characteristics and body composition play an important role in achievements in most sports, however, there are still no specific recommendations on anthropometry-body composition for women in futsal. The research aimed to determine differences in anthropometric characteristics and body composition between elite female futsal players, members of the best futsal clubs in Montenegro, North Macedonia, and Croatia. The sample of respondents consisted of 30 elite female futsal players. The first sub-sample consisted of players from Montenegro (n=11, 19.18±2.44 yrs), the second sub-sample consisted of players from North Macedonia (n=10, 18.10±3.03 yrs), and the last sub-sample was consisted by players from Croatia (n=9, 20.33±5.17 yrs). Anthropometric characteristics and body composition were evaluated by a battery of 11 variables: body height (BH), body mass (BM), triceps skinfold (TS), biceps skinfold (BiS), back skinfold (BS), abdominal skinfold (AS), upper leg skinfold (UlS), lower leg skinfold (LlS), body mass index (BMI), fat percentage (FP), and muscle mass percentages (MP). Based on the ANOVA and post-hoc tests, findings showed that have differences between the groups in 4 anthropometric parameters such body height, body mass, abdominal skinfold, and muscle mass percentages. Female futsal players from Montenegro and Croatia have significantly higher results in parameters such as body height, body mass, and muscle mass percentages. While futsal players from Montenegro have significantly higher abdominal skinfold values than Croatia. The results suggest that there is some difference in anthropometry-body composition between groups of female futsal players, but for more complete conclusions an analysis should be performed on a larger sample of high-level female futsal players. However, despite that, this research made a significant contribution because it was the first to identify differences in anthropometric characteristics and body composition among elite female futsal players from different countries in this part of the world. In this way, it has opened the doors for further research into the anthropometry of female futsal players in this region.

Keywords: *female, professional futsal players, morphological characteristics, body fat percentages*

Introduction

The International Association Football Federation (FIFA) has authorized the five-a-side indoor soccer game known as futsal, which has been played for more than 80 years and has gained popularity all over the world (Komici, Verderosa, D'Amico, Parente, & Guerra, 2023). One of the fastest growing indoor sports in the globe during the past ten years has been futsal. In the mentioned game, two teams of five play-

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ers—four on-court players, plus the goalkeeper - play against one another (Spyrou, Ribeiro, Ferraz, Alcaraz, & Travassos, 2023). In contrast to soccer, which uses 11 athletes per team, this game has a pivot (attacker), a fixed (defender), and two wings (left and right). Futsal is played with considerable intensity on a smaller field (20x40 m) for a shorter amount of time (2x20 min) (Queiroga, Cavazzotto, Portela, Tartaruga, & Silva, 2019). In addition, the performance is the impact of combining several high-intensity intermittent actions (such as accelerations, ion, decelerations, and direction changes) to fulfill certain tactical requirements (Atakan, Karavelioğlu, Harmancı, Cook, & Bulut, 2019).

Various sporting disciplines have researched body composition and physical fitness (Komici et al., 2023). However, little is known about the anthropometric traits and body composition of competitive elite female futsal players, likewise, there are still no specific recommendations on anthropometry-body composition for women in futsal. Multiple techniques were used to examine body composition parameters. Body composition and anthropometric measurements are made for a variety of reasons, including tracking athletes' growth and development and identifying changes in body composition due to nutritional therapies and physical exercise (Mendoza-Muñoz, Adsuar, Pérez-Gómez, Muñoz-Bermejo, & Carlos-Vivas, 2020).

When compared to less competitively successful teams, better teams had much lower percentages of body fat (Arnason, Sigurdsson, Gudmundsson, Holme, & Bahr, 2004). It is necessary to take into account that the athlete's body composition may affect all physical performance measures. Furthermore, the fat percentage can affect sprint time and body power as measured by the vertical jump. (Castillo, Martínez-Sanz, Penichet-Tomás, Sellés, & Sospedra, 2022).

However, despite being an indoor form of the world's most popular sport, there have been little research on the morphological structure of female participants. Some authors linked female players' anthropometric characteristics to key futsal skills (ball control, ball sprint speed, passing, dribbling, and shooting), concluding that there was a direct negative relationship between fat percentage and the ability to dribble and control the ball at maximum speed, with the ability to dribble and control the ball at maximum speed being worse the higher the fat percentage (Kooshaki & Nikbakht, 2014). On the contrary, the findings imply that in order to compete at the highest level in women's futsal, they should be a greater weight, taller, and have a higher BMI than their non-elite counterparts (Gómez-Campos, Vidal-Espinoza, Muñoz-Muñoz, Vasquez, Portugal, & Bolaños, 2023). The previous findings revealed that there is no association between body combination parameters (height, weight, body fat percentage, and body mass index) and basic skills like control and shooting. However, there is a negative and substantial association between body fat percentage and dribble skill, which means that increasing this component reduces subject performance (Kooshaki & Nikbakht, 2014).

Although there is some information on the anthropometry of female futsal players, it is not consistent, so it is essential to investigate the anthropometric characteristics of female players. Furthermore, there is no research on the anthropometry of female futsal players in this region, especially no study that has compared the anthropometry of female futsal players from different countries in this region. Based on the aforementioned, the aim of this research was to determine differences in anthropometric characteristics and body composition between elite female futsal players, members of the best futsal clubs in Montenegro, North Macedonia, and Croatia.

Method

Participants

The sample of respondents consisted of 30 elite female futsal players. The first sub-sample consisted of players from Montenegro (n=11, 19.18±2.44 yrs), the second sub-sample consisted of players from North Macedonia (n=10, 18.10±3.03), and the last sub-sample was consisted by players from Croatia (n=9, 20.33±5.17). The participants provided written consent, and the study was conducted in accordance with the Helsinki Declaration.

Measurements

Anthropometric assessments were conducted following the guidelines of the International Biological Program (Eston & Reilly, 2009). Anthropometers, calipers, and measuring tape were employed for the morphological measurements. A Tanita body fat scale (model BC-418MA) was utilized to assess body composition. Anthropometric characteristics and body composition were evaluated by a battery of 11 variables: body height (BH), body mass (BM), triceps skinfold (TS), biceps skinfold (BiS), back skinfold (BS), abdominal skinfold (AS), upper leg skinfold (UlS), lower leg skinfold (LlS), body mass index (BMI), fat percentage (FP), and muscle mass percentages (MP).

Statistics

Descriptive statistics were utilized to represent the data as mean values and standard deviations for each variable. To assess disparities in anthropometric characteristics and body composition among the three groups of futsal players, a discriminative parametric approach was employed, involving ANOVA and post-hoc testing, with a statistical significance threshold set at $p<0.05$. Data analysis was carried out using SPSS 26.0 software (Chicago, IL, USA).

Results

Based on the descriptive statistics (Table 1), it was noticeable that Macedonian female futsal players had slightly lower body height (161.28±6.96 cm) and weight (54.84±4.60 kg) compared to players from Montenegro (169.24±3.87 cm, 63.26±6.21 kg) and Croatia (173.22±6.11 cm, 62.12±4.09 kg). The highest numerical values of BMI were observed in Montenegrin players (22.04±1.66) compared to Macedonian (21.10±1.84) and Croatian (20.69±1.20) female futsal players. On the other hand, the distribution of fat percentages was highest among Macedonian female futsal players (20.48±4.98), followed by players from Montenegro (19.22±3.95), and the lowest percentage was observed in Croatian players (18.09±3.61). Meanwhile, muscle mass values were similar between Montenegrin (28.81±1.97) and Croatian (28.79±2.11) female futsal players, and slightly lower in Macedonian players (24.61±1.72).

Based on the ANOVA with Post-hoc test (Table 1), differences were observed among the groups in anthropometric parameters. It was found that the female futsal players from Montenegro and Croatia had significantly higher values in body height (0.000) and body weight (0.002) compared to the group of Macedonian female futsal players. Additionally, significantly higher values were found in the Montenegrin female futsal players compared to Croatian female futsal players in abdominal skinfold (0.025). In the parameter of muscle mass, the Montenegrin and Croatian female futsal players achieved significantly higher values than the group of Macedonian female futsal players (0.000). However, there were no significant differences observed in other anthropometric characteristics among these three groups of female futsal players.

Table 1. Descriptive data and ANOVA with post-hoc test of futsal players from different clubs

	Montenegro Mean±SD	Macedonia Mean±SD	Croatia Mean±SD	p	Post-hoc
Age	19.18±2.44	18.10±3.03	20.33±5.17	.420	/
Body height	169.24±3.87	161.28±6.96	173.22±6.11	.000*	I>II, III>II
Body weight	63.26±6.21	54.84±4.60	62.12±4.09	.002*	I>II, III>II
Triceps skinfold	13.25±2.99	11.99±3.99	13.27±3.89	.669	/
Skinfold of the back	12.54±2.67	10.70±2.08	11.19±1.87	.174	/
Biceps skinfold	10.42±2.26	7.95±2.17	8.82±2.38	.056	/
Abdominal skinfold	15.55±5.88	11.14±3.93	9.92±3.27	.025*	I>III
Upper leg skinfold	12.11±4.06	10.15±4.07	11.20±3.80	.539	/
Lower leg skinfold	15.19±4.59	16.03±3.04	15.13±3.38	.840	/
Body mass index	22.04±1.66	21.10±1.84	20.69±1.20	.172	/
Fat percentage	19.22±3.95	20.48±4.98	18.09±3.61	.478	/
Muscle mass	28.81±1.97	24.61±1.72	28.79±2.11	.000*	I>II, III>II

Legend: Mean - Arithmetic mean; SD - Standard deviation; * - significant difference between groups.

Discussion

This study aimed to identify differences in anthropometric characteristics and body composition among female futsal players from different countries. Based on ANOVA and post-hoc tests, it was observed that groups of elite female futsal players from Montenegro and Croatia had significantly higher results in parameters such as body height, body mass, and muscle mass percentages compared to the group of futsal players from Macedonia. However, futsal players from Montenegro had significantly higher abdominal skinfold values than those from Croatia. There were no significant differences in other parameters among the groups. The results suggest that there are some differences in anthropometry and body composition among groups of elite female futsal players.

By examining the obtained values, it is noticed that futsal players from Macedonia have a body height similar to the average height of players from Spain and Brazil, ranging from 161.5 to 164.1 cm (Kravchychyn, Silva, & Machado, 2013; Rubio-Arias et al., 2015; Teixeira et al., 2019; Queiroga et al., 2019). However, these values are lower compared to the average height of players from Montenegro (169.2 cm) and especially Croatia (173.2 cm) in this study. Additionally, Macedonian players have a lower average body mass (54.8 kg), which is less than the values of futsal players from Montenegro (63.3 kg) and Croatia (62.1 kg), and it is approximately similar to the values of players from Spain and Brazil (58.5-62.2 kg) (Kravchychyn et al., 2013; Rubio-Arias et al., 2015; Teixeira et al., 2019; Queiroga et al., 2019; Castillo et al., 2022). When considering BMI as a measure of the degree of nutrition among futsal players, similar values are observed for all three groups (20.7-22.04), which is consistent with the results of other studies (22.3-22.6; Kravchychyn et al., 2013; Queiroga et al., 2019; Castillo et al., 2022). Among other measures, it is worth noting the thickness of abdominal skinfold, which is 15.6 mm in players from Montenegro, corresponding to data from the study (Castillo et al., 2022), while it is significantly lower in futsal players from Macedonia and Croatia (11.1 mm and 9.9 mm, respectively).

The percentage of body fat, as an objective indicator of adipose tissue, is 18.09% for Croatia, 19.22% for Montenegro, and 20.48% for Macedonia. These results are approximately in line with values for professional female futsal players from Spain and Brazil (18.8-21.0%), although slightly higher values were reported in one study (22.4%; Queiroga et al., 2019), and significantly higher values were found in Spanish professional futsal players (27.1%; Rubio-Arias et al., 2015). On the other hand, the percentage of muscle mass varies from 24.6% in Macedonian futsal players to 28.8% in Croatian and Montenegrin players, which is significantly lower compared to a study (Castillo et al., 2022) where participants had 38.4% muscle mass. This may indicate possible differences in the way muscle mass is measured between these studies.

It should be noted that most research in these countries has focused on assessing the anthropometric characteristics and body composition of male soccer and futsal players (Arifi, Bjelica, & Masanovic, 2019; Gardasevic, Bjelica, Corluka, & Vasiljevic, 2019; Cerkez, Bjelica, Katanic, & Corluka, 2023; Katanic, Bjelica, & Milosevic, 2023), with very few studies focusing on women in these sports. This represents a limitation when comparing the given results. However, the obtained values of anthropometric parameters are roughly consistent with results achieved in anthropometric studies on female soccer players from this region (Stanković, Đorđević, Lilić, & Hadžović, 2022; Stankovic, Capric et al., 2023), as well as other female athletes (Milic et al., 2017; Katanic, Bjelica, & Covic, 2022) from these areas.

The main findings of this study indicate that the groups of female futsal players from Montenegro and Croatia are significantly taller and heavier than the group of futsal players from Macedonia. These findings align with the assertions of Gomez-Campos and colleagues (2023), who emphasize that female futsal players aiming to compete at the highest level should have slightly higher body weight, greater height, and a higher BMI compared to their non-elite counterparts. Montenegro players had significantly higher values in abdominal skinfold compared to Croatian futsal players. Additionally, both Montenegro and Croatia player groups achieved significantly higher values in muscle mass percentages compared to the group of Macedonian futsal players. There were no significant differences in other parameters among the groups. These results cannot be directly compared to other studies because this is the first study evaluating anthropometric parameters in female futsal players in this specific region. However, some studies that assessed differences in anthropometric parameters based on residential and regional characteristics did not find differences among participants from similar regions (Gardasevic, Bjelica, Corluka, & Vasiljevic, 2019; Ilic, Vitasovic, Katanic, Rakocevic, & Vasileva, 2023). In contrast, some studies did find differences in certain anthropometric parameters based on place of residence (Katanic et al., 2023). Nevertheless, drawing conclusions can be challenging due to the different samples of participants, so further anthropometric research, especially on female athletes from these countries, is needed in the future.

The previous findings have revealed that there is no association between body composition parameters (height, weight, body fat percentage, and body mass index) and fundamental skills such as ball control and shooting. However, there is a negative and significant correlation between body fat percentage and dribbling skill, indicating that an increase in this component reduces the subject's performance (Kooshaki & Nikbakht, 2014). In comparison to less successful teams, better teams have a much lower body fat percentage (Arnason, Sigurdsson, Gudmundsson, Holme, & Bahr, 2004). It is known that body fat percentage can affect sprint time and lower limb explosive power (Castillo et al., 2022). Therefore, in line with this, Kooshaki and Nikbakht (2014) found a direct negative relationship between body fat percentage and the ability to control, dribble, and handle the ball at maximum speed. Additionally, the results have shown a negative correlation between agility and the fat component, as well as a positive correlation between the muscle component and aerobic capacity, agility, speed, and explosiveness (Stankovic, Capric et al., 2023). These findings underscore the importance of maintaining optimal body composition in female futsal players, as it can significantly impact their performance in various aspects of the game.

It should be emphasized that this is one of the rare studies in this field, and the only one that has examined the differences in anthropometry among elite female futsal players in this region. Therefore, it may serve as a stimulus for future researchers to conduct more detailed investigations into the anthropometric status of professional female futsal players in the region. Particularly, considering that, based on a systematic analysis (Katanić, Ugrinić, & Ilić, 2019), anthropometric characteristics and body composition have been identified as among the most significant parameters in football, it can be expected that anthropometric characteristics could be a crucial factor in futsal as well.

One of the limitations of this study pertains to the small sample size, especially within subgroups, making it challenging to generalize the obtained results. Therefore, future studies should aim to research a larger sample of female futsal players, considering position-specific anthropometric parameters. Additionally, alongside anthropometric measurements, incorporating specific motor tests could provide insights into the overall morpho-motor status of female futsal players.

Conclusion

The results indicate that there are differences in anthropometry and body composition among groups of elite female futsal players from different countries. It is known that an effective training program depends on understanding the anthropometric parameters and body structure of professional futsal players, as well as the demands placed on them directly during matches (Stanković, Đorđević et al., 2023). Therefore, the practical implications of this study would be to provide significant information about the anthropometry of female futsal players, which would be beneficial for practitioners and coaches in their work.

However, for more comprehensive conclusions, an analysis should be conducted on a larger sample of high-level female futsal players. Nevertheless, this research has made a significant contribution as it was the first to identify differences in anthropometric characteristics and body composition among elite female futsal players from different countries in this part of the world. In this way, it has paved the way for further research into the anthropometry of female futsal players in this region.

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Conflict of Interest
The author declares that there is no conflict of interest.

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ORIGINAL SCIENTIFIC PAPER

Asymmetry in Ball Kicking Speed between Preferred and Non-preferred Leg in Young Soccer Players

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Abstract

Kicking ability is one of the most important skills in soccer. It is very important to have a good kicking ability with both feet. The main goal of this research was to determine a ball kicking asymmetry (KA) between preferred and non-preferred leg in young soccer players for the U-15 and the U-17 age category. The second goal was to determine differences in kicking asymmetry (KA) between players of different playing positions. 183 young players in U-15 category as well as 87 players from U-17 age category were tested with simple soccer specific test to evaluate kicking velocity (km/h) with preferred and non-preferred leg using Pocket radar. Testing took place on the artificial grass during dry and warm weather and ball was kicked while stationary at the 11-m spot. Results obtained with Student's T-test showed that differences in the maximum kicking velocity with the preferred and non-preferred leg were statistically significant ($p \leq 0.05$) in both age groups (90.2±8.8 km/h /78.5±9.8 km/h for U-15) and (102.7±6.4 km/h /89.5±9.4 km/h for U-17) which confirmed KA is present. Additionally, univariate analysis of variance (ANOVA) showed that there were significant differences in KA between different playing positions in U-17 group (KA=14.57/9.72/14.1 for defenders, midfielders and attackers, respectively). Soccer kicking can be a quality indicator for assessing the soccer kicking ability. By assessing kicking ability, relevant results can be obtained in a fast, and efficient way and can be utilized in the selection process of young talents.

Keywords: *kicking asymmetry, instep kick, soccer, youth players, playing positions*

Introduction

Football is one of the most popular sports in the world, played by more than 200 million people, including professionals and amateurs (Dvorak et al., 2000). It is a sport that is both extremely energetically and physically demanding due to various accelerations, decelerations, sliding on the ball and hitting (Reilly et al., 2000; Marques et al., 2011). In order to win games, the objective of football is to score as many goals as possible. Number of shots on goal is previously highlighted as a predictor for wining (Lago-Peñas & Lago-Ballesteros, 2011; Bjelica, Popović, & Petković, 2013). Ball kicking is one of the most important skills in football (Bačvarević et al., 2012). There are a few basic types of shot but instep kick is the fastest kick in football and is often used as the main tool for scoring goals (Nunome et al., 2002; Arpinar-Avsar & Soylu, 2010; Sterzing et al., 2009). Faster kicks have higher chance to be successful (Dorge et al., 2002; Marković, Dizdar, & Jarić, 2006; Sinclair et al., 2014). Measuring the speed of the kick is a simple, fast and affordable way to assess the quality of a soccer kick, which is desirable for the sport. Today, football is played faster than ever and there is no time to adjust on better leg, therefore players should have both legs equally developed. Many coaches believe that it is desirable for players to be able to use either foot equally well and that excessive one-sidedness would be a disadvantage. Some studies suggest that it could be advantageous for players to possess a good technique and abil-

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ity to kick the ball equally with both feet (Mclean & Tumilty, 1993; Grouios et al., 2002; Nunome et al., 2006; Bjelica, Popović, & Petković, 2013). Asymmetry in ball kicking speed (KA-kicking asymmetry) should be as low as possible to have as much chance for success as possible. Therefore, maximal kicking velocity with both the preferred and non-preferred leg must be an important objective for soccer coaches (Sterzing et al., 2009). Some authors (Rodríguez-Lorenzo et al., 2016) suggest kicking deficit as the percentage of the difference between the maximum ball velocity each player achieves by the non-preferred leg kick in relation to the preferred leg kick. Due to its popularity and extreme competition, it is of great interest for clubs and coaches to assess as much attributes from young players as possible. That way, training program and selection process of players are more complete and there is less chance to make mistakes. Several studies examined ball velocity with preferred and non-preferred leg (Bjelica, Popović, & Petković, 2013; Rađa, Erceg, & Grgantov, 2016; Maly et. al., 2018). Rodríguez-Lorenzo et al. (2016) did research paper with age groups (G-14) and (G+14) regarding kicking deficit (KD) between preferred and non-preferred leg which was first such finding. Findings about kicking asymmetry of young soccer players of different age groups such as pioneers (U-15) and cadets (U-17) are scarce. To the best of authors' knowledge, there is limited or none scientific information regarding positional differences in KA between defenders, midfielders and attackers. Therefore, the main goal of this research was to determine if there is a kicking asymmetry (KA) in the kick speed of U-15 and U-17 young football players. Additionally, differences in kicking asymmetry (KA) between different playing positions were evaluated.

Methods

Participants

Two hundred and sixty male participants that play in Croatian youth soccer leagues were recruited for participation in the present investigation. Written consent for participation in this study was obtained from the subjects' parents/guardians after being thoroughly informed about the purpose, benefits, and the potential risks of this study. Consent forms were specifically approved by the "The Ethical Committee of the Faculty of Kinesiology" approval number: 2181-205-02-05-23-018 (Split, Croatia). This committee approved the entire study design, which was conducted according to the ethical standards of the 1964 Helsinki Declaration and its subsequent amendments. Inclusion criteria to participate in the study were: i) participation in at least 85% of the training sessions, ii) regularly participating in the previous competitive seasons, iii) having a valid sport medical certification, and iv) being healthy (no pain or injury) and clear of any drug consumption. All players had Croatian Soccer Federation identity card signed and were fully healthy and medically examined by a local sport specialist doctor. Participants refrained from drinking caffeine-containing beverages for 24 hours and did not eat for 2 hours prior to testing in order to reduce any possible interference with the experiment.

Design and procedures

This study is a cross-sectional investigation with two main objectives: to determine ball kicking asymmetry with dominant and non-dominant leg, and to determine kicking asymmetry between players of different playing positions. Participants were divided according to different age groups (U-15, U-17). Effective playing time was not taken into consideration. The current research took place in June at the end of competition season 2016/2017. Each participant completed all trials in the same time period of the testing day and under the same climate conditions (4–7 p.m., 25.6±0.8˚C temperature, and 36.3±2.5% relative humidity). Participants were asked to avoid any stressful activity during testing or between training sessions.

Measurements

Asymmetry coefficient was calculated with formula (Rodríguez-Lorenzo et al., 2016):

$$KA=\left(\frac{KVdomMax-KVnodomMax}{KVdomMax}\right)*100$$

KVdomMax refers to kicking velocity with dominant (preferred) leg (IKPL), while KVnodomMax refers to kicking velocity with nondominant (non-preferred) leg (IKNL). Anthropometric data were measured with a portable stadiometer (SECA, Leicester, UK; for height) and an electronic scale (HD-351, Tanita, Arlington Heights, USA; for body mass). Testing protocol included a standard warm-up of 45 minutes (with 50% of theoretical maximal heart rate [220-age in yrs] as target value). Warm-up included sequences of 10 minutes of jogging with and without the ball as well as 10 minutes of dynamic stretching with a strong focus on leg and abdominal muscles. During the last 15 minutes, approaching testing and for familiarization purpose, participants were passing and shooting with the instep kick using dominant and non-dominant leg, alternatively. Players slowly increased their kicking speed as warm-up progressed as well as kicking distance. Testing took place on the artificial grass during dry and warm weather and ball was placed on the 11-m spot. Participants were wearing their own soccer shoes and the balls used were Jabulani football (Adidas, Germany; 69.0±0.2 cm in circumference and 440±0.2 g in mass). Participants shot the ball three times with instep kick using dominant and non-dominant leg, alternatively, which makes a total of 6 shots. Fastest kick per each type/leg was considered for further analysis. Players were instructed to shoot the ball from 11m as fast as they could and straight to the center of the goal. After each player would shoot, he was instructed to go to the end of the line to avoid any potential influence of fatigue. That way, every player had at least 3 minutes of rest between repeated shots. During the tests, a sport scientist was involved to provide better control and manage tasks. He had to give instruction: "Ready–Set–Go", so at "Go" the player started the running kick, while another sport scientist took the measures with a pocket radar (Pocket Radar, Inc. Santa Rosa, California), with ±2 km/h accuracy, 1m behind the goal at ball height during the kick.

Statistical analysis

Basic descriptive statistics were calculated and namely as means or average score (AS), standard deviation (SD), range–minimum and maximum results (Min., Max.) for the anthropometric status of the participants. Systematic bias of kicking performance variables was determined by using 1-way analysis of variance (ANOVA) for repeated measures, with Fischer LSD post-hoc test for eventual significant comparisons. The kicking asymmetry (KA) was calculated and conclusions were drawn about differences between players of different playing positions. To evaluate the magnitude of differences, the partial eta squared (η2) effect size was calculated. Threshold values to interpret the effect size were >0.01 (small effect), 0.06 to 0.13 (medium effect), >0.14 (large effect). Figures with a P-value of less than 0.05 were regarded as statistically significant. All statistical analyses were carried out using the commercial Statistica software version 13.0 (Dell Inc., Round Rock, TX USA).

Results

The current study included 260 young football players, 173 of which were U-15 players and 87 of which were U-17 players. Anthropometric data are presented in Table 1. Body height and weight for U-15 group was 167.22±9.80 cm and 54.24±10.42 kg respectively. Body height and weight for U-17 group was 177.32±6.63 cm 67.22±7.81 kg respectively. Body mass index for U-15 category was 19.21±2.05 and 21.34±1.88 for U-17 age group. Training age for U-15 participants was 6.53±1.74 age and 8.66±1.42 age for U-17 participants.

Table 1. Basic anthropometry of participants

	U-15 (N=173)		U-17 (N=87)	
	MEAN±SD	Range	MEAN±SD	Range
BH	167.22±9.80	145.05-194.30	177.32±6.63	155.60-195.50
BM	54.24±10.42	30.80-82.60	67.22±7.81	37.20-83.70
BMI	19.21±2.05	13.90-25.70	21.34±1.88	15.36-26.37
TA	6.53±1.74	1.00-9.00	8.66±1.42	2.00-11.00

Legend: BH-body height, BM-body mass, BMI-body mass index, TA-training age, MEAN-arithmetic mean, SD- standard deviation, Range- minimum and maximum results.

Table 2. T-test, differences ball speed with preferred and non-preferred leg

		MEAN±SD	t	df	p
U-15	IKPL (km/h)	90.2±8.8	22.96	172	0.00
	IKNL (km/h)	78.5±9.8			
U-17	IKPL (km/h)	102.7±6.4	14.97	86	0.00
	IKNL (km/h)	89.5±9.4			

Legend: IKPL-instep kick preferred leg, IKNL- instep kick non-preferred leg, MEAN- arithmetic mean, SD- standard deviation, t- t value, df- degree of freedom, p- p value.

Results obtained with Student's T-test (table 2) showed that differences in the maximum kicking velocity with the preferred and non-preferred leg were statistically significant (p≤0.05) in both age groups (90.2±8.8 km/h / 78.5±9.8 km/h for U-15) and (102.7±6.4 km/h /89.5±9.4 km/h for U-17) which confirmed kicking asymmetry (KA) was present. Furthermore, univariate analysis of variance ANOVA was used (presented in table 3 and table 4). With p=0.023 and η2=0.086 for U-17 age group ANOVA showed that there were statistically significant differences and medium effect size in KA between different playing positions (KA=14.57/9.72/14.1 for defenders, midfielders and attackers, respectively). There were no significant differences in KA between playing positions for U-15 age group. Range of results for KA (1.02-43.24/0.94-47.37 for U-15 and U-17 respectively) suggest quite a diverse ability to kick the ball with both feet.

Table 3. Kicking asymmetry (KA), ANOVA - differences between playing positions

	KA MEAN±SD	Range	CI	F	p	η2
U-15	12.94±7.46	1.02-43.24	6.75-8.34	0.145	0.865	0.002
U-17	12.79±7.68	0.94-47.37	6.68-9.03	3.930	0.023	0.086

Legend: KA – kicking asymmetry, MEAN- arithmetic mean, SD- standard deviation, Range - minimum and maximum results, CI – confidence intervals, F- F value, p- value, η2- partial eta squared.

Table 4. Fischer LSD post hoc, differences between positions for U-17 players

	(d) KA=14.57	(m) KA=9.72	(a) KA=14.10
(d)		0.01	0.82
(m)	0.01		0.04
(a)	0.82	0.04	

Legend: d-defenders, m-midfielders, a-attackers, KA – kicking asymmetry.

Using Post-Hoc (presented in table 4 and figure 1), statistically significant differences were found between midfielders and defenders (p=0.01) and between midfielders and attackers (p=0.04). There were no statistically significant differences between attackers and defenders (p=0.82). Within figure 1 it is visible that KA averages (in km/h) of midfielders are significantly lower than the ones from attackers and defenders.

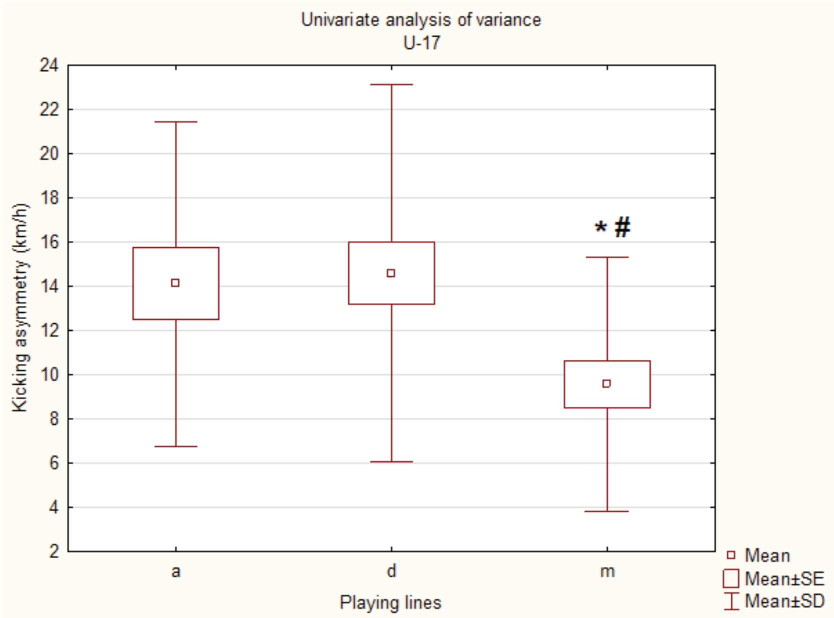

FIGURE 1. Differences between playing positions (U-17)
* significant differences from attackers; # significant differences from defenders

Discussion

Results showed that there were statistically significant differences in maximum kicking velocity with the preferred and non-preferred leg for both groups which confirmed KA is present. Also, there were statistically significant differences between different playing positions for U-17 age group, but there were no significant differences in KA between playing positions for U-15 group.

It is important to note, that to the best of our knowledge, there were very few scientific papers regarding asymmetry in ball velocities with preferred and non-preferred leg in soccer, especially for young players. Previous studies focused primarily on preferred / non-preferred leg asymmetries regarding strength and flexibility. However, we believe that implementation of simple soccer specific tests such as the one done in this research could improve knowledge regarding the mentioned problem. In this research, significant differences in ball velocities between preferred and non-preferred leg were obtained which confirmed KA. These findings correlate with some previous studies (Nunome et al., 2006; Rodríguez-Lorenzo et al., 2016). KA results from this study 12.94±7.46 group for U-15 and 12.79±7.68 for U-17 group are lower values than the ones found in Rodríguez-Lorenzo et al. (2016) for elite youth level players from Spain (15.31±7.32/15.83±7.88 for group -14 and group +14, respectively). Interestingly, Oliveira, Barbieri and Gonçalves (2013) found differences in kick distance between dominant and non-dominant leg for senior soccer players and for active men. This could suggest differences in ball speed when kicking the ball with preferred and non-preferred leg. Also, in this research statistically significant differences in KA between different playing positions for U-17 players were found. As mentioned previously, differences in kicking asymmetry between positions is a topic yet to be fully explored scientifically. However, several studies tested the differences between playing positions in ball velocities (Rađa, Erceg, & Grgantov 2016; Arafat, Rickta, & Mukta 2020). Rađa, Erceg and Grgantov (2016) found differences in ball speed with instep kick and with side foot kick with both feet for different playing positions. Defenders performed significantly worse than midfielders with preferred and with non-preferred leg, however, KA was not calculated. In research done by Arafat, Rickta and Mukta (2020) there were no significant differences in kicking velocities between defenders, midfielders and attackers for senior players (82.14/79.77/79 km/h, respectively).

As presented in table 3, there were no statistically significant differences in KA between different playing positions for U-15 age group. Possible reasons for these findings could be that there is much less player specialization at that age and each player probably plays more than one position. That way, it is possible that there are still a lot of players that will change position or have some other roles in the team as time passes by. On the other hand, U-17 players are closer to professional status and probably already solely play at their natural or best positions. Midfielders had the smallest asymmetry in ball speed between legs (KA=9.72) in U-17 age group. That was somewhat expected because of the game role and assignments they do during the game. Also, selection of players is such that one of the most desired abilities for midfielders is actually ball kicking ability with both feet. Either while passing the ball or trying to score the goal, midfielders are often in the different areas of the pitch and need to possess good kicking ability with both feet. On the other hand, defenders and attackers had similar kicking asymmetry results (KA=14.57/14.1, respectively). This type of results suggests that attackers and defenders have much more one footedness than midfielders. Whether it is because of the selection process or because of the game situations they often perform during the match, it seems that attackers and defenders have less adaptation of ball kicking with non-preferred leg than midfielders. Kicking asymmetry is one of the relatively unexplored and a new possible way of looking at kicking ability which could further enhance our understanding of soccer. This study, as presented offers another way of assessing players' skill and potential talent in youth soccer. Unfortunately, this study had a few limitations. Players were divided in only three playing positions (defenders, midfielders and attackers) without goalkeepers, whereas it would be better in future research to divide them with even more specific classification with full backs and wingers. Future research should also include younger players in youth academies as well as senior players in order to make conclusions regarding KA throughout players soccer development and career. That way, scientists could detect certain occurrences of KA in different age groups and soccer coaches would be able to make appropriate changes in training routines regarding scientific guidelines.

Conclusion

Kicking asymmetry (KA) between preferred and non-preferred leg using soccer instep kick is evident and seems to remain similar in different age groups (U-15 and U-17). These results suggest that KA is possibly present during whole soccer development. Midfielders demonstrated a higher level of bilateral kicking ability when compared to other playing positions in older group of respondents (U-17), which suggests a certain level of positional specialization. Soccer kicking can be a quality indicator for assessing the soccer kicking ability. By assessing kicking ability with both feet and subsequently KA, relevant results can be obtained in a fast, and efficient way and can be utilized in the selection process of young talents.

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Conflict of Interest
The authors declare that there is no conflict of interest.

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SportMont

ORIGINAL SCIENTIFIC PAPER

How Many Trials Is Enough To Access Balance On The Biodex Balance System?

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Abstract

While the Biodex Balance System (BBS) proved to be a reliable apparatus for testing balance, there is lacking evidence on the number of familiarization trials the subject must undergo before being tested, especially when lower stability levels are used on the BBS. The goal of this study was to determine after how many attempts the result of the balance test on the BBS stabilize. The sample consisted of 66 boys (13.42±0.50 years old) and 57 girls (13.39±0.49 years old) not active in sports. The subjects underwent the Postural Stability Test three times and the Overall Stability Index variable of the BBS was examined. In the analysis the results were separately examined for boys and girls using the t-test. Significant differences were found in all three measurements for boys while for girls there was no significant difference between the second and third measurement. The results of this study indicate that, for boys of this age, the results did not stabilize through the three trials conducted which could mean a need for more than three attempts so the negative effects of familiarization could be avoided. For girls it seems that the result stabilized at the second trial which could indicate that only one trial is needed for them to familiarize themselves with the test. These findings can be important for future studies with a similar female sample to save time and conduct just one test trial before the actual testing, since testing balance on the BBS requires a significant amount of time.

Keywords: *familiarization, learning effects, balance, adolescents, testing, t-test*

Introduction

Although there is no universal definition of balance and postural control among scholars, one mechanical definition reads that the state of an object with the resultant forces acting upon it equal to zero can be defined as balance (Pollock, Durward, Rowe, & Paul, 2000). Human balance, simply, could be therefore defined as the person's ability to not fall (Pollock et al., 2000). Postural control could be defined as the act of maintaining, achieving or restoring a state of balance during activity (Pollock et al., 2000). Strong evidence shows that the postural control and balance system develops during childhood and adolescent age, where young children before the age of 6 mostly rely on their visual and vestibular system and later more towards a somatosensory and vestibular control (Wälchli et al., 2018). In todays children it has become increasingly important to address insufficiencies in balance and postural control to prevent musculoskeletal pathologies (Azevedo, Ribeiro, & Machado, 2022). The training of balance proved to be effective in improving dynamic balance as well as maintaining static posture with applied oscillations in athletes and nonathletes (Zech et al., 2010). In clinical testing, a widely spread tool for assessing balance and postural control is the Biodex Balance System (BBS) that can provide relevant measures of the aforementioned motor skills (Cug & Wikstrom, 2014). The BBS proved to be a reliable and acceptable tool for clinical screening of balance capabilities (Hinman, 2000). However, some studies indicate a need for familiarization in one session testing on the BBS to mitigate learning effects that cause a disruption in the data interpretation (Cug & Wikstrom, 2014; Garcia et al., 2017). The amount of familiarization needed on the BBS varies from one author to another, where authors used

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two familiarization trials on twenty one healthy female adults aged 22.8±1.0 (Pereira et al., 2008), three familiarization trials on twenty sedentary university students (11 male aged 22±2.24 years old and 9 female aged 20.88±1.16 years old) (Cug & Wikstrom, 2014) and five familiarization trials on nineteen healthy university students aged 24.4±4.2 years old (Schmitz & Arnold, 1998). Given a lower stability level on the BBS, learning effects may be seen even at six trials on ten healthy non-athletic adults aged 25.0±1.74 years old (Bagheri, Sarmadi, & Arani, 2012). Authors of this paper could not find previous studies related to the effects of familiarization on the BBS in the population observed in this research. Previous studies also seem to be inconsistent regarding the amount of familiarization trials needed on different populations when testing balance on the BBS. In relation to this, the aim of this research was to study the effects of familiarization on the BBS in adolescents during early and ongoing puberty.

Methods

Participants

All of the participants in this study (N=123; 66 males with an average age of 13.42±0.50 years old and 57 females with an average age of 13.39±0.49 years old) are students of the 7th and 8th grades (13-14 years of age) in Croatian elementary schools in Split. The participants that are included in this study have their legal guardians consent to participate in the study. The study was approved by the ethics committee of the University of Split, Faculty of Kinesiology (Approval number: 2181-205-02-05-22-0021; Approval date: 24.04.2022.).

Measurements

The measurement apparatus used for assessing balance was the BBS, which studies proved to be a reliable tool for assessing balance (Cachupe, Shifflett, & Wughalter, 2001; Parraca et al., 2011) and the observed variable for balance on the BBS was the Overall Stability Index (OSI) using the Postural Stability Test (PST). Due to the nature and the aim of this study (regarding familiarization) it is important that the participants were not familiar with the BBS and the specific balance test. The goal of the PST for the participants was to maintain balance on the dynamic platform. More specifically, the goal was to have the dot displayed on the BBS screen (which was the live representation of the projection of their center of mass on the platform) to stay in the center with as little deviation from it as possible. When the participants first step on the BBS platform, they were instructed to have a comfortable shoulder width stance with as a symmetrical stance as it was possible to achieve. The measurer then blocks the view of the display from the participants and adjusts the participants forward or backwards so that the dot on the BBS display is in the center. In this process it is important that the participant moves forward to correct the starting stance instead of leaning. When this was achieved, the measurer then lets the participants see the display in front of them and explains the goal of the PST. The participants went through three trials of the PST each lasting 20 seconds and with 10 seconds of pause in between trials. The stability level was selected to be 9 out of 12 (1 being the least stable and 12 being the most stable). All the three trials were done barefoot. Anthropometric measurements of body height and body mass were measured three times and the arithmetic mean of all three measurements was used as the representative body height and body mass of each participant.

Statistics

Descriptive statistics was applied to the variables: body height, body mass and OSI. The parameters observed for these variables were: number of valid participants, arithmetic mean, standard deviation, minimum, maximum, Kolmogorov Smirnov test, p-value. The t-test statistical analysis for dependent samples was applied to measure the statistical differences between each of the three trials (OSI1, OSI2 and OSI3). Significance level was set to 0.05. All statistical analyses were done separately on male and female subjects using the STATISTICA 14 by StatSoft (Europe) GmbH.

Results

The results of the descriptive statistical analysis (Tables 1 and 2) describe the participants in regards to their body height, and body mass. The number of valid male participants in this study is 66 (N=66) with the average body height of 170.48±7.80 centimeters, and average body mass of 61.64±13.5 kilograms. The number of valid female participants in this study is 57 (N=57) with the average body height of 165.99±5.35 centimeters, and average body mass of 56.28±9.97 kilograms. The Kolmogorov-Smirnov normality test indicates a normal distribution for the male participants in their body height and body mass ($p>0.20$), while the variable OSI is not normally distributed ($p<0.01$). The same can

Table 1. Descriptive Statistics for Boys

V	N	Mean+SD	MIN	MAX	KS	p
BH (cm)	66	170.48±7.80	151.00	185.33	0.07	p>0.20
BM (kg)	66	61.64±13.51	36.30	94.50	0.10	p>0.20

Note. V – variable; N – number of valid participants; Mean – arithmetic mean; SD – standard deviation; MIN – minimum; MAX – maximum; KS – Kolmogorov Smirnov test; p – p-value; BH - body height; BM - body mass.

Table 2. Descriptive Statistics for Girls

V	N	Mean+SD	MIN	MAX	KS	p
BH (cm)	57	165.99±5.35	152.00	178.50	0.10	p>0.20
BM (kg)	57	56.28±9.97	32.40	83.00	0.13	p>0.20

Note. V – variable; N – number of valid participants; Mean – arithmetic mean; SD – standard deviation; MIN – minimum; MAX – maximum; KS – Kolmogorov Smirnov test; p – p-value; BH - body height; BM - body mass.

be observed with the female subjects in regards to the normal distribution of body height and body mass (p>0.20) as well as for the variable OSI (p<0.10).

The results of the t-test statistical analysis (Tables 3 and 4) show the differences between the three trials on the PST (observing the OSI variable), separately for male and female participants. Statistically significant differences can be observed in all the comparisons between trials in male and female participants except between the second and third trial in the female participant group (p=0.123).

Table 3. Statistical Analysis (t-test) for the Boys

V	N	Mean+SD	T	P	CI-95	CI+95
OSI1	66	1.27±1.18	2.889	0.005	0.042	0.228
OSI2		1.14±0.98				
OSI2	66	1.14±0.98	2.261	0.027	0.009	0.148
OSI3		1.06±0.81				
OSI1	66	1.27±1.18	3.571	0.001	0.094	0.333
OSI3		1.06±0.81				

Note. V – variable; N – number of valid participants; Mean – arithmetic mean; SD – standard deviation; T - t-test result; P – p-value; CI-95 – confidence interval of -95%; CI+95 - confidence interval of +95%; OSI1 - overall stability index (first trial); OSI2 - overall stability index (second trial); OSI3 - overall stability index (third trial).

Table 4. Statistical Analysis (t-test) for the Girls

V	N	Mean+SD	T	P	CI-95	CI+95
OSI1	57	0.90±0.40	2.313	0.024	0.012	0.170
OSI2		0.81±0.36				
OSI2	57	0.81±0.36	1.565	0.123	-0.017	0.136
OSI3		0.75±0.26				
OSI1	57	0.90±0.40	3.608	0.001	0.067	0.235
OSI3		0.75±0.26				

Note. V – variable; N – number of valid participants; Mean – arithmetic mean; SD – standard deviation; T - t-test result; P – p-value; CI-95 – confidence interval of -95%; CI+95 - confidence interval of +95%; OSI1 - overall stability index (first trial); OSI2 - overall stability index (second trial); OSI3 - overall stability index (third trial).

Discussion

When observing the results and the variable OSI, it is important to identify that the number representing OSI and test performance are inversely correlated. Meaning, the lower the OSI the better the performance on the test because OSI represents displacement from a level platform position in degrees. It is somewhat clear that the female participants have better performance overall on the PST in comparison to the boys. This is confirmed when observing the t-test analysis (the first and also on average the worst trial for the female participants is 0.90±0.40 while for the male participants the third and on average the best attempt is 1.06±0.81). Although, previous a study suggests small differences between genders in balance performance in adolescents and account for the differences being due to other factors as well (Valtr, Psotta & Abdollahipour, 2016). For the male participants significant differences were shown across all three trials. This indicates that the results on the PST did not stabilize and the male participants may have not been familiarized with the test by the third trial (test performance was significantly improved after each trial). Authors in previous research in different motor skills have acknowledged the utilization of familiarization trials and its importance in minimizing disturbance of data caused by this effect (Glaister et al., 2009). Utilization of familiarization trials has shown to be most beneficial in testing motor skills such as agility, muscular endurance and balance (Coledam & de Oliveira, 2020). Moreover, more research is needed on how many trials are enough for the results to stabilize on the BBS using the PST for the male population aged 13-14 years old on different stability levels on the BBS since in this study three trials were not enough for the results to stabilize and as of the knowledge of the authors of this paper, no studies have examined the familiarization effects on the BBS in this population group. For the female participants it seems that after only one trial the results on the PST stabilize. The reason for these gender differences can only be speculated. In regards to this, the authors of this paper recommend that future studies take into consideration if there are significant differences between variables or groups for choosing the appropriate method of condensing the results for statistical analysis. Further research should also take into consideration the learning effects on the BBS with lower stability levels since this was not observed in this paper. The amount of familiarization trials varies in different studies, regarding balance and the BBS, with a different population observed and different platform instability levels used (Schmitz & Arnold, 1998; Pereira et al., 2008; Bagheri et al., 2012; Basar et al., 2012; Cug & Wikstrom, 2014). Finding studies with a similar population using the same instability level on the BBS has shown to be difficult because BBS has 12 different instability levels as well as a static level. Older versions of the BBS have 8 levels of instability and not 12 as the newer models (Schmitz & Arnold, 1998; Cachupe et al., 2001). Studies using the BBS tend to not give information if the subjects are already familiar with the apparatus or is this their first time being exposed to it (Inanir, Cakmak, Hisim & Demirturk, 2014; Sung & Kim, 2018).

Some studies specify they used familiarization trials on the BBS but do not specify exactly how many trials and in what length (Hobbs, 2008).

Limitations of the study

The small number of measurement trials in men may represent a limitation of this study, as it was not possible to obtain stabilization of the results through (only) three trials. Therefore, it cannot be concluded how many trials men of this age need to familiarize themselves with this test.

For future studies, the authors of this study recommend taking these findings into account when exploring balance on the BBS for the male population as they may disrupt the validity of the results if the learning effects are not taken into consideration. Also, it would be useful to investigate after how many attempts the effects of familiarization are not significant any more in men. Furthermore, differences in body height and mass between genders could have also played a factor since the morphological characteristics seem to have influence over balance results (Greve, Alonso, Bordini, & Camanho, 2008; Ku, Abu Osman, Yusof & Wan Abas, 2012). These differences could be seen when observing Table 1 and Table 2, but those characteristics were not included in the analysis since this is not in the scope of this study so further studies could take this into consideration when setting up their research.

Conclusion

To summarize, the BBS has shown to be a reliable tool for assessing balance and postural control, but authors should not underestimate the importance of familiarization with the BBS when subjects are exposed to it for the first time. This paper focused on discovering if familiarization effects are present when testing on the BBS and how much of an effect it could have on the results. If studies involving the PST on the BBS do not take familiarization into consideration, the interpretation of results could be inaccurate, in other words, relations between balance and some other variable(s) are questionable. The female participants had better performance on the BBS and only one familiarization trial seems to be enough for them to get stable results on the BBS. These findings could be important for future studies involving a similar female sample to save time since testing and using the BBS can be a long and complicated process. Results for the male participants did not stabilize so more research is needed to conclude on how many familiarization trials are needed for the results to stabilize. Further, the authors of this paper recommend future studies to first see if there are significant differences between measurements and depending on that should choose the appropriate method of condensing the results.

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Conflict of Interest

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ORIGINAL SCIENTIFIC PAPER

Determine the Occurrence of Latent Myofascial Trigger Points Through Binary Logistic Regression Model in Sportsmen

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Abstract

This study aimed to ascertain the prevalence of latent myofascial trigger points (L-MTrPs) among male athletes. Ninety participants were randomly included in the study. The presence of L-MTrPs was identified based on Simon's criteria, leading to the categorization of participants into two groups: L-MTrPs (n=45) and non-myofascial trigger points (N-MTrPs) (n=45). Pain pressure thresholds (PPT) of muscles were quantified using a pressure algometer, while force production was measured with HUMAC NORM isokinetic. The range of motion (ROM) for knee flexors and extensors was determined using Kinovea software. These parameters were treated as independent variables, whereas L-MTrPs and N-TrPs were considered dependent variables. Binary Logistic Regression (Enter Method) was employed for data analysis. Model fitness was evaluated through standard error computation, Wald's $\chi 2$, and odds ratios with a 95% confidence interval. Model adequacy was assessed via the likelihood ratio, Cox & Snell (R2), Nagelkerke (R2), and Hosmer and Lemeshow tests. The results effectively predicted L-MTrPs and N-MTrPs, revealing that force production, PPT, and ROM were the most significant predictive variables for L-MTrPs. In conclusion, reduced force production, lower PPT, and restricted ROM were indicative of L-MTrPs. Consequently, regular evaluation of muscular strength, PPT, and ROM are recommended for athletes to prevent the development of L-MTrPs.

Keywords: *binary logistic, myofascial pain syndrome, peak torque, range of motion, regression model, athletes*

Introduction

Myofascial Pain Syndrome (MPS) is a musculoskeletal ailment responsible for the sensation of pain (Duarte et al., 2021). This type of pain is characterised by the presence of hyperirritable regions known as trigger points (TrPs). These hyperirritable regions, specifically termed myofascial trigger points (MTrPs), are located within taut groupings of skeletal muscle fibers (Bethers et al., 2021; Das & Jhajharia, 2022b). These are tender, firm nodules measuring 3–6 mm in diameter (Donnelly & Simons, 2019). The research findings indicate that there is a notable difference in the occurrence of myofascial pain between males (37%) and females (65%) (Xia et al., 2017; Sabeh et al., 2020). Moreover, clinically, MTrPs are categorized into two types, namely active and latent MTrPs (L-MTrPs) (Jiménez-Sánchez et al., 2021). MTrPs that do not elicit pain are commonly referred to as L-MTrPs (Ge & Arendt-Nielsen 2011; Cygańska et al., 2022), whereas constant discomfort is a result of active-MTrPs (A-MTrPs) (Ibrahim et al., 2021). Musculoskeletal injuries are very common, and research in the fields of sports and exercise has shown that these are the most frequent kinds of injuries (Haser et al., 2017; Lee et al., 2020). Moreover, scientific investigation indicates that myofascial pain constitutes a substantial percentage (approximate-

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ly 85%) of musculoskeletal pain from injuries (Wheeler, 2004), As a consequence, scholars have recommended that MPS be taken into consideration as a possible cause of musculoskeletal pain (San-Antolín et al., 2020). Research indicates that athletes are susceptible to developing MTrPs over time (Fousekis & Kounavi, 2016; Kisilewicz et al., 2018; Das et al., 2022). Long-term or inconsistent training, low-load repetitive muscular activity, chronic and acute mechanical and electrical damage, persistent stress, and prolonged ischemia can all damage myofibrils and promote the formation of L-MTrPs (Ge & Arendt-Nielsen, 2011), It requires a thorough evaluation and a customised treatment plan (Barbero et al., 2019). The presence of MTrPs in the myofascial system is linked with reduced flexibility and strength of the affected muscles, as well as an increased sensation of muscle tightness (Fousekis & Kounavi, 2016; Haser et al., 2017). Athletes who are injured experience a decline in physical performance (San-Antolín et al., 2020; San-Antolín-Gil et al., 2022). Therefore, it becomes very important to identify MTrPs and take proper action. A correct diagnosis is the first step to treatment, and diagnostic accuracy depends on test reliability. Reliability pertains to the level of agreement among examiners when conducting the same test on the same patients (Das & Jhajharia, 2022a). There are a number of technologies that have been utilised to detect MTrPs, including microdialysis, biopsies (Do et al., 2018), imaging techniques by ultrasonography (Elbarbary et al., 2023), electromyography (Chattrattrai et al., 2023), and a digital pressure algometer (Karpuz et al., 2023). Among them, the most cost-friendly tool is the pressure algometer. This device is used by various researchers and they confirm that it has high reliability (Escalona-Marfil et al., 2020). This instrument measures the pain pressure threshold (PPT) of muscles. From the review of literature, it was observed that L-MTrPs reduced range of motion (ROM), muscular strength (Walsh et al., 2019), and PPT (Rodríguez-Jiménez et al., 2022). Therefore, these factors are believed to be crucial to determining the existence of L-MTrPs. Despite the fact that a great number of studies on MPS and MTrP have been published, only a few have been published regarding the symptoms of L-MTrPs. In addition, L-MTrPs can have a negative impact on daily life and sports performance (Shah et al., 2015; Das & Jhajharia, 2022b; Das et al., 2023), therefore, L-MTrPs is perceived as a serious health problem by researchers (Ribeiro et al., 2018). A review of related literature suggests that MTrPs reduce muscular strength (Das & Jhajharia, 2022b) and flexibility (Das et al., 2023) which affect the daily activity of sports person as well as sedentary people. Therefore, the purpose of this study was to identify that losing muscular strength, flexibility, and low PPT can be the symptoms of L-MTrPs. From the past research evidence, it was found that most of the studies were conducted on A-MTrPs, and very few studies are available on L-MTrPs (Duarte et al., 2023). Li and his colleagues reported in their systematic review article that a number of studies failed to properly report the MTrP diagnostic criteria (Li et al., 2020). Therefore, the objective of the study was to find out the symptoms of L-MTrPs. To fulfil the purpose of this study, a binary logistical regression approach was used. The findings of the present study will help the athletes detect the early symptoms of L-MTrPs, and that will help them prevent and take proper care of myofascial layer.

Methods

Participants

For this study, 90 national-level players were randomly selected on the basis of their sports participation and sports experience from Madhya Pradesh, India, and the research was carried out at the Exercise Physiology Laboratory of Lakshmibai National Institute of Physical Education in Gwalior, India. In this study, a cross-sectional comparative research design was used. The diagnostic criteria proposed by Simons (Donnelly & Simons, 2019) were utilized to identify L-MTrPs. (1) a band of muscle that is physically taut; (2) a spot that is tender and overly sensitive; (3) the compression of MTrP elicits the reproduction of referred pain; (4) and the jump sign when compressed (Zuil-Escobar et al., 2015). The inclusion criteria for the L-MTrPs group were based on three requirements. Firstly, the presence of L-MTrPs in the hamstring and quadriceps muscle groups located in the lower limbs. Secondly, the subjects had to be male athletes who played a sport that involved jumping, sprinting, twisting, turning, acceleration, and deceleration as essential components. Additionally, these athletes had to play their sport competitively, which was defined as participating a minimum of twice per week in an organized training or match situation for a team that competed in an official league or cup. Lastly, all subjects were collegiate athletes who were currently in the competition phase of their sport. On the other hand, the inclusion criteria for the non-L-MTrPs group were the absence of a palpable taut band in the muscles. The exclusion criteria for this study were multifaceted. Firstly, subjects were excluded if they were currently experiencing any injury or illness, including any systemic muscular or neural disease, or any lower limb or lower back injury in the previous three months. Secondly, the study excluded subjects who had recently been diagnosed with or treated for fibromyalgia, suffered from vascular or neural conditions, or received treatment for MTrPs (active or latent). Subjects who met the inclusion criteria were explained the study and were included in the sample only if they agreed to participate. Purposive sampling was used to select the first group, which consisted of 45 subjects diagnosed as positive for L-MTrPs. The second group consisted of subjects who were negative for MTrPs. To ensure an equal sample size and better statistical inference, 45 non-MTrPs subjects were included in this study. A skilled

Table 1. Participants' basic characteristics

Parameter	L-MTrPs (N-45)	Non-TrPs (N-45)
Subject characteristics	Mean ± SD	Mean ± SD
Age (Years)	20.67±2.17	20.55±2.28
Height (Centimetre)	161.11±5.81	161.20±5.56
Weight (Kilogram)	58.41±5.26	59.84±5.71

L-MTrPs: Latent Myofascial Trigger Points; Non-TrPs: Non-Trigger Points; N: Sample Size;SD: Standard Deviation physiotherapist supervised the test. Written consent was obtained from the participants' parents or legal guardians. The ethics committee of the Lakshmibai National Institute of Physical Education approved this study. It was conducted according to the latest version of the Declaration of Helsinki (approval number: 392/1346/27, 22 February 2023).

Table 1 demonstrates the basic characteristics of L-MTrPs Group's age 20.67±2.17 years, height 161.11±5.81cm, and the weight 58.41±5.26kg, whereas non-TrPs age 20.55±2.28 years, height 161±5.56cm, weight 59.84±5.71kg (Table 1). The mean age, height, and weight were found to be statistically same for both the group (p>0.05). All the participants were actively engaged in competitive sports.

Instruments

The evaluation of PPT was performed using a digital pressure algometer (FPX 25 Wagner Instruments, Greenwich, CT, USA) in accordance with the methodology reported by Cygańska et al. (2022). It has been reported that this device has high reliability (Castien et al., 2021; Cygańska et al., 2022). The measurement of force production by peak torque (Newton/meter2) in knee tests was performed using the HUMAC NORM isokinetic machine (Computer Sports Medicine, Inc., Stoughton, MA, USA). It has been reported that the reliability of this machine ranges between 0.74-0.89, indicating its consistent and dependable performance in measuring the peak torque (Habets et al., 2018). Kinovea® version 0.9.5 software was used by the researchers to measure the ROM. The software was found to be a valid and reliable tool for measuring accurately at distances up to 5 meters from the object. It is available for free download at (https://www.kinovea.org/) (Puig-Diví et al., 2019; Fernández-González et al., 2020).

Pain Pressure Threshold Examination

The pain test utilised algometric measurement, with the pain threshold defined in lbs/cm2 and the pressure applied at a constant speed. The PPT was determined as the threshold point at which pressure sensations transitioned to pain (Ortega-Santiago et al., 2020). The measurement of the pain threshold for particular muscles was conducted, and the identification of the locations of the L-MTrPs was established as mentioned by Cygańska et al. (2022). A device at a 90° angle to the skin surface measured the quadriceps and hamstrings in a supine and prone position, respectively (Fig 1), consistently beginning at the locations on the right side. When the test point initially hurt, the individual was instructed to say "STOP." Before measuring the actual sites, a trial measurement was taken on the subject's forearm muscles to ascertain their pain threshold. Each measurement was analysed by the same researcher and separated by five minutes. The following sequence was used to test each subject's muscle groups: initially, the right and left quadriceps muscle groups (Vastus lateralis, Rectus femoris, and Vastus Medialis), then the right and left hamstring muscles (Bicep femoris, semitendinosus, and semimembranosus). A physical therapist with over a decade of experience oversaw the measurement process. A 30-second break was provided between each measurement, and the mean of three trials was computed for analysis. This approach to assessing PPT has been proven to demonstrate strong intra- and inter-examiner consistency (Chesterton et al., 2007).

FIGURE 1. Evaluation of Latent Myofascial Trigger Points

Muscular Strength Measurement

After algometric measurements, the CSMi HUMAC NORM isokinetic dynamometer was used to measure the participants' lower limb force production. Knee flexion and extension muscle strength were measured while participants were positioned on the testing table with stabilisation straps and a horizontal pad over their thighs. The trunk was supported by the backrest of the table. Peak isokinetic concentric knee extension and flexion torque of both legs were evaluated at an angular velocity of 180°/s velocity. The knee extension and flexion contractions were performed through a range of 0-90° (full extension is defined as 0 degree). All participants were instructed to complete three submaximal trials at a given angular velocity for familiarisation and warmup. Peak isokinetic concentric knee extension and flexion torque were measured with the right and left legs at 180°/s extension. All the participants performed five maximal repetitions of knee extension and flexion at a selected angular velocity. A break of at least 3 minutes was given when the machine setting was changed for the opposite leg. The order of testing was randomised for the

FIGURE 2. Evaluation of Muscular Strength

right and left legs. Verbal encouragement and visual feedback were given by the investigator to all participants to help them concentrate on the quality of their movements. The greatest peak torque (Nm) for knee extension and flexion was calculated automatically by the HUMAC NORM System and served as the outcome measure (Tasmektepligil, 2016).

ROM Examination

A GoPro 9 action camera recorded each participant's knee joint from the sagittal and frontal axis profiles. Markers, both passive and reflective, were placed on the greater trochanter, the external femoral condyle, and the lateral malleolus to measure the angular displacement of the knee joint as seen from the side (Silva et al., 2018; Fernández-González et al., 2020). A tripod-mounted camera was 80 cm high and 1.5 m from participants. To maintain camera-to-participant distance, the tripod was placed on floor tape. Before the tests, everyone did a five-minute warm-up. The knee of each subject was positioned near the edge of the table while they lay prone on it (Fig 3). The individual was instructed to bend their knee as much as possible before extending it to determine the angle of knee flexion. The unfolded leg was measured as knee extension, and a closer to 0° angle was considered a good extension ROM. All videos were imported onto a laptop and analysed using Kinovea software. The greater trochanter, external femoral condyle, and lateral malleolus were marked in the Kinovea. An angle was placed in the external femoral condyle. One line went through the humerus bone and ended at the greater trochanter (stationary arm), while the other went through the tibia bone and ended at the lateral malleolus (movable arm). Angles were used to describe the two lines' intersection (Das et al., 2023).

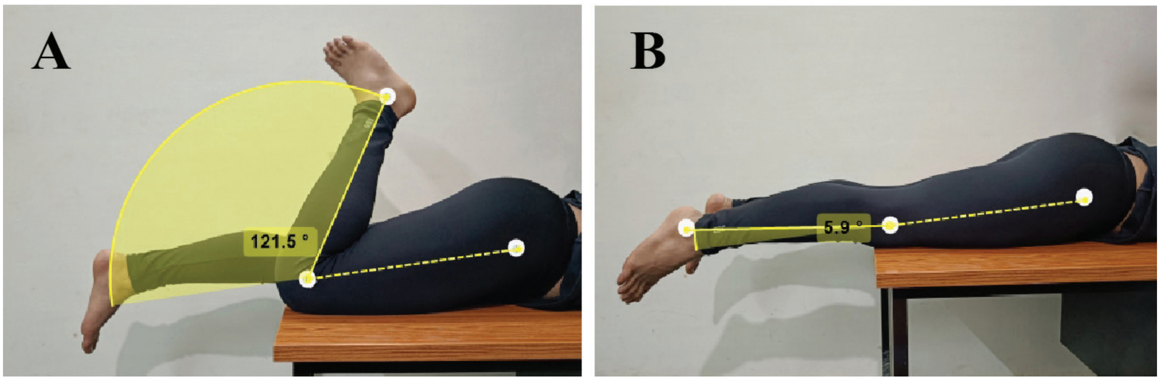

FIGURE 3. Knee Range of Motion (A) Flexion, (B) Extension

Statistical Analyses

The statistical analysis was performed using IBM SPSS (version 26.0.0) and involved binary logistic regression (Entre Method) to analyze the relationship between L-MTrPs and N-MTrPs (dependent variables) and PPT, force production, and ROM (independent variables). The analysis included calculation of, β, standard error β, Wald's χ2, odds ratio with a 95% confidence interval. Model evaluation was conducted using the likelihood ratio test, Cox & Snell (R2), and Nagelkerke (R2) tests, and the goodness of fit test was evaluated using the Hosmer & Lemeshow test. Additionally, the observed and predicted frequencies by the regression model were calculated with a cut-off of 0.50. The level of statistical significance was set at $p \leq 0.05$.

Result

Table 2 demonstrates the descriptive statistics of average force production, average PPT, and average ROM of keen flexion and extension in the L-MTrPs and non-TrPs groups. And there was

Table 2. Descriptive Statistics of selected variables reported for L-MTrPs and Non-TrPs Group

	Group	N	Mean ± S. D	S. E	p-value
Avg. Right leg Force Production	L-MTrPs	45	152.36 ± 21.73	3.23	0.013*
	non-TrPs	45	163.07 ± 18.04	2.69	
Avg. Left leg Force Production	L-MTrPs	45	165.23 ± 19.63	2.92	0.033*
	non-TrPs	45	175.64 ± 25.60	3.81	
PPT Right leg	L-MTrPs	45	21.07 ± 2.85	0.42	0.000*
	non-TrPs	45	25.04 ± 1.49	0.22	
PPT Right leg	L-MTrPs	45	20.95 ± 2.31	0.34	0.000*
	non-TrPs	45	24.91 ± 1.62	0.24	
Avg. Flexion ROM	L-MTrPs	45	136.11 ± 2.08	0.31	0.021*
	non-TrPs	45	137.03 ± 1.62	0.24	
Avg. Extension ROM	L-MTrPs	45	2.38 ±.80	0.11	0.028*
	non-TrPs	45	2.02 ±.76	0.11	

Avg: Average; ROM: Range of Motion; S. D: Standard Deviation; S. E: Standard Error;

a statistically significant difference found as the p-value is <0.05.

Model summary (Table 3) shows that the 82.5% change in the criterion variable can be accounted to the predictor variables in this model.

Table 3. Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	37.937a	.619	.825

a. Estimation terminated at iteration number 8 because parameter estimates changed by less than .001.

Table 4. Hosmer and Lemeshow Test

Step	Chi-square	df	Sig.
1	3.193	8	.922

The hosmer and lemeshow test is also a test of model fit. The hosmer-lemeshow statistic indicates a poor fit if the significance value is less than 0.05. Here the model adequately fit the data, as the p-value is .922 (>0.05; Table 4).

Table 5 shows the overall predictive accuracy is 91.1% now that we have our predictors in the model. The model predicted

Table 5. Classification Tablea

Observed			Predicted: TrPs: Non-TrPs	Predicted: TrPs: LTrPs	Predicted: Percentage Correct
Step 1	TrPs	Non-TrPs	41	4	91.1
		L-MTrPs	4	41	91.1
	Overall Percentage				91.1

a. The cut value is .500

Table 6. Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I.for EXP(B) Lower	95% C.I.for EXP(B) Upper
Step 1a	Avg. Right leg Force Production	-0.06	0.03	4.92	1	0.026*	0.93	0.88	0.99
	Avg. left leg Force Production	-0.02	0.02	1.48	1	0.223	0.97	0.93	1.01
	PPT Right leg	-0.51	0.15	10.66	1	0.001*	0.59	0.44	0.81
	PPT Left leg	-0.83	0.24	11.35	1	0.001*	0.43	0.26	0.70
	Avg. Flexion ROM	-0.72	0.34	4.46	1	0.035*	0.48	0.24	0.94
	Avg. Extension ROM	1.49	0.74	4.08	1	0.043*	4.45	1.04	18.99
	Constant	142.56	54.35	6.87	1	0.009*	8.224E+61		

Avg.: Average; ROM: Range of Motion

for L-MTrP (91.1%) and for non-TrPs (91.1%). from Block 0 that without considering any of our predictors, the likelihood or probability of a correct prediction was 50%; thus, our predictors certainly contributed to successful prediction.

The significant predictors were observed, such as average right leg force production (OR 0.93; 95% CI: 0.88–0.99), PPT right leg (OR 0.59; 95% CI:0.44–0.81), PPT left leg (OR 0.43; 95% CI: 0.26–0.70), average flexion ROM (OR 0.48; 95% CI: 0.24–0.94) and average extension ROM (RO 4.45; 95%CI: 1.04-18.99), as their p-values were less than 0.05. However, average left leg force production (p-value 0.22) was not significant predictors in the logistic regression model (Table 6).

From the table summary, we found that this model was 82.5% fit, Hosmer and Lemeshow test also confirmed that this model was good as the p-value is >0.05. That means average force production, PPT, and ROM were predicting L-MTrPs. Therefore, if the reduction in force production, lower PPT, and restriction of ROM occur, those are symptoms of L-MTrPs. In this study, it was hypothesised that force production, PPT, and ROM were the significant predictor variables of the L-MTrPs. According to the model this hypothesis was accepted.

Discussion

The objective of this research endeavour was to identify pertinent variables directly associated with the prediction of L-MTrPs. To achieve this objective, the study employed binary logistic regression analysis. Force production, PPT, and ROM were considered independent variables, and L-MTrPs and non-TrPs were considered dependent variables. Drawing from an extensive review of related literature, it became evident that L-MTrPs exert a negative influence on muscle force production (Kim et al., 2017; Das & Jhajharia, 2022b). A critical finding from this investigation was the prominence of PPT as a diagnostic marker for L-MTrPs. Research evidence suggests PPT is a reliable source to identify A-MTrPs (Valera-Calero et al., 2023). Notably, several researchers have called for further elucidation on this aspect (Doraisamy & Anshul, 2011). Additionally, L-MTrPs were observed to adversely impact joint ROM, this phenomenon has been consistently underscored by a variety of authors (Charles et al., 2019; Walsh et al., 2019; Öztürk et al., 2022). For instance, a study by Girasol et al. investigated the correlation between L-MTrPs and ROM in the upper trapezius, revealing a reduction in ROM (Girasol et al., 2018). Similarly, other studies highlighted L-MTrPs induced limitations in ankle planter-flexion and dorsi-flexion ROM due to gastrocnemius muscle TrPs (Benito-de-Pedro et al., 2020). Consequently, the evident connection between these variables and L-MTrPs emerges as significant. Consistent with prior research, a lower pain threshold has been linked to muscular abnormalities. Notably, a discrepancy in PPT greater than 2 kg/cm2, in comparison to the identical muscles on the opposing side, is indicative of the presence of L-MTrPs (Park et al., 2011; Cordeiro et al., 2021; Das et al., 2022). This firmly establishes PPT as a robust diagnostic parameter for L-MTrPs, a conclusion bolstered by our own statistical analysis. The precise pathophysiology underpinning myofascial pain remains somewhat enigmatic. Studies indicate that oxidative stress, inflammation, and glial cell activity, particularly astrocytes within the central nervous system, contribute to the persistence of pain signals, culminating in MTrPs (Widyadharma, 2020). Nociceptive chemicals released in response to tissue injury or inflammation are pivotal in perpetuating pain in MTrPs. These chemicals sensitise nerve fibers, inducing pain perception. Biochemical accumulation within MTrPs, localised regions of muscle stiffness and discomfort, includes neurotransmitters, neuropeptides, cytokines, and inflammatory mediators. According to Simons' hypothesis, abnormal endplate activity prompts the release of elevated acetylcholine levels, triggering a cascade. Calcium channels open, and calcium binding to troponin prompts muscle fiber contraction. Inadequate ATP supply sustains contraction near abnormal endplates, leading to an energy crisis, heightened metabolic demands, reduced blood flow, hypoxic conditions, and polarization. This energy crisis may trigger the release of neuroreactive substances and metabolic byproducts (e.g., bradykinin, substance P, and serotonin), sensitising peripheral nociceptors (Shah et al., 2015). Beyond PPT, athletes should not overlook the significance of force production and ROM, as research underscores their pronounced influence by L-MTrPs (Walsh et al., 2019; Cao et al., 2021; Yeste-Fabregat et al., 2021). The exact mechanisms behind these effects remain a subject of exploration. Emerging hypotheses, on the other hand, suggest that L-MTrPs could be a source of muscle dysfunction, causing fatigue, muscle tightness, shortened sarcomeres, and different patterns of activation. These dynamics result in suboptimal muscle contraction, diminished force production, and compromised ROM (Bagcier et al., 2022; Schneider et al., 2022). Consequently, athletes displaying reduced force production and ROM may be predisposed to L-MTrPs. The implications of this study indicate that athletes may unwittingly develop L-MTrPs, potentially impairing performance. It's worth noting that untreated L-MTrPs can even carry greater consequences (Celik & Mutlu, 2013). Therefore, athletes are advised to prioritise proper maintenance of their fascial structures to mitigate such risks.

Strength of the study

This investigation demonstrates novelty and innovation, as the majority of prior research has primarily focused on A-MTrPs and their associated symptoms, with limited inquiry into L-MTrPs within the sports domain. Our study has yielded substantial findings that can serve to aid athletes in averting the transformation of L-MTrPs into A-MTrPs, thus preserving and optimizing their physical performance.

Limitations

The limitations of this study are that it has an exclusive focus on the hamstring and quadriceps muscle groups, as well as its restriction to athletes performing at the national level.

Conclusion

The findings of the study suggest that if an athlete has lower force production, lower PPT in muscles, and restrictions in ROM, that indicates the occurrence of L-MTrPs. Therefore, athletes should take proper care of the muscle fascial structure. Coaches and physical trainers should include routine assessments of the force production, PPT, and ROM of the athletes. Sports scientists and physiotherapists should conduct more studies in this aspect and explore this area. Future studies are recommended for other muscle groups and focus on specific athletic disciplines.

Conflicts of Interest:
The authors did not report any potential conflicts of interest.

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ORIGINAL SCIENTIFIC PAPER

Effect of Concurrent Strength and Endurance Training on Distance Running Performances in Well-Trained Athletes

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Abstract

Athletes competing in distance competitions have used a combination of aerobic and anaerobic training approaches to train and enhance the performance-determining elements. Nevertheless, few studies have reported data related to the effect of concurrent training on well-trained distance (3,000 m – 10,000 m) runners. Because of limited evidence available for this population, this study aimed to investigate the effect of concurrent strength and endurance training on distance running performance. A randomized study was conducted. Thirty-nine distance runners (16.62±0.71 years) were randomly assigned into the endurance training group (ETG; n=13), strength training group (STG; n=13), and concurrent training group (CTG; n=13). The 12 weeks of training in which each group trained 3 times a week. The participants were tested on 1RM squat test, push-up test, VO2 max, and 5-km time trial. Findings showed that STG significantly higher than ETG enhancements on 1RMsquat (p<0.001) and push-up (p<0.001) and STG significantly higher than CTG enhancements on 1RM squat (p<0.001), push up (p=0.045). ETG results were significantly better than those obtained by STG on VO2 max (p=0.002) and 5-km time trial (p=0.004). Finally, the improvements obtained by CTG were significantly higher than those attained by ETG on 1RM squat (p<0.001), push-up (p<0.001); VO2 max (p<0.001) and 5-km time trial (p=0.002). In conclusion, performing 12-week concurrent training program improves performance variables that can be obtained with strength and endurance training in long-distance running. Athletes can acquire strength and endurance adaptations by engaging in concurrent training regimens.

Keywords: *aerobic training, muscle strength, physiological performance, resistance training*

Introduction

Distance running performance is the consequence of a complex interaction of physiological and physical factors and it is dominated by combinations of strength, speed, endurance, flexibility, and coordination (Blagroveet al., 2018). Athletes competing in distance events have improved performance-determining factors using a combination of aerobic and anaerobic training methods (Berryman et al., 2017). Furthermore, due to the nature of the competition schedule or training time available distance running often performs a combination of endurance training and strength training on the same day (Enright et al., 2017). There is growing evidence that concurrent strength and endurance training improves running performance more than endurance training alone, even though the factors that determine distance running performance have historically been established by aerobic running training (Blagrove, Howatson, & Hayes, 2018).

According to a thorough systematic evaluation, concurrent strength and endurance training has a moderately positive impact on middle and long-distance time trials up to 10 km (Blagrove et al., 2018). The effect of concurrent training has been widely investigated by researchers. Some of them provide strong evidence that after concurrent training intervention muscle hypertrophy, strength, and power adaptations

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were mostly attenuated, compared with those after isolated strength training stimuli (Tsitkanou et al., 2017). Conversely, Boullosa et al. (2020), and Ferrari et al. (2021) stated that combining training strength and cardiorespiratory fitness with strength in a training cycle could increase performance more than single-mode training.

In the word of Piacentini et al. (2013), endurance athletes benefit from concurrent strength and endurance training because the rate of force production, one of the key factors of endurance performance, is critical for running. There have been numerous research investigations looking into the function of maximal oxygen consumption (VO2max) in distance running. In heterogeneous groups, research has demonstrated a substantial correlation between VO2max and middle-distance (800 m, r=0.75) and long-distance (marathon, r=0.78) performance (Ingham et al., 2008; Beattie et al., 2017). Another study also showed eight weeks of concurrent strength and endurance training has beneficial effects on musculoskeletal power, maximal oxygen uptake, and the record level of running time (Saud & Nabia, 2016). Significantly, the majority of past studies on concurrent training have concentrated on confirming the compatibility of concurrent strength and endurance training. And research findings are sometimes ambiguous in this area, and it is still unclear how useful strength training is for endurance athletes. Consequently, this study investigated the effects of 12 weeks of concurrent strength and endurance training on the performance of long-distance running. This study expected that concurrent training may have significant effect on long-distance running performance. So, this study examined the effect of strength, endurance, and concurrent training on the performance of long-distance running.

Methods

Study design and participants

A randomized study was conducted. Using random sampling select 39 well-trained distance runners (3,000 m – 10,000 m) from the Tilili athletics center. After being informed of the benefits and potential risks of the investigation, participants were randomized into three groups based on their 5km running time; Endurance training group (ETG) (n=13; 16.7±0.8 years; 57.77±2.29 kg (kilogram); 1.72±0.05 m (meter)), Strength training group (STG) (n=13; 16.62±0.86 years; 56.42±2.01 kg; 1.72±0.05 m) and Concurrent training group (CTG) (n=13; 16.55±0.686 years; 56.96±2.24 kg; 1.717±0.06 m). Participants in the study had an average training background of at least 2 years.

All experimental procedures were ratified by the Bahir Dar University, Sports Academy Research Ethics Committee (No. ERC 01/2022). In addition, all participants in this research had to provide written informed consent.To investigate the effect of concurrent strength and endurance training intervention on strength qualities, physiological indicators, and a 5km time record of long-distance running; the researchers collected quantitative data through appropriate field tests measures such as the Cooper VO2 max test, push-up test, 1RM squat test, and 5km time trial.

Training protocol

Athletes were kept in a training log, which included any physical activity done outside of the training program. All training sessions were coached by an experienced coach. Each strength session lasted approximately sixty minutes. The training program lasted 12 weeks, with study participants attending three non-consecutive sessions per week. The training interventions took place from the beginning of September to the end of November 2022.

Strength training

The training sessions included squat, leg curl, triceps extension, bench press, calf raise, trunk extension, pulldown, hurdle hops, extended bounds, and uphill running performed. There are rests between sets and between exercises. This training program is performed by STG. A more detailed description of the strength training program is presented in Table 1.

Table 1. Training Methodology that Was Used Within STG

Week	Training Parameters							
	Exercises	I (% 1RM)	S (No)	Rep	Di (Km)	Incl (%)	RTS (′)	RTR (′)
1	Squat, Leg Curl, Triceps Extension	60	4	14			2	
2	Squat, Bench Press, Calf Raise	65	4	12			2	
3	Trunk Extension, Leg Curl, Calf Raise	70	5	12			2	
4	Squat, Leg Curl, Pulldown	75	5	10-8			3	
5	Squat, Bench Press, Calf Raise	80	4	8-6			3	
6	Squat, Bench Press, Calf Raise	90	4	8			3	
7	Squat + Hurdle Hops + 2′30″ Running at 100% Of MAS	80	6	6+10			5	
8	Squat + Hurdle Hops + 2′30″ Running	82	6	5+10			5	
9	Squat + Extended Bounds (Cover 50 M Alternating Legs by Doing the Lowest Possible Number of Strides) + 2′15″ Running.	84+105	6	4			5	
10	Squat + Extended Bounds (Cover 50 M Alternating Legs by Doing the Lowest Possible Number of Strides) + 2′ Running	86+110	6	3			5	
11	Uphill Running	115	3	5	0.2	6	10	3
12	Uphill Running	120	2	5	0.2	6	10	3

Notes: 1RM: One-repetition maximum; Di: Distance; Incl: Inclination; Km: Kilometer;MAS:Maximume average speed; No: Number, Rep: Repetitions; RTR:Resting time between reps; RTS: Resting time between sets; S: Sets; STG: strength training group

Endurance training

The main components of endurance training (see table 2) were continuous exercise lasting 45–50 minutes, Fartlek for 40–60 minutes, and reputation training. The amount of time devoted to endurance training and how this time was distributed throughout the training zones were the same across endurance-only training groups and concurrent training groups.

Table 2. Training Methodology that Was Used Within ETG

Week	Training Parameters					
	Exercises	I (b.p.m)	Du (′)	Rep (No)	Di(km)	RT (′)
1	Continuous Training	130–140	45			
2	Continuous Training	140–144	50			
3	Continuous Training	145–150	45			
4	Fartlek Training.	115–160	50			
5	Fartlek Training.	115–160	55			
6	Fartlek Training.	115–160	60			
7	Extensive Interval Training (Long Intervals)	155–160	3	10		2
8	Extensive Interval Training (Long Intervals)	160–165	2	10		2
9	Extensive Interval Training (Long Intervals)	165–170	1	14		2
10	Repetition Training	180	3	5		8
11	Repetition Training	185	2	5		6
12	Competition	100		1	5	

Notes: b.p.m: beat per minute; Di: distance; Du: Duration; ETG: Endurance training group;I: Intensity of training; Km: kilometer; No: Number; Rep: Repetition; RT: Resting Time

Concurrent training

The concurrent training group (CTG) performed both strength and endurance programs on the same day, in which the endurance sessions are performed first and after 8-hour rest followed by the strength session. The training program of CTG included the same strength exercises as STG and ETG.

Assessments

To reduce the impact of extraneous factors and tidal variation on outcome measurements, many control procedures were put in place. The participants were instructed to continue living their regular lives, eating their regular meals, and refraining from using any nutritional supplements during the intervention time. After 12 weeks of training, each participant completed a familiarization before physical and physiological tests. After a 48-hour rest period, the lead researcher gave a thorough explanation of the testing procedures. Then before and after the 12-week training program, some anthropometric and physiological parameters were assessed (pretest and posttest). Before beginning the tests, a warm-up was carried out; running for 10 minutes at 60% of one's theoretical maximum heart rate, followed by five minutes of joint mobility activities which serve as the general warm-up.

The participant's body weight, age, and height were all taken into consideration. A Seca digital column scale, model 769, was used to measure body weight (Hamburg, Germany). 1RM squat test was used to assess lower body maximum strength. Participants were instructed to maintain a naturally upright trunk position throughout the assessment. Both hands were securely gripping the bar, which was also being supported by the shoulders. The performer's legs were parallel to the ground when the test began with their knees bent 90 degrees. The subjects then stood up straight with their legs completely extended. Each subject needed between two and four tries to determine their 1RM. Only completed efforts were recorded. Between each trial, there was a three-minute break.

An other test was push-up test; a standard push-up beginning with the hands and toes on the floor, the body and legs in a straight line, feet spaced slightly apart, and the arms outstretched, shoulder width apart, and at a right angle to the torso. The individual lowers their body while maintaining a straight back and knees to a predefined point, to another object, or until their elbows are at a 90-degree angle, then raises their arms back up to their starting positions. The test continues until tiredness until they can no longer perform them in rhythm, or until they have completed the required number of pushups (Saud & Nabia, 2016).

To determine the VO2 max, Cooper's 12-minute run test was employed. Cooper's 12-minute run test should be applied with the newly calculated norm as a viable approach for accurate, correct, and precise assessment of cardiorespiratory fitness in terms of VO2 max (Bandyopadhyay, 2015). The subjects were run for a total of 12 minutes on a 400-meter circular track. The number of completed laps was counted, and the finish line was marked. By dividing the number of complete laps by 400 and adding the distance (in meters) of the final incomplete loop, one may compute the total distance (in meters) traveled in 12 minutes. The following equation was used to predict the VO2 max and convert the distance in meters to kilometers. VO2 max (ml · kg−1 · min−1) = (22.351 × distance covered in kilometers) − 11.288 (Meredith & Welk, 2010).

The running test of choice was a 5-kilometer time trial. Only dry circumstances and winds below 2 m/s were used during testing (Karsten et al., 2016). A 5-kilometer run on a 400-meter outdoor running track was performed after a 10-minute warm-up at a self-determined pace. Participants were instructed to give it their all during the 5-km time trial. Completion times were noted to the nearest second.

Statistical analysis

Descriptive statistical analysis was made to display means (M) and standard deviation (SD). Levene's test was used to confirm the homogeneity of variances, and the Shapiro-Wilk test was performed to compare the normality of the variables. To confirm the performance differences between groups, a one-way ANOVA test was used. A two-way repeated-measure ANOVA was then conducted to evaluate the training effects on the physical and performance characteristics between groups (ETG vs. STG vs. CTG) and within groups (pre-test vs. post-test). When statistically significant p-values were found, post hoc multiple comparisons with Bonferroni correction were used to identify those differences (group-by-time interaction effect or main effects of time or group). To check the consistency between the pre-test and post-test measures, the interclass correlation coefficient (ICC) was calculated for each of the examined performance factors. Low ICC=0.49, moderate ICC=0.50, high ICC=0.75, and remarkable ICC=0.9 were used to interpret ICC data (Koo & Li, 2016). In addition, Cohen's effect size was used to determine the size of treatment effects within groups (ES). The four categories of ESs: are no significant (0.25), minor (0.25-0.50), moderate (0.50-1.0), and large (>1.0) (Cohen, 2013). The defined level of significance was p<0.05. Data were statistically analyzed using the IBM SPSS V.26 computing application (IBM Corp., Armonk, NY, USA).

Results

Thirty-nine individuals completed the study without mentioning any harm they may have suffered as a result of the intervention training session. According to the Shapiro-Wilk test and Levene's test, all variables appeared to have a normal and homogenous distribution. Moreover, the ICC values for all three groups were higher than 0.9 between the pre-test and post-test for all parameters. Results of comparative analysis (one-way ANOVA) among ETG, STG, and CTG at baseline (see Table 3) revealed that there were no statistically significant differences before the start of the training program.

Table 3. Baseline characteristics of participants in each training group

Variables	Group			F	P
	ETG (n=13) M±SD	STG (n=13) M±SD	CTG(n=13) M±SD		
Age (y)	16.54±0.52	16.62± 0.86	16.69±0.85	0.131	0.877
Height (m)	1.72±0.061	1.72±0.05	1.71±.04	0.153	0.859
Weight(kg)	57.76±2.29	56.42±2.00	56.96±2.24	1.249	0.299
1RMSquat (No)	53.00±1.78	53.00±1.78	54.00±1.78	1.368	0.267
1RMPush up (No)	23.77±1.96	23.31±2.097	23.85±1.77	0.290	0.750
Vo2max (ml.kg^{-1}.min^{-1})	68.38±1.23	68.40±1.19	68.37±1.089	0.002	0.998
5km time trial	1010.38±15.21	1010.02±14.90	1009.95±14.73	0.003	0.997

Note: M±SD=mean ± Standard deviation, ETG= Endurance Training group, STG= Strength training group, CTG= Concurrent training group, y= year, m= meter, kg= kilogram, ml.kg^{-1}.min^{-1}= milliliters per kilogram per minute, M= Male, F= Female, n= number of participants, No=number; P=statistical significance.

As for the within-subject comparisons, the two-way repeated ANOVA revealed (showed in Table 4), STG significantly improved between the pre-and post-tests in 1RM squat (p<0.001; d=8.40), 1RM push-up (p<0.001; d=7.26), VO2max (p=0.0185; d=0.19) and 5 km time trial (p=0.001; d=0.14). The effect size of these improvements was small in the case of VO2 max and 5-km time trial, and large for 1RM squats and push-ups. Similarly, ETG significantly improved between

Table 4. Results were obtained by the Three Experimental Groups in the Pre-and Post-Test in all the Variables Assessed

Variable	Group	PT (M± SD)	POT(M±SD)	P-value	Cohen's d
1RMSquat(Kg)	STG	53.00±1.78	69.31±2.09	<0.001	8.40
	ETG	53.00±1.78	59.23±1.96	<0.001	3.33
	CTG	54.00±1.78	64.15±1.2	<0.001	6.69
Push Up (No)	STG	23.31±2.09	38.38±2.06	<0.001	7.26
	ETG	23.77±1.96	28.85±2.04	<0.001	2.32
	CTG	23.85±1.77	34.00±1.78	<0.001	5.72
Vo2Max(mL/kg/min)	STG	68.40±1.19	68.19±1.01	0.0185	0.19
	ETG	68.38±1.24	71.24±0.78	<0.001	2.76
	CTG	68.37±1.08	74.22±1.45	<0.001	4.58
5km Time Trial (sec)	STG	1010.01±14.9	1012.66±13.4	0.011	0.14
	ETG	1010.38±15.2	974.69±9.9	<0.001	2.78
	CTG	1009.9±14.73	936.38±17.2	<0.001	4.53

Notes. 1RM: One-repetition maximum; CTG: Concurrent training group; ETG: Endurance Training group; M±SD: Mean ± Standard deviation; POT: Post-Test; PT: Pre-Test; STG: Strength training group the pre-and post-tests in the following parameters:1RM squat ($p<0.001$; $d=3.33$), push-up ($p<0.001$; $d=2.32$), VO2 max ($p<0.001$; $d=2.76$) and 5-km time trial ($p<0.001$; $d=2.78$). The effect size of these improvements was large in the case of all variables. In addition, CTG significantly improved its results between the pre-and the post-tests in the following variables; 1RM squat ($p<0.001$; $d=6.69$), push-up ($p<0.001$; $d=5.72$), VO2 max ($p<0.001$; $d=4.58$) and 5-km time trial ($p<0.001$; $d=4.53$). The effect size of these improvements was large in the case of all variables.

Then, the two-way repeated ANOVA (see Table 5) showed that there was a group-by-time interaction effect for 1RM squat, push-up, VO2 max, and 5-km time trial ($F_{(2-36)}=90.946$, $p<0.001$, $F_{(2-36)}=543.257$, $p<0.001$, $F_{(2-36)} = 138.841$, $p=0.001$ and $F_{(2-36)}=175.685$, $p<0.001$, respectively). The 1RM squat, 1RM push-up, VO2max, and 5-km time trial all showed a main effect of time ($F_{(1-36)}=1255.794$, $p<0.001$, $F_{(1-36)}=6652.971$, $p<0.001$, and $F_{(1-36)} =458.374$, $p<0.001$, respectively). The 1RM squat ($F_{(2-36)}=36.508$, $p<0.001$), push-up ($F_{(2-36)}=18.345$, $p<0.001$), VO2 max ($F_{(2-36)}=26.579$, $p<0.001$), and 5-km time trial ($F_{(2-36)}=26.155$, $p<0.001$) were the last exercises to show a main impact of the group.

Table 5. Between-Subjects Comparisons of all the Variables Assessed

Variable	Main Effect of Time		Main Effect of Group		Group x Time Interaction Effect	
	F (1-36)	P	F (2-36)	P	F (2-36)	P
Body weight	385.714	<0.001*	1.635	0.209	428.464	<0.001*
1RMSquat	1255.79	<0.001*	36.508	<0.001*	90.946	<0.001*
Push up	6652.97	<0.001*	18.345	<0.001*	543.257	<0.001*
Vo2max	364.653	<0.001*	26.579	<0.001*	138.841	<0.001*
5km Time trial	458.374	<0.001*	26.155	<0.001*	175.685	<0.001*

Notes. F: Variation between sample means/variation within the samples; p: Level of statistical significance; VO2max: Maximum oxygen consumption; *p<0.05

Table 6: Bonferroni post hoc comparison

	Group(I)	Group(J)	d= (I-J)	CI 95 (Up	LB)	P-value
1RMSquat	ST	ETG	5.04*	3.55	6.53	0.000
		CTG	2.08*	0.59	3.57	0.004
	CTG	ETG	2.96*	1.47	4.45	0.000
1RMPush up	STG	ETG	4.54*	2.65	6.43	0.000
		CTG	1.92*	0.03	3.81	0.045
	CTG	ETG	2.62*	0.73	4.50	0.004
Vo2max	ST	ETG	-1.52*	-0.48	-2.55	0.002
		CTG	-3.00*	-1.96	-4.03	0.000
	CTG	ETG	1.49*	2.52	-0.45	0.003
5km Time trial	STG	ETG	18.80*	5.54	32.05	0.003
		CTG	38.17*	24.91	51.42	0.000
	CTG	ETG	-19.37*	-32.62	-6.11	0.002

Note: VO2max= Maximum oxygen consumption, CI95 (Up-LB) = 95% Confidence Interval (Upper Bound-Lower Bound), p = Level of statistical significance, * = The mean difference is significant at the 0.05 level, d = Mean Difference, (I-J) =Group(I)-Group(J).

Discussion

Based on the statistics applied; our results confirm: concurrent training improves distance running performance. Finding out whether concurrent training saw the greatest gains in muscle strength after concurrent training compared to endurance training only (CTG) and strength training only (STG) was one of the study's main objectives. After 12 weeks of training, the STG and CTG groups had significant increases in lower-body and upper-body strength as seen by the group's post-training values for the 1RM squat and 1RM push-up tests. Only the marks acquired by endurance training only (ETG) were considerably lower than those attained by STG and CTG in the post-test. STG and CTG both saw an improvement in their scores between the pre-and post-test. Given that strength training and endurance training are combined in the same training sessions, this may mean that adding strength training to endurance training offers extra benefits. In line with this study, Saud and Nabia's (2016) findings showed that concurrent training obtained greater improvements in 1RM squats and push-ups. Similar to this, according to Prieto and Sedlacek (2022), the concurrent training did not experience the interference effect because there were no significant differences between strength training and concurrent training in the post-test results for the 1RM squat. They also improved their marks in the resistance training group and the 1RM squat between the pre-and post-tests.

The current findings additionally demonstrated that maximal strength (i.e., 1RM) increased in STG and CTG while

ETG saw only minor changes as a result of the intervention. Similarly, Sousa et al. (2018) studies conducted with trained distance runners also found gains in maximal strength with concurrent training. The gain in upper body strength for the push-up test in the STG and CTG groups was also significantly different. Results from previous studies by Vikmoen et al. (2017) are consistent with those from the current study. The findings of the current study are in line with those of Vikmoen et al. (2017), and Sousa et al. (2017).The effectiveness of concurrent training to enhance the 1RM squat and push up was confirmed in two instances (Prieto & Sedlacek, 2022).

VO2 max is the most important physiological performance indicator in distance running (Beattie et al., 2014). The trainability of the VO2 max variable could be conditioned by genetic factors (Williams et al., 2017). In this study, the ETG (68.38±1.24→71.24±0.78 ml.kg-1.min-1) and CTG (68.37±1.08→74.22±1.45 ml.kg-1.min-1) obtained significant improvements. It follows that these enhancements are likely training-specific adaptations. Also, the fact that ETG's benefits were not greater than CTG's shows that no interference effect has taken place. These findings are in agreement with studies by Patoz et al. (2021). However, in a systematic review, Blagrove et al. (2018) verified that the implementation of concurrent training programs has no impact on VO2 max. Few studies found significant improvements in VO2 max after the exclusive practice of strength training. In this regard, only three out of the 17 studies analyzed by Ozaki et al. (2013) were used to review the research on the benefits of strength training on boosting VO2 max. Thus, the disparity between studies may exist because there is a genetic predisposition to VO2 max improvement (Williams et al., 2017). They also discovered that increasing VO2 max becomes more challenging the more intense the exercise (Ozaki et al., 2013). In the other study, by Berryman et al. (2019), strength training was added to the sprint interval training intervention, as was a concurrent regimen consisting of a sprint protocol. While VO2 max was considerably improved only in the concurrent group, there were no variations in maximal force between interventions, indicating that both situations improved.

Finally, regarding the 5-km time trial, both ETG (1010.38±15.2→974.69±9.9) and CTG (1009.9±14.73 → 936.38±17.2) improved this variable. This improvement resulted from the training methods specifically designed to enhance the running time of distance running. In reality, a faster race time would be a better indicator of enhanced performance (Sedano et al., 2013). Likewise, the results attained by CTG were significantly better than those achieved by STG and ETG. Theoretically, CTG decreased by a mean of 73.52 seconds. Furthermore, as expected, STG did not significantly improve the 5-km time trial (1010.01±14.9 seconds →1012.66±13.4 seconds). In this occurrence, we believe that the lack of endurance training accounts for the lack of appreciable progress. These results are corroborated by Beattie et al., (2014), which showed that adding a strength training program to an endurance training regimen greatly improved performance during a 5km time trial.

Finally, it is important to discuss the study's strengths and limitations. Regarding the strengths, there are two important features. First, strength training group was added along with concurrent and single-mode endurance training group. This situation helped confirm the changes endurance runners can achieve with strength training and identify whether interfering effects in CTG might be present. Second, equal weekly training sessions were administered to each of the three experimental groups. It should be noted that in numerous prior research, the concurrent group conducted two or three additional strength sessions per week compared to the endurance training group, indicating that a different number of sessions was used.

As for the limitations, First, the sample size was small. Second, if the intervention period had been longer, the three experimental groups may have seen more gains.

Conclusion

In conclusion, concurrent training program that was implemented improved distance running performance significantly. This study demonstrated that adding strength training to a running program significantly improved strength, VO2 max, and 5km record times. Although speculative, the enhanced lower limb strength levels in the CTG group appear to be the primary cause of the trend of improvement, which may have improved elements like stride length.

Practical Applications

The practical implications would relate to: Athletes can acquire strength and endurance adaptations by engaging in concurrent training regimens. To do this, however, a minimum of nine hours between training sessions is required to avoid severe interferences, or 24 hours to more fully ensure that the adaptations won't be attenuated (Baldwin, 2022); The strength training program must be adjusted to the athlete's goals, level of fitness, and prior training experience with regard to the training load and the type of strength that must be produced; Long-term interventions of more than 12 weeks might be used.

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Conflict of interest

The authors declare that there are no conflicts of interest.

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REVIEW PAPER

Bridging the Gap between the Body and the Machine: Embodied Learning with Interventional Brain Computer Interfaces?

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Abstract

This review aims to understand how the body and the brain interact with different brain computer interfaces (BCI) and to analyze the implications of these tools on embodied learning in the educational field. Through a theoretical approach a review of the literature is developed by studying the relationship between the body, the brain and BCI. To conduct this research, the keywords "embodied learning", "cognition", "digital learning", "body", "brain-computer interface" were used in Pubmed, Frontiers, Google Scholar and Researchgate. There are multiple concepts related to digitization and they can vary from owning digital tools such as computers, phones, virtual reality devices to even using interventional BCI. BCI are being reported safe and are capable of reversing physical and cognitive disabilities. The impact of these tools is variable according to their nature, the environmental factors linked to their use, and the condition of the brain and body while using them. With the massive development of technology nowadays many interrogations are coming into surface about the relationship between the human and the machine, and at what level the digital world will be able to interfere with our lives and integrate our bodies.

Keywords: *brain augmentation, digital learning, neuroscience, education*

Introduction

Brain computer interfaces were established in 1929 with the invention of electroencephalography (EEG) that allowed the detection of brain activity and translating it into electrical signals (Spüler, 2017). But it wasn't until 1973 that Jacques J. Vidal invented in his paper "Toward direct brain-computer communication", the term Brain-Computer-Interface (BCI) (Rebolledo-Mendez et al, 2009). BCIs can be classified according to their external technical implementation to (open-loop: recording) or closed-loop (recording and stimulation), or their internal implementation to non-invasive or invasive (Saha et al., 2021). The 2020 horizon of brain neural computer interaction and the European commission of BCI use in research coordination identified 6 application themes for BCIs as follows: Restore, Improve, Replace, Enhance, Supplement, and use as a Research tool (Saha et al., 2021). BCIs use in cognition lies under the umbrella term of brain augmentation (Jangwan et al., 2022). Some BCIs have been proven to interfere with cognition and thus learning. The term embodied cognition means that the body is crucial for cognition. However, with the invention of BCIs this relationship is questionable regarding the possibility that BCIs can replace or complement the body function in cognitive learning (Serim et al., 2023). The use of BCIs in educational settings is recent and still limited, yet the understanding of how these tools can interfere with physical and cognitive capacities is important. Given the inevitable increase in technologies' implementation in educational settings it is important to describe the underlying theoretical perspectives and recent pedagogical and neuroscientific research findings on the use of BCIs and their impact on embodiment as a physical and cognitive learning phenomenon.

It is in this theme that this research aims to understand

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how the body and the brain interact with different brain computer interfaces (BCI). We will also analyze the implications of these tools on embodied learning (EL) in the educational field.

Discussion

There are multiple concepts related to digitization and they can vary from owning digital tools such as computers, phones, virtual reality devices to even using interventional BCIs. Some BCIs are being reported safe and capable of reversing physical and cognitive disabilities. The impact of these tools is variable according to their nature, the environmental factors, and the condition of the brain and body.

Through a theoretical approach a narrative review of the literature is developed by studying the relationship between the body, the brain and BCIs. To conduct our research, we used the keywords "embodied learning", "cognition", "digital learning", "body", "brain-computer interface" in Pubmed, Frontiers, Google Scholar and Researchgate.

This review will start by discussing embodied cognition and embodied learning (EL) from a neuroscientific point of view. Then describe the different available BCIs used for cognitive brain augmentation and learning. And finally identify the principal theoretical and empirical impacts of using BCIs on EL.

Neuroscientific basis of embodied cognition and EL

Embodiment is a phenomenon that explains how the brain-body interaction with the environment generates intelligent behavior. Cognitive hypotheses of embodied cognition include: 1. Replacement hypotheses that highlight the role of sensory-motor contingencies induced by movement within an environment in controlling behavior. 2. Constitution hypotheses stating that cognitive systems built in the brain extend to the body and environment. And that bodily and environmental cues can be part of the memory system. 3. Influence hypotheses that describe a bidirectional interaction between the body and the brain in cognition. In this context, physical states of the body can alter cognition and cognitive rehearsal training can improve procedural skills. 4. Conceptualization hypotheses postulating that sensorimotor networks stimulation induces concepts' creation that are fundamental building blocks for grounded cognition (Matheson & Barsalou, 2018). Sensorimotor experiences generate information linked to all types of concepts; abstract and concrete (Harpaintner et al., 2020). Conceptualization hypotheses are the most commonly used in embodied cognitive neuroscience. Mirror neurons theory is based on conceptualization and implies that similar neural activation firing occurs when we perform an action, and also when we observe another person performing it (Caramazza et al., 2014). Language as a social communication tool also re-enhances embodied experiences by reactivating sensory-motor networks' clusters and cognitive processes (Macedonia, 2019). These concepts within the brain are represented through neural networks or modality specific systems that respond with high specificity to the different modalities of sensory or motor stimulation and that are organized hierarchically. These hypotheses highlight the importance of sensorimotor stimulation in the generation and quality of embodied cognition and have been confirmed by many neuroimaging studies (Matheson & Barsalou, 2018; Harpaintner et al., 2020). In figure 1 we present a simplified illustration of the basics underlying embodied cognition hypotheses.

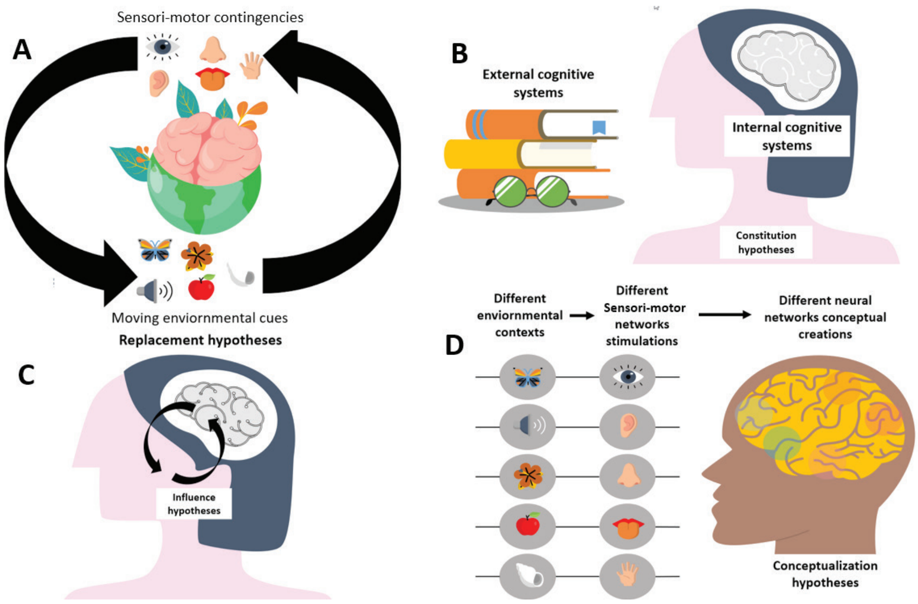

FIGURE 1. Simplified Illustration of the ground of embodied cognition hypotheses
Legend A. Replacement hypotheses: kinetics environmental cues induce sensori-motor contingencies that stimulate embodied cognition and thus control behavior accordingly. B. Constitution hypotheses: there are both external and internal cognitive systems. C. Influence hypotheses: the body and brain are interdependent in generating cognition. D. conceptualization hypotheses: different environmental contexts activate different sensori-motor networks that generate different neural conceptual cognitive networks.

BCIs use in embodied cognition focuses mainly on conceptualization hypotheses. BCIs like transcranial magnetic stimulation (TMS) and magnetic encephalography (MEG) helped understand the role of modality specific stimulation in establishing brain grounded cognition (Matheson & Barsalou, 2018), and how contextualization can modulate motor action and behavior.

Three major themes are the basis of future research in grounded cognitive neuroscience (Matheson & Barsalou, 2018): The first is associative processes that help generate predictions based on bidirectional feedback of Hebbian learning rule. Hebbian learning postulates that synaptic plasticity is at its maximal function when both the presynaptic and postsynaptic neurons are activated (Munakata & Pfaffly, 2004). Sensory-motor features are thus modulated by networks of neurons called "maps" among which some are "controllers". These latter are formed and activated based on variable experiences that generate different associative weights guiding intelligent behavior (Matheson & Barsalou, 2018). The second is network dynamics. In fact, this phenomenon implies that brain cognitive function is based on connectomes and clusters of concepts that are characterized by their dynamic distribution, degree of activation, and the modality of physical and environmental stimulation (situatedness). Network dynamics are thus context dependent (Matheson & Barsalou, 2018). The third theme is representation. Representation is a foundation of classical construct cognitivism that depends on semantic content and implies a structural aspect that requires 4 elements: homomorphism between the target and the internal state, causal connection between them, the possibility of decoupling, and their role in action control (Piccinini, 2018).

To summarize, there 4 four main hypotheses that tried to explain embodiment and these are replacement, constitution, Influence and conceptualization hypotheses. They all affirm the important role of the body and external environment in cognitive embodiment processes. This interaction is dynamic and complementary at different levels. This dynamicity and interaction occur hypothetically through associative processes, network dynamics or representation.

Definition and use of BCIs in cognition and education
BCIs and brain augmentation

BCIs used to improve cognition and motricity fall under the umbrella term of brain augmentation (Jangwan et al., 2022). BCI tools are classified according to their external technical implementation to (open-loop: recording) or closed-loop (recording and stimulation), or their internal implantation to non-invasive or invasive (Saha et al., 2021). In the group of invasive BCIs (IBCI) a new term minimally-invasive has emerged describing interventional tools that do not require a craniectomy or do not enter in a direct contact with the brain parenchyma and that can be of temporary use with a possibility of safe removal, some of these tools include functional ultrasonography (fUS), the layer 7 cortical interface (Ho et al., 2022), and endovascular Stentrode (Mitchell et al., 2023).

Non-invasive BCIs can be subdivided into recording and stimulation tools (Saha et al., 2021). Brain augmentation interventions can require one or multiple BCIs (Jangwan et al., 2022). Brain augmentation in the classroom experiments used EEG (Rebolledo-Mendez et al, 2009; Spüler, 2017; Caitlin et al, 2017), EEG combined to virtual reality (VR) and intelligent tutoring systems (ITS) (Tremmel, 2019) or functional near infrared spectroscopy (fNIRS) (Watanabe et al., 2016; Oku et al, 2021) (Table 2). Other tools have potential cognitive benefits and are still limited to laboratory research or clinical settings like transcranial magnetic stimulation (TMS) using magnetic stimulation (Grau et al., 2014; Rao et al., 2014), and transcranial electrical stimulation (tES) and transcranial direct current stimulation (tCDS) using electric stimulation (Dockery et al., 2009; Coffman et al., 2014; Heth & Lavidor, 2015; Younger et al., 2016). Other tools like neuro-prosthetics are often used for restoring deficient senses (Wegemer, 2019). Table 1 summarizes the different BCIs according to their external and internal technical implementation.

Many BCIs are used in medicine for neurorehabilitation in patients with cognitive and motor disabilities (Jangwan et al., 2022). Their use has extended recently to robotics and healthy humans to serve in physical brain augmentation (Jangwan et al., 2022). Brain augmentation can occur through the use of physical, biochemical or behavioral strategies (Jangwan et al., 2022). Invasive BCIs' use is still limited to laboratory research and clinical settings for patients with neurological and psychiatric disorders (Zhao et al., 2023). Motion-based video games using computers and VR have been proved useful in EL efficacy by increasing academic performance and student engagement (Howison et al., 2011; Abrahamson & Lindgren, 2014; Verkijika & De Wet, 2015; Cook et al, 2016; Sullivan, 2018; Kosmas et al., 2019). Recording, stimulation and hybrid BCIs have been used in the classroom, computational neuroscience research, and clinical settings in patients with neuro-

Table 1. BCI tools according to their external and internal technical implementation.

		BCIs	References
Recording	Non- invasive	EEG, MEG, fMRI, fNIRS, PET	(Jangwan et al., 2022)
	Invasive	Electrodes: ECog, ICRT, Neuralink, iMEA	(Jangwan et al., 2022) (Saha et al., 2021)
	Minimally invasive	FUS Layer 7 cortical interface Endovascular: Stentrode	(Soloukey et al., 2023) (Ho et al., 2022) (Mitchell et al., 2023)
Stimulation	Non- invasive	tES, TMS, tDCS	(Jangwan et al., 2022)
	Invasive	Electrodes: ICST, ICMS, DBS, Neuralink,, iMEA close-loop-DBS	(Jangwan et al., 2022) (Saha et al., 2021)
	Minimally invasive	FUS Layer 7 cortical interface Endovascular: Stentrode	(Soloukey et al., 2023) (Ho et al., 2022) (Mitchell et al., 2023)

Legend: EEG: electro-encepahlography, MEG: Magnetic encepaholgraphy, fMRI: functional magnetic resonance imaging, fNIRS: functional near infrared spectroscopy, PET: positron-emission tomography, ECog: Electrocorticography, ICRT: intracortical recording, iMEA: intracortical microelectrode array, tES transcranial electrical stimulation, TMS: Transcranial magnetic stimulation, FUS: transcranial focused ultrasound stimulation, tDCS: transcranial direct current stimulation, ICST: intracranial stimulation, ICMS: intracortical micro-stimulation, DBS: deep brain stimulation.

logical and psychiatric disorders (Table 2). These tools showed promising results in neurorehabilitation, brain augmentation, and also personalized learning through the measurement of cognitive load (Table 2). The integration of BCIs in educational laboratory research is still scarce, and this may be due to their high cost (Vourvopoulos & Badia, 2016), the absence of a clear policy regarding their use in research education and the ethical consideration related to brain augmentation in the school environment (Zeng et al., 2021). With the new rising of minimally invasive BCIs, many questions are rising regarding the future of this human machine relationship and the evolution of homosapiens to homosapiens technologicus (Zehr, 2015). Table 2 summarizes classroom Kinetic virtual games to study the relationship between EL and BCIs reported useful in cognitive research in the classroom, laboratory research and clinical settings.

This section summarizes the classification of BCIs according to their technical external and internal implementation and the results of actual research on their use in brain cognitive augmentation and education. The rapid evolution of technology is bringing about breakthroughs in cognitive science evolution and opening up perspectives about the use of BCI in education to achieve a maximal and personalized learning efficacy. Nevertheless, the implementation of BCIs in schools should be planned ahead according the possible personal and social benefits and drawbacks. Their impact on brain health on the short and long term should be considered. In the next section we will discuss the dual relationship between BCIs and embodied learning and the controversies related to their impact on the body and embodied learning.

Theoretical and empirical concepts of embodied learning related to BCIs' use

Many BCIs have been used in research of embodied learning. To understand how BCIs interfere with EL we need to understand how the body participates in learning and the impact of these mechanistic aspects of BCIs on the body.

How the body participates in learning?

The body and language learning

Macedonia et al, showed that using fMRI brain network mapping in 31 right handed German natives helped identify enhanced linguistic performance by combining audio and visual stimulation by using metaphorical gestures with words in second language learning (Macedonia, 2019). Significant activation correlations ($p<0.05$) during sensorimotor and audiovisual tasks observations were found in the left fusiform gyrus, right superior temporal gyrus, right cerebellar vermis, right and left precentral gyri, and right and left inferior parietal lobules (Macedonia, 2019). In fact, during speech processing the brain uses a multi-sensory Hub for audiovisual information pairing. This hub includes areas in the posterior superior temporal sulcus/gyrus, or the superior parietal lobule (Gonzales et al., 2021). PET and fMRI studies have shown that abstract concepts of the amodal symbolic verbal system are mediated by the middle and superior temporal gyri, and left inferior frontal gyrus (Harpaintner et al., 2020). Which again highlights the role of sensorimotor modalities in abstract language processing.

The body and sciences learning

School experiments enhanced mathematic concepts learning by combining speech and gestures, and a link between counting and fingers (Macedonia, 2019). This association has been proven by functional MRI studies and seem to be linked to Hebbian learning mechanisms (Macedonia, 2019). The theory of grounded and embodied mathematical cognition (GEMC) implies that gestures are fundamental in learning sciences (Nathan & Walkington, 2017). Nathan et al showed in a study of 120 participants that mathematical intuition depended on body actions (Church et al., 2017). Smith et al on the other hand could prove that metaphorical arm gestures helped understand better geometric angle concepts (Smith et al., 2014). Studies in brain lesioned patients have shown that the motor cortex participates in processing numerical concepts, social interactions and mental processes (Harpaintner et al., 2020).

The body and sports education

Recent literature have demonstrated that physical activity and sports use embodied learning to improve motor and creative skills acquisition (Ravn, 2022). Embodied learning research in sports science helped identify that the intensity of physical exercise and situational variability induce motivation and improve attention, executive functions, and empathy (Ceciliani, 2018). Engström identified that movement enhanced expressing creativity during dance performance (Engström et al., 2018). Other studies have shown that mountain biking skills required embodiment experience shaped through environmental interaction (Ravn, 2022). The use of body-machine interfaces allowed a sophisticated and detailed analysis of the relationship between movement and emotional stimulation and the mirror neuron system (Grodal, 2009; Lim & Ku, 2018). A BCI-based action observation game in 15 healthy sujbects showed a significant stronger activation of the mirror neuron system (Lim & Ku, 2018). Other studies identified a relationship between specific situational race performance simulations and enhanced learning through perception-action and imagery skills (Bedir & Erhan, 2021).

BCIs and embodied classroom learning

Embodied cognition based on conceptualization and mirror neurons theory has been used in the classroom and has shown positive outcomes in terms of efficacy of learning based on motor actions, imitation, increased recall and comprehension (Sullivan, 2018; Macedonia, 2019). The challenges observed in learning in online-classroom could be explained by the low solicitation of embodied grounded cognition that requires motor and gesture-based cognitive stimulation (Sullivan, 2018). Within the context of education neuroscience, learning outcomes depend on instruction embodiment and its degree of sensitivity. Digital kinetic based learning tools using BCIs were used in the classroom and have proven efficacy in learning outcome and cognitive functions (Kosmas et al., 2019; Macrine & Fugate, 2021). BCIs that served in grounding embodied classroom learning include motion-based or kinetic games (Kosmas et al., 2019; Sullivan, 2018), functional brain imaging tools (fMRI/PET) (Harpaintner et al., 2020), and EEG coupled to VR or ITS or fNIRS (Rebolledo-Mendez et al., 2009; Spüler et al., 2017; Oku & Stato, 2021).

New brain to brain interfaces are short-cutting the necessity of body-to-body interactions for communication and individuals can communicate with computers and other indi-

Table 2. Kinetic virtual games studying the relationship between EL and BCIs reported useful in cognitive research in the classroom, laboratory research and clinical settings

BCI	Invasive (I) /Non-invasive (NI)	Healthy (H)/ Unhealthy (UH) participants	Use in the classroom (CL)/ laboratory (LAB)/ Clinical setting (CS)	Recording/Stimulation	References
			Motion based virtual games to assess embodied learning in the classroom		
Kinect Sensor	NI	H	CL	Embodied learning in physics education Engaging Statistics and research methods in psychology	(Sullivan, 2018)
Kinematics project	NI	H	CL	Mathematical imagery trainings	(Howison et al., 2011)
Other video games	NI	H	CL	Reduce math anxiety	(Verkijika & De Wet, 2015)
MEteor	NI	H	CL	Astronomy education	
Uniboxit/ Lexis	NI	H	CL	Attention, working memory and language	(Kosmas et al., 2019)
Instructional avatar	NI	H	CL	Role of using gestures in mathematics learning	(Cook et al., 2016)
			Brain imaging, recording and stimulation tools in ground cognition		
fMRI	NI	H	LAB	Visual and motor abstract concepts' grounding Social cognition	(Harpaintner et al., 2020) (Parvizi & Kastner, 2018)
fMRI/PET	NI	H	LAB	Role of perceptual system in concrete concepts and verbal system in abstract concepts	(Wang, Conder, Blitzer, & Shinkareva, 2010)
EEG	NI	N/A I H H	LAB LAB CL CL	Humanoid robotic control Attention Attention cognitive work load	(Choi & Jo, 2013) (Cinel et al., 2019) (Abrahamson & Lindgren, 2014) (Rebolledo-Mendez et al, 2009) (Spüler et al., 2017, Caitlin et al., 2017)
EEG+ VR/ ITS	NI	H H H I	LAB CL CL CS	Emotion detection, and decision making Brain painting in virtual reality settings (art) Neuro-ergonomics, measuring work load Motor and cognitive rehabilitation	(Winslow et al., 2016) (Botrel et al., 2015) (Tremmel et al., 2019) (Vourvopoulos et al., 2016) (Arpaia et al., 2020)
MEG	NI	H	LAB	Social cognition	(Acar et al., 2013)
fNIRS+EEG	NI	N/A H H	LAB CL CL	Robotic control Attention Language learning	(Sereshkeh et al., 2019) (Oku et al., 2021) (Watanabe et al., 2016)
TMS+ EEG	NI	H (Animal) I	LAB CS	Brain to brain interaction Learning, attention, perception, memory, and decision making Treat pain, depression and psychotic disorders	(Rao et al., 2014) (Grau et al., 2014) (Coffman et al., 2014) (Lefaucheur et al., 2014), (Brunoni et al., 2016)
tES	NI	H	LAB	Learning, attention, perception, memory, and decision making Treat pain, depression and psychotic disorders	(Coffman et al., 2014) (Lefaucheur et al., 2014), (Brunoni et al., 2016)
tDCS	NI	I/H	LAB	Reading difficulties Sight word efficiency Memory enhancement Complex problem solving	(Heth & Lavidor, 2015) (Younger et al., 2016) (Cinel et al., 2019) (Dockery et al., 2009)
CLDA	NI	H (Animal)	LAB	Visuomotor learning	(Orsborn et al., 2014)
ECog	I	H	LAB	Social cognition, theory of the mind default mode network Working memory	(Tan et al., 2022) (Zhang & Jacobs, 2015)

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Table 2. Kinetic virtual games studying the relationship between EL and BCIs reported useful in cognitive research in the classroom, laboratory research and clinical settings

BCI	Invasive (I) /Non-invasive (NI)	Healthy (H)/ Unhealthy (UH) participants	Use in the classroom (CL)/ laboratory (LAB)/ Clinical setting (CS)	Recording/Stimulation	References
ICMS	I	H (Animal) I	LAB CS	Brain to brain interaction Perceptions (tactile and visual) restitution	(Rao et al., 2014)
DBS Closed loop DBS	I	I I I I	CS	Obsessive compulsive disorder, movement disorders, Post-traumatic stress disorder Insomnia Severe depression	(Dougherty, 2018) (Meeres et al., 2022) (Castillo et al., 2020) (Guidetti et al., 2021)
Neuralink	I	I/H	LAB	Potential in motor and cognitive rehabilitation, and brain augmentation	(Musk, 2019)
FUS	Minimally I	H (Animal)	LAB	Movement planning	(Norman et al., 2021)
L7CI	Minimally I	I	LAB	Potential cognitive and motor rehabilitation	(Ho et al., 2022)
Stentrode	Minimally I	I	LAB	Cognitive computer control in motor disabled patients	(Mitchell et al., 2023)

viduals through brain signals (Hildt, 2019). This phenomenon can have big implications on the necessity to learn from body movement and gestures, besides other ethical issues like individual brain autonomy, the possibility of brain-hacking and cognitive bias (Hildt, 2019). Decreasing or suppressing physical solicitation can harm the learning process, since silencing body movements and environmental interactions can hinder human skills and activities.

This review discusses the neuroscientific theories of EL and embodied cognition. Conceptualization hypotheses are mainly used to explain EL phenomena and the dynamics linking the brain-body-environment interaction through sensorimotor networks. Then provided an updated summary of the available invasive and non-invasive BCIs used for cognitive brain augmentation and learning. Given the limited literature on the use of BCIs in classrooms, the use of non-invasive tools, primarily EEGs linked to VR, artificial intelligence ITS, or fNIRS, has been identified. These tools have proven promising outcomes in terms of attention, language learning, and cognitive work load control. Finally, an attempt was made to decipher the primary theoretical and empirical impacts of using BCIs on EL based on an understanding of the role of the body in the learning processes. The role of the body in language learning, science, and physical education is highlighted. And how technology-based tools helped understand the complex relationship between movement, perception, emotion and cognition. It could also be identified through recent literature that the use of BCIs has expanded from being cognitive, motor and brain augmentation tools to brain-to-brain and brain-to-machine interaction platforms, which can either exclude or limit the role of the body in interpersonal interactions. This contribution outlines the principal theoretical frameworks in scientific literature about BCIs and EL and highlights the enormous potential of BCIs in learning and cognitive augmentation. Although empirical studies about BCIs' use in education are scarce, a bigger interest should be given and translational studies need to be implemented from laboratory to classroom settings to analyze their potential educational implications. With the massive and fast development of technology nowadays many interrogations are coming into surface about the relationship between the human and the machine, and at what level the digital world will be able to interfere with our lives and integrate our bodies.

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The authors declare that there is no conflict of interest.

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REVIEW PAPER

Speed and Agility Training in Female Soccer Players - A Systematic Review

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Abstract

Female soccer players performs between 1350-1650 activity changes, along with jumping, accelerating and decelerating. The ability to repeat these actions identically in competition are essential for success in female soccer. Hence, the study aim was to summarize relevant literature on the effects of speed and agility training in female soccer players. Literature identification were conducted according to the PRISMA guidelines and in multiple databases (Google Scholar, PubMed, Scopus, Cochrane Library, ProQuest, EBSCOhost and Science Direct). Based on the pre-defined inclusion criteria (year of publication (2003-2022), full-text study published in English, the experimental study that had included healthy and injury-free female soccer players as participant sample) database search have identified 23502 potential studies. In the end, a total of seven full-text studies were included, with a total of 165 female participants. There were a variety of experimental programs, such as resisted, assisted and traditional sprint training, high-speed treadmill, speed and agility trainings, and repeated agility and strength group, along with their comparison with strength training group. Likewise, different types of duration, intensity and frequency were observed and resulted overall speed and agility improvements in female soccer players. Authors can conclude that only with well prepared and organized program, especially in pre-season, female soccer players should be able to improve important and specific factors, in order to achieve desired aim and result in terms of speed and agility.

Keywords: *football, sprint, quickness, power performance*

Introduction

Women's soccer has advanced significantly in terms of play, finance and media in recent years and as a result, the demands for women's soccer as team sport have risen sharply (Peeters & Elling, 2005). Nevertheless, it's growing popularity, female athletes are subjected to higher training volumes and competition demands than ever before, necessitating a better understanding of female athletes' performance changes in order to design effective training programs (Datson et al., 2004). Changes in the movement mechanism of the arms or legs can influence linear action such as acceleration and velocity. Thus, the ability to develop speed quickly (acceleration) is an important component for supporting performance in a variety of sporting activities (Azmi & Kusnanik, 2018). Speed, agility, along with the quickness (SAQ) exercises covers the entire training intensity spectrum and it is a very small percentage that can be improved due to heredity (Szabo, Neagu, & Sopa, 2020). What is more, acceleration and sprint performance is associated with maturity status (Murtagh et al., 2020).

According to the match statistics, female soccer players covers 9-12km during the game (Mohr, Krustrup, Andersson, Kirkendal, & Bangsbo, 2008), with as much as 8-12% of that being high-intensity running or sprinting (Rampinini et al., 2007). The average sprint duration is between 2-4sec. and occurs during crucial moments of the soccer game, with the vast majority of sprint displacements being less than 20m (Andrašić et al., 2021). Furthermore, fe-

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male soccer players performs between 1350-1650 activities changes, including dribbling passing and tackling (Munro & Herrington, 2011). Jumping, accelerating, decelerating, different sprints with changes of direction and the ability to repeat these actions identically in competition are essential for success in team sports like soccer (Stankovic et al., 2022). Moreover, mentioned explosive actions, like tackling, jumping, changing directions (COD) and sprinting, have a direct impact on the outcome of the result (Loturco et al., 2022). Therefore, explosive strength of the lower extremities is one of the most important motor characteristics (Prvulović, Martinović, Kostić, & Katanić).

There are numerous studies that compare the level of speed between competition levels, age categories, as well as the relationship between reactive speed, COD speed and individual movement speed (Ates, 2018; Andrašić et al., 2021; Stankovic et al., 2022). Despite the research mentioned above, there is currently little scientific evidence to shows effects of speed and agility training in female soccer players (Paradis, 2003; Upton, 2011; Shalfawi et al., 2013a; Shalfawi et al., 2013b; Mathisen & Danielsen, 2014; Mathisen & Svein, 2015; Page et al., 2021). Paradis et al. (2003) reported that the SAQ program improved power, speed and agility, but not strength in young soccer players. In addition, Shalfawi et al. (2013a) reported significant improvements in repeated agility training and repeated sprint training in elite female soccer players, with no significant differences between groups in any of the measured variables. Furthermore, two studies (Mathisen & Danielsen, 2014; Mathisen & Svein, 2015) found that short sprint bouts at maximum effort had a significant effect on agility performance in adolescent female soccer players. On the other hand, Shalfawi et al. (2013b), reported that agility and repeated sprint training had no significant effects in well-trained elite female soccer players.

To the authors' knowledge, there are a few studies that have analyzed the effects of speed and agility training in female soccer players. However, no study has been conducted that summarizes the literature in women's soccer. As a result, the purpose of this study is to summarize relevant literature on the effects of speed and agility training in female soccer players.

Materials and Methods

Literature Identification

PRISMA guidelines (Page et al., 2021; Rethlefsen et al., 2021) were used for the search and analysis of the studies. Furthermore, a multiple database identification was carried out, such as Google Scholar, PubMed, Scopus, Cochrane Library, ProQuest, EBSCOhost and Science Direct.

For study identification in mentioned databases, the multiple keywords (combination are separately) were used: („speed enhancement "OR „quickness "OR „soccer speed "OR „agility enhancement "OR „agility" OR „soccer agility" OR „SAQ" OR „mechanical stress" OR „physical stress") AND („soccer" OR „football" OR „female soccer" OR „female football" „team sport" OR „collective sport" OR „female team sport"). The study identification and data extraction were examined separately, by a total of two authors (M.S. and D.Dj.). Then, each author had to cross-examine the identified studies, and considered if the study is eligible for further analysis or not.

Furthermore, a descriptive method was used for obtained data examination, whereas all titles, abstracts and full-text articles were reviewed for eventual study inclusion in the systematic review. After detailed identification process, studies were considered to be relevant and included, only if they met the pre-defined inclusion criteria.

Inclusion Criteria

Each study had to meet the following inclusion criteria: year of publication (2003-2022), full-text study published in English, the experimental study that had included healthy and injury-free female soccer players as participant sample. In addition, there were no exclusion criteria in terms of years of training nor experience or rank (elite, sub elite, amateur, etc.).

Exclusion Criteria

The studies were not included if they have realized before 2003, published studies in other language than English, studies with male or mixed gender participants, studies where full-text possibility was unable, the studies that have included supplements usage and studies where experimental program was influenced on other parameters beside physical performance.

Bias Risk Assessment

The study quality and the potential risk of bias was assessed and determined by the PEDro scale (de Morton, 2009). Assessment were carried out by two authors, separately. The author's concordance was calculated using k-statistics data to examine the complete text, to determine relativity and bias risk. In case of disagreement, the provided data was evaluated and finalized by a third reviewer, independently. The concordance between reviewers was k=0.93.

Data Extraction

The necessary information was extracted from the studies, using Cochrane Consumer and Communication Review Group's. The main study characteristics were: first author and year of publication, age, sample size, experimental intervention program (type, duration, frequency and training duration), measured outcomes and study results.

Results

Study Quality

According to Maher et al. (2003), a PEDro scale points has to be awarded in order to identify the study quality. Further, if the study has gained between 0-3 points, the study will be classified with poor quality, 4-5 points with fair quality, 6-8 points with good quality and 9-10 points with excellent quality. Same authors have also stated that 8-11 points are optimal. In studies that have included in the final analysis, three studies have classified with fair quality, while rest of three studies with good quality. Table 1 presents PEDro scale total results.

Selection and Characteristics of Studies

Based on the database study identification, a total of 23502 were identified. At the main beginning, 522 duplicate studies were excluded, whereas a total of 22980 studies were further taken into consideration. According to the pre-defined inclusion criteria, 78 were excluded by review-

Table 1. PEDro scale for cross-sectional studies

	Criterion											
Study	1	2	3	4	5	6	7	8	9	10	11	Σ
Paradis et al. (2003)	Y	N	N	Y	N	N	N	Y	Y	Y	Y	4
Upton (2011)	Y	Y	Y	Y	N	N	N	Y	Y	Y	Y	7
Jonhson et al. (2013)	Y	Y	Y	Y	N	N	N	Y	Y	Y	Y	8
Shalfawi et al. (2013)	Y	Y	Y	Y	N	N	N	Y	Y	Y	Y	7
Shalfawi et al. (2013)	Y	Y	Y	Y	N	N	N	Y	Y	Y	Y	7
Mathisen et al. (2014)	Y	N	N	Y	N	N	N	Y	Y	Y	Y	5
Mathisen et al. (2015)	Y	N	N	Y	N	N	N	N	Y	Y	Y	4

Legend: 1—eligibility criteria; 2—random allocation; 3—concealed allocation; 4—baseline comparability; 5—blind subject; 6—blind clinician; 7—blind assessor; 8—adequate follow-up; 9—intention-to-treat analysis; 10—between-group analysis; 11—point estimates and variability; Y—criterion is satisfied; N—criterion is not satisfied; Σ—total awarded points.

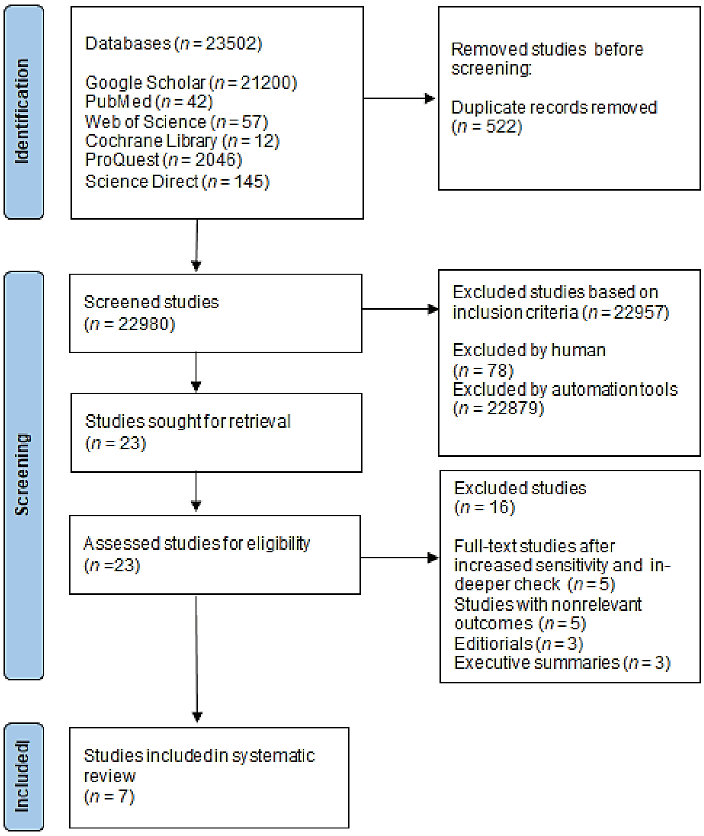

FIGURE 1. PRISMA flow chart of study identification

ers and 22879 were excluded by automation tools, whereas 23 studies were assessed for eligibility. Additional 16 studies were excluded based on in-deeper check, non-relevant outcomes, editorials and executive summaries. In the end, seven studies were included in the systematic review (Figure 1.).

Table 2 presents studies that have included in the systematic review based on pre-defined criteria.

There were a total of 165 female participants. The youngest participant was 13 years old (Mathisen et al. 2014), while the oldest was 21 years old (Shalfawi et al., 2013a). A total of 2 studies have presented 3 groups only (without control group) (Upton, 2011; Shalfawi et al., 2013a; Shalfawi et al., 2013b) and 4 studies have presented one experimental and one control group (Paradis et al., 2003; Mathisen et al., 2014; Mathisen et al., 2015). Experimental program duration variated from 4-10 weeks, whereas variables variated from running speed in all studies, RSA (Shalfawi et al., 2013a; Shalfawi et al., 2013b), agility (Shalfawi et al., 2013a; Shalfawi et al., 2013b; Mathisen et al., 2014; Mathisen et al., 2015) and explosive strength (Paradis et al., 2003; Shalfawi et al., 2013a; Shalfawi et al., 2013b). Only one study have had examining the influence on the Yo-Yo IR1 test (Shalfawi et al., 2013a) and one on the high-speed treadmill (Shalfawi et al., 2013b).

There were a variety of experimental programs, such as resisted, assisted and traditional sprint training (Upton, 2011), high-speed treadmill (Johnson et al., 2013), speed and agility trainings (Paradis et al., 2003), repeated agility and strength group (Shalfawi et al., 2013a), along with their comparison with strength training group (Shalfawi et al., 2013b).

Table 2. Studies that have included in the systematic review

First author and year of publication	Participants		Duration (weeks)	Program (type, intensity frequency, training duration)	Measured outcomes	Results		
	Age (Years)	Number and groups						
Paradis et al. (2003)	13.07 ±0.59	SA-19 C-13	6	SA–training program 2x a week C–regular soccer training	T-test 40yd CMJ LJ	SA T-test ↑* 40yd ↑* CMJ ↑* LJ ↑		C T-test ↓ 40 yd ↓ CMJ ↑ LJ ↑
Upton (2011)	19.6 ±0.9	N-27 AST-8 RST-9 TST-10	4	AST–supramaximal efford 20yd + 20yd deceleration to jog, 10x assisted sprint, 3min rest RST–20yd + 20yd maximal effort sprint + 20yd deceleration to jog, 10x resisted sprint, 3min rest TST–20yd sprint + 20yd deceleration to jog, 10x maximal efford sprint, 3min rest	5, 15, 25, 40 yd	AST 5yd ↑* 10yd ↑* 25yd ↑* 40yd ↑*	RST 5yd ↔ 10yd ↓ 25yd ↑ 40yd ↑*	TST 5yd ↓ 10yd ↑ 25yd ↑ 40yd ↑
Jonhson et al. (2013)	16.6 ±1.19	SPO-11 HST-13	6	HST-3 series 10min; 2 min warm-up at 0% incline at 6–8 mph; training-maximal speeds ranged from 18 to 22 mph. 10sec sprints with 40-60sec rests at a maximum incline of 10%; 5sec sprints with 20-40-sec	40-Yard sprint Isometric Strength (flexor, extensors)	HST 40yd ↑* Flexor ↔ Extensors ↑*		SPO 40yd ↓ Flexor ↓ Extensors ↑
Shalfawi et al. (2013)	21.2 ±2.6	RAG-8 RSG-9	8	RAG–2x4 agility run, 120sec. recovery between exercises, 10min recovery between sets, Intensity=95-100% first 5 weenks, rest of 3 was 100% RSG–2x(5-9)x40m, 90sec. recovery between exercises, 10min recovery between sets, Intensity=95-100% first 5 weeks, rest of was 100%	40m sprint 40m agility CMJ RSA-10x40m YY1	RAG 40m sprint ↑ 40m agility ↑* CMJ ↑ RSA ↑* YY1 ↑*		RSG 40m sprint ↑* 40m agility ↑ CMJ ↑* RSA ↑* YY1 ↑*
Shalfawi et al. (2013)	19.4 ±4.4	N-20 RAG/RSG STG	10	RAG–2-4 sets, 1min. recovery between exercises, 10min. recovery between sets, 100% intensity RSG–2-5 sets of 4-5x40m, 90sec. rest between exercises, 10min. rest between sets, Intensity=95-100% first 4 weeks, rest of was 100% STG–leg press, squat jump, nordic hamstring, leg extension, cable hip flexion and extension	SJ CMJ RSA-7x30m 40m sprint 40m agility Bt	RAG/RSG SJ ↑ CMJ ↑ RSA ↓ 40m sprint ↓ 40m agility ↑ Bt ↑*		STG SJ ↑* CMJ ↑ RSA ↑ 40m sprint ↓ 40m agility ↓ Bt ↑*
Mathisen et al. (2014)	13.6 ±0.2	E-13 C-13	8	E–32 short-burst sprints 10min warm-up, 50min short-burst running LIN or COD sprints (40-90sec. rest) (once a week in addition to 2 regular trainings) per week C–regular soccer training	10m sprint 20m sprint Agility	E 10m sprint ↑* 20m sprint ↑* Agility ↑*		C 10m sprint ↔ 20m sprint ↑ Agility ↓
Mathisen et al. (2015)	15.5 ±0.7	E-10 C-9	8	E–32 short-burst sprints 10min warm up, 45 min short-burst running LIN or COD sprints (60-90sec. rest) (once a week in addition to 2 regular trainings) per week C–regular soccer training	10m sprint 20m sprint Agility	E 10m sprint ↑* 20m sprint ↑* Agility ↑*		C 10m sprint ↔ 20m sprint ↑ Agility ↓

Legend: N–total number of participants, E–experimental group, C–control group, SA–speed and agility, yd–yards, T-test–agility T-test, CMJ–countermovement jump, SJ–squat jump, LJ–long jump, RST–resisted sprint training, AST–assisted sprint training, TST–traditional sprint training, RAG–repeated agility group, RSG–repeated sprint group, STG–strength training group, RSA–repeated sprint ability, COD–change of direction, LIN–linear sprint, YY1–Yo-Yo IR1 test, Bt–Beep test, SPO–soccer practice-only, HST–high-speed treadmill, *–significant result, ↑–result improved, ↓–result decreased, ↔–result maintained.

Discussion

The study aim was to summarize relevant literature on the effects of speed and agility training in female soccer players. The main study findings are seven studies that have presented various types of speed and agility training, with different types of duration, intensity and frequency that have resulted overall speed and agility improvements in female soccer players.

Increased step frequency and reduced ground contact time have a positive effect on maximum speed as well as the result of reduced acceleration time (Mero et al., 1992; Paradis, 2003). Kyröläinen, Avela and Komi (2005) have found that during the acceleration phase of sprinting, maximal integrated electromyographic (EMG) activity is greater than during the constant velocity period, indicating that this is the moment when the sprinter's neural activation is greatest. A significant increase in muscle force development in initial acceleration in the AST group occurred in the first 5yd (4.6m) of the sprint, while the RST group had the greatest increase in speed during the 15 to 25yd (13.7 to 22.9m), and as it was hypothesized acceleration increased significantly ($p<0.001$) over a 4 week period (Upton, 2011). Repeated linear sprint training improves intermittent running ability more than agility training, while repeated agility training improves specific agility improvement and both groups on the RSA test (10x40m) with 95% maximum running speed finished with 97% in the post-test (Shalfawi et al., 2013a). In that regard, similar results were also presented elsewhere (Tønnessen et al., 2011). Above mentioned indicates the ability to achieve repeated sprints close to maximal intensity.

Since it was observed only moderate improvements ($d=0.8$) in the RAG/RSG group, as well as trivial to negative in agility performance in the STG group (Shalfawi et al., 2013b), these results are not in accordance with Dupont et al. (2004), who have observed improvements in RSA. These soccer players have performing one repeated sprint session and one aerobic training session each week, in addition to one game and 8-10 normal soccer training sessions during the season. As a result, a physical conditioning program must be carefully balanced with regular soccer training (Morgans, Orme, Anderson, & Drust 2004). A carefully constructed training program for one set of skills may impede the development of other vital attributes and vice versa (Jalilvand, 2015). It is also recognized that the constant stress, along with the strength and conditioning program, can create a „chronic catabolic environment" for the neuromuscular system. Because these studies were done in-season, this setting may result in modest or no changes in other physical characteristics (Kraemer et al., 2004). Hence, an additional physical fitness program must be well planned and balanced together with regular soccer training, especially during in-season period.

According to Yap and Brown (2000), female training regimens are identical to males training protocols, as women's training programs have improved significantly over the years. Mathisen & Danielsen (2014) have resulted a significant increase (6.2%) in agility performance in a 8 week LIN and COD program, which is consistent with findings Pettersen and Mathisen (2012). Although initial acceleration and short sprint are reported to be more difficult to improve than maximal speed (Meilan & Malatesta, 2009), this study also shows a significant improvement in the acceleration phase (5.1%) in the 10m sprint and (3.5%) in 20m sprint. Furthermore, results from other study (Mathisen et al., 2015), with a bit older participants have revealed 10m straight sprint (4.1%), 20m straight sprint (3.2%) and agility performance (5.2%) improvement. Since growth and maturation could increase sprint performance (Vescovi et al., 2011), maturity status has a crucial role in modulating the response to speed exercises (Malina et al., 2004).

Likewise, women go through a biological process during the menstrual cycle, where hormone levels rise and fall (Keay et al., 2021). Julian et al. (2017) have highlighted that there could be a performance decreases during the mid-luteal phase where hormones were contrasted in the peak phase of the menstrual cycle and this decreases was not found in jumping or sprint performance. Hence, in order to examine and analyze how the phases of the menstrual cycle affect physical performance, it is necessary to take into consideration the specificity of sport (Mkumbuzi et al., 2021). But further investigation is needed.

The strength of this study lies in the fact that it is the only systematic review on the topic of speed and agility training in women's soccer. This is especially significant given that the majority of soccer training research focuses on men. Additionally, the paper has provided valuable guidelines for the training of female soccer players. Thus, the practical implications would involve the implementation of various training programs that have shown positive effects on speed and agility qualities in female soccer players.

As far as the study limitations, there is some. First, they have taken into consideration studies that have dealt with regular speed and agility, but not reactive. Second, they did not take into consideration the anterior cruciate ligaments (ACL) condition in the participants' sample, which can be an important factor for both speed and agility. Hence, future studies can include the mentioned medical state for both future experimental studies and systematic reviews.

Conclusion

Since the speed is about 95% congenital, the same can be relatively enhanced. On the other hand, agility is not congenital as speed, but it can be more influenced. Hence, only with well prepared and organized program, especially in pre-season, female soccer players should be able to improve these important and specific factors, in order to achieve desired aim and result.

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Conflict of Interest
The authors declare that there is no conflict of interest.

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REVIEW PAPER

Effectiveness of Inspiratory Muscle Training among Chronic Obstructive Pulmonary Disease Patients: A Systematic Review

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Abstract

Inspiratory muscle training (IMT) has been shown to be effective for some types of illnesses or conditions, including respiratory problems, cardiac disorders, and neuromuscular disorders. Reduced exercise capacity and health-related quality of life, weakness of respiratory muscles, and complaints of dyspnea are all frequent among COPD patients. This study aimed to investigate whether the use of IMT can alter the consequences of COPD in patients and provide an updated understanding of the effects of IMT in COPD patients. The specific objectives of the study were to evaluate the effectiveness of IMT on respiratory muscle strength, exercise tolerance, dyspnea, and health-related quality of life in COPD patients. Included databases were the Physiotherapy Evidence Database (PEDro), Google Scholar, and PubMed. A preliminary selection of papers was produced using inclusion/exclusion criteria. Subsequently, a final selection was made based on the quality of the studies assessed using the PEDro scale. For the realization of this systematic review, seven studies were involved. According to the evidence gathered from the literature review, IMT therapy has been shown to improve respiratory muscle strength, exercise tolerance, perception of dyspnea, and health-related quality of life in COPD patients. It is necessary to conduct studies using high-quality, evidence-based data to draw more definitive conclusions about the effectiveness of IMT in COPD patients. Future research should examine whether these improvements are applicable to all COPD patient groups. Additionally, it should explore the long-term effectiveness of these therapeutic improvements and determine which IMT treatment protocol provides the most significant clinical benefits.

Keywords: *respiratory muscle strength, exercise tolerance, dyspnea perception, quality of life*

Introduction

Chronic Obstructive Pulmonary Disease (COPD) is a preventable and treatable condition characterized by airflow limitation. Although it primarily affects the lungs, it can also have serious systemic effects (Viegi et al., 2007).

It is increasingly recognized that many COPD patients have co-morbidities that significantly impact their survival and quality of life. Chronic and worsening dyspnea, coughing, and sputum production are common symptoms of COPD (Vogelmeier et al., 2017). The most significant symptom of COPD is shortness of breath. Patients are increasingly dependent on their accessory muscles to maintain breathing, particularly during physical activities (Calverley & Georgopoulos, 2006).

COPD results in reduced physical activity, which leads to deconditioning, mainly due to difficulty breathing. This results in a greater fear of exertion and a decreased interest in social and physical activities, which can trap the patient in a vicious cycle that worsens isolation and depression and lowers their quality of life (Corhay et al., 2014).

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Improvements in exercise capacity, a reduction in perceived dyspnea severity, an increase in health-related quality of life, and a decrease in hospitalizations and hospital stays are all advantages of pulmonary rehabilitation (PR) for COPD patients (Gloeckl et al., 2013). Patients with stable COPD may benefit from a PR program, as may those who have recently experienced an exacerbation. All individuals experiencing systemic effects from COPD are suitable candidates for PR. The four primary components of PR are education, nutritional assistance, psychological support, and exercise therapy (Corhay et al., 2014).

Respiratory muscle therapy (RMT) is a technique that focuses on enhancing respiratory muscle performance through targeted exercises. Particularly, inspiratory muscle training (IMT) has been found to improve respiratory muscle performance and may reduce exertional dyspnea (Pereira et al., 2019). IMT has been shown to be effective for some types of illnesses or conditions, including respiratory problems, cardiac disorders, and neuromuscular disorders (Dsouza et al., 2021). Although there is research showing a positive impact of IMT in patients with COPD, there is still a need for systematization of knowledge to draw clear conclusions regarding the effect of IMT on respiratory muscle strength, exercise tolerance, dyspnea perception, and the impact on the quality of life of patients with COPD.

Reduced exercise capacity, weakened respiratory muscles, and complaints of dyspnea are all common among COPD patients, leading to a decrease in health-related quality of life (Tkáč et al., 2007). This study aimed to investigate whether the use of IMT can alter the consequences of COPD in patients and provide an updated understanding of the effects of IMT in COPD patients. The specific objectives of the study were to evaluate the effectiveness of IMT on respiratory muscle strength, exercise tolerance, dyspnea, and health-related quality of life in COPD patients.

Methods

Search strategy

For the realization of this literature review are followed the PRISMA guidelines for conducting systematic reviews and meta-analyses, as outlined by Moher et al. (2009). For the purpose of performing this systematic review, the following databases were included: Physiotherapy Evidence Database (PEDro), Google Scholar, and PubMed. Studies conducted within twenty years from 2000 to 2020, were considered. The search was conducted from January to June 2023. Keywords such as respiratory muscle strength, exercise tolerance, dyspnea, and quality of life have been used. A preliminary selection of papers was produced using inclusion criteria based on the information provided in the titles and abstracts. Then, a final selection was made according to an individual critical assessment of the quality of the studies. After identifying potential articles for inclusion in the review, the full texts were thoroughly read to select those that met all inclusion criteria. These selected articles became part of the systematic review. All the phases involved in the search process are illustrated in Figure 1. The evaluation was conducted using the PEDro scale, which evaluates the validity of each study (Table 1).

Inclusion criteria

The inclusion criteria for this study were Randomized controlled trial (RCT) studies, studies with adult participants, studies with diagnosed COPD patients, free studies, studies with specified measures, studies that include IMT therapy, and English-language studies.

Exclusion criteria

Case studies, literature review studies, and studies conducted prior to 2000 were excluded from the analysis.

Results

Quality of the Studies

To assess the methodological quality of the studies included in the systematic review, the PEDro scale developed by the Centre for Evidence-Based Physiotherapy was utilized. The evaluation of the methodological quality was done using the PEDro scale, which rates the studies as outstanding (9–10), good (6–8), fair (4–5), or poor (<4). This systematic review includes seven studies, of which four were of high quality, one demonstrated good quality, and two indicated poor quality, as shown in Table 1. The quality score is based on expert consensus and is intended to eliminate the risk of bias.

Selection and Characteristics of Studies

A total of 134 studies were found by searching through database resources and reviewing reference lists. Finally, the

Table 1. Quality Assessment Using the PEDro Scale

	Items by number on the PEDro Scale											
Article	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**Total**
Minoguchi et al. (2002)	N	Y	N	N	N	N	N	N	N	Y	Y	3
Magadle et al. (2007)	N	Y	N	Y	Y	N	Y	Y	N	Y	Y	7
Petrovic et al. (2012)	Y	Y	N	Y	N	N	N	N	N	Y	Y	4
Elmorsi et al. (2016)	Y	N	N	Y	N	N	N	N	N	Y	Y	3
Dellweg et al. (2017)	Y	Y	Y	Y	Y	N	Y	Y	N	Y	Y	8
Wang et al. (2017)	Y	Y	Y	Y	N	N	Y	Y	Y	Y	Y	8
Langer et al. (2018)	N	Y	Y	Y	Y	N	Y	Y	N	Y	Y	8

Note. N: criterion not fulfilled; Y: criterion fulfilled; 1: eligibility criteria were specified; 2: subjects were randomly allocated to groups or to a treatment order; 3: allocation was concealed; 4: the groups were similar at baseline; 5: all subjects were blinded; 6: all therapists were blinded; 7: all assessors were blinded; 8: measures of at least one key outcome were obtained from over 85% of the subjects who were initially allocated to groups; 9: intention-to-treat analysis was performed on all subjects who received the treatment or control condition as allocated; 10: the results of between-group statistical comparisons are reported for at least one key outcome; 11: the study provides both point measures and measures of variability for at least one key outcome; total score: each satisfied item (except the first) contributes 1 point to the total score, yielding a PEDro scale score that can range from 0 to 10.

systematic review comprised a total of seven full-text research articles. Figure 1 depicts the selection procedure for the study. Table 2 lists the seven studies that were part of the systematic review. The majority of scientific research articles selected for in-depth analysis contained one experimental group and one control group. Table 2 shows the findings of the study.

Table 2. Summary Table of the Studies Included in the Systematic Review

Author / Year	Population	Experimental Group	Control Group	Measurements	Results
Minoguchi et al. (2002)	N=12, COPD patients, <78 years of age.	For four weeks, the patients received IMT (two sessions of 10 minutes of training each day at 30% of PImax).	For four weeks, the patients received respiratory muscle stretching (three sessions of five patterns, four times each, daily) for 4 weeks.	Respiratory muscle strength, exercise tolerance.	Muscle strength increased in the IMT treatment group (from 66.1±5.9 to 79.1±6.5 cmH2O; p=0.002), Exercise tolerance increased significantly in the IMT group (from 386±21 m to 412±18 m; p=0.041).
Magadle et al. (2007)	N=34, COPD patients, <68 years of age.	Patients were treated with general reconstructive exercises for 12 weeks and then treated with IMT for 6 months.	Patients were treated with general exercise for 12 weeks and then with sham IMT for 6 months at a load known not to improve inspiratory muscle function.	Respiratory muscle strength, dyspnea, quality of life.	There was a significant increase in respiratory muscle strength in the experimental group (from 66±4.7 to 78 ± 4.5 cm H2O; p<0.01). Dyspnea decreased in the experimental group (from 20.2±0.4 to 14.9±0.3 total Borg score; p<0.001). This was accompanied by a significant improvement in the quality-of-life questionnaire scores.
Petrovic et al. (2012)	N=20, patients with COPD stage II or III, <70 years of age.	IMT treatment sessions were performed by subjects once a day, seven days a week, for eight weeks.	Daily treatment of the inspiratory muscles with strength and endurance exercises was performed by the subjects for eight weeks.	Respiratory muscle strength, dyspnea.	IMT improves respiratory muscle strength (from 7.75±0.47 kPa to 9.15±0.73 kPa, respectively; p< 0.001) and the perception of dyspnea (from 5.0±1.0 to 4.0±1.1; p<0,01).
Elmorsi et al. (2016)	N=60, patients with moderate to very severe COPD, <62 years of age.	Group A received IMT and exercise for the peripheral muscles at a rate ranging from 30 to 60% of their inspiratory pressure.	Only peripheral muscle exercise was administered to Group B. Group C did not receive any care.	Respiratory muscle strength, exercise tolerance, dyspnea, quality of life.	IMT offers COPD extra advantages in terms of respiratory muscle strength and exercise capacity. However, this improvement did not translate into additional improvements in dyspnea and quality of life compared with what is achieved by peripheral muscle exercise training alone.
Dellweg et al. (2017)	N=29, patients with COPD stage III or IV, <74 years of age.	For four weeks, IMT was administered once a day during the workdays.	For four weeks, sham IMT was administered once a day during the workdays.	Respiratory muscle strength, exercise tolerance	IMT improves exercise capacity and increases respiratory muscle strength and power.
Wang et al. (2017)	N=81, patients with stable COPD, <40 years of age.	For eight weeks, IMT was combined with a CET.	For eight weeks, there were two treatment groups: the CET group used a cycle ergometer, whereas the control group was treated with free walking.	Respiratory muscle strength, exercise capacity, dyspnea, quality of life.	The experimental group showed improvements in respiratory muscle strength, exercise capacity, dyspnea, and quality of life (p<0.05).
Langer et al. (2018)	N=20, patients with stable COPD, <77 years of age.	For eight weeks, the IMT group worked out twice a day at a training intensity of 40 to 50% of Pimax.	The treatment involved performing 30 breaths twice to three times a day for four to five minutes each, seven days a week for eight weeks.	Respiratory muscle strength, dyspnea.	IMT improves respiratory muscle strength and dyspnea (p<0.05).

Note: CET=cycle ergometer, PImax=maximum inspiratory, kPa=kilopascal, cmH2O=centimeters of water, m=meter.

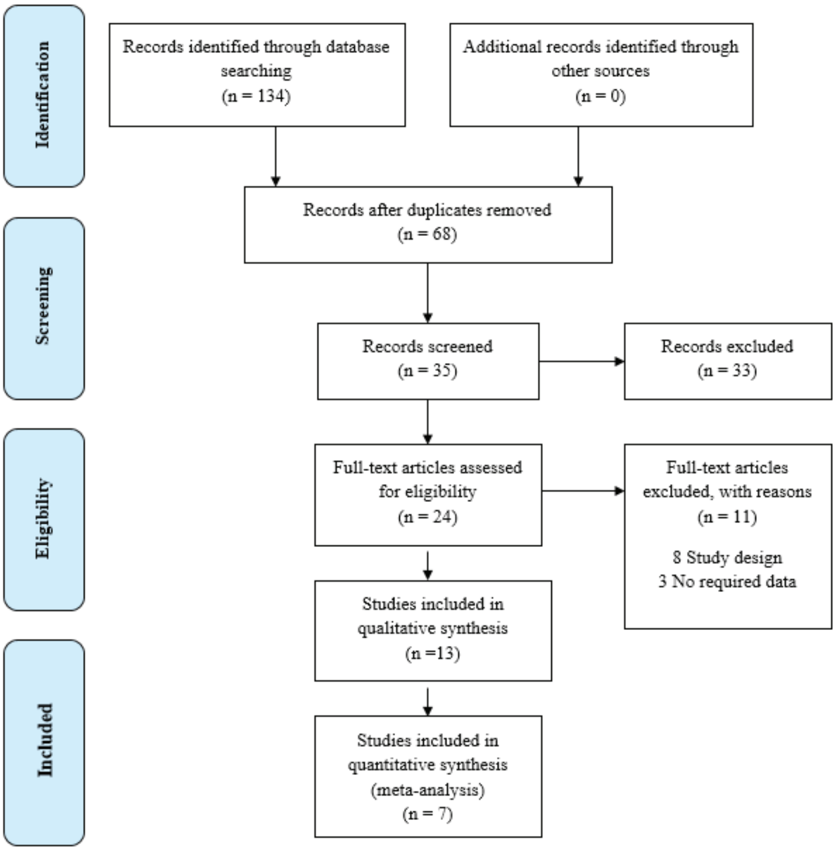

FIGURE 1. PRISMA Study flow diagrams

Discussion

Respiratory muscle strength

In most studies, respiratory muscle strength was measured using the maximum inspiratory pressure (PImax) and maximal expiratory pressure (PEmax). According to the study by Wang et al. (2017), the group that received treatment with a cycle ergometer and IMT showed a substantial increase in Pimax and PEmax (P<0.05). As per Dellweg et al. (2017) study, only patients in the IMT group who were determined to have increased inspiratory muscular strength, as measured by PImax, showed significant improvement. Additionally, at the end of weeks two and four, these patients also demonstrated a decrease in maximum negative inspiratory pressure. As reported by Petrovic et al. (2012), after 8 weeks of daily IMT, the IMT training group exhibited a significantly greater degree of inspiratory muscle performance. PImax increased by 18% (P<0.001). Elmorsi et al. (2016) found that IMT combined with peripheral muscle exercise training enormously improved PImax and PEmax (P= 0.000). Based on the study conducted by Minoguchi et al. (2002), respiratory muscle strength, as measured by PImax, showed a notable increase in the IMT treatment group (p=0.002). As specified by Langer et al. (2018), measures of respiratory muscle strength in the IMT group demonstrated significant post-intervention changes (P<0.05). According to Magadle et al. (2007), after three months of training, there was a substantial difference in the PImax in the IMT training group, and this difference was maintained after a further three months of training.

Exercise tolerance

In the majority of the studies, exercise tolerance was measured using the 6-minute walk test (6 MWT). As indicated by Wang et al. (2017), exercise capacity was evaluated using the 6-minute walk test (6MWT) and a maximal exercise test. The group that received cycle ergometer + IMT treatment indicated substantial increases in the 6MWT from baseline (p<0.05). As evidenced by Elmorsi et al. (2016), 6MWD significantly increased in the IMT+peripheral muscles exercise treatment group (p=0.000). Six-minute walking distance considerably increased following treatment in the group receiving IMT, as shown by Minoguchi et al. (2002) (p=0.041). According to Dellweg et al. (2017), the 6-minute walk test (6MWT) showed noticeable improvement in the IMT group by week 2. By the end of the study, patients in the IMT group had walked a longer distance than those in the sham IMT group.

Dyspnea

The Borg Scale was used to measure dyspnea perception in the majority of the studies. Petrovic et al. (2012) study revealed that the level of dyspnea assessed with the BORG Scale showed a notable decrease (p<0.01). As reported by Magadle et al. (2007), the perception of dyspnea in the IMT treatment group significantly decreased (p<0.001). As stated in the study conducted by Langer et al. (2018), dyspnea in the IMT group improved after 8 weeks, despite no significant changes in ventilation, breathing pattern, or lung function. The relief of dyspnea was accompanied by a decrease in diaphragm activation compared to the maximum level. The study conducted by Wang et al. (2017) utilized the COPD Assessment Test (CAT) and the modified mMRC (Modified Medical Research Council) Dyspnea Scale to assess the severity of dyspnea. The results showed that the group that underwent cycle ergometer + IMT treatment experienced a significant reduction in dyspnea compared to the control group (p<0.05). The findings of Elmorsi et al. (2016) found that the group receiving IMT + peripheral muscle exercise therapy had a considerably decreased dyspnea rate, as measured by mMRC (p=0.001).

Quality of life

In most studies, the measurement of health-related quality of life was conducted using the St. George's Respiratory Questionnaire (SGRQ). According to Wang et al. (2017), the health-related quality of life, as measured by SGRQ, in the group that received cycle ergometer + IMT therapy showed substantial improvements in quality of life, depression, and anxiety (p<0.05). Treatment with IMT+ peripheral muscle exercises significantly improved SGRQ-C questionnaire domains, as shown by Elmorsi et al. (2016) (p=0.000). The SGRQ score in the IMT treatment group reportedly decreased steadily, according to Magadle et al. (2007). At the end of the sixth month of training, the difference between the IMT-treated group and the sham IMT group became significant (p<0.05).

Findings from studies that were not included in the review

Numerous studies have investigated the effects of IMT on muscle strength, and the majority have demonstrated that treatment with IMT significantly increases the strength of respiratory muscles in COPD patients (Geddes et al., 2005; Geddes et al., 2008; Gosselink et al., 2011; Lötters et al., 2002; Beaumont et al., 2018; Schultz et al., 2018). Regarding the effect of IMT on exercise tolerance, several studies have confirmed our findings that IMT increases exercise capacity in COPD patients (Riera et al., 2001; Lötters et al., 2002; Geddes et al., 2008; Shoemaker et al., 2009; Gosselink et al., 2011; Beaumont et al., 2018). Other research studies have corroborated our findings regarding the reduction in dyspnea perception following the treatment of COPD patients with IMT (Riera et al., 2001; Lötters et al., 2002; Geddes et al., 2005; Geddes et al., 2008; Shahin et al., 2008; Shoemaker et al., 2009; Gosselink et al., 2011; Beaumont et al., 2018). The effectiveness of IMT in improving health-related quality of life has been confirmed in several studies (Riera et al., 2001; Gosselink et al., 2011; Beaumont et al., 2018).

Strength of the study

The study provides detailed information to explain the effects of IMT in patients with COPD. The data collection was cost-effective, the data are relatively easy to analyze, and also the data are very consistent, accurate, and reliable. Another strength of the study is that the findings can be generalized.

Limitations of the study

As with any scientific research or literature review, we encountered various limitations when producing this article. Firstly, the studies had different treatment protocols, including variations in the type of IMT used and its association with other methods, even within the experimental groups. The diversity of comparability and reliability would be higher in studies with homogeneous groups that follow standardized treatment procedures. Secondly, conducting studies on patients in the same stages of COPD, where each objective mentioned above is studied separately for distinct classes, would yield significant statistical results. This would help estimate the IMT for each stage of COPD. Thirdly, due to financial constraints, we have been unable to acquire many valuable articles that are relevant to our research goals. As a future research direction, several studies can be conducted to provide useful evidence on the effectiveness of IMT in different stages of the disease. Additionally, standardized IMT treatment protocols can be developed for patients with COPD.

Conclusions

This systematic review has highlighted the efficacy of IMT in improving outcomes for patients with COPD. According to the evidence gathered from the literature review, IMT therapy has been shown to improve respiratory muscle strength, exercise tolerance, perception of dyspnea, and health-related quality of life in COPD patients. It is necessary to conduct studies using high-quality, evidence-based data to draw more definitive conclusions about the effectiveness of IMT. Future research should examine whether these improvements are applicable to all COPD patient groups. Additionally, it should explore the long-term effectiveness of these therapeutic improvements and determine which IMT treatment protocol provides the most significant clinical benefits.

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Conflict of interest
The authors declare that there are no conflicts of interest.

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REVIEW PAPER

Physical Education and Embodied Learning: A Review

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Abstract

Recent neuroscientific research highlighted interesting interactions between superior cognitive functions and the sensorimotor system. Overcoming the traditional views of the philosophy of mind and cognitive sciences, current empirical evidence shows that bodily states are the basis of information processing and that incarnation contributes to various aspects of mental phenomena. Starting from this theoretical framework, the purpose of this contribution is to outline, through exposure to the most significant discoveries in the world of neuroscience, the importance of Embodied Cognition as an emerging vision that considers the cognitive processes deeply rooted in the interaction of the body with the world. After careful selection and analysis of studies on neuroscience applied to teaching, the study focused on those who experienced in the school context embodied-teaching approaches aiming to promote transversal learning through movement and corporeality in action and interaction. The analysis highlights the strengths and critical points of the studies implemented in recent decades that focus on the integration and transversality of the body in learning. This is to understand whether embodied learning environments involving all spheres of personality can foster perception, knowledge, and conscious action in teaching and learning processes. An important result from the overview is the potential of theory in different educational environments and disciplines. The contemporary theoretical framework highlights the great potential of corporeality and physical education as actors in learning, but at the same time place in the foreground the need to experiment and disseminate new teaching approaches and perspectives.

Keywords: *neuroscience, corporeality, embodied learning, physical education, teaching-learning process, transversal skills*

Introduction

The body dimensions have taken on a central role in learning nowadays (Bresler, 2013). This highlights its educational, social, and inclusive potential and brings about the necessity to analyze the interweaving between body movement and learning in terms of efficiency and quality. The body can convey information that, if thoroughly investigated and understood, can enrich the knowledge about the dynamics of decision-making processes, both in individual and social contexts. It is within Embodied Cognition that these studies are rooted; the theoretical frameworks focusing on bodily states and the consequent sensorimotor interactions with the outside world helped better understand the link between the environment and the different cognitive states and effects (Farina, 2021). Within this perspective, Embodied Cognition represents one of the most emerging scientific approaches in the field of Educational Neuroscience (Fischer, 2009; Gomez Paloma, 2009; Fischer, 2010) and the awareness of this new vision of the body is the basis of the research conducted in this field (Ledoux, 2002).

The birth of embodied cognitive science, or embedded cognitive science, dates back to the late 1980s. Embodiment implies that the mind is no longer independent from the body, however, inscribed in it. Nowadays, the embodiment approach is getting more attention and is defined as "incarnation". Research in the field of education science can fuel new neuro didactics that focus both on the complexity of the body as a bio-dynamic entity in formation and considers unfolding the mechanisms underlying our perception of the living reality. The adoption of heuristic and synergistic approaches alone can not define global objectives (Rivoltella, 2012; Caruana et al., 2016; Shapiro et al., 2019).

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But, they can shape and define helpful research tools to better understand through practical experiences the complexity and uniqueness of the individual (Gomez Paloma, 2016). Embodied cognition considers that human cognition is fundamentally grounded in sensory-motor processes and our body's morphology and internal states (Ionescu et al., 2014). In particular, it is important to consider the essential role that body movements can play in achieving high-quality education for children (Chandler et al., 2015). Numerous studies have shown that body movements can have substantial positive effects on children's cognition, learning, and academic achievement (Tomporowski et al., 2008; Bartholomew et al., 2011; Donnelly et al., 2011; Fedewa et al., 2011; Erickson et al., 2015). Based on their review study, Tomporowski et al. (2008) concluded that exercise is a simple and promising method of improving the aspects of children's mental functioning that are central to their cognitive development. For effective teaching to improve skills, children must take action, experiment and actively participate in classes. Numerous scientific evidence emphasizes the important role of the body and movement in cognitive development even in childhood (Thelen et al., 1994; Robertson et al., 2009). Research on embodied co-cognition has radically changed the conception of how action, perception, and cognition relate and interact with each other. Sensory and cognitive abilities, as well as their interactions, are more fragile during the developmental age, thus offering a unique window on the emergence of embedded cognitive effects and age-related differences. The gestures that children and adults make when they speak are a revealing window into the processes of cognitive change: the body represents, in an incarnate context, a source of basic information, understanding, and cognition (Samuelson, 2011). Neuroscience focuses its attention on the study of mind and body, which are fundamental aspects, especially in terms of individualized and personalized teaching. Such learning implies a commitment of mind and body, biological and psychological factors. The representation of one's own body is the foundation of feelings and thought: it is indispensable for subjectivity (Blaesi et al., 2010). The motor system allows the subject a complete understanding. The physical body and movement are involved in processes related to cognition in schools such as language and logical-mathematical learning and, at the same time, play a fundamental role in the development of emotional and interpersonal skills (Gallese, 2005). Embodied learning should also be the cornerstone of physical education programs (Thorburn et al., 2017), as classes during motor activity class are engaged, not only physically, but also cognitively, facilitating the development in children of key cognitive skills, in particular executive functions (D'Anna, 2023).

In light of this evidence, the aim of this paper was to analyze embodiment from studies on applied neuroscience in education. Studies have been selected based on their experimental aspect of embodied-based teaching approaches in the school environment, using transversal learning through movement and corporeality in action and interaction. Starting from the premise that the key principle of embodied cognition offers unprecedented opportunities to exploit the differences in learning processes (Boat et al., 2022), and is necessary for creating innovative teaching methodologies.

Methods

An analysis of the scientific literature was conducted by analyzing studies published with the following keywords: Physical Education, Corporeality, Learning, Education, Embodied Cognition, Neuroscience, Embodied Cognition Approach. In particular, studies have been analyzed that highlight how corporeity plays a decisive role in the learning processes and the design of training courses focused on the use of the body and movement as a mediator of knowledge.

Discussion

The body in Embodiment theories, relies on perception and action to act as a mediator for the learning process. It thus overcomes the classical scientific frameworks that consider it as a mere evaluation object used to acquire the dignity of a subject of cognition (Sgambelluri, 2020). Post-constructivist hypotheses (Rivoltella, 2012) investigated the relationship between the environment, the body, and cognition and studied the role of these interactions in generating Embodied Cognition (Merleau Ponty, 2002; Gomez Paloma, 2016). This theory centralizes the role of the body in cognitive implementation. And thus considers knowledge and cognition as active processes rooted within the body and its biological dimensions. This reformulation of the body as a protagonist in knowledge conceptualization and as a communication tool with oneself, others, and the environment significantly influences the neuroscience of cognition. This argument explains that every form of knowledge and cognition is "embodied" and passes necessarily through bodily experiences (Gola et al., 2023).

This review examines the recent available theoretical and experimental studies performed in identifying the role of embodiment in classroom learning. Then we will describe the advances in cognitive neuroscience research explaining the role of brain-body interaction and its role in learning.

Theoretical concepts of embodiment

The reference framework also includes recent educational contributions from neuroscience considering the body as an integral part of learning that can not be separated from the brain (Gomez Paloma, 2016). Through this concept, it is precisely within the body that there is an interrupted activity of exchange, information processing, and storage. Neuroscientific studies (Fisher, 2007; Bruer, 2016) are revisiting the theories and methodologies that support teaching and study design where education can transfer this interaction between emotions, mind, and body. New educational strategies are centering the body and consider movement as an active element for cognitive processes (Sibilio, 2017). This embodied cognition perspective demonstrates that cognition is grounded in bodily interactions with the environment and culture and that abstract concepts are tied to the body's sensory and motor systems (Leung et al., 2011).

The principles of embodiment are methodologically opposite to Cartesian visions in the way that the brain is not the center of cognitive processing but is however part of a more complex machinery including the body and the external environment. Not only that, the body has a primary role in integrating and generating cognitive processes. It is, therefore, an interactive "tool" that allows observation, analysis, and collection of external information through sensorimotor networks and can serve for cognitive and behavioral adaptation through communication with external cues (Palmiero, 2018).

The body is a tool that serves for integrating cognition to experience a subject during learning. In this context, the environment is the place where transformation takes place: meanings are generated based on external elements and representations; the body interacts with its surroundings through sensorimotor networks (Bersalou, 2008; Alibabi et al., 2012). From this, ideas emerge, and cognitive processes are generated within very complex neurofunctional networks and paradigms. For example, bio-psycho-social cues and neurodiversity rely on holistic approaches, whose dimensions require an interaction between the living system and its surroundings, linking its biological, psychological, and social dimensions. The different theoretical approaches that have investigated the relationship between body cognition and movement from a plurality of perspectives have recognized the centrality of the body-kinesthetic dimension in the mechanisms of construction of knowledge, creating the conditions for a reconsideration of the potential learning, expressive, and communicative of the body in the didactic action. Specifically, the results of research in the field of Embodied Cognitive Science (ECS) have enriched the teaching of further reflections on vicarious processes and embodied simulation (Gallese et al., 1998; Rizzolatti et al., 2007) that can contribute to a better understanding of the complexity of the individual and, in a broad sense, of the actors involved in the teaching-learning process (Sibilio et al., 2022).

Experimental studies about embodiment in classroom learning

The neuroscience research (Francesconi et al., 2012; Ceciliani, 2018) has intensified in the most recent years, especially those concerning the correlation between brain, cognitive, and motor activities, which is more evident in younger subjects. The development of cognitive processes and representations influences the baby's movements in his first moments of life. Indeed, the kinaesthetic-bodily intelligence is manifested in a child through exploratory actions and the ability to reason in a certain situation, and it represents the first expression of learning to learn (life skill), a transversal competence expected by the European community in the training of citizens. This intelligence, among other things, is now supported by neuroscience, embodied cognition theories, and the consequent impact on embodied education, with an increasing focus on motor involvement in the development of the mind. Initial research on metacognition, that is, children's awareness of what they know and how they can use it and current research on Executive Functions highlight the importance of physical activity promotion (Bransford., 1999). Sports and motor activities provide the involvement of the totality of the individual by activating connections between consciousness and emotion, between the body and the surrounding environment. As a result, through this balance, the subject can govern his or her movements. As a consequence, cognition controls the emotion, so that the subject can choose which action to perform, making him move from input information to action (D'anna et al., 2023).

Many studies (Montagnoli et al., 2018) highlighted the vital role of the body in the learning process. Research about mathematics education showed that students are confronted with complex situations that hindered their reasoning and problem-solving skills. This increased the daily burden on teachers in managing their learning processes, administrative responsibilities, and their risk of failure. Mathematical knowledge, including oral counting, enumeration, and numerical representation, develops in parallel with motor skills. These latter seem to influence comprehension and interfere with the degree of mastery of learning mathematical concepts. Math learning, especially in primary school, should be perceptual-motor centered (Lakoff et al., 2005). Active observation allows the student to contextualize their learning according to the environment and the teacher to constantly grasp the different aspects and nuances of the various situations within the classroom. Another study showed positive learning outcomes by combining juggling exercises with multiplication tables (Van den Berg et al., 2019). Other studies have shown the role of gestures in learning mathematical concepts (Alibali et al., 2012). This beneficial effect of gestures' use in learning has been also demonstrated in other domains, like science, and foreign language education. Romano et al. (2000) have shown that playing and playful dimensions increase children's motivation, and attention, and improve their strategies of free discovery during learning. Literature also demonstrated that motor activity outside school was linked to better learning outcomes and that increasing the number of hours of physical education in school settings improved academic performance (Latino et al., 2020). Regardless of the effects of both modes of physical activity on cognition and academic achievement, it is widely recognized that they contribute to the healthy and better acquisition of motor skills in children.

Another classroom study (Have et al., 2018) implemented physical activity in the classroom as a facilitating tool for teaching mathematics, while analyzing the impact of this intervention on body mass index, aerobic fitness, and the level of physical activity. This research showed an improvement in academic performance in math after the integration of physical activity. In addition, this strategy helped maintain and consolidate a healthy lifestyle in students. Math symbolic writing difficulties and theoretical concept creation in children were facilitated by using gestures. Communication through touching objects, pointing towards them, and using fingers for counting were some of the practical gesture-based tools in body-centered learning approaches. This phenomenon shows how ideas are generated progressively through physical action. After recognizing external objects, children learn to connect the idea of the numbers of similar objects with the quantity without touching them and thus learn to simultaneously count "one, two, three, etc".

This body-environment interaction seems critical in learning (Soto-Johnson, 2018). There is a linear neural correlation between sensorimotor physical operations and arithmetic operations (Radford, 2011). In addition, an embedded spatial education can improve fundamental cognitive abilities, and spatial thinking, which are necessary for learning mathematics. Fostering the development of such skills from primary school seems also to contribute to a knock-on effect, increasing student interest and success in science-technology disciplines throughout education (Burte et al., 2017).

Peppler (2017) has recently described how counting originated in the real world: We count sheep by observing them. If our eyes are shut or the sheep are not where we are, we retrieve a concrete image of the animals to count them. We don't think of abstract living beings. Instead, we visualize sheep and not dogs the way we have experienced them. Furthermore, during counting, we use the body to support the cognitive task by counting on our fingers. We explain math basic operations (addition, subtraction, division, and multiplication) to children by putting real-world things together, taking them apart from each other, cutting them, and so on. All this, we do use our bodies. Even illiterate persons who cannot read or write numbers can perform the operations by referencing the task to real-world objects.

In two experiments by Crollen and Noël (2015), 5 to 9-year-old children had to accomplish one-target, two-target counting tasks, and additions. While doing this, participants had no constraints, interfering hand movements, and interfering foot movements to deal with. Hand movements disrupted counting more than foot movements. These results suggest a connection between our fingers and counting. Considering that in childhood the acquisition of numerical skills is tightly connected with finger counting, these results do not surprise. In an fMRI study, Tschentscher et al. (2012) presented adults' digits from 1 to 9 only visually while they lay quietly in the scanner. Depending on the participants' counting habits (left- or right-starters), hemodynamic activity in the contralateral motor cortex was observed when the numbers were presented. This neuroscientific evidence supports the view of embodied mathematical knowledge. The connection between motor activity in the brain and counting with fingers is connected to Hebbian learning mechanisms. They apply during learning and connect cognitive mathematical operations with finger movements performed while counting (Sato et al., 2008).

Advances in cognitive neuroscience research explaining the role of brain-body interaction and its role in learning

In recent years, psychological and neuroscientific research (Fischer et al.,2007) has shown that cognitive functions depended on action and perception and thus were experience-based. The body and its movement can through senses like vision, touch, and hearing internalize conscious concepts and representations of the body and identity; these can also serve in learning, intelligent behavior, and stimulating memory and emotional intelligence (Fischer et al.,2007). From this perspective, embodied cognitive research can have important implications for education because it highlights an approach to learning that passes through whole-body engagement (Gallagher et al., 2015).

Research (Dini, 2022) on learning through playing shows that sensorimotor stimulation is essential in promoting learning. Motricity in children contributes to learning through instinctive and innate representations that they tend to have with the world and their surroundings. These representations are incorporated by their senses and the concepts change depending on the constantly variable environmental motion-based elements. This can explain how representations of numbers, letters, and geometric shapes can occur by using the body and various environmental instruments, such as beams, balls, clubs, and musical tools. And how students are more likely able to solve arithmetic tasks with body-centered teaching approaches (Magistro et al., 2022; Lovecchio, 2022). Embedded learning is configured as an effective teaching mode that can involve all participants: embodied cognition as a "modus" can help build students' knowledge and shape their educational success (Paloma et al., 2016).

The central role of experience, the importance of the body acting in mathematical thought, and the multiple possibilities and modes related to experience are three key aspects that have brought the individual, and the researchers, closer to embodied mathematical cognition (Hall et al.,2012). There is little research on how physical activity interferes with learning. Some studies (Alibabi et al.,2012; Cook et al., 2016) associated math and geometry education with physical gameful activity and concluded that this combination resulted in better geometric acquisition skills of right angles, rectangles, and squares. This observation demonstrates a synergy between the cognitive and motor components within the nervous system in modulation and shaping behavior. Learning is a holistic process and is not just limited to memorization and thinking. Embodiment in learning depends on coordinating, cognitive, emotional, perceptual, and motor skills to guide and enrich our interaction with the environment and society (Pesce et al., 2019). Applying body-centered teaching methods can facilitate the acquisition of cognitive tools students may require in educational subjects based on conceptualization and critical thinking (Hraste et al., 2018).

The observations in this study highlight the importance of early implementation of motor activity education and shaping school curricula to use body-centered approaches (Donnelly, 2016). Recent research showed a positive correlation between mathematical knowledge and motor functions in children aged between 3 and 10 years old; this association was justified in previous studies by the theory of decision-making (Iannello et al., 2007). This theory implies that the execution of motor skills requires ahead-of-action and analysis-oriented decision-making reasoning. A statistical analysis of learning mathematical concepts and motor skills showed positive outcomes embedded in teaching on development (Rio et al., 2015), highlighting again that cognitive processes are deeply rooted in the interaction of the body with the world.

Future directions

In the scientific panorama, specifically educational neuroscience, are an emerging area bringing together researchers in cognitive and developmental neuroscience, education psychology, education theory, and other related disciplines to explore the interactions between biological and educational processes. This is a recognized field of research that makes transdisciplinarity its specific episteme, which tends to combine the principles of research evidence and recognized scientific rigor, with ecological paradigms such as classroom situations in which students and teachers act (Gola et al.,2023).

Motor activity to bring health benefits and development of executive functions such as attention, perception, decision-making, and problem-solving must be regularly practiced given the close link between body activity and cognitive processes (Paloma, 2013). t is necessary to adopt, during the hours of Physical Education, teaching that involves a dual approach, qualitative and quantitative, engaging children in recreational contexts with activities of moderate intensity and vigorous. The qualitative aspect favors, in the child, the acquisition of active lifestyles and healthy eating habits that allow for to prevention of numerous health problems while, the quantitative, consolidates the idea that physical activity is fundamental in everyday life (Ceciliani, 2018).

Osgood-Campbell (2015) states that future research should show evidence of the link between sensorimotor action and cognition in classroom activities and, specifically, should examine the improvement of specific academic skills such as language comprehension, mathematics, and scientific thinking. More so, future work should investigate the impact of sensory-motor abilities on language acquisition and comprehension (Willems et al., 2012). Studies should focus on teaching and learning investigating how designers can build new understandings of embodied mathematical cognition in learning environments (Hall et al., 2012).

The practical didactic applications of the embodied cognition theory, applied to motor activity and movement, can influence positively the development of cognitive functions above all through aerobic exercise with high coordination and mental commitment (Best, 2010). This approach should be used both in school and out of school it can be an excellent methodology to bring children closer to sport through cognitive involvement and motivation (Tosi, 2022).

Conclusion

In this paper, it is highlighted how learning requires a dynamic interaction of motor and cognitive skills. The efficacy of learning should not only focus on cognitive functions but also the role of experience and body involvement and its potentiality. Neuroscience research on the developing brain shows that physical activity and interaction with the environment are linked to better neuroplasticity and that experiences, and participatory learning leads to better learning outcomes. It is necessary to overcome the current idea that closed classroom school designs are necessary for promoting efficient learning, since they often exclude social and body-centered environmental interactions. Effective learning requires involving the student in doing by generating dual mental and physical stimulation. New educational cultures setting the student as a protagonist of his knowledge and encouraging his/her active participation in educational curricula seems necessary. The theory of embedded learning presented in this paper can be translated into educational research by combining theoretical knowledge with motor skills. Starting from experience to achieve skills and knowledge favors body-centered learning investigations. And can also help decipher the individual differences in interaction with the environment in generating cognition and thinking processes. The fundamental innovative aspect is the role of action in learning, understood not only as a simple execution of a movement but as an articulated interaction with the surrounding world capable of capturing and collecting as much information as possible.

Future research can focus lay the foundations for the construction of touchscreen tablets, and related applications, able to create opportunities for students to solve problems of motor action designed specifically to give rise to targeted protocols that in turn can help the reflective side. In short, touchscreen applications have the potential to be meeting places for action, perception, and cognition (Duijzer et al., 2017). Currently, there is no defined protocol to be able to implement an embodied and effective teaching about learning mathematical concepts, however, it is clear that it is necessary to become aware of the role that the body plays in learning. In general, these studies concluded that physically active learning improved results in mathematics. The applications of Embodied Cognition in this field of teaching allow this theory to be taken as a theoretical lens, since it allows the use of gestures as a demonstration of evidence, as, through the latter, students learn linear algebra, differential equations, complex variables, and more. The activation of what is involved in embodied knowledge is in itself indicative of a high, in some way conscious, elaboration of what is happening in the surrounding environment (Tosi, 2022).

Although there is little scientific evidence, the picture outlined highlights a strong interest on the part of educational research for the experimentation of protocols and teaching models that can amplify learning in terms of generalization and memorization.

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Revised September 2019

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When preparing the final version of the manuscripts, either NEW or REVISED authors should strictly follow the guidelines. Manuscripts departing substantially from the guidelines will be returned to the authors for revision or, rejected.

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SM honors six-weeks for an initial decision of manuscript submission.

Authors should submit the manuscripts as one Microsoft Word (.doc) file.

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Number (Arabic numerals) the pages consecutively (centering at the bottom of each page), beginning with the title page as page 1 and ending with the Figure legend page.

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Body Composition of Elite Soccer Players from Montenegro

Original Scientific Paper

Elite Soccer Players from Montenegro

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E-mail: sportmont@t-com.me

Word count: 2,946

Abstract word count: 236

Number of Tables: 3

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One work by one author
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Nepocatych, S., Balilionis, G., & O'Neal, E. K. (2017). Analysis of dietary intake and body composition of female athletes over a competitive season. *Montenegrin Journal of Sports Science and Medicine, 6*(2), 57-65. doi: 10.26773/mjssm.2017.09.008

Duffield, R., & Marino, F. E. (2007). Effects of pre-cooling procedures on intermittent-sprint exercise performance in warm conditions. *European Journal of Applied Physiology, 100*(6), 727-735. doi: 10.1007/s00421-007-0468-x

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Journal article (online; electronic version of print source):

Williams, R. (2016). Krishna's Neglected Responsibilities: Religious devotion and social critique in eighteenth-century North India [Electronic version]. *Modern Asian Studies, 50*(5), 1403-1440. doi:10.1017/S0026749X14000444

Journal article (online; electronic only):

Chantavanich, S. (2003, October). Recent research on human trafficking. *Kyoto Review of Southeast Asia, 4*. Retrieved November 15, 2005, from http://kyotoreview.cseas.kyoto-u.ac.jp/issue/issue3/index.html

Conference paper:

Pasadilla, G. O., & Milo, M. (2005, June 27). *Effect of liberalization on banking competition*. Paper presented at the conference on Policies to Strengthen Productivity in the Philippines, Manila, Philippines. Retrieved August 23, 2006, from http://siteresources.worldbank.org/INTPHILIPPINES/Resources/Pasadilla.pdf

Encyclopedia entry (print, with author):

Pittau, J. (1983). Meiji constitution. In *Kodansha encyclopedia of Japan* (Vol. 2, pp. 1-3). Tokyo: Kodansha.

Encyclopedia entry (online, no author):

Ethnology. (2005, July). In *The Columbia encyclopedia* (6th ed.). New York: Columbia University Press. Retrieved November 21, 2005, from http://www.bartleby.com/65/et/ethnolog.html

Thesis and dissertation:

Pyun, D. Y. (2006). *The proposed model of attitude toward advertising through sport.* Unpublished Doctoral Dissertation. Tallahassee, FL: The Florida State University.

Book:

Borg, G. (1998). *Borg's perceived exertion and pain scales*: Human kinetics.

Chapter of a book:

Kellmann, M. (2012). Chapter 31-Overtraining and recovery: Chapter taken from Routledge Handbook of Applied Sport Psychology ISBN: 978-0-203-85104-3 *Routledge Online Studies on the Olympic and Paralympic Games* (Vol. 1, pp. 292-302).

Reference to an internet source:

Agency. (2007). Water for Health: Hydration Best Practice Toolkit for Hospitals and Healthcare. Retrieved 10/29, 2013, from www.rcn.org.uk/newsevents/hydration

2.5. Tables

All tables should be included in the main manuscript file, each on a separate page right after the Reference section.

Tables should be presented as standard MS Word tables.

Number (Arabic) tables consecutively in the order of their first citation in the text.

Tables and table headings should be completely intelligible without reference to the text. Give each column a short or abbreviated heading. Authors should place explanatory matter in footnotes, not in the heading. All abbreviations appearing in a table and not considered standard must be explained in a footnote of that table. Avoid any shading or coloring in your tables and be sure that each table is cited in the text.

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2.5.1. Table heading

Table heading should be written above the table, in Title Case, and without a full stop at the end of the heading. Do not use suffix letters (e.g., Table 1a, 1b, 1c); instead, combine the related tables. *See* example:

- ✓ **Table 1.** Repeated Sprint Time Following Ingestion of Carbohydrate-Electrolyte Beverage

2.5.2. Table sub-heading

All text appearing in tables should be written beginning only with first letter of the first word in all capitals, i.e., all words for variable names, column headings etc. in tables should start with the first letter in all capitals. Avoid any formatting (e.g., bold, italic, underline) in tables.

2.5.3. Table footnotes

Table footnotes should be written below the table.

General notes explain, qualify or provide information about the table as a whole. Put explanations of abbreviations, symbols, etc. here. General notes are designated by the word *Note* (italicized) followed by a period.

- ✓ *Note.* CI: confidence interval; Con: control group; CE: carbohydrate-electrolyte group.

Specific notes explain, qualify or provide information about a particular column, row, or individual entry. To indicate specific notes, use superscript lowercase letters (e.g. [a, b, c]), and order the superscripts from left to right, top to bottom. Each table's first footnote must be the superscript [a].

- ✓ [a]One participant was diagnosed with heat illness and n = 19.[b]n =20.

Probability notes provide the reader with the results of the texts for statistical significance. Probability notes must be indicated with consecutive use of the following symbols: * † ‡ § ¶ || etc.

- ✓ *P<0.05,†p<0.01.

2.5.4. Table citation

In the text, tables should be cited as full words. *See* example:

- ✓ Table 1 (first letter in all capitals and no full stop)
- ✓ ...as shown in Tables 1 and 3. (citing more tables at once)
- ✓ ...result has shown (Tables 1-3) that... (citing more tables at once)
- ✓ ....in our results (Tables 1, 2 and 5)... (citing more tables at once)

2.6. Figures

On the last separate page of the main manuscript file, authors should place the legends of all the figures submitted separately.

All graphic materials should be of sufficient quality for print with a minimum resolution of 600 dpi. SM prefers TIFF, EPS and PNG formats.

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Figures and figure legends should be completely intelligible without reference to the text.

The price of printing in color is 50 EUR per page as printed in an issue of SM.

2.6.1. Figure legends

Figures should not contain footnotes. All information, including explanations of abbreviations must be present in figure legends. Figure legends should be written bellow the figure, in sentence case. *See* example:
- ✓ **Figure 1.** Changes in accuracy of instep football kick measured before and after fatigued. SR – resting state, SF – state of fatigue, *p>0.01, †p>0.05.

2.6.2. Figure citation

All graphic materials should be referred to as Figures in the text. Figures are cited in the text as full words. *See* example:
- ✓ Figure 1
  - × figure 1
  - × Figure 1.
  - ✓ ….exhibit greater variance than the year before (Figure 2). Therefore…
  - ✓ ….as shown in Figures 1 and 3. (citing more figures at once)
  - ✓ ….result has shown (Figures 1-3) that... (citing more figures at once)
  - ✓ ….in our results (Figures 1, 2 and 5)... (citing more figures at once)

2.6.3. Sub-figures

If there is a figure divided in several sub-figures, each sub-figure should be marked with a small letter, starting with a, b, c etc. The letter should be marked for each subfigure in a logical and consistent way. *See* example:
- ✓ Figure 1a
- ✓ …in Figures 1a and b we can…
- ✓ …data represent (Figures 1a-d)…

2.7. Scientific Terminology

All units of measures should conform to the International System of Units (SI).

Measurements of length, height, weight, and volume should be reported in metric units (meter, kilogram, or liter) or their decimal multiples.

Decimal places in English language are separated with a full stop and not with a comma. Thousands are separated with a comma.

Percentage	Degrees	All other units of measure	Ratios	Decimal numbers
✓ 10%	✓ 10º	✓ 10 kg	✓ 12:2	✓ 0.056
× 10 %	× 10 º	× 10kg	× 12 : 2	× .056

Signs should be placed immediately preceding the relevant number.		
✓ 45±3.4	✓ p<0.01	✓ males >30 years of age
× 45 ± 3.4	× p < 0.01	× males > 30 years of age

2.8. Latin Names

Latin names of species, families etc. should be written in italics (even in titles). If you mention Latin names in your abstract they should be written in non-italic since the rest of the text in abstract is in italic. The first time the name of a species appears in the text both genus and species must be present; later on in the text it is possible to use genus abbreviations. *See* example:

✓ First time appearing: *musculus biceps brachii*

Abbreviated: *m. biceps brachii*

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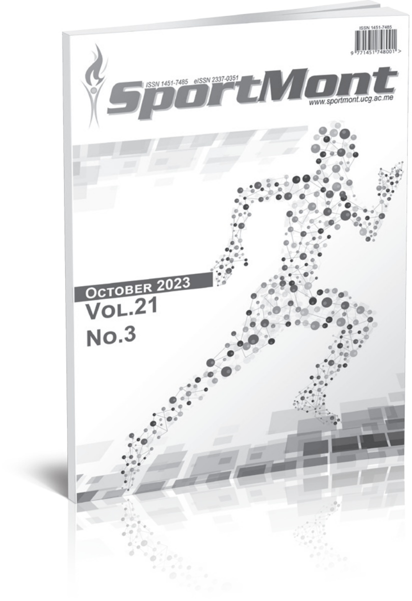

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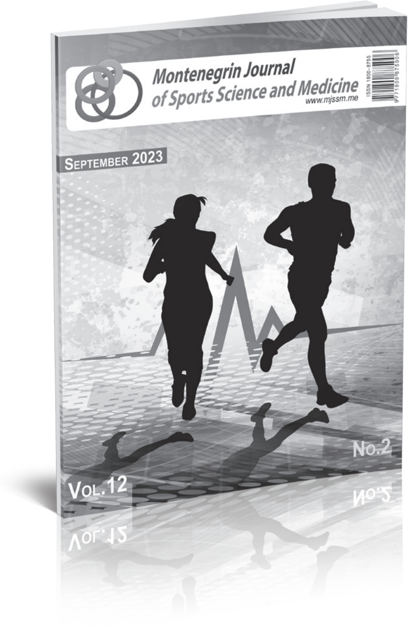

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MJSSM is published biannually, in September and March of each year. MJSSM publishes original scientific papers, review papers, editorials, short reports, peer review - fair review, as well as invited papers and award papers in the fields of Sports Science and Medicine, as well as it can function as an open discussion forum on significant issues of current interest.

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MONTENEGRIN SPORTS ACADEMY

Founded in 2003 in Podgorica (Montenegro), the Montenegrin Sports Academy (MSA) is a sports scientific society dedicated to the collection, generation and dissemination of scientific knowledge at the Montenegrin level and beyond.

The Montenegrin Sports Academy (MSA) is the leading association of sports scientists at the Montenegrin level, which maintains extensive co-operation with the corresponding associations from abroad. The purpose of the MSA is the promotion of science and research, with special attention to sports science across Montenegro and beyond. Its topics include motivation, attitudes, values and responses, adaptation, performance and health aspects of people engaged in physical activity and the relation of physical activity and lifestyle to health, prevention and aging. These topics are investigated on an interdisciplinary basis and they bring together scientists from all areas of sports science, such as adapted physical activity, biochemistry, biomechanics, chronic disease and exercise, coaching and performance, doping, education, engineering and technology, environmental physiology, ethics, exercise and health, exercise, lifestyle and fitness, gender in sports, growth and development, human performance and aging, management and sports law, molecular biology and genetics, motor control and learning, muscle mechanics and neuromuscular control, muscle metabolism and hemodynamics, nutrition and exercise, overtraining, physiology, physiotherapy, rehabilitation, sports history, sports medicine, sports pedagogy, sports philosophy, sports psychology, sports sociology, training and testing.

The MSA is a non-profit organization. It supports Montenegrin institutions, such as the Ministry of Education and Sports, the Ministry of Science and the Montenegrin Olympic Committee, by offering scientific advice and assistance for carrying out coordinated national and European research projects defined by these bodies. In addition, the MSA serves as the most important Montenegrin and regional network of sports scientists from all relevant subdisciplines.

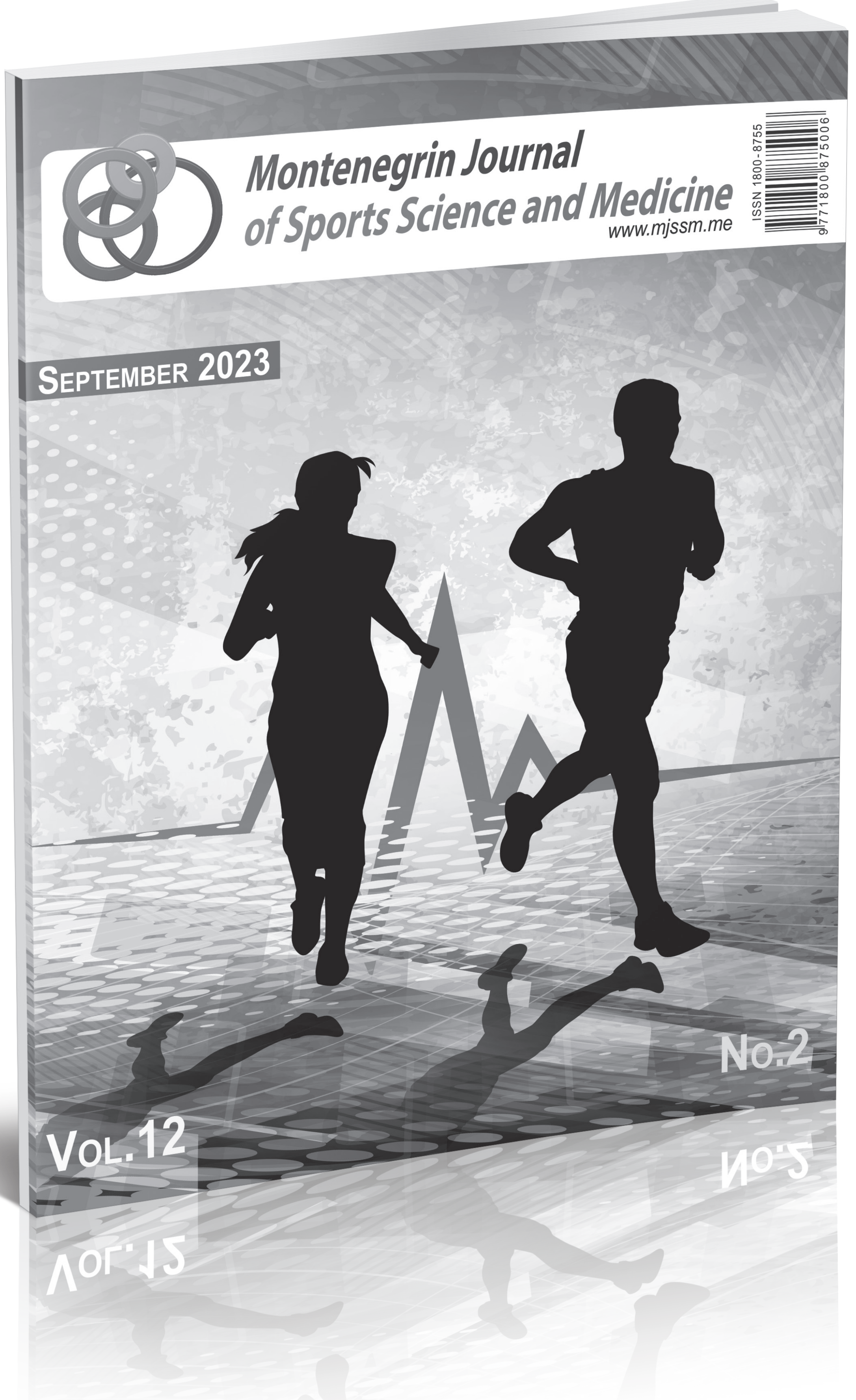
The main scientific event organized by the Montenegrin Sports Academy (MSA) is the annual conference held in the first week of April.

Annual conferences have been organized since the inauguration of the MSA in 2003. Today the MSA conference ranks among the leading sports scientific congresses in the Western Balkans. The conference comprises a range of invited lecturers, oral and poster presentations from multi- and mono-disciplinary areas, as well as various types of workshops. The MSA conference is attended by national, regional and international sports scientists with academic careers. The MSA conference now welcomes up to 200 participants from all over the world.

It is our great pleasure to announce the upcoming 21th Annual Scientific Conference of Montenegrin Sports Academy “Sport, Physical Activity and Health: Contemporary Perspectives” to be held in Dubrovnik, Croatia, from 18 to 21 April, 2024. It is planned to be once again organized by the Montenegrin Sports Academy, in cooperation with the Faculty of Sport and Physical Education, University of Montenegro and other international partner institutions (specified in the partner section).

The conference is focused on very current topics from all areas of sports science and sports medicine including physiology and sports medicine, social sciences and humanities, biomechanics and neuromuscular (see Abstract Submission page for more information).

We do believe that the topics offered to our conference participants will serve as a useful forum for the presentation of the latest research, as well as both for the theoretical and applied insight into the field of sports science and sports medicine disciplines.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